

SOMATOM Definition

Replacement of Parts Gantry

Instructions for replacement or installation of parts

Document Version

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1.1 Safety information

Only qualified and system-trained service staff is allowed to perform CT system installation, service, maintenance and quality assurance. Ensure that the most recent version of the service documentation is available.

To avoid any risk of injury to persons and/or damage to the system read and observe the **General Safety Notes** (TD00-000.860.01.xx.xx). Please read and observe the **Product-specific safety notes** (C2-028.860.01.xx.xx) which include very important safety-related information as well as information about the handling of screws and nuts, use of Loctite and instructions for torque wrenches. You will find the following information in the "Product-specific safety notes."

- Safety information related to the method of risk management (→ General safety warnings / C2-028.860.01).
- General safety information such as the handling of service documentation, CT system training request, X-ray protection, electrical protection, service tools... (→ General safety information / C2-028.860.01).
- General safety information about working in the gantry with the power off/on (→ Working in the gantry with gantry power off / C2-028.860.01).
- Use of the patient table as lifting device for gantry part replacement (→ Usage of patient table as lifting device / C2-028.860.01).
- Laser products (→ Laser Products / C2-028.860.01).
- Attachment of screws and nuts, use of Loctite, adjustment instructions for the torque wrench (→ Screw Connections / C2-028.860.01).

1.1.1 Safety warnings

The following safety information is an extract of the information described in the **Product-specific safety notes** (C2-028.860.01.xx.xx) (→ Safety information / Page 16). It is mandatory to read and observe the safety information described below. The information is valid for all replacements described in these instructions.



CAUTION

[hz_serdoc_F13G07U01M01]

Handling parts of the system that may have come into contact with patients may lead to infection caused by blood-borne pathogens.

Infection caused by blood-borne pathogens!

- » Take appropriate precautions against exposure to blood-borne pathogens (e.g. wear gloves).

**WARNING**

[hz_serdoc_F13G01U13M02]

High voltages and/or mechanical movements (e.g. gantry rotation) may lead to accidents and injuries when opening system covers.

Risk of accident and injury!

- » Only authorized service personnel is allowed to open system covers.

**WARNING**

[hz_serdoc_F13G01U05M03]

Dangerous voltages are present when the system is switched on. Dangerous voltages may be present even when the system is switched off due to capacity power.

Risk of electric shock!

- » Observe power-off instructions and discharge wait times (allow at least five minutes discharge time after the last scan for all involved HV and UDC/UPS parts).
- » Secure the system against unintended switch-on (e.g. block breaker and/or mark breaker against switch on).
- » Ensure via measurements that all voltages are switched off.
- » Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry.

**WARNING**

[hz_serdoc_F13G05U01M03]

Loose parts in the gantry due to incorrect assembly may cause injury and damage during rotation. Failure to observe the guidelines specified in the service documentation may lead to serious injury or loss of life.

Risk of injury and damage!

- » Only calibrated tools are allowed when performing service activities.

**CAUTION**

[hz_serdoc_F13G01U01M08]

Unexpected gantry rotation during replacement of parts.

Risk of accident and injury!

- » Secure the gantry with the "Rotation safety bolt" before replacing any parts in the rotating part of the gantry.
- » Do not remove the "Rotation safety bolt" until all parts are reinstalled and secured.

**WARNING**

[hz_serdoc_F13G05U01M02]

Loose parts in the gantry caused by incorrect assembly may cause injury and damage during rotation. Failure to observe the guidelines specified in the service documentation may lead to serious injury or loss of life.

Risk of injury and damage!

- » The guidelines in the service documentation must be strictly followed.

**WARNING**

[hz_serdoc_F13G05U01M04]

Unfixed parts/cables during gantry rotation.

Risk of accident and injury!

- » Make sure that all parts and cables are mounted properly before starting rotation.

**CAUTION**

[hz_serdoc_F13G02U01M04]

Flammable sprays.

Risk of inflammation!

- » Only use approved and recommended cleaning agents as described.

1.2 Information for switching off the gantry power

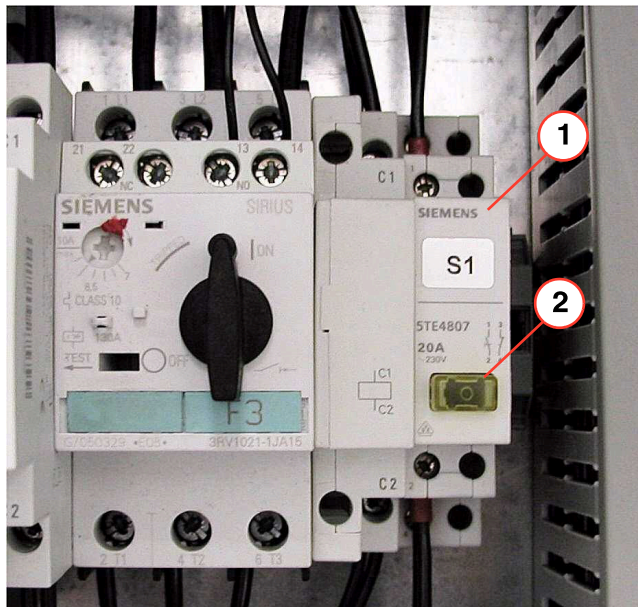
- Switch off system (Click <**SYSTEM**> and select <**END**> in the user main menu to switch off the system automatically after shut down). Additionally, switch off the gantry power using service push button **S1** in the PDC-A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.



Read and observe the safety information in the “Product-specific safety notes” prior to performing service work in the gantry. If the gantry power is switched off for service work, always secure it against accidental switch on.

1.3 Service Push-Button S1 in PDC-A

Fig. 1: Service push-button S1



- (1) Service push-button S1
- (2) Push-button

1.4 Safety bolts

- Front safety bolt -> located at bottom right gantry front

Fig. 2: Rotation safety lock - unlock position

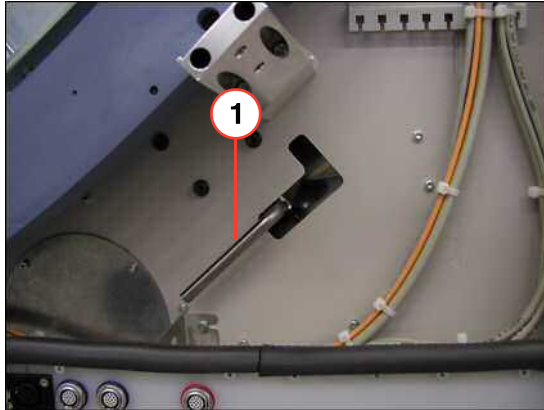
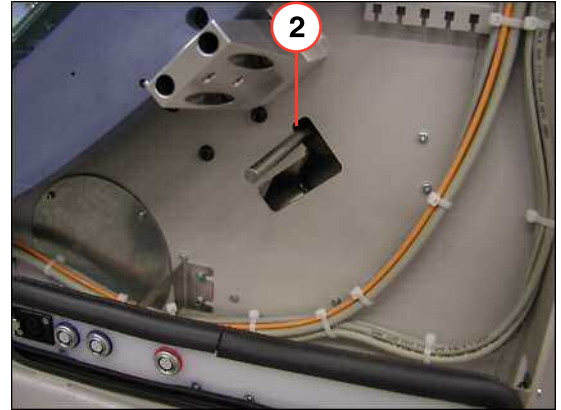
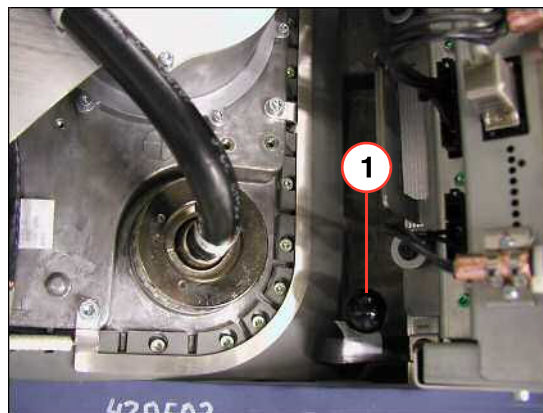


Fig. 3: Rotation safety lock - lock position



- Unlock position of gantry front safety bolt (item 1)
- Lock position of gantry safety bolt (item 2).
- **Rear safety bolt**

Fig. 4: Rotation safety lock - back



- Rear rotation safety bolt (item 1) located between XGR A and HV tank A at the rear of the gantry.
- Lift bolt and turn it 90 degrees to bring it into the lock or unlock position.
 - » **Make sure that the bolt is really at the correct (lock/unlock) position. If the bolt accidentally moves into the lock position during rotation, the gantry may be damaged.**

- Functional principle of rear safety bolt

Fig. 5: Back safety block - unlock position

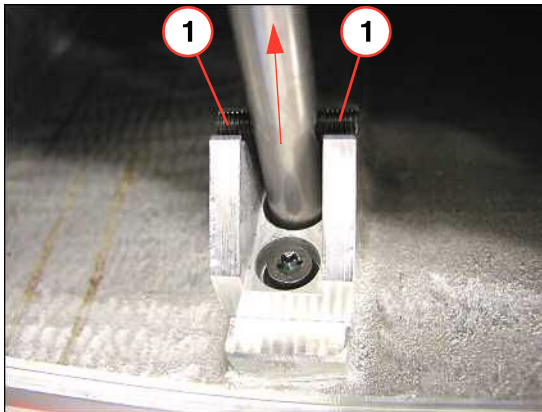
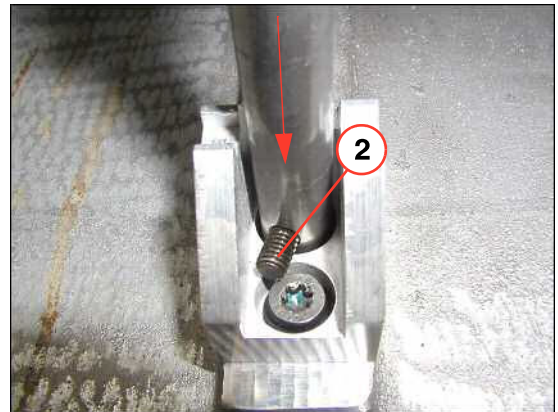


Fig. 6: Back safety block - lock position



- Unlock position of gantry front safety bolt (left image, item 1). From lock position (item 2), lift the bolt in the direction of the arrow and rotate it 90 degrees to bring the bolt into the unlock position (item 1).
- Lock position of gantry safety bolt (right image, item 2). From unlock position (item 1), lift the safety bolt and rotate it 90 degrees to bring it into the lock position (item 2).

1.5 Data ring protection

- Installing the data ring protection

Fig. 7: Data ring protection



- Install the 2 data ring protections (e.g. item 1) before working in the gantry front
 - 2 Plexiglass protections are necessary to cover a larger area on the data ring
 - Attach them to the stationary data ring
 - **Do not forget to remove the protections before you start the gantry rotation!**

1.6 Lifting support

- Installing the lifting support



If there is an MPT table for your system, make sure that you install the standard table top first. The lifting support only fits to the standard table top.

Fig. 8: Lifting device

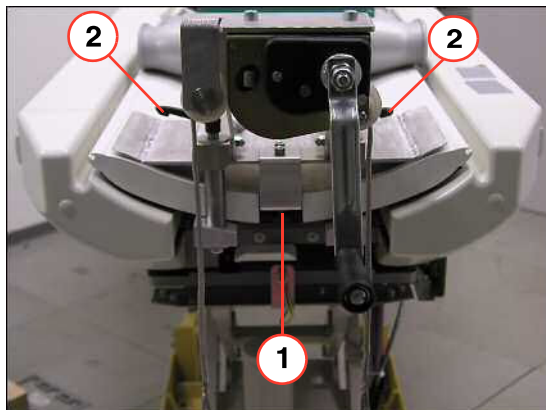
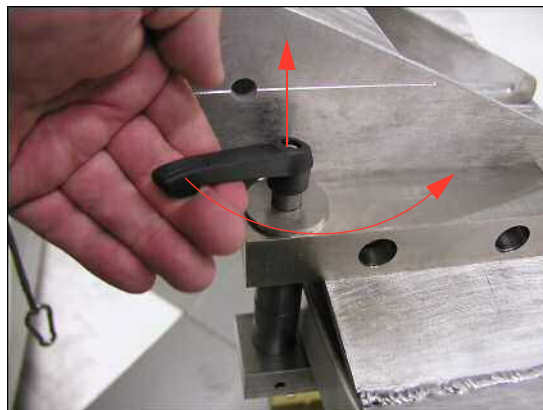


Fig. 9: Lifting device



- Slide the lifting support into the receptacle at the head end of the patient table (item 1) until it is firmly engaged.
- Secure the lifting support to the tabletop using the 2 handles (item 2). Lift and turn the handle (right image, arrows) until the support is tightened to or released from the tabletop.

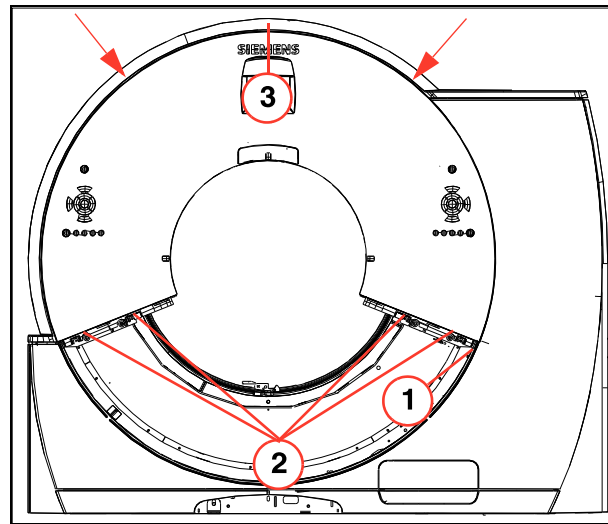
1.7 Opening/closing the top front ring segment cover



Two types of top front ring segments are available at the systems. A removable segment was installed at the beginning of system delivery. This type will be replaced by a liftable type of front ring segments. The handling of both types is described next.

- Prepare top front ring segment for removal/lifting.

Fig. 10: Front top ring segment



- Disconnect all cable connections at right gantry front (item 1).
- Remove the 4*M13 nuts (item 2).
- Loosen the 2 securing locks (arrows, 1/4 turn) using a 5 mm hexagonal wrench key.
- Pull top of ring segment out from its positioning clamps (item 3).
- **Only removable front top ring segment:** Remove the top front ring segment with two persons.
 - » Reinstall removable ring segment and attach it to the gantry by reversing the order of removal

- Lift the top front ring segment to service position

Fig. 11: Lifting/closing the front segment

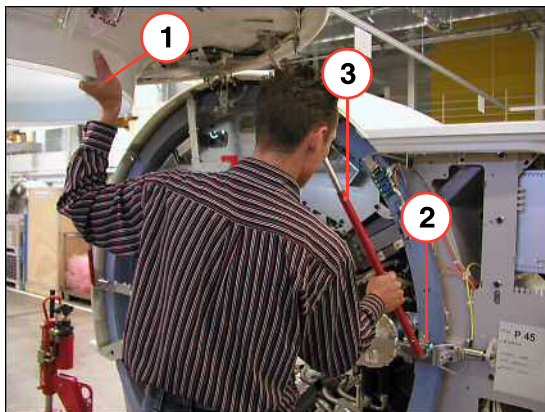
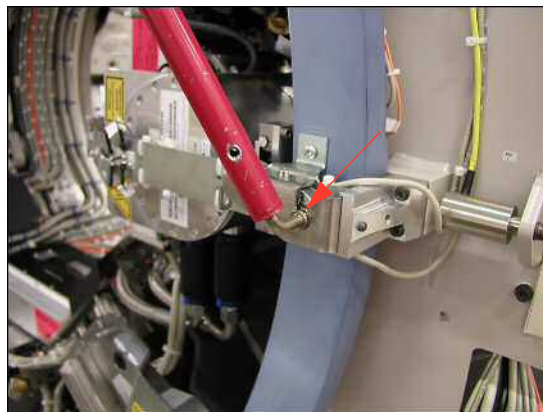


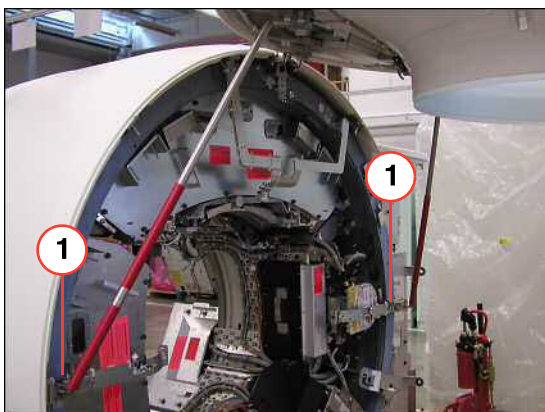
Fig. 12: Supporting stand lock



1. Lift top segment with one hand (item 1).
2. Remove supporting stand from its parking position and extend it until it is blocked (item 3).
3. Secure the supporting stand to the lock (item 2 / right image, arrow).

» Perform worksteps 2 and 3 for the left and the right supporting stand

Fig. 13: Lifted front segment secured by 2 supporting stands



- Close the top front ring segment from service position

Fig. 14: Lifting/closing the front segment

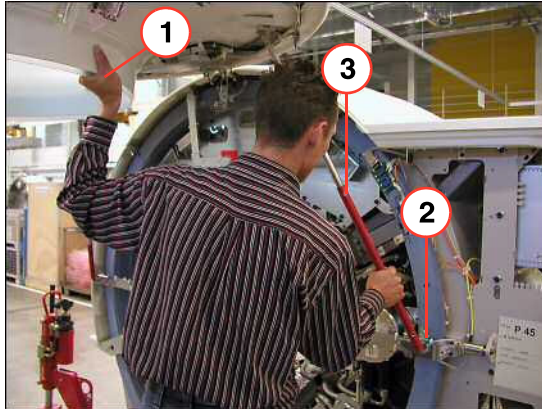
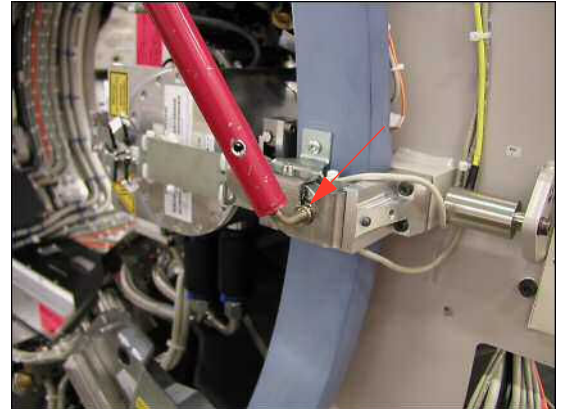
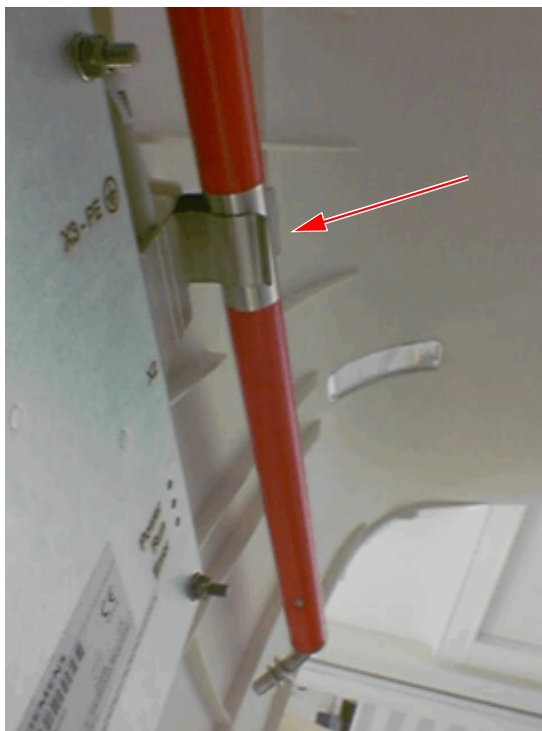


Fig. 15: Supporting stand lock



1. Lift top segment with one hand (item 1).
2. Release the supporting stand from its lock (item 2 / right image, arrow).
3. Minimize the supporting stand by pressing the lock (item 3) and bring it to its parking position at the cover.

Fig. 16: Supporting stand in park position



- » Perform worksteps 2 and 3 for the left and the right supporting stand
 - » Not properly parked supporting stands may led to severe damage of components inside the gantry
4. Close cover and attach it to the gantry in the reverse order of removal.

1.8 Tools

Normally, a standard tool kit is required for performing service as described in this document. If special tools are needed, they are described in the relevant replacement description under the "Tools" item.



CAUTION

[hz_serdoc_F13G01U02M01]

Only approved service tools should be used during replacement and troubleshooting.

Risk of injury!

- » **Use only approved tools.**

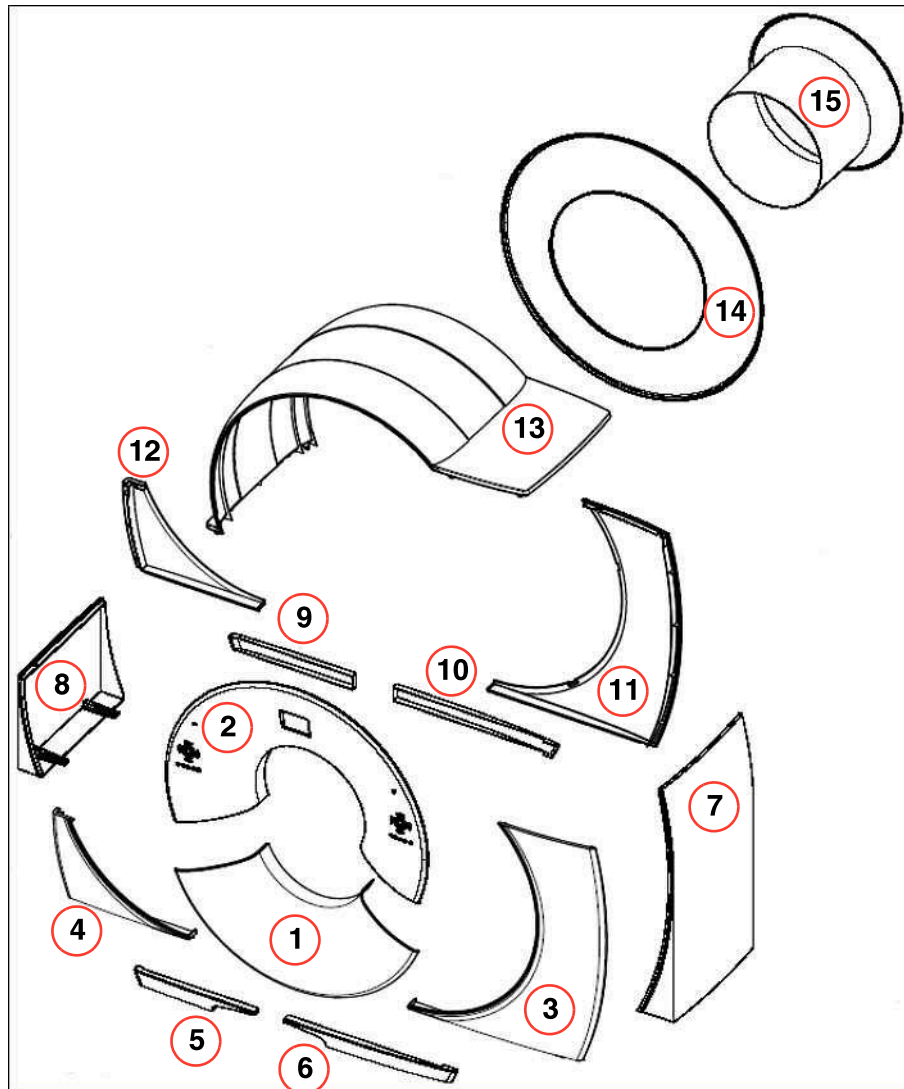


Torque wrenches must be checked for accuracy on a regular basis.

2.1 Overview

The figure below shows the location of the gantry covers.

Fig. 17: Cover, overview



- The gantry covers include the following components:

- » **Line of sight from the front**

- 1- Bottom front ring segment
- 2 -Top front ring segment
- 3 - Right front cover
- 4 - Left front cover
- 5 -Left front baseboard
- 6 -Right front baseboard
- 7 - Right side cover
- 8 - Left side cover
- 9 - Left rear baseboard

- 10 - Right rear baseboard
- 11 - Right rear cover
- 12 - Left rear cover
- 13 - Hood
- 14 - Rear ring cover
- 15 - Funnel

2.2 Bottom front ring segment

2.2.1 Prerequisites

2.2.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.2.1.2 Time and man power

- 1 person
- Estimated total repair time: n.a.

2.2.1.3 Tools and auxiliary equipment

- Standard service kit

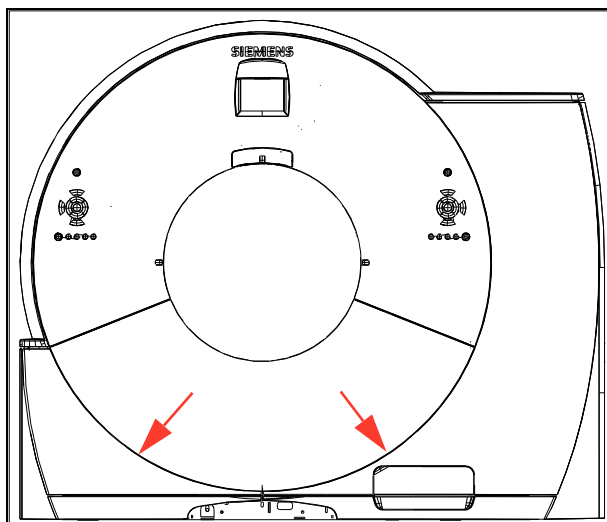
2.2.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.

2.2.3 Removing

1. Remove bottom ring segment.

Fig. 18: Front bottom ring segment



- Loosen the 2 securing locks (arrows, 1/4 turn) using a 5 mm hexagonal key.
- Remove the bottom ring segment by pulling it straight out of the gantry.

2.2.4 Installation

1. Install bottom ring segment in the reverse order of removing it.

2.2.5 Startup

2.2.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.2.5.2 Software

n.a.

2.2.5.3 Tune-up

n.a.

2.2.5.4 Testing

- Rotate gantry at slow speed. Listen for abnormal noise in the gantry.
 - » If there is any unusual noise, remove the cover again and check/eliminate the source of this noise.

2.2.6 Final steps

n.a.

2.3 Top front ring segment (removable)

2.3.1 Prerequisites

2.3.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the “General” section of this document and/or the “Product specific safety notes”.

2.3.1.2 Time and man power

- 1 person
- Estimated total repair time: 0.5 hours

2.3.1.3 Tools and auxiliary equipment

- Standard service kit

2.3.2 Preliminary steps

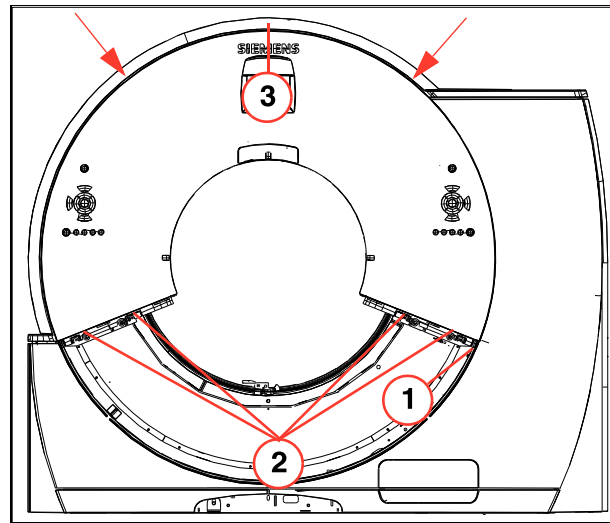
1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove bottom ring segment. Refer to (→ Removing / Page 32) bottom ring segment.

2.3.3 Removing

1. Use strips of tape to secure the plastic ring to the funnel (this step makes it easier re-installing the front ring segments).

2. Remove top ring segment .

Fig. 19: Front top ring segment



- Disconnect all cable connections at right gantry front (item 1).
- Remove the 4*M13 nuts (item 2).
- Loosen the 2 securing locks (arrows, 1/4 turn) using a 5 mm hexagonal wrench key.
- Pull top of ring segment out from its positioning clamps (item 3).
- Remove the top ring segment.
 - » **The plastic ring should stay in its original position because of the tape.**

2.3.4 Installation

1. In case of replacing the top ring segment, perform the following steps before installing it:
 - a) Remove the 2 panels, graphic display and MV45 from the old cover.
 - b) Install 2 panels, graphic display and MV45 at the new cover.
 - c) Insert plugs and install ground connections at their original position.
2. Install top front ring segment in the reverse order of removing it.
3. Install bottom front ring segment in the reverse order of removing it.
4. Remove tape.

2.3.5 Startup

2.3.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.3.5.2 Software

n.a.

2.3.5.3 Tune-up

n.a.

2.3.5.4 Testing

- Rotate gantry at slow speed. Listen for unusual noise in the gantry.
 - » If there is any unusual noise, remove the cover(s) again and check/eliminate the source for noise.
- Test panels, graphic display and communication function.

2.3.6 Final steps

n.a.

2.4 Top front ring segment (liftable)

2.4.1 Prerequisites

2.4.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.4.1.2 Time and manpower

- 1 person
- Estimated total repair time: 0.5 hours

2.4.1.3 Tools and auxiliary Equipment

- Standard service kit

2.4.2 Preliminary Steps

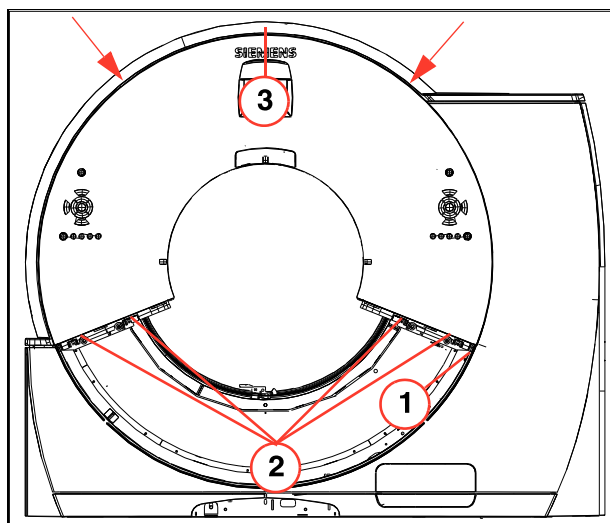
1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove bottom ring segment. Refer to (→ Removing / Page 32) bottom ring segment.

2.4.3 Removing

1. Use strips of tape to secure the plastic ring to the funnel (this step makes it easier re-installing the front ring segments).

2. Prepare the top front ring segment for removal.

Fig. 20: Front top ring segment



- Disconnect all cable connections at right gantry front (item 1).
- Remove the 4*M13 nuts (item 2).
- Loosen the 2 securing locks (arrows, 1/4 turn) using a 5 mm hexagonal wrench key.
- Pull top of ring segment out of its positioning clamps (item 3).

3. Lift the top front ring segment to service position

Fig. 21: Lifting/closing the front segment

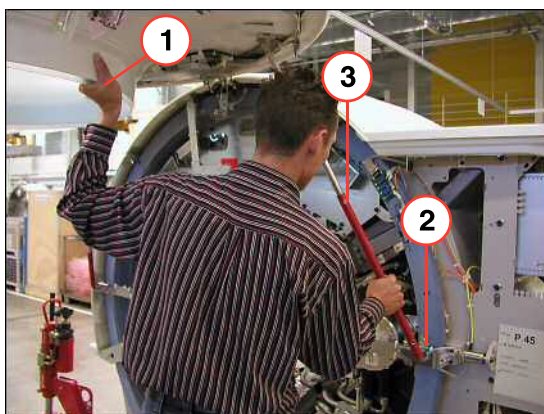
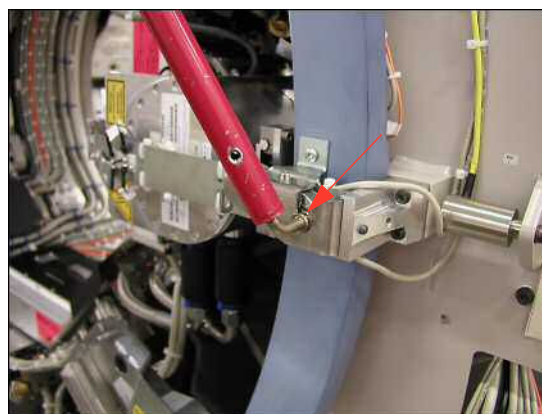


Fig. 22: Supporting stand lock



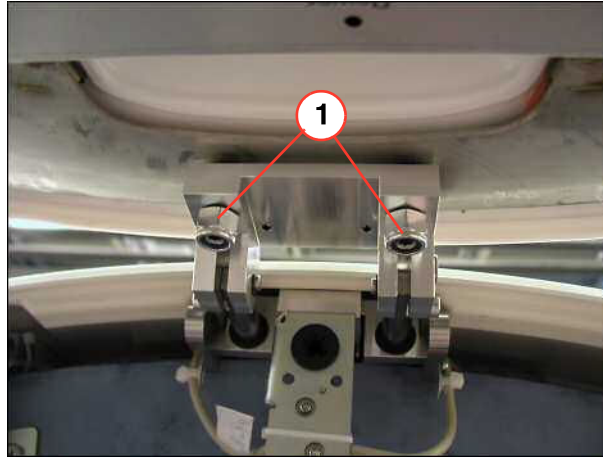
1. Lift top segment with one hand (item 1).
2. Remove supporting stand from its parking position and extend it until it is blocked (item 3).
3. Secure the supporting stand to the lock (right image. arrow).
 - » Perform worksteps 2 and 3 for left and right supporting stand

4. Prepare the gantry front top ring segment for removal



Two persons are required to remove the front top ring segment. One person to hold the cover, and the other person to remove the nuts.

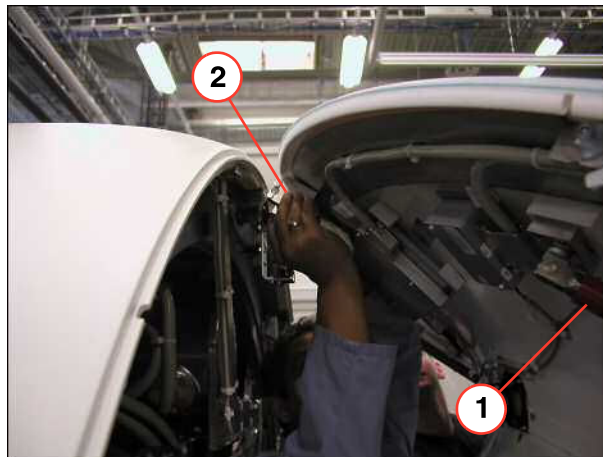
Fig. 23: Holding screws, cover



- Loosen the 2 (M12) nuts by a few turns, until the self-securing (blue) parts of the nuts no longer touch the 12 mm bolts. Do not remove the nuts completely!

5. Remove the gantry front top ring segment.

Fig. 24: Removing the cover



- Bring the supporting stands to their parking position (item 1) at the cover, while holding the cover.
- Hold the cover with 2 persons and remove the 2 (M12) nuts (item 2) completely
- Remove the cover from the gantry

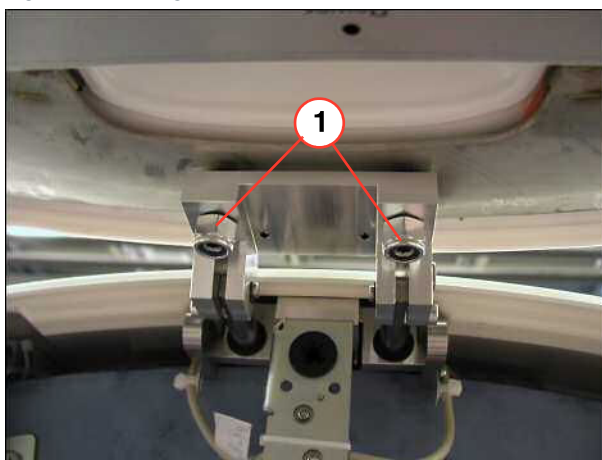
2.4.4 Installation

1. In case you are replacing the top ring segment, perform the following steps before installing it:
 - a) Remove the 2 panels, graphic display and MV45 from the old cover.
 - b) Install 2 panels, graphic display and MV45 at the new cover.
 - c) Insert plugs and install ground connections at their original position.
2. Install the gantry top front ring segment in the reverse order of removing it



Two persons are required to install the front top ring segment. One person to hold the cover, and the other person to install the nuts.

Fig. 25: Holding screws, cover



- Insert the two bolts into the holding frame of the cover. Install the 2 (M12) nuts and tighten them with a torque of **45 Nm**.
 - » **To keep the cover lifted during the above step, it is advised to use the two supporting stands as holders.**
 - Close/attach the cover and reconnect the cable connections
3. Install bottom front ring segment in the reverse order of removing it.
 4. Remove tape.

2.4.5 Startup

2.4.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.4.5.2 Software

n.a.

2.4.5.3 Tune-up

n.a.

2.4.5.4 Testing

- Rotate gantry at slow speed. Listen for unusual noise in the gantry.
 - » If there is any unusual noise, remove the cover(s) again and check/eliminate the source for noise.
- Test panels, graphic display and communication function.

2.4.6 Final Steps

n.a.

2.5 Funnel

2.5.1 Prerequisites

2.5.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.5.1.2 Time and man power

- 1 person
- Estimated total repair time: 0.5 hours

2.5.1.3 Tools and auxiliary equipment

- Standard service kit

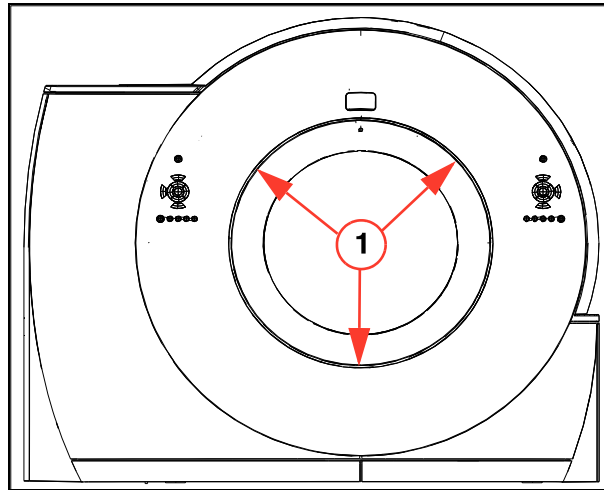
2.5.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove bottom front ring segment. Refer to (→ Removing / Page 32) bottom front ring segment.
4. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
5. Remove plastic ring.

2.5.3 Removing

1. Remove funnel .

Fig. 26: Funnel



- Loosen the 3 securing locks (item 1, 1/4 turn) using a 5 mm hexagonal wrench key.
- Remove funnel.

2.5.4 Installation

1. Install the funnel in reverse order of removing it.
2. Install plastic ring onto the funnel. Secure the plastic ring to the funnel with tape strips.
3. Install/close top front ring segment in the reverse order of removing/opening it.
4. Install bottom front ring segment in the reverse order of removing it.

2.5.5 Startup

2.5.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.5.5.2 Software

n.a.

2.5.5.3 Tune-up

n.a.

2.5.5.4 Testing

- Rotate gantry at slow speed. Listen for unusual noise in the gantry.
 - » If there is any unusual noise, remove the cover(s) again and check/eliminate the source for noise.

2.5.6 Final steps

n.a.

2.6 Plastic ring

2.6.1 Prerequisites

2.6.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.6.1.2 Time and man power

- 1 person
- Estimated total repair time: n.a.

2.6.1.3 Tools and auxiliary equipment

- Standard service kit

2.6.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove bottom front ring segment. Refer to (→ Removing / Page 32) bottom front ring segment .

2.6.3 Removing

1. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
2. Remove plastic ring

2.6.4 Installation

1. Install covers in the reverse order of removing them.

2.6.5 Startup

2.6.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.6.5.2 Software

n.a.

2.6.5.3 Tune-up

n.a.

2.6.5.4 Testing

- Rotate gantry at slow speed. Listen for unusual noise in the gantry.
 - » If there is any unusual noise, remove the cover(s) again and check/eliminate the source for noise.

2.6.6 Final steps

n.a.

2.7 Rear ring cover

2.7.1 Prerequisites

2.7.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.7.1.2 Time and man power

- 1 person
- Estimated total repair time: n.a.

2.7.1.3 Tools and auxiliary equipment

- Standard service kit

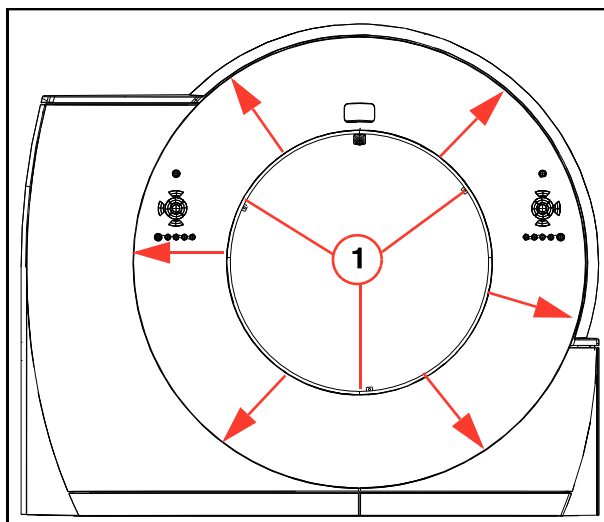
2.7.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove bottom front ring segment. Refer to (→ Removing / Page 32) bottom front ring segment.
4. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
5. Remove plastic ring
6. Remove funnel. Refer to (→ Removing / Page 43) funnel.

2.7.3 Removing

1. Remove rear ring.

Fig. 27: Back ring cover



- Remove the 3 torx screws (item 1) using a TX30 torx bit.
- Loosen the 6 securing locks (arrows, 1/4 turn) using a 5 mm hexagonal wrench key.
- Remove the plug connection of the rear ring, which is located at the upper left side of the ring (approx. 10 o'clock position) if applicable.
- Lift rear ring cover and remove it.

2.7.4 Installation

1. In case of replacing the ring cover, perform the following steps before installing it:
 - a) Remove the 2 panels and MV45 from the old cover.
 - b) Install 2 panels and MV45 at the new cover.
 - c) Insert plugs and install ground connections at their original position.
2. Lift the rear ring cover into the corresponding metal holders and secure the ring, following the removal instructions in reverse order.
3. Install the funnel in the reverse order of removing it.
4. Install plastic ring onto the funnel. Secure the plastic ring to the funnel with tape strips.
5. Install/close top front ring segment in the reverse order of removing/opening it.
6. Install bottom front ring segment in the reverse order of removing it.

2.7.5 Startup

2.7.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.7.5.2 Software

n.a.

2.7.5.3 Tune-up

n.a.

2.7.5.4 Testing

- Rotate gantry in slow speed. Listen for unusual noise in the gantry.
 - » If there is any unusual noise, remove the cover(s) again and check/eliminate the source for noise.
- Test panels, graphic display and communication function.

2.7.6 Final steps

n.a.

2.8 Right side cover

2.8.1 Prerequisites

2.8.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.8.1.2 Time and man power

- 1 person
- Estimated total repair time: n.a.

2.8.1.3 Tools and auxiliary equipment

- Standard service kit

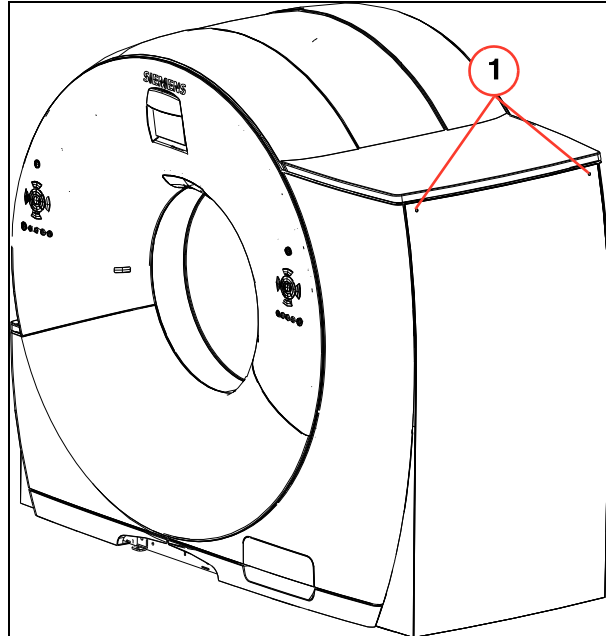
2.8.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.

2.8.3 Removing

1. Remove right side cover .

Fig. 28: Right side cover



- Loosen the 2 securing screws (item 1).
- Pull upper side cover out from its positioning clamps.
- Remove side cover.

2.8.4 Installation

1. Install side cover in the reverse order.

2.8.5 Startup

2.8.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.8.5.2 Software

n.a.

2.8.5.3 Tune-up

n.a.

2.8.5.4 Testing

n.a.

2.8.6 Final steps

n.a.

2.9 Right front cover

2.9.1 Prerequisites

2.9.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.9.1.2 Time and man power

- 1 person
- Estimated total repair time: n.a.

2.9.1.3 Tools and auxiliary equipment

- Standard service kit

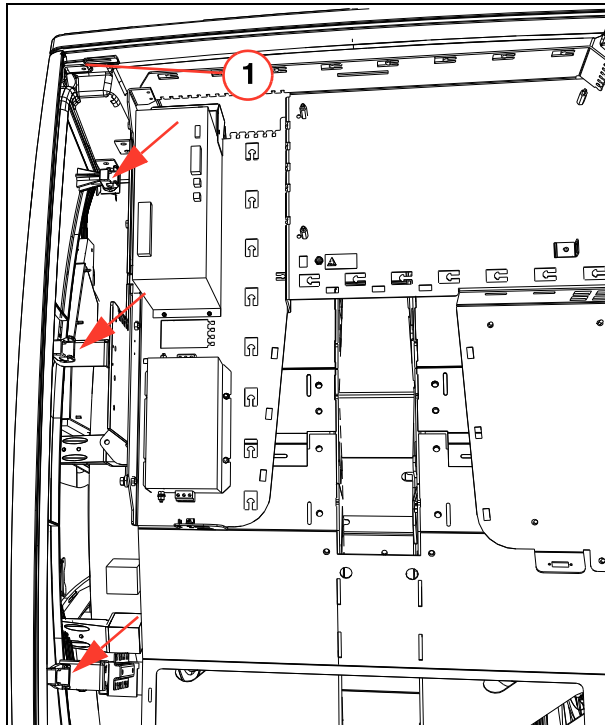
2.9.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove right side cover. Refer to (→ Removing / Page 51) right side cover.

2.9.3 Removing

1. Remove right front cover .

Fig. 29: Right front cover

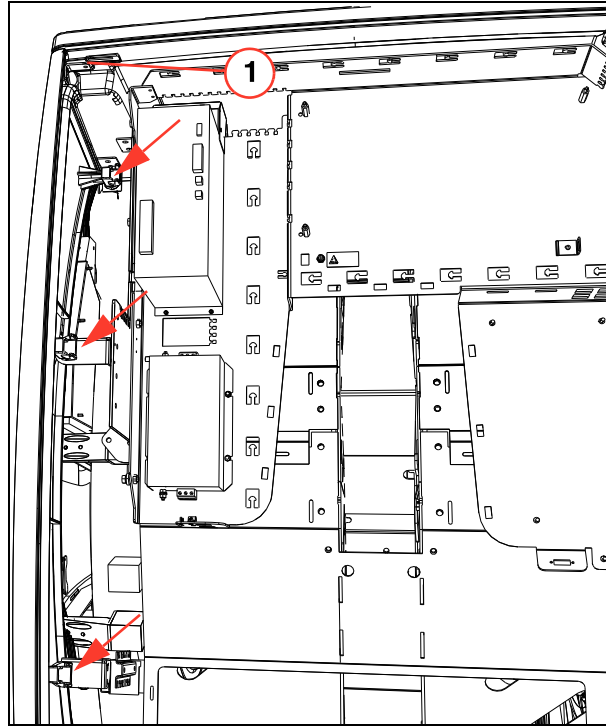


- Remove the securing bracket (item 1).
- Pull out right front cover .

2.9.4 Installation

1. Install right front cover in the reverse order of removing it .

Fig. 30: Right front cover



- Slide in the cover on the guide rail using it and the 3 cover pins as a guide until it is aligned correctly (arrows).
- Tighten the M6 *mm holding screw using a torque of xx Nm.

2. Install the right side cover in the reverse order of removing it.

2.9.5 Startup

2.9.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.9.5.2 Software

n.a.

2.9.5.3 Tune-up

n.a.

2.9.5.4 Testing

n.a.

2.9.6 Final steps

n.a.

2.10 Right rear cover

2.10.1 Prerequisites

2.10.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.10.1.2 Time and man power

- 1 person
- Estimated total repair time: n.a.

2.10.1.3 Tools and auxiliary equipment

- Standard service kit

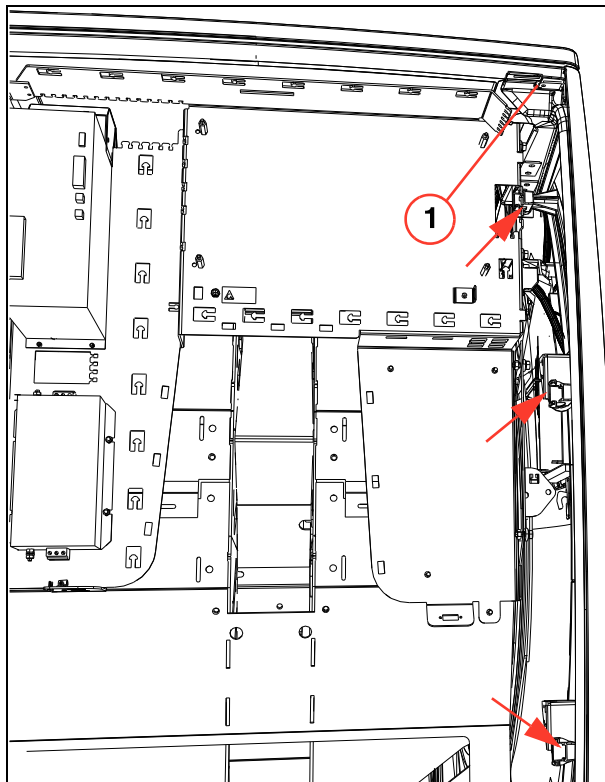
2.10.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove right side cover. Refer to (→ Removing / Page 51) right side cover.

2.10.3 Removing

1. Remove right rear cover.

Fig. 31: Right back cover

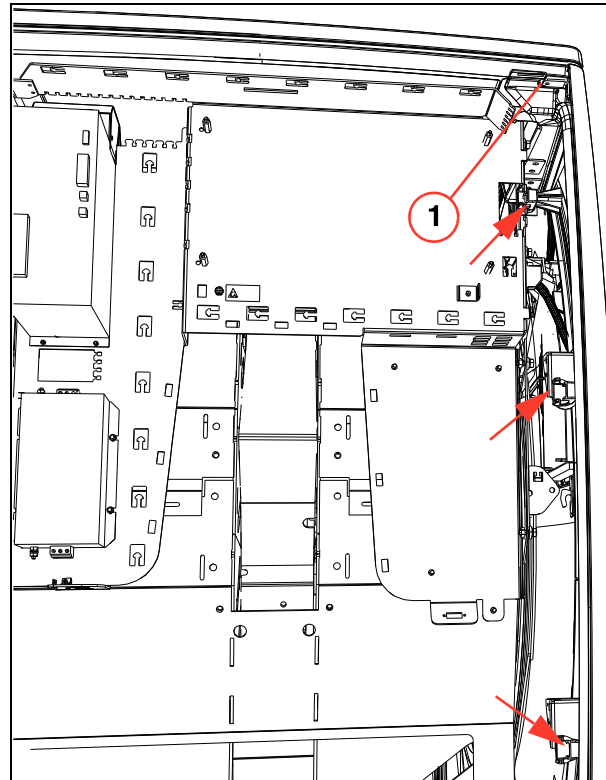


- Remove the securing bracket (item 1).
- Pull out the rear cover.

2.10.4 Installation

1. Install right rear cover.

Fig. 32: Right back cover



- Slide in the cover on the guide rail using it and the 3 cover pins as a guide until it is aligned correctly (arrows).
- Tighten the M6*16mm holding screw using a torque of xx Nm.

2.10.5 Startup

2.10.5.1 Mechanical (Gantry Covers)

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch F2, F3, F5, F6 breakers to "0" position and then to "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.10.5.2 Software

n.a.

2.10.5.3 Tune-up

n.a.

2.10.5.4 Testing

n.a.

2.10.6 Final steps

n.a.

2.11 Left side cover

2.11.1 Prerequisites

2.11.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.11.1.2 Time and man power

- 1 person
- Estimated total repair time: n.a.

2.11.1.3 Tools and auxiliary equipment

- Standard service kit

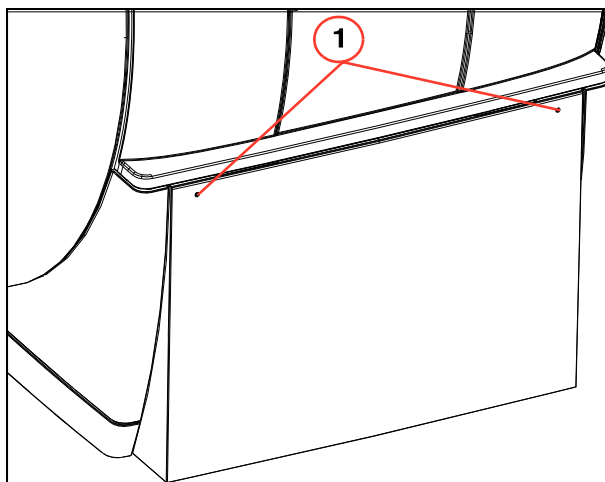
2.11.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.

2.11.3 Removing

1. Remove left side cover.

Fig. 33: Left side cover



- Loosen the 2 securing screws (item 1).
- Pull bottom of side cover out from its positioning clamps.
- Remove side cover.

2.11.4 Installation

1. Install side cover in the reverse order.

2.11.5 Startup

2.11.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.11.5.2 Software

n.a.

2.11.5.3 Tune-up

n.a.

2.11.5.4 Testing
n.a

2.11.6 Final steps
n.a.

2.12 Left front cover

2.12.1 Prerequisites

2.12.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.12.1.2 Time and man power

- 1 person
- Estimated total repair time: n.a.

2.12.1.3 Tools and auxiliary equipment

- Standard service kit

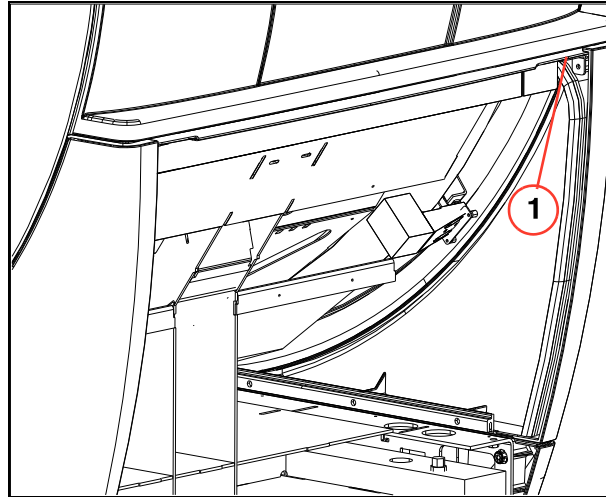
2.12.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove left side cover. Refer to (→ Removing / Page 62) left side cover.

2.12.3 Removing

1. Remove left front cover .

Fig. 34: Front left cover



- Remove the securing bracket (item 1).
- Pull out left front cover .

2.12.4 Installation

1. Install left front cover in the reverse order of removing it.
2. Install left front side cover in the reverse order of removing it.

2.12.5 Startup

2.12.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.12.5.2 Software

n.a.

2.12.5.3 Tune-up

n.a.

2.12.5.4 Testing

n.a.

2.12.6 Final steps

n.a.

2.13 Left rear cover

2.13.1 Prerequisites

2.13.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the “General” section of this document and/or the “Product specific safety notes”.

2.13.1.2 Time and man power

- 1 person
- Estimated total repair time: n.a.

2.13.1.3 Tools and auxiliary equipment

- Standard service kit
- Add above label information to the spare parts order. This assures you that the new parts will arrive with the correct labels.



All system labels are located on the left rear cover. The spare parts will include all necessary labels already placed on the cover. To ensure that all necessary labels are affixed to the spare parts, it is mandatory that the existing label information (from all label part numbers, serial number date) is sent with the order.

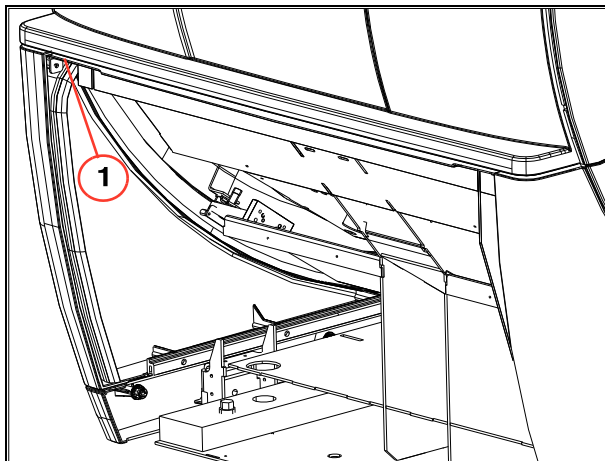
2.13.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove left side cover. Refer to (→ Removing / Page 62) left side cover.

2.13.3 Removing

1. Remove left rear cover.

Fig. 35: Back left cover



- Remove the securing bracket (item 1).
- Pull out the rear cover.

2.13.4 Installation

1. Install the cover in the reverse order.
2. In case of replacement, assure that the labeling is correctly affixed on the part (if necessary, check the old part).

2.13.5 Startup

2.13.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.13.5.2 Software

n.a.

2.13.5.3 Tune-up

n.a.

2.13.5.4 Testing

Check that all system labels are placed on the new cover (in case of exchange)

2.13.6 Final steps

n.a.

2.14 Right/left front base board

2.14.1 Prerequisites

2.14.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.14.1.2 Time and man power

- 1 person
- Estimated total repair time: n.a.

2.14.1.3 Tools and auxiliary equipment

- Standard service kit

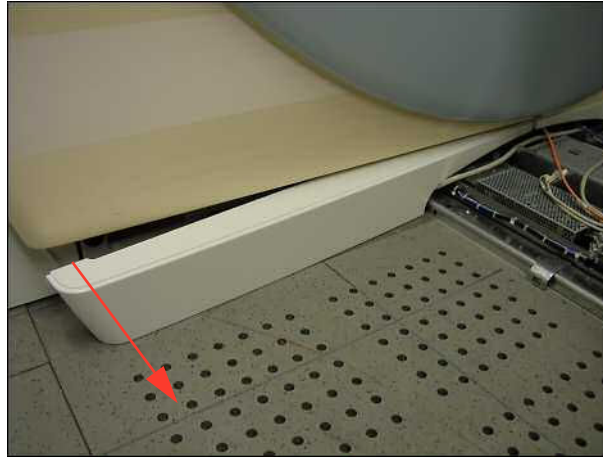
2.14.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove the PHS electronic cover. Refer to (→ Removing / C2-028.841.02) PHS electronic cover.

2.14.3 Removing

1. Remove front base cover.

Fig. 36: Front base cover



- Remove base cover by pulling it at the end of the cover (arrow).

2.14.4 Installation

1. Install base cover in the reverse order.

2.14.5 Startup

2.14.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.14.5.2 Software

n.a.

2.14.5.3 Tune-up

n.a.

2.14.5.4 Testing

n.a.

2.14.6 Final steps

n.a.

2.15 Right/left rear baseboard

2.15.1 Prerequisites

2.15.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.15.1.2 Time and man power

- 1 person
- Estimated total repair time: n.a.

2.15.1.3 Tools and auxiliary equipment

- Standard service kit

2.15.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.

2.15.3 Removing

1. Remove base cover

Fig. 37: Back left base cover



Fig. 38: Back left base cover



1. Loosen base cover by pulling on it at the end (left image) .
2. Remove the cover by pulling it with both hands (arrows) out of the gantry.
 - » The cover may be damaged if it is just pulled out at the end of the cover!

2.15.4 Installation

1. Install the cover in the reverse order.

2.15.5 Startup

2.15.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.15.5.2 Software

n.a.

2.15.5.3 Tune-up

n.a.

2.15.5.4 Testing

n.a.

2.15.6 Final steps

n.a.

2.16 Front panels right/left

2.16.1 Prerequisites

2.16.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the “General” section of this document and/or the “Product specific safety notes”.



If the right or left panel with revision level 01 needs to be replaced, then replace both panels in parallel. Other revision levels can be replaced separately.

2.16.1.2 Time and man power

- 1 person
- Estimated total repair time: 1.0 hour.

2.16.1.3 Tools and auxiliary equipment

- Standard service kit

2.16.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove bottom front ring segment. Refer to (→ Removing / Page 32) bottom front ring segment.
4. Use strips of tape to secure the plastic ring to the funnel (this step makes it easier reinstalling the front ring segments).
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).

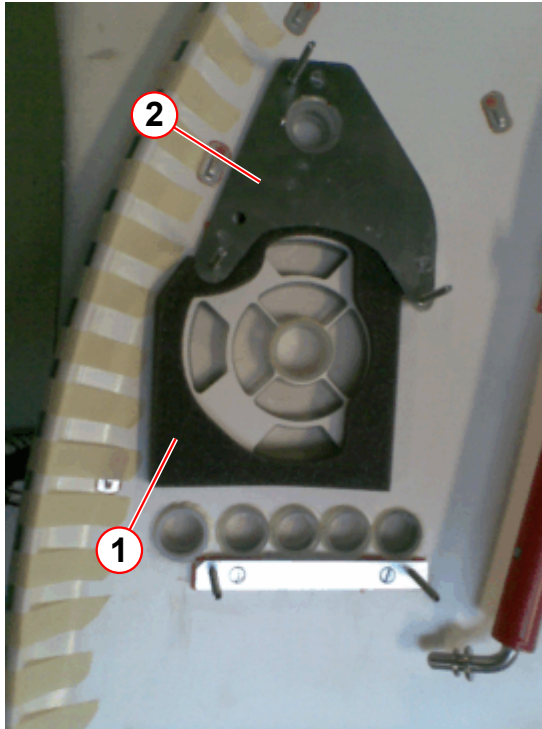
2.16.3 Removing

1. Disconnect plugs at the panel and cut ties to release the cable.

2. Remove the 5 nuts and remove the panel.
3. If the plastic pad located between panel and cover is visibly damaged, remove the pad as well.

2.16.4 Installation

Fig. 39: Front panel



- (1) Foam rubber gasket
(2) Mounting plate

1. Install plastic pad in the reverse order of removing it (if applicable). After installation, check the correct position of the pad.
 - » **Install the foam rubber gasket (enclosed in spare part kit) before the panel is installed. (Push the foam rubber gasket under the mounting plate).**
2. Install panel in the reverse order.
3. Install/close top front ring segment in the reverse order of removing/opening it.
4. Install bottom front ring segment in the reverse order of removing it.

2.16.5 Startup

2.16.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.

2. Switch on the system (**SYSTEM/ON**) at the control box.

2.16.5.2 Software

- Go to the Service home menu, select Options -> Service -> Local Service
 - In the service software menu select **TuneUp** -> **FRU Replace**
 - Select the corresponding part replaced and click GO
- Note:** If the part you replaced is not in the list, press go and continue. The firmware check will be performed anyhow
- Follow the output instructions



If the firmware update aborts with an error and the firmware status is NOT ok, perform the firmware update again.

2.16.5.3 Tune-up

n.a.

2.16.5.4 Testing

- Check the function of the panel

2.16.6 Final steps

n.a.

2.17 Rear panels, right/left

2.17.1 Prerequisites

2.17.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.17.1.2 Time and man power

- 1 person
- Estimated total repair time: 1.5 hour.

2.17.1.3 Tools and auxiliary equipment

- Standard service kit

2.17.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove bottom front ring segment. Refer to (→ Removing / Page 32) bottom front ring segment.
4. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
5. Remove plastic ring
6. Remove funnel. Refer to (→ Removing / Page 43) funnel .
7. Remove the rear ring cover. Refer to (→ Removing / Page 48) rear ring cover.

2.17.3 Removing

1. Disconnect plugs at the panel and cut ties to release the cable.
2. Remove the 5 nuts and remove the panel.

3. If the plastic pad located between panel and cover is visibly damaged, remove the pad as well.

2.17.4 Installation

1. Install plastic pad in the reverse order of removing it (if applicable). After installation, check the correct position of the pad.
2. Install the panel in reverse order.
3. Install the ring cover. Refer to (→ Installation / Page 48) ring cover.

2.17.5 Startup

2.17.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.17.5.2 Software

- Go to the Service home menu, select Options -> Service -> Local Service
 - In the service software menu select **TuneUp -> FRU Replace**
 - Select the corresponding part replaced and click GO
- Note:** If the part you replaced is not in the list, press go and continue. The firmware check will be performed anyhow
- Follow the output instructions



If the firmware update aborts with an error and the firmware status is NOT ok, perform the firmware update again.

2.17.5.3 Tune-up

n.a.

2.17.5.4 Testing

- Check the function of the panel

2.17.6 Final steps

n.a.

2.18 Graphic display

2.18.1 Prerequisites

2.18.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.18.1.2 Time and man power

- 1 person
- Estimated total repair time: 1.0 hour.

2.18.1.3 Tools and auxiliary equipment

- Standard service kit

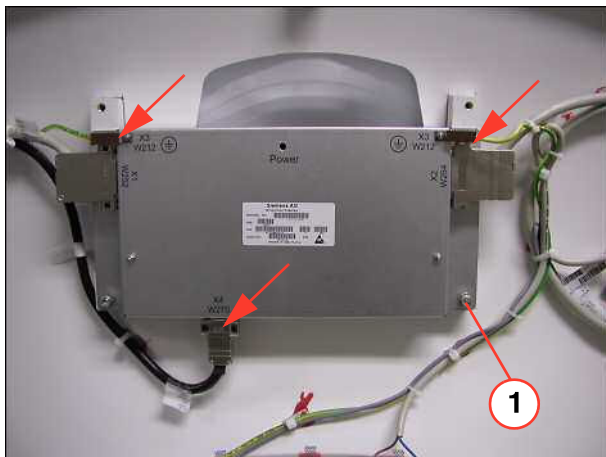
2.18.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove bottom front ring segment. Refer to (→ Removing / Page 32) bottom front ring segment.
4. Use strips of tape to secure the plastic ring to the funnel (this step makes it easier reinstalling the front ring segments).
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).

2.18.3 Removing

1. Remove the graphic display.

Fig. 40: Removing the graphic display



- Disconnect plugs and ground connection at the graphic display (arrows) and cut ties to release cable, if necessary.
- Remove the 4 holding nuts (item 1) and remove the display.

2.18.4 Installation

1. Install the graphic display in the reverse order.
 - » **Don't forget to remove the protection foil from the graphic display before installing it.**
2. Install/close top front ring segment in the reverse order of removing/opening it.
3. Install bottom front ring segment in the reverse order of removing it.

2.18.5 Startup

2.18.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.18.5.2 Software

n.a.

2.18.5.3 Tune-up

n.a.

2.18.5.4 Testing

- Check the function of the display

2.18.6 Final steps

n.a.

2.19 Front loudspeaker/microphone/MV45

2.19.1 Prerequisites

2.19.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.19.1.2 Time and man power

- 1 person
- Estimated total repair time: 1.0 hour.

2.19.1.3 Tools and auxiliary equipment

- Standard service kit

2.19.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove bottom front ring segment. Refer to (→ Removing / Page 32) bottom front ring segment.
4. Use strips of tape to secure the plastic ring to the funnel (this step makes it easier reinstalling the front ring segments).
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).

2.19.3 Removing

Description to remove the loudspeaker

1. Disconnect plugs at the loudspeaker.
2. Remove the 4 holding nuts and remove the speaker

Description to remove the microphone

1. Remove cable connection to MV45. Cut ties as necessary.
2. Remove the 4 holding nuts and remove the microphone.

Description to remove the MV45 unit

1. Disconnect plugs and cut ties to release cable, if necessary.
2. Remove the 4 holding screws and remove the MV45 unit.

2.19.4 Installation

1. Install the speaker/microphone/MV45 in the reverse order.
2. Install/close top front ring segment in the reverse order of removing/opening it.
3. Install bottom front ring segment in the reverse order of removing it.

2.19.5 Startup**2.19.5.1 Mechanical**

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.19.5.2 Software

n.a.

2.19.5.3 Tune-up

n.a.

2.19.5.4 Testing

- Check the function of the changed part.

Fig. 41: Control box



- Press the control box buttons (item 1/2) for the function test.

2.19.6 Final steps

n.a.

2.20 Rear loudspeaker or microphone or MV45

2.20.1 Prerequisites

2.20.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

2.20.1.2 Time and man power

- 1 person
- Estimated total repair time: 1.0 hour.

2.20.1.3 Tools and auxiliary equipment

- Standard service kit

2.20.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove bottom front ring segment. Refer to (→ Removing / Page 32) bottom front ring segment.
4. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
5. Remove plastic ring
6. Remove funnel. Refer to (→ Removing / Page 43) funnel.
7. Remove rear ring. Refer to (→ Removing / Page 48) rear ring.

2.20.3 Removing

Description to remove the loudspeaker

1. Disconnect plugs at the loudspeaker and cut ties to release cable, if necessary.

2. Remove the 4 holding nuts (item 1) and remove the speaker.

Description to remove the microphone

1. Cut ties to release cable to MV45.
2. Remove the 4 holding nuts (item 1) and remove the microphone.

Description to remove MV45

1. Remove all plugs (item 1) and cut ties to release cable, if necessary.
2. Remove the 4 holding screws (item 1) and remove the MV45 unit.

2.20.4 Installation

1. Install the speaker or microphone or MV45 in the reverse order of removing it.
2. Install the ring cover. Refer to (→ Installation / Page 48) ring cover.
3. Install/close top front ring segment in the reverse order of removing/opening it.
4. Install bottom front ring segment in the reverse order of removing it.

2.20.5 Startup

2.20.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

2.20.5.2 Software

n.a.

2.20.5.3 Tune-up

n.a.

2.20.5.4 Testing

- Check the function of the changed part.

Fig. 42: Control box



- Press the control box buttons (item 1/2) for the function test.

2.20.6 Final steps

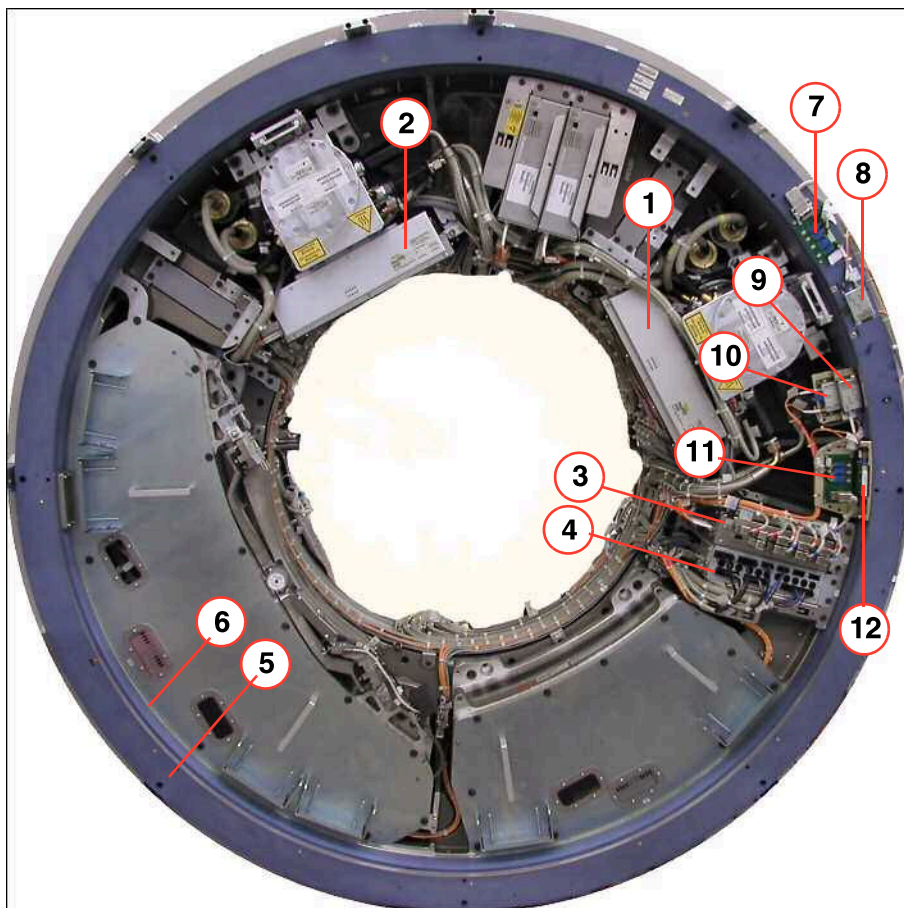
n.a.

3.1 Overview

This chapter describes the replacement of all parts located in the gantry front, except for XRS and DMS.

The figure below shows the location of the visible parts, located in the gantry front and described in this chapter.

Fig. 43: Overview gantry - front parts



Parts in gantry front:

- Tube collimator A (item 1)
- Tube collimator B (item 2)
- UMAR (item 3)
- Power distribution, rotating (PDR) (item 4)
- Stationary Data Ring (item 5)
- Rotating Data Ring (item 6)
- Wiring interface, stationary (item 7)
- Stationary Transmitter (fastlink data) (item 8)
- Rotating Transmitter (fastlink data) (item 9)
- Rotating Transmitter (DMS data) (item 10)
- Wiring interface, rotating (item 11)

- Rotating Receiver (fastlink data) (item 12)

3.2 Tube Collimator (TCO)

- The following instruction is applicable for:
 - Tube collimator Z40-A, part no. 77 41 700
 - Tube collimator Z40-B, part no. 77 41 940

3.2.1 Prerequisites

3.2.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

3.2.1.2 Time and Man Power

- 1 Person
- Estimated total repair time: 1.5 hours.

3.2.1.3 Tools and auxiliary Equipment

- Standard tool kit
- TX40 Torx bit.

3.2.2 Preliminary steps

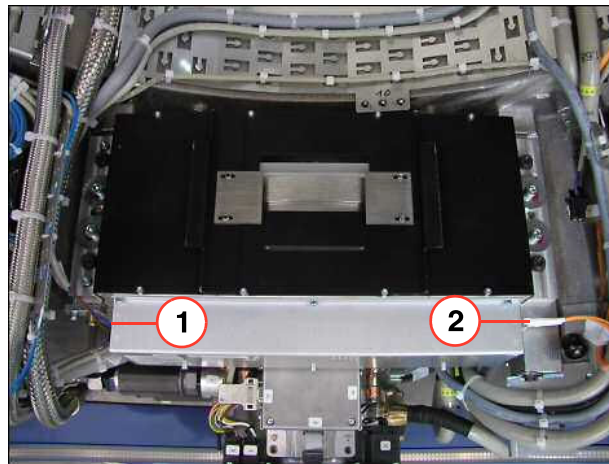
1. Allow at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube, XTC ... may reach up to 130 degree Celsius.
2. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
4. Remove the gantry bottom front ring segment (→ Removing / Page 32).
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Remove the the Plexiglas ring.

7. Turn the gantry by hand until the collimator to be replaced is in the 6 o'clock position.
8. Secure against unintended rotation by using the front safety bolt (→ Safety bolts / Page 21).
9. Install the data ring protection (→ Data ring protection / Page 23).

3.2.3 Removing

1. Disconnect the cable connections.

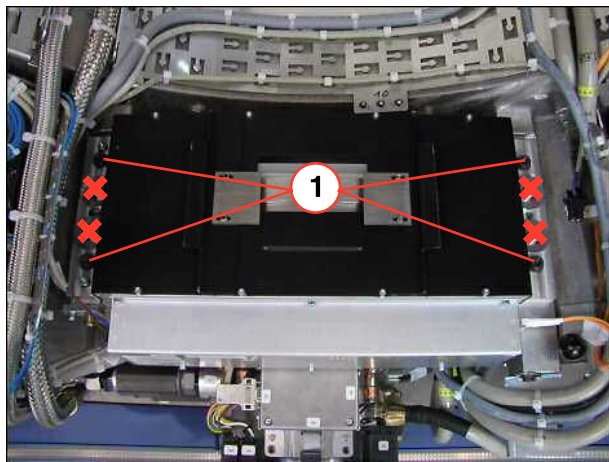
Fig. 44: Example of tube collimator B



- Disconnect the following cables:
 - On the left side, power connector X10 (item 1)
 - On the right side, fast link connector X11 (item 2)
 - Remove cable ties as needed.

2. Remove the screws securing the tube collimator.

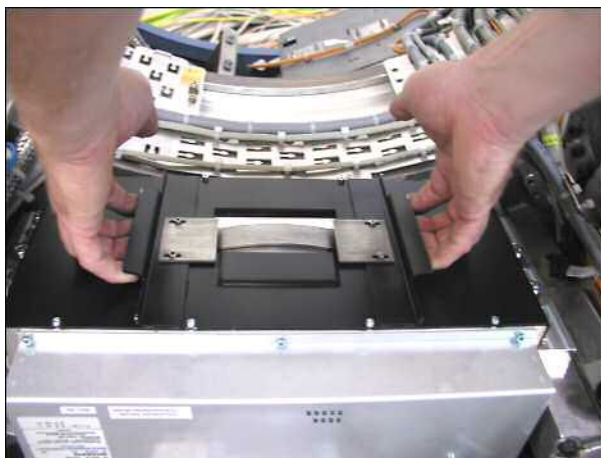
Fig. 45: Attachment screws - tube collimator



- Remove the four M8*30 mm Torx bolts (Item 1) using a T40 Torx key.
 - » **Discard the Torx bolts !!**
 - » **Do not touch the 4 M8 screws secured with red lacquer (crosses) !!!**

3. Remove the tube collimator

Fig. 46: Removing the tube collimator



- Remove the tube collimator from the gantry using the 2 latches as a handle.

3.2.4 Installation

1. Installation is in the reverse order of removal.
2. Tighten the four M8*30mm Torx bolts using a T40 Torx key with a torque of **27 Nm**.



Install the collimator with the new enclosed Torx bolts !!

3. Connect all cables according to the original configuration.

3.2.5 Start-up

3.2.5.1 Mechanical

1. Remove the data ring protections.
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.

3.2.5.2 Software

- Go to the Service home menu, select Options -> Service -> Local Service
- In the service software menu select **TuneUp -> FRU Replace**
- Select the corresponding part replaced and click GO

Note: If the part you replaced is not in the list, press go and continue. The firmware check will be performed anyhow
- Follow the output instructions



If the firmware update aborts with an error and the firmware status is NOT ok, perform the firmware update again.

3.2.5.3 Tune-up

1. Run the tune-up sequence as described under **Local Service -> Tune Up -> FRU replacement**.
2. Close gantry covers when instructed during the guided tour or after performing the guided tour.



Disable gantry rotation using S301 when performing mechanical adjustments inside the gantry during tune-up!

3.2.5.4 Testing

n.a

3.2.6 Final steps

n.a

3.3 Universal Master Rotation (UMAR)

3.3.1 Prerequisites

3.3.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

3.3.1.2 Time and Man Power

- 1 person
- Estimated total repair time: 2.0 hours.

3.3.1.3 Tools and auxiliary equipment

- Standard service kit

3.3.2 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove the gantry bottom front ring segment (→ Removing / Page 32).
4. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
5. Remove the Plexiglas ring (→ Removing / Page 34).
6. Turn the gantry by hand until the Straton MX tube - A is in the 6 o'clock position (UMAR approximates the 7 o'clock position).
7. Secure against unintended rotation by using the front safety bolt (→ Safety bolts / Page 21).
8. Install the data ring protection (→ Data ring protection / Page 23).

3.3.3 Removing

1. Replace metal cover located over UMAR.

2. Disconnect all connectors necessary to remove the UMAR.
3. Remove the ground connection (XGND).
4. Remove the four holding screws to remove the UMAR assembly.
 - » **Observe ESD precautions!**

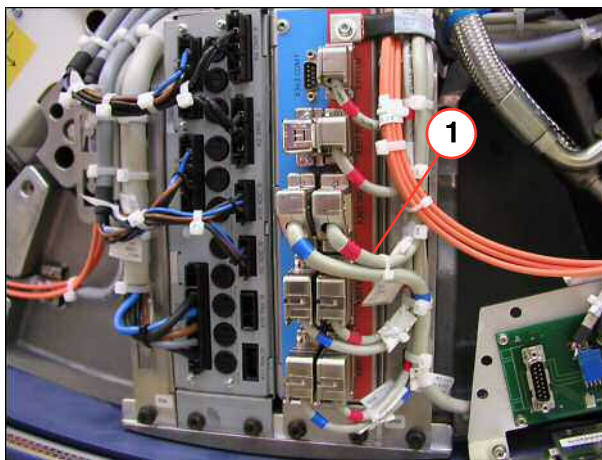
3.3.4 Installation



With revision 04, two filter adapters are installed on plug X304 and plug X305 on UMAR. With revision > 04, these two filter functions are located **inside** the UMAR. For this reason, it is not allowed to install the two filter adapters (from rev. 04) additionally on UMAR, rev. >04 (plug X304/X305). Ignoring this information may lead to unusual CAN bus problems!

1. Install the new UMAR assembly by reversing the order for removing it.
2. Remove Plug X306 BAC from old UMAR and install it on the new UMAR assembly.
3. Tighten the 4 M6 * 16 mm holding screws using **14 Nm**.
4. Ensure that all connectors are connected properly and are secure.
5. Install cable ties

Fig. 47: UMAR cabling



- Ensure correct positioning of cables and cable ties (e.g. item1)



Wrong routed/secured cables might have contact to the gantry covers during rotation. Route the cables as narrow to the UMAR assembly as possible.

6. Tighten the ground connection (XGND) M4 attachment screw using **1.8 Nm**.

3.3.5 Start-up

3.3.5.1 Mechanical

1. Remove the data ring protections.
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.

3.3.5.2 Software with VA11

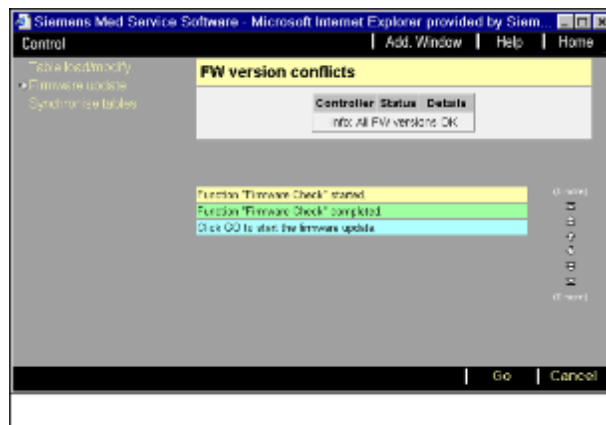
1. Automatic controller FW load will take place after switch-on
 - » **Wait until the system tells that FW loading is finished**
2. Perform a **restore** of site-specific tables
 - Select in **Local Service** -> **Backup/Restore**
 - > **Command**: Restore
 - > **Drive**: Select drive with stored tune-up tables
 - > **Archive**: select most current tune-up file
 - > **Groups**: select "Service\SiteSpecific\TuneUp"
 - » Site-specific tables are stored in UMAS. After replacement of UMAS, the tables must be restored into the new UMAS

3.3.5.3 Software from VA20 onwards

1. Go to the Service home menu, select Options -> Service -> Local Service
2. In the service software menu select **TuneUp** -> **FRU Replace**
3. Select the corresponding part replaced and click GO
 - Accept the pop-up message with OK
4. Click GO to start the Auto Complete function
5. The AutoComplete check displays the status of the steps to be performed
6. Check if the firmware status is "unknown" after replacing UMAS/UMAR.
 - If the firmware status **IS** "unknown," continue with step 7
 - If the firmware status **IS NOT** "unknown," continue with step 8
7. Open the Window Explorer and go to C:\Somaris\tools\bin\
 - Double-click the **Update_VA11A_FW.bat** file
 - The firmware update starts
(UMAR will take 300s)

- During the firmware update the SysOn LED at the control box is blinking.
Wait until the LED no longer blinks indicating that the firmware update is finished.
- 8. In the service software menu select **Control -> Firmware Update**
- 9. The firmware check starts, and the following appears:

Fig. 48: Firmware check - successful



- » If the firmware version status is OK and "Function Firmware Check is completed" is displayed, the firmware is successfully loaded.
- » If the firmware version status is not OK, click GO to start the firmware update again.

3.3.5.4 Tune-up

n.a

3.3.5.5 Testing

1. Perform a test scan
2. Install all covers which have been removed before.

3.3.6 Final steps

n.a

3.4 Power Distribution Rotating (PDR)

3.4.1 New PDR from the manufacturer TDK-Lambda

From November 2010 onwards, a second supplier (TDK-Lambda) has been established. The replacement procedures for both PDRs (ASTEC and TDK-Lambda) are the same.

Differences between the ASTEC PDR and the TDK-Lambda PDR:

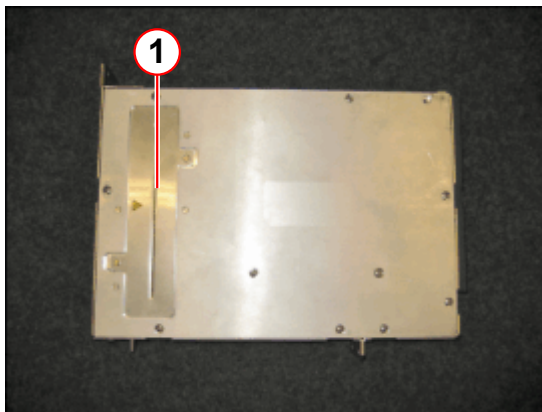
Connector X16:

The TDK-Lambda PDR no longer has an X16 CAN connector. The CAN connection PDR_X16 to UMAR_X307 was never used. If cable W558 is present, disconnect plug X16 from the ASTEC PDR and attach it to the cable harness with a cable tie. Cable W558 will no longer be used.

Fuses:

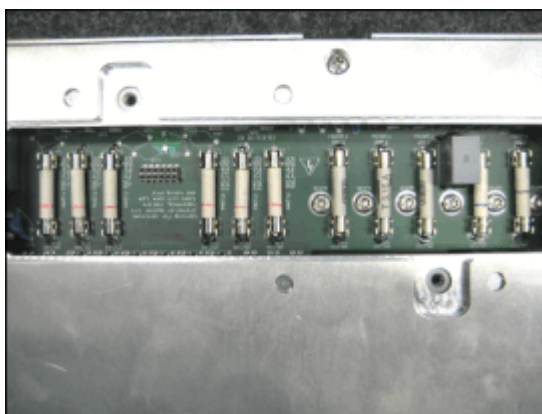
To access the fuses of the TDK-Lambda PDR, the PDR must be removed from the gantry.

Fig. 49: Top view of the TDK-Lambda PDR



(1) Fuse cover

Fig. 50: Fuses of the TDK-Lambda PDR



3.4.2 Prerequisites

3.4.2.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

3.4.2.2 Time and Man Power

- 1 person
- Estimated total repair time: 1.0 hours.

3.4.2.3 Tools and auxiliary Equipment

- Standard service kit

3.4.3 Preliminary steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove the gantry bottom front ring segment (→ Removing / Page 32).
4. Turn the gantry by hand until the Straton MX tube - A is in the 6 o'clock position (UMAR approximates the 7 o'clock position).
5. Secure against unintended rotation by using the front safety bolt (→ Safety bolts / Page 21).
6. Install the data ring protection (→ Data ring protection / Page 23).

3.4.4 Removing

1. Replace metal cover located over PDR.
2. Disconnect all connectors at the PDR.
3. Disconnect the ground connection.
4. Remove the four holding screws and remove the PDR assembly.

3.4.5 Installation

1. Install the new PDR assembly by reversing the order for removing it.
2. Tighten the 4 M6*16mm holding Torx screws using **14 Nm**.
3. Tighten the ground connection with a torque of **6 Nm**
4. Ensure that all connectors are connected properly and are secured.

3.4.6 Start-up

3.4.6.1 Mechanical

1. Remove the data ring protections.
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.

3.4.6.2 Software

n.a

3.4.6.3 Tune-up

n.a

3.4.6.4 Testing

- Check LED status on PDR.
 - » After stand-by status of system, all LED's on the PDR must be switched on.
- Perform a test scan.
- Install all covers which have been removed before.

3.4.7 Final steps

n.a

3.5 LPS unit (system serial number < 60048)

3.5.1 Prerequisites

3.5.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the “General” section of this document and/or the “Product specific safety notes”.

3.5.1.2 Time and Man Power

- 1 person
- Estimated total repair time: 2 hours.

3.5.1.3 Tools and auxiliary Equipment

- Standard service kit
- 1 screw M 14*65 mm (previously used to secure the gantry drum during transport)

3.5.2 Preliminary steps

1. Allow at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube, XTC ... may reach up to 130 degree Celsius.
2. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
4. Remove the gantry bottom front ring segment (→ Removing / Page 32).
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Remove the Plexiglas ring.
7. Turn the gantry by hand until the tube collimator A is in the 6 o'clock position.
8. Secure against unintended rotation by using the front safety bolt (→ Safety bolts / Page 21).

9. Install the data ring protection (→ Data ring protection / Page 23).

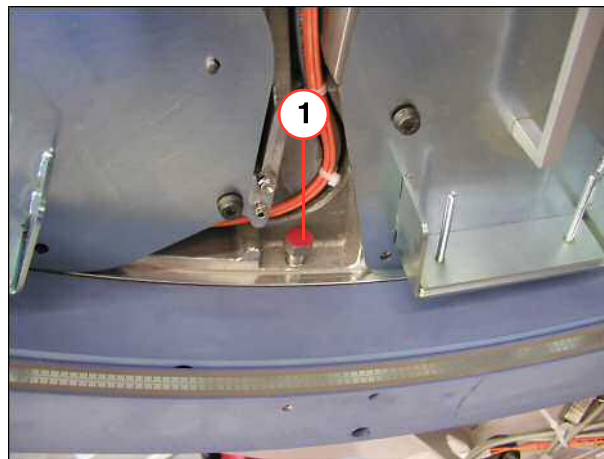
3.5.3 Removing

1. Remove tube collimator A. Refer to (→ Removing / Page 93) tube collimator.
 - » **Below the tube collimator, a service opening is located in the gantry drum.**
2. Install gantry rotation blocking screw .



In the above-mentioned serial number range, the safety bolt cannot be used to block the gantry rotation during LPS replacement. Instead a screw M14*65 mm (previously used to secure the gantry drum during transport) is used to block the gantry rotation.

Fig. 51: Gantry attachment screw

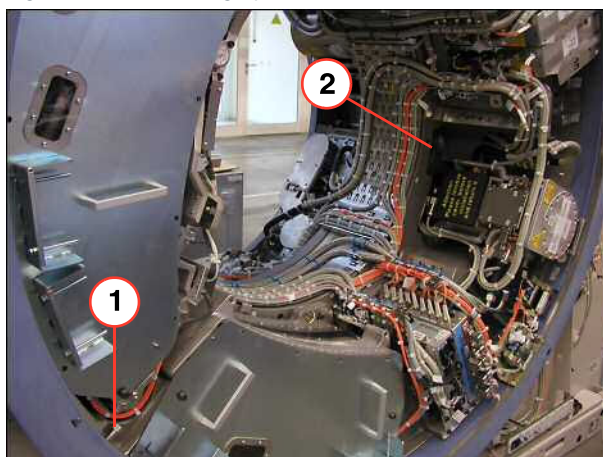


- Install the rotation blocking screw M 14*65 mm between DMS_A and DMS_B but don't tighten the screw at this moment!
3. Position gantry drum.



Since the tube collimator is removed, the gantry is no longer balanced. Be careful when moving the rotating part of the gantry.

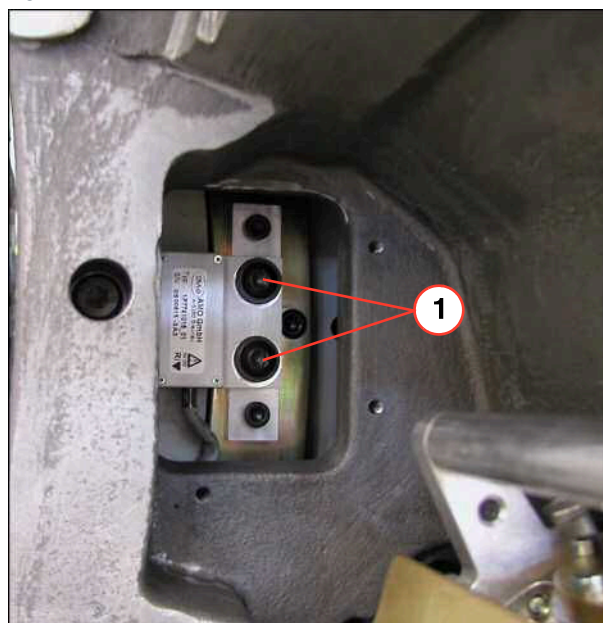
Fig. 52: LPS exchange position



- Unlock the rotation safety bolt
- Rotate the service opening slowly close to the 3 o'clock position until the LPS unit appears (item 2).
- Secure gantry rotation using the securing screw (item 1). Tighten the screw until gantry rotation is blocked (try to turn the gantry).

4. Remove the LPS unit.

Fig. 53: LPS unit



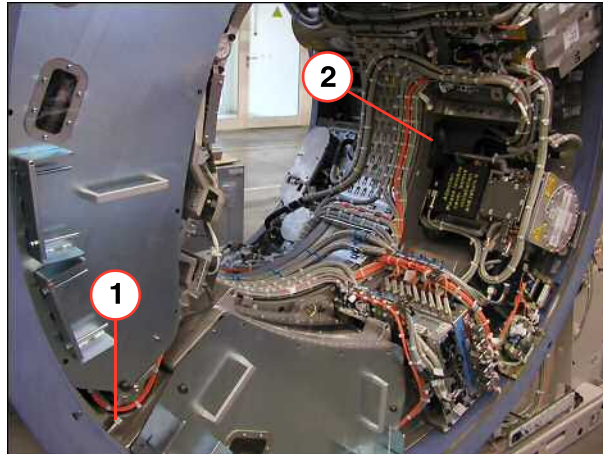
- Remove the 2 holding screws (item 1) and remove the LPS unit.
- Move LPS unit carefully out of the service opening

5. Position gantry drum.



Since the tube collimator is removed, the gantry is no longer balanced. Be careful when moving the rotating part of the gantry.

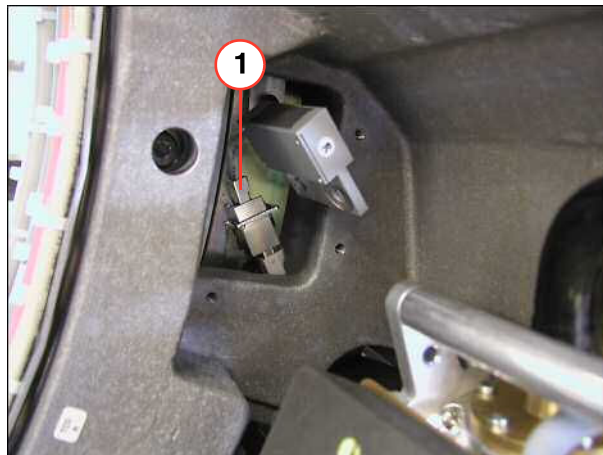
Fig. 54: LPS exchange position



- Open the securing screw (item 1) slowly and turn the gantry carefully until the service opening (item 2) is positioned over the LPS unit plug.
- Secure gantry rotation using the securing screw (item 1). Tighten the screw until gantry rotation is blocked (try to turn the gantry).

6. Disconnecting the LPS plug.

Fig. 55: LPS unit, removable

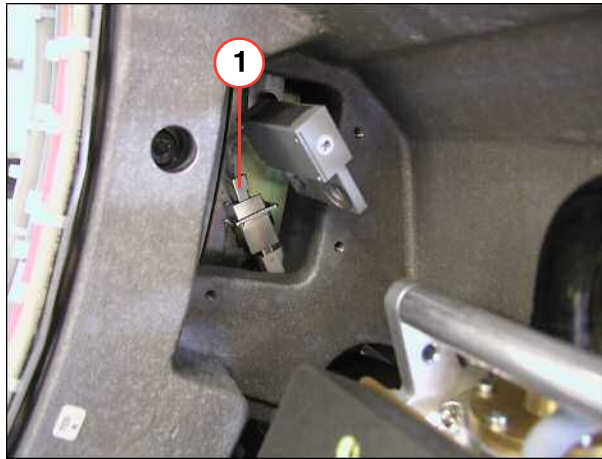


- Disconnect connector (item 1) and remove LPS unit.

3.5.4 Installation

1. Connect the LPS plug.

Fig. 56: LPS unit, removable



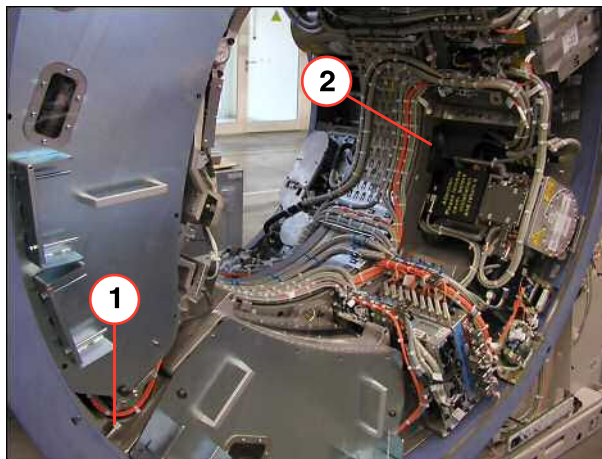
- Plug in the connector (item 1).

2. Position gantry drum.



Since the tube collimator is removed, the gantry is no longer balanced. Be careful when moving the rotating part of the gantry.

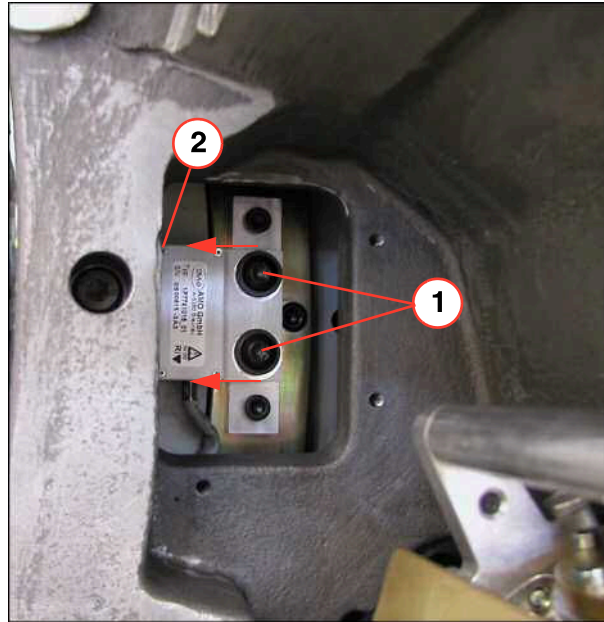
Fig. 57: LPS exchange position



- Open the securing screw (item 1) slowly and turn the gantry carefully until the service opening (item 2) is positioned over the LPS unit mounting place.
- Secure gantry rotation using the securing screw (item 1). Tighten the screw until gantry rotation is blocked (try to turn the gantry).

3. Installing the LPS unit.

Fig. 58: LPS unit adjustment



- Install the 2 holding screws of the LPS unit (item 1), but do not tighten them at this time.
- Install the enclosed adjustment foil (delivered with the part) between the LPS unit and the measuring tape (item 2).
- Press the LPS unit against the foil (arrows) and tighten the 2 M8*14mm holding screws (item 1) using a torque of **4 Nm**.
- Remove the foil carefully.
- Remove securing screw M14*65 mm.

4. Reinstall tube collimator A. Refer to (→ Installation / Page 94) tube collimator.

3.5.5 Start-up

3.5.5.1 Mechanical

1. Remove the data ring protections.
2. Rotate the gantry by hand and check for unusual noise.
3. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
4. Switch on the system (**SYSTEM/ON**) at the control box.

3.5.5.2 Software

n.a

3.5.5.3 Tune-up

1. Run the tune-up sequence as described under **Local Service -> Tune Up -> FRU re-placement**.
2. Close gantry covers when instructed during the guided tour or after performing the guided tour.



Disable gantry rotation using S301 when performing mechanical adjustments inside the gantry during tune-up!

3.5.5.4 Testing

n.a

3.5.6 Final steps

n.a

3.6 LPS unit (system serial number > 60047)

3.6.1 Prerequisites

3.6.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

3.6.1.2 Time and Manpower

- 1 person
- Estimated total repair time: 2 hours.

3.6.1.3 Tools and auxiliary Equipment

- Standard service kit

3.6.2 Preliminary Steps

1. Allow at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube, XTC ... may reach up to 130 degree Celsius.
2. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
4. Remove the gantry bottom front ring segment (→ Removing / Page 32).
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Remove the Plexiglas ring (→ Removing / Page 34).
7. Turn the gantry by hand until the tube collimator A is in the 6 o'clock position.
8. Secure against unintended rotation by using the front safety bolt (→ Safety bolts / Page 21).
9. Install the data ring protection (→ Data ring protection / Page 23).

3.6.3 Removing

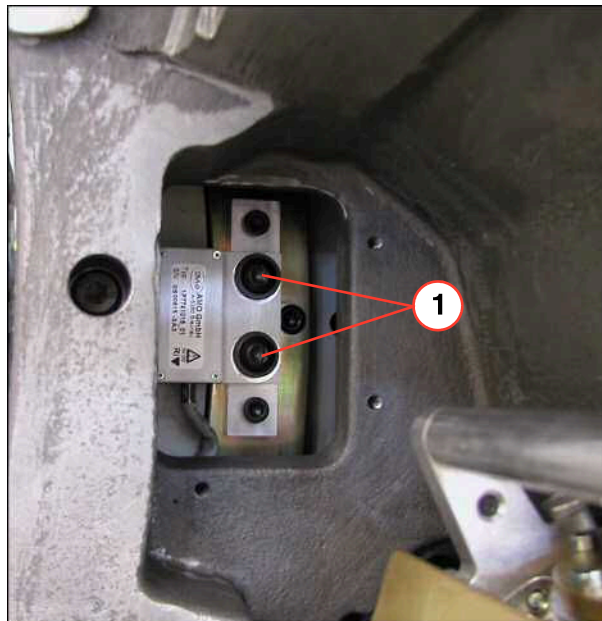
1. Remove tube collimator A. Refer to (→ Removing / Page 93) tube collimator.
 - » **Below the tube collimator, a service opening is located in the gantry drum.**
2. Unlock the rotation safety bolt and rotate the service opening slowly until the LPS unit appears (approximates the 3 o'clock position).



Since the tube collimator is removed, the gantry is no longer balanced. Be careful when moving the rotating part of the gantry.

3. After the correct position has been reached, secure against unintended rotation by using the front safety bolt.
4. Remove the LPS unit.

Fig. 59: LPS unit



- Remove the 2 holding screws (item 1) and remove the LPS unit.
 - Move LPS unit carefully out of the service opening
5. Unlock the rotation safety bolt and rotate the service opening slowly upwards until the LPS connector appears (approximates the 4 o'clock position).

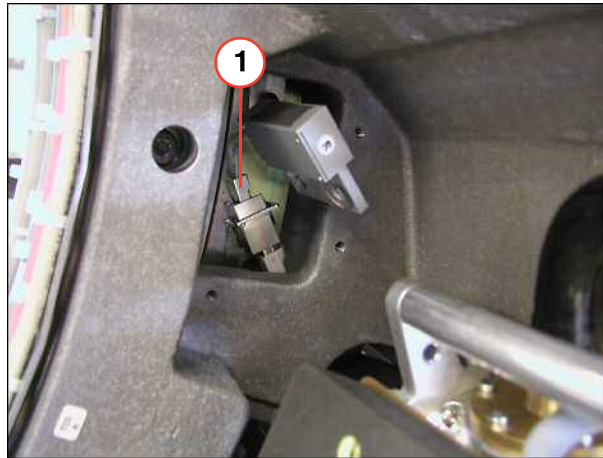


Since the tube collimator is removed, the gantry is no longer balanced. Be careful when moving the rotating part of the gantry.

6. After the correct position has been reached, secure against unintended rotation by using the front safety bolt.

7. Disconnecting the LPS plug.

Fig. 60: LPS unit, removable

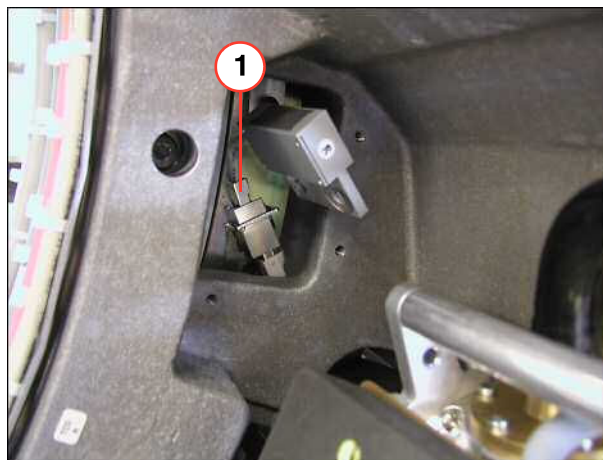


- Disconnect connector (item 1) and remove LPS unit.

3.6.4 Installation

1. Connect the LPS plug.

Fig. 61: LPS unit, removable



- Plug in the connector (item 1).
2. Unlock the rotation safety bolt and rotate the service opening slowly upwards until the LPS unit mounting place appears (approximates the 3 o'clock position).

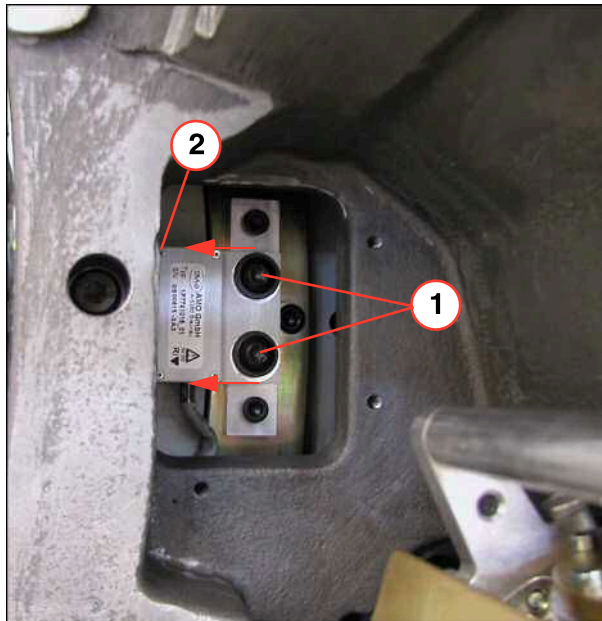


Since the tube collimator is removed, the gantry is no longer balanced. Be careful when moving the rotating part of the gantry.

3. After the correct position has been reached, secure against unintended rotation by using the front safety bolt.

4. Installing the LPS unit.

Fig. 62: LPS unit adjustment



- Install the 2 holding screws of the LPS unit (item 1), but do not tighten them at this time.
- Install the enclosed adjustment foil (delivered with the part) between the LPS unit and the measuring tape (item 2).
- Press the LPS unit against the foil (arrows) and tighten the 2 M8*14mm holding screws (item 1) using a torque of **4 Nm**.
- Remove the foil carefully.

5. Reinstall tube collimator A. Refer to (→ Installation / Page 94) tube collimator.

3.6.5 Start-up

3.6.5.1 Mechanical

1. Remove the data ring protections.
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.

3.6.5.2 Software

n.a

3.6.5.3 Tune-up

1. Run the tune-up sequence as described under **Local Service -> Tune Up -> FRU re-placement**.
2. Close gantry covers when instructed during the guided tour or after performing the guided tour.



Disable gantry rotation using S301 when performing mechanical adjustments inside the gantry during tune-up!

3.6.5.4 Testing

n.a

3.6.6 Final steps

n.a

3.7 Stationary light markers (lateral)

3.7.1 Prerequisites

3.7.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

3.7.1.2 Time and Man Power

- 1 person
- Estimated total repair time: 1.5 hours.

3.7.1.3 Tools and auxiliary equipment

- Standard service kit

3.7.2 Preliminary Steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove the gantry bottom front ring segment (→ Removing / Page 32).
4. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
5. Remove the Plexiglas ring (→ Removing / Page 34).

3.7.3 Removing

1. Remove the lateral markers

Fig. 63: Lateral light marker, left

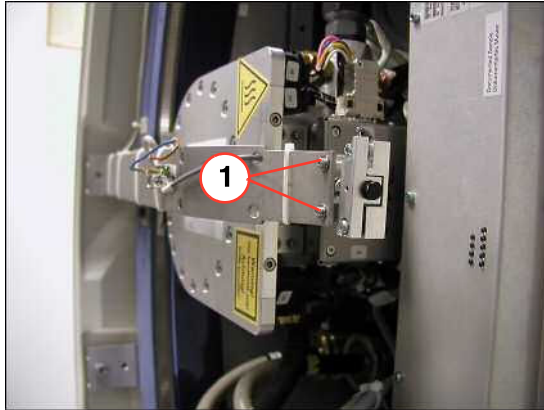
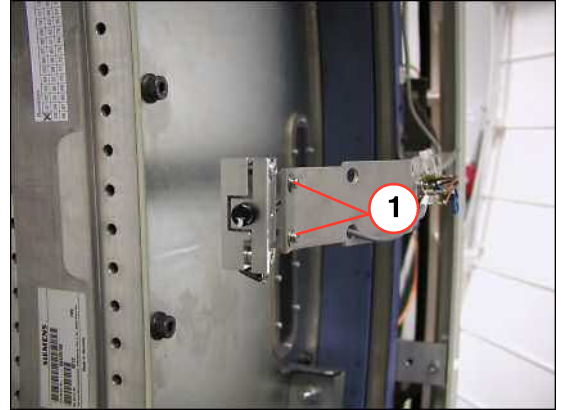


Fig. 64: Lateral light marker, right



1. Remove cable ties as needed.
2. Disconnect the connector .
3. Remove the 2 securing M4 Torx screws (item 1) and remove the light marker .

3.7.4 Installation

1. Installation is in the reverse order of removal.
2. Tighten the holding screws with a torque of **3 Nm**.
3. Connect all cables according to the original configuration.

3.7.5 Start-up

3.7.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7 and F11 in the PDC A
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

Adjustment of the lateral Light Markers

1. Adjustment screws of lateral markers

Fig. 65: Lateral light marker adjustment

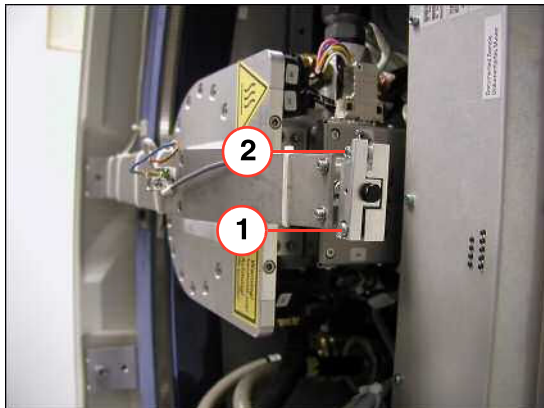
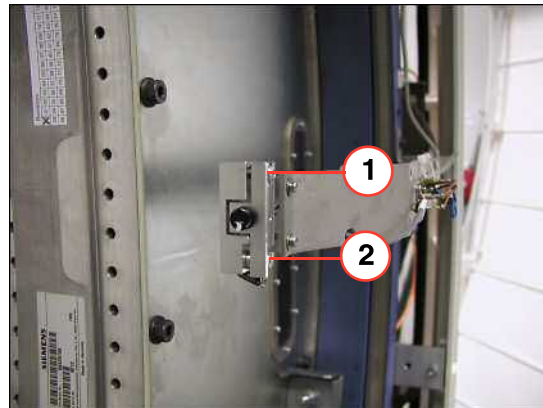


Fig. 66: Lateral light marker adjustment



Adjust the position of the marker via the 2 adjusting M4 Torx screws (item 1/ 2)

1. Adjust the light marker angle with the adjustment M4 Torx screw (item 1).
2. Adjust the light marker position with the adjustment M4 Torx screw (item 2).

2. Adjustment of the lateral markers

1. Install the lifting support at the tabletop (the 2 left/right lines on the support are used for lateral angle adjustment).
2. Switch on the light marker using the light marker switch on the rear ring cover.
3. Adjust the tabletop (height, position). Use the untouched marker beam as a reference and adjust the table until the beam is aligned to the line on the lifting support.
4. Adjust the light marker beam of the exchanged marker until the beam is aligned to the other line on the lifting support. For alignment, use the adjustment screws at the light marker mentioned before.
5. After the adjustment, remove the lifting support.
6. The light marker adjustment is finished.

3.7.5.2 Software

n.a

3.7.5.3 Tune-up

n.a

3.7.5.4 Testing

Both light markers must hit the opposite light marker outlet.

3.7.6 Final steps

n.a

3.8 Stationary light markers (sagittal)

3.8.1 Prerequisites

3.8.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

3.8.1.2 Time and Man Power

- 1 person
- Estimated total repair time: 1.5 hours.

3.8.1.3 Tools and auxiliary Equipment

- Standard service kit

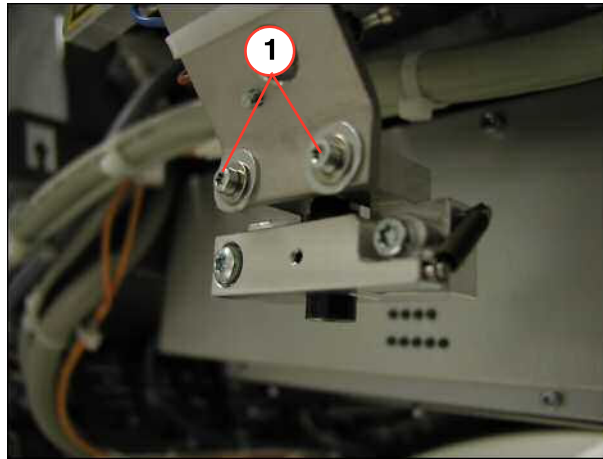
3.8.2 Preliminary Steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove the gantry bottom front ring segment (→ Removing / Page 32).
4. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
5. Remove the Plexiglas ring (→ Removing / Page 34).

3.8.3 Removing

1. Remove the sagittal light marker.

Fig. 67: Removing the sagittal light marker



- Remove cable ties as needed.
- Disconnect the connector .
- Remove the 2 securing M4 Torx screws (item 1) and remove the light marker.

3.8.4 Installation

1. Installation is in reverse order of removal.
2. Tighten the screws with a torque of **3 Nm**.
3. Connect all cables according to the original configuration.

3.8.5 Start-up

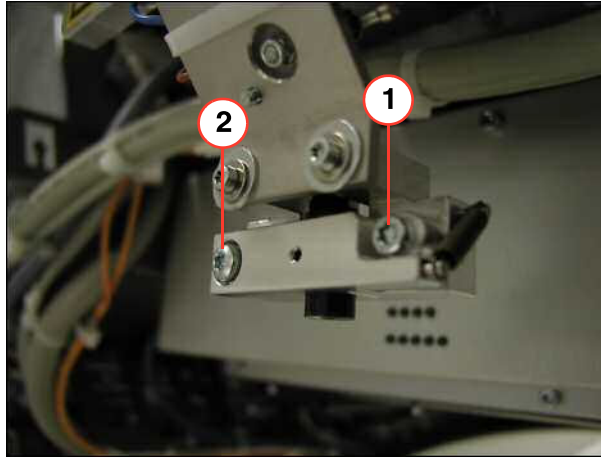
3.8.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7 and F11 in the PDC A
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

Adjustment of the sagittal Light Marker

1. Adjustment screws of the sagittal light marker.

Fig. 68: Sagittal light marker

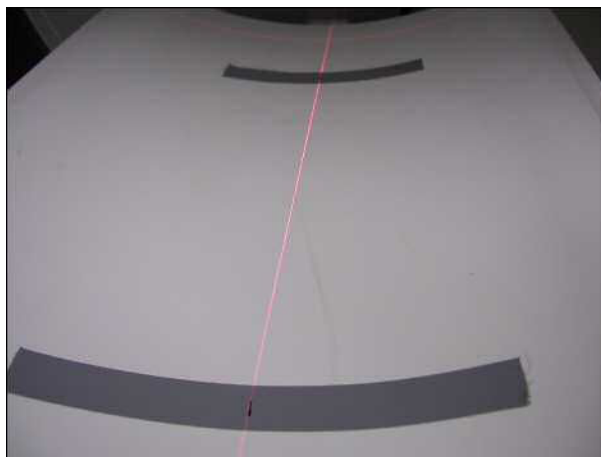


Adjust the position of the marker using the 2 adjusting M4 Torx screws (item 1/2)

- Adjust the light marker position with the adjustment M4 Torx screw (item 1).
- Adjust the light marker angle with the adjustment M4 Torx screw (item 2).

2. Adjusting the sagittal marker

Fig. 69: Sagittal light marker adjustment



1. Place 2 tape strips (distance approx. 80 cm) on the tabletop and mark the center of the tabletop with these 2 positions
2. Switch on the light marker using the light marker switch on the rear ring cover.
3. Adjust the sagittal light marker (angle and position) using the marks on the tabletop as a reference. The light marker is correctly adjusted if the light marker beam is aligned with the tabletop marks.

3.8.5.2 Software
n.a

3.8.5.3 Tune-up
n.a

3.8.5.4 Testing
n.a

3.8.6 Final steps
n.a

3.9 Stationary Light Markers (transaxial)

3.9.1 Prerequisites

3.9.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

3.9.1.2 Time and Man Power

- 1 person
- Estimated total repair time: 1.5 hours.

3.9.1.3 Tools and auxiliary equipment

- Standard service kit

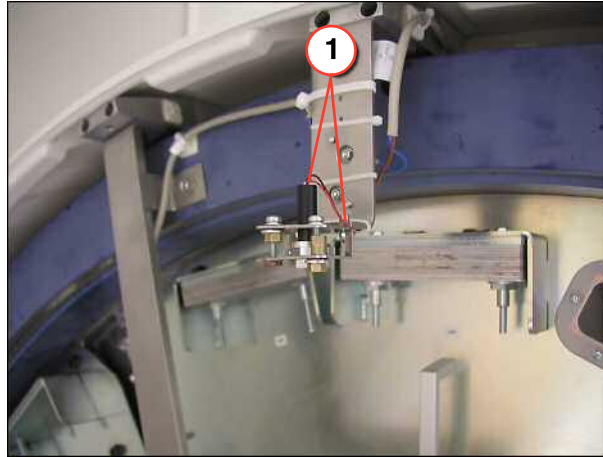
3.9.2 Preliminary Steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove the gantry bottom front ring segment (→ Removing / Page 32).
4. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
5. Remove the Plexiglas ring (→ Removing / Page 34).

3.9.3 Removing

1. Remove the transaxial light marker.

Fig. 70: Transaxial light marker



- Remove cable ties as needed.
- Disconnect the connector .
- Remove the 2 securing M4 Torx screws (item 1) and remove the light marker.

3.9.4 Installation

1. Installation is in reverse order of removal.
2. Tighten the screws with a torque of **3 Nm**.
3. Connect all cables according to the original configuration.

3.9.5 Start-up

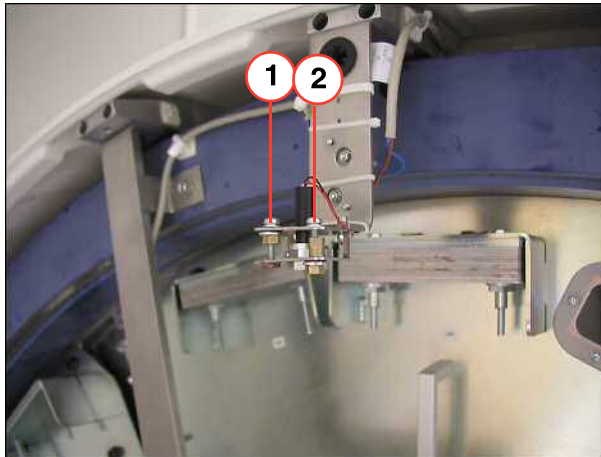
3.9.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7 and F11 in the PDC A
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

Adjustment of the transaxial Light Marker

1. Adjustment screws of the transaxial light marker.

Fig. 71: Transaxial light marker



Adjust the position of the marker via the 2 adjusting screws (item 1/ 2)

- Adjust the light marker angle with the adjustment screw (item 1).
- Adjust the light marker in the Z-position with the adjustment screw (item 2).

2. Adjustment of the transaxial light marker

Fig. 72: Transaxial light marker adjustment

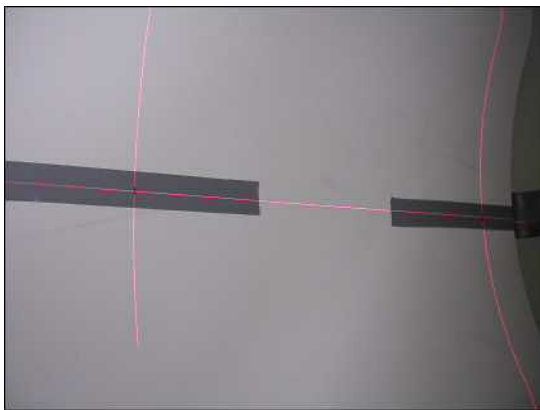


Fig. 73: Transaxial light marker adjustment



1. Switch on the light marker using the light marker switch on the rear ring cover.
2. Raise the table height to approx. 110 and feed the tabletop into the scan field. Place some tape on the tabletop and mark the position of the rotating markers on the tabletop.
3. Adjust the transaxial marker so that it is 342 mm from the rotating marker.
4. Adjust the transaxial marker so that it is in parallel 342 mm in distance to the rotating gantry.

3.9.5.2 Software
n.a

3.9.5.3 Tune-up
n.a

3.9.5.4 Testing
n.a

3.9.6 Final steps
n.a

3.10 Rotating Light Markers

3.10.1 Prerequisites

3.10.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

3.10.1.2 Time and Man Power

- 1 person
- Estimated total repair time: 1.5 hours.

3.10.1.3 Tools and auxiliary Equipment

- Standard service kit

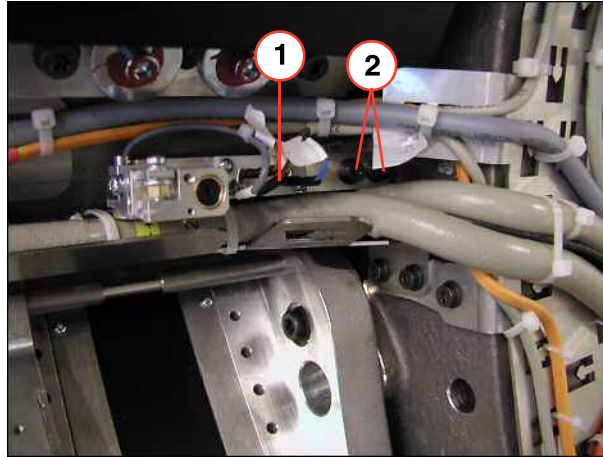
3.10.2 Preliminary Steps

1. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
2. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove the gantry bottom front ring segment (→ Removing / Page 32).
4. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
5. Remove the Plexiglas ring (→ Removing / Page 34).

3.10.3 Removing

1. Removing the light marker

Fig. 74: Light marker, rotating



- Remove the connectors at the light marker (item 1).
- Remove the two attachment bolts M6* 20 mm (item 2) and remove the complete light marker assembly with the mounting bracket.

3.10.4 Installation

1. Install the light marker in the reverse order. Tighten the attachment bolts with a torque of **10 Nm**.

3.10.5 Start-up

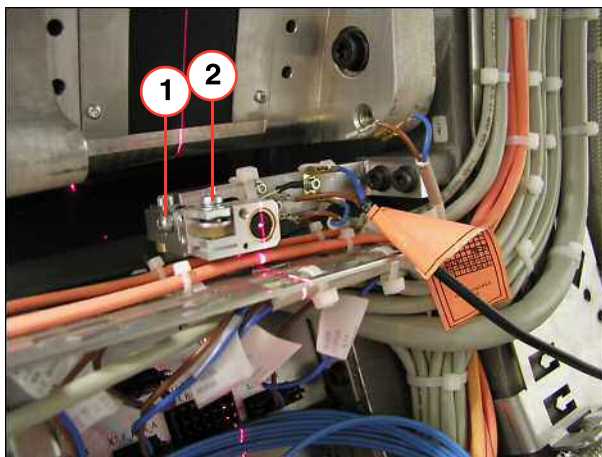
3.10.5.1 Mechanical

1. Switch on breakers F2, F3, F5, F6, F7 and F11 in the PDC A
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
2. Switch on the system (**SYSTEM/ON**) at the control box.

Adjustment of the rotating Light Markers

1. Adjustment screws of rotation light marker
 - » The adjustment screws for all 4 rotating light markers are similar.

Fig. 75: Rotating light marker adjustment



Adjust the position of the marker using the 2 adjusting screws (item 1/ 2) using a TX20 L-key

- Adjust the light marker angle with the adjustment screw (item 1).
- Adjust the light marker position with the adjustment screw (item 2).

2. Adjustment of the rotating markers

Fig. 76: Rotating light marker angle

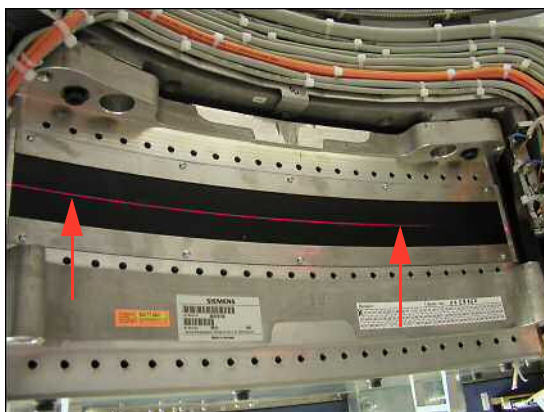
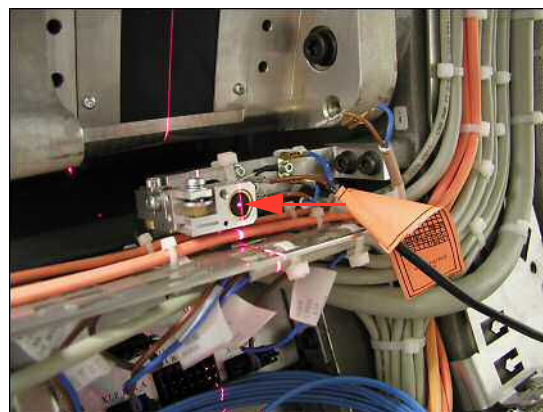


Fig. 77: Rotating light marker position



1. Switch on the light marker using the light marker switch on the rear ring cover.
2. The changed light marker has to be adjusted using the 3 other light marker beams as a reference.
 - Adjust the angle of the light marker according the other 3 light marker lines (see left image).
 - Adjust the position of the light marker using the opposite LM as a reference (see right image).
 - » Finally only one line should be seen when all 4 light markers are switched on.

3.10.5.2 Software
n.a

3.10.5.3 Tune-up
n.a

3.10.5.4 Testing
n.a

3.10.5.5 Final steps
n.a

3.11 Stationary Data Ring (with removable top front ring segment)

3.11.1 Prerequisites

3.11.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the “General” section of this document and/or the “Product specific safety notes”.



Follow ESD - rules when touching data transmission parts

3.11.1.2 Tools and auxiliary Equipment

- Standard tool kit with extensions.
- Alignment pins (delivered with spare part)
- Alignment jigs (delivered with the system)
- Torque wrench
- Loctite 243

3.11.1.3 Time and Man Power

- 2 persons for removing and installing the data ring
- Estimated total repair time: 2.0 hours.

3.11.2 Preliminary Steps

1. Position the tube lifting device at the front end of the patient table and switch on the light markers.
2. Mark the positions of the lateral light markers at the lifting device using strips of tape and a pen. **Write down the value for the table height. Read it off the gantry display**

3. Mark the positions of the sagittal and transaxial light markers at the patient table using strips of tape and a pen. **Write down the value for the horizontal table position. Read it off the gantry display**



During the procedure, all stationary light markers have to be removed. As a result, it is necessary to mark the light marker positions and write down the table positions for adjusting the light markers later on.

4. Move the table with the installed tube lifting device carefully out of the gantry.
5. Allow at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Note that some system components may be very hot, i.e., the Straton tube may reach 130 degree Celsius
6. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down)
7. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
8. Disable gantry rotation using rotation enable switch S301.
9. Remove the bottom front ring segment (→ Removing / Page 32)
10. Remove the top front ring segment from gantry (→ Removing / Page 34)
11. Turn the gantry by hand until **DMS A** is in the 6 o'clock position
12. Secure against unintended rotation by using the front safety bolt (→ Safety bolts / Page 21)

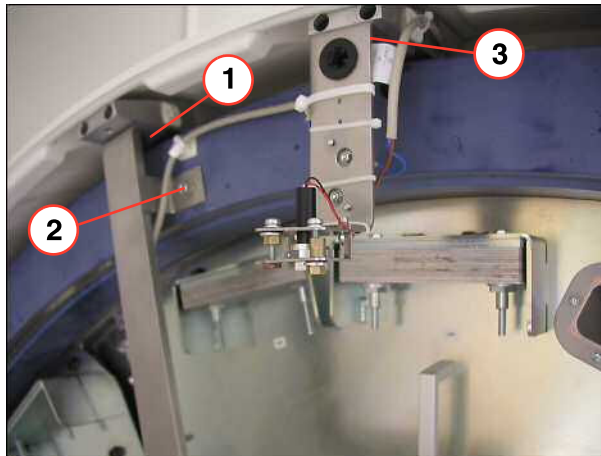
3.11.3 Removing

1. Remove the 4 stationary light marker holding brackets including the light markers.



The following image shows the brackets of the sagittal (item 1/2) and transaxial (item 3) light markers. Removal of the lateral light markers is similar to the sagittal one.

Fig. 78: Light markers, transaxial and sagittal



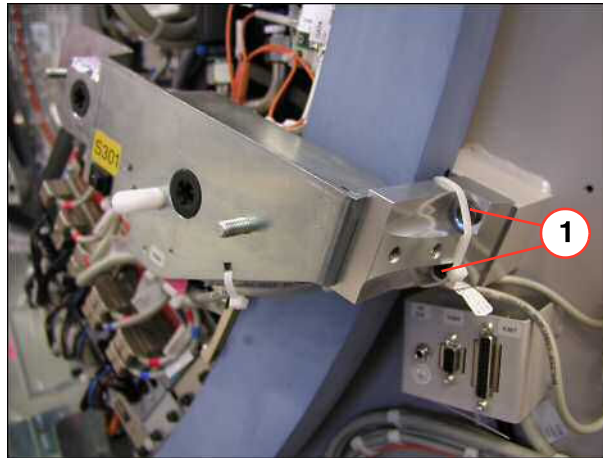
- Remove all cables on the transaxial light marker. Remove cable ties as necessary. Remove the bracket with the transaxial light marker (item 3)
- Remove the bracket with the sagittal light marker by removing the screws (item 1 and item 2) at both sides. Remove cable ties as necessary and lift the whole bracket to the gantry top. Secure the bracket with some tape to gantry top.
- Remove the 2 brackets with the lateral light markers in the same way as the sagittal light marker. Remove cable ties as necessary and carefully secure the brackets with some tape or cable ties to the gantry frame.



Carefully handle the brackets with the light markers and do not misadjust the light markers on the bracket!

3. Remove the brackets holding the cover. (The image below shows the right bracket.)

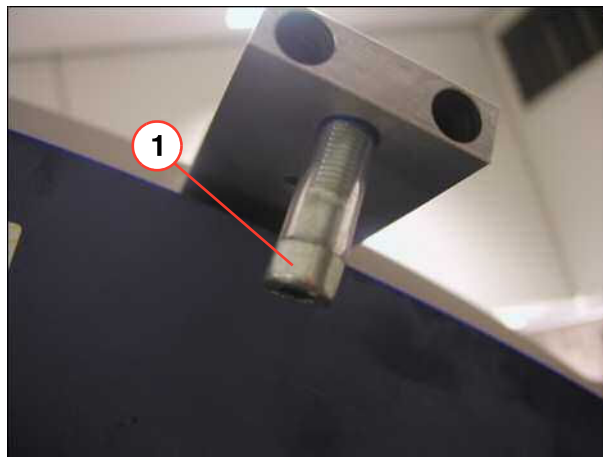
Fig. 79: Removing the holding bracket, right cover



- Remove the 2 (M6) screws from the right bracket (item 1) using a Torx key T30
- Remove cable ties as necessary and carefully secure the bracket with some tape or cable ties to the gantry frame.
- Remove the 2 (M8) screws from the left bracket using a Torx key T40 and an extension.

4. Remove the bolts securing the cover

Fig. 80: Bolt



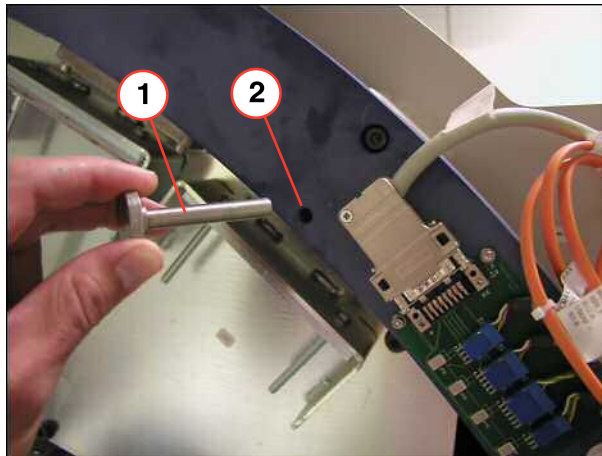
- Remove the 4 bolts (e.g.item1) from the gantry using a 6 mm Allen key

5. Inserting the guide pins



DMS A has to be in the 6 o'clock position during the entire procedure. The guide pins have to be installed through the holes of the stationary data ring to the rotating data ring. If the pins do not fit in the rotating ring, wiggle the the rotating part slightly.

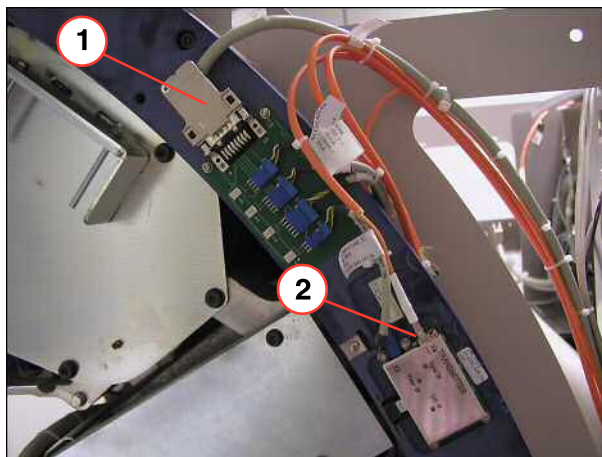
Fig. 81: Guide pins



- Insert the 4 guide pins (e.g. item 1) in the 4 holes (e.g. item 2) of the stationary data ring.

6. Remove the connectors

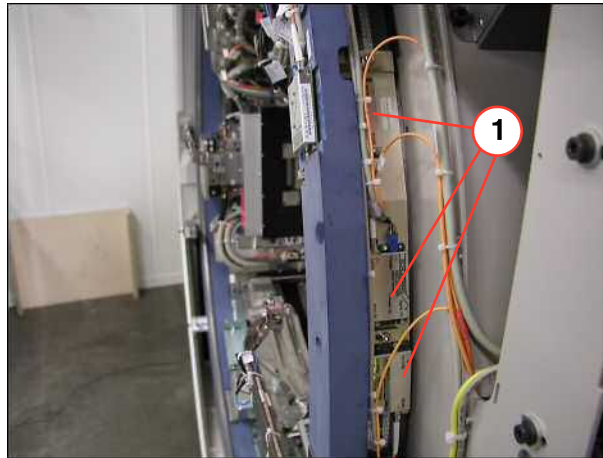
Fig. 82: Connectors, wiring interface, and transmitter



- Remove all connectors from the stationary wiring interface (item 1)
- Remove all connectors from the stationary transmitter (item 2)

7. Remove the connectors at the triple receiver assembly

Fig. 83: Connectors at receivers

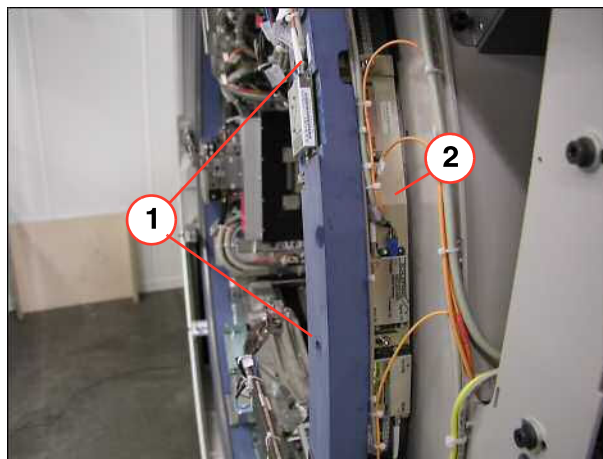


- Remove all connectors from the stationary receivers (item 1)

8. Remove the cable ties as necessary

9. Remove the triple receiver assembly (do not yet install it at the new ring)

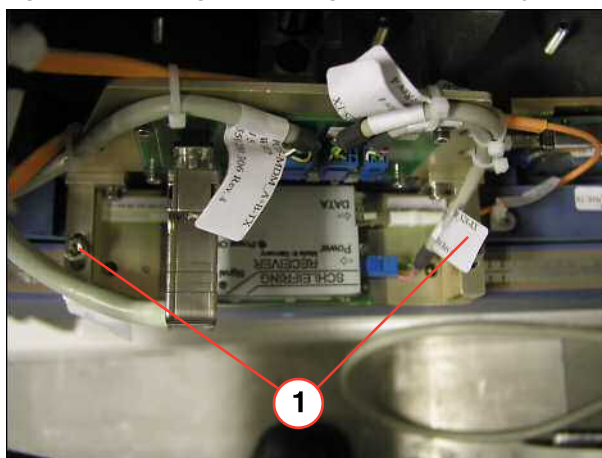
Fig. 84: Removing the triple receiver assembly



- Loosen the 2 (M5) screws (adjustment screws for triple receiver Z-position) on top of the tripple receiver assembly (item 2). (Necessary to get enough space for removal)
- Remove the triple receiver assembly (item 2) by removing the 2 (M4x20) screws using a Torx key T20 (item 1)

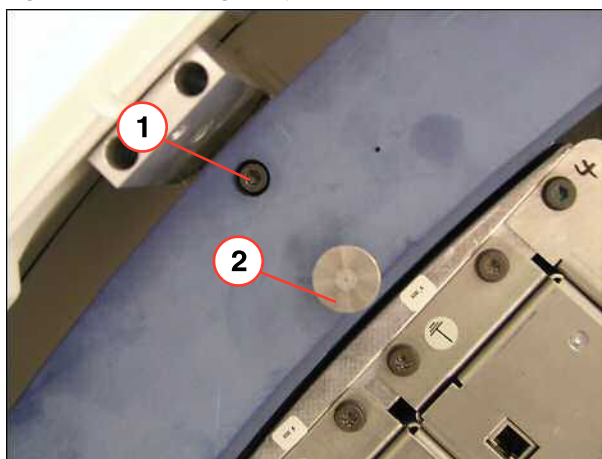
10. Remove the rotating receiver assembly (to prevent damage)
(if not already performed during the instructions for "Rotating Data Ring")

Fig. 85: Removing the rotating receiver assembly



- Disconnect cables and cable ties as necessary.
 - Remove the receiver assembly by removing the 2 screws (item 1)
11. Remove the stationary transmitter and install it at the new data ring. Refer to the sections on "Removing/Installing the Stationary Transmitter" (→ Removing / Page 177) (not necessary if the rotating data ring has to be replaced)
 12. Remove the stationary wiring interface and install it at the new data ring. Refer to sections on "Removing/Installing the wiring interface" (→ Removing / Page 195) (not necessary if the rotating data ring has to be replaced)
 13. Remove the stationary data ring

Fig. 86: Screw and guide pin



- Remove the 8 (M6x35) holding screws (e.g. item 1) using a Torx key T30, while holding the ring.
- Take the whole ring with 2 persons and move it out slightly on the guide pins (e.g. item 2)

3.11.4 Installation

1. Lift the data ring to the gantry front and insert the guide pins in the ring.



Position the data ring with the holes for the triple receiver assembly at approx. 2 o'clock. Refer to the holes in the stationary data ring for orientation

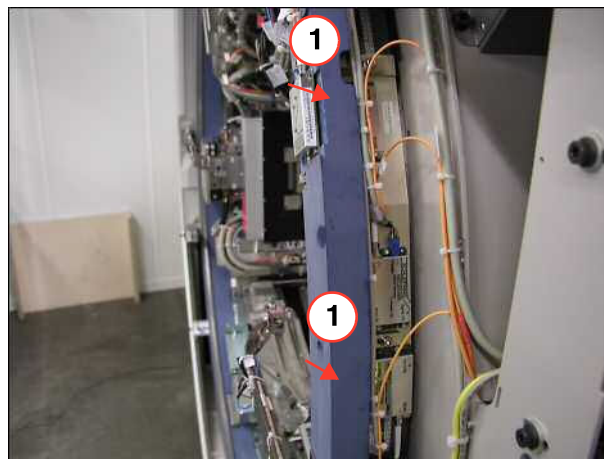
2. Install the new data ring using the guide pins. Be careful not to damage or scratch the surface of the antenna.
3. Use Loctite 243 and install the 8 (M6x35) screws while holding the ring. Tighten the screws with a torque of **10Nm**
4. Remove the 4 guide pins.



Do not forget to remove the guide pins!

5. Install the triple receiver assembly

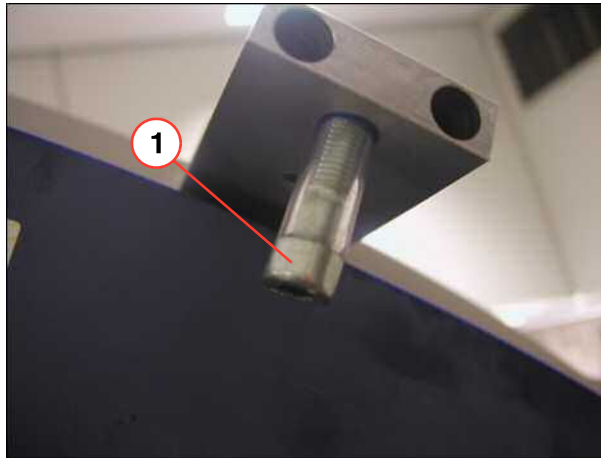
Fig. 87: Installing the triple receiver assembly



- Carefully mount the triple receiver assembly to the stationary antenna ring.
 - Move the triple receiver assembly out (item 1) as much as possible (long holes in data ring) and tighten the 2 screws by hand. (Adjustment later on)
6. Connect the cables to the wiring interface. Connect the transmitter and receivers according to the original configuration.
 7. Install the rotating receiver assembly and tighten the 2 screws by hand. (Adjust later on). Connect the cables according to the original configuration.
 8. Install cable ties where you removed them before

9. Install the bolts securing the cover

Fig. 88: Bolt



- Install the 4 bolts (e.g.item 1) using a 6 mm Allen key and Loctite 243
- Tighten the bolts unless the plastic part contacts the gantry

10. Install the 2 cover holding brackets (mounting for ring cover left and right).

- Tighten the M8 screws at the left bracket with a torque of **31 Nm**
- Tighten the M6 screws at the right bracket with a torque of **14 Nm**

11. Install the 4 stationary light marker holding brackets including the light markers.

- Tighten the M8 screws attaching the brackets to the gantry frame with a torque of **31Nm**
- Tighten the M6 screws attaching the brackets to the stationary data ring with a torque of **10Nm**

12. Reconnect the cables to the transaxial light marker

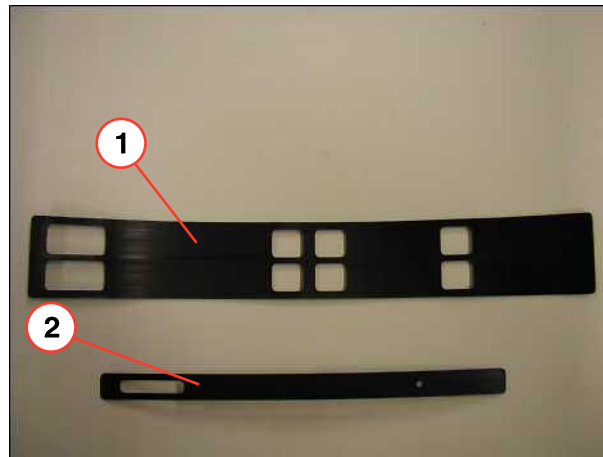
3.11.5 Start-up

3.11.5.1 Mechanical

1. Unlock the safety bolt for gantry rotation

2. Use alignment jigs to verify and adjust the distance of the transmitter carriers to the data ring

Fig. 89: Alignment jigs



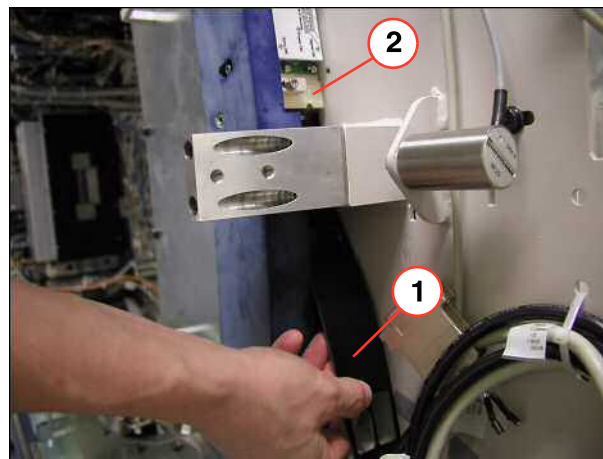
- The large jig (item 1) is used to verify and adjust the distance between the stationary receiver carrier and the rotating data ring
- The small jig (item 2) is used to verify and adjust the distance between the rotating receiver carrier and the stationary data ring

3. Adjust the triple receiver assembly (distance to ring)



The adjustment is correct if the alignment jig can be removed without using force. During the adjustment procedure, rotate the gantry by hand and check at various gantry positions. The alignment jig has to fit all gantry positions.

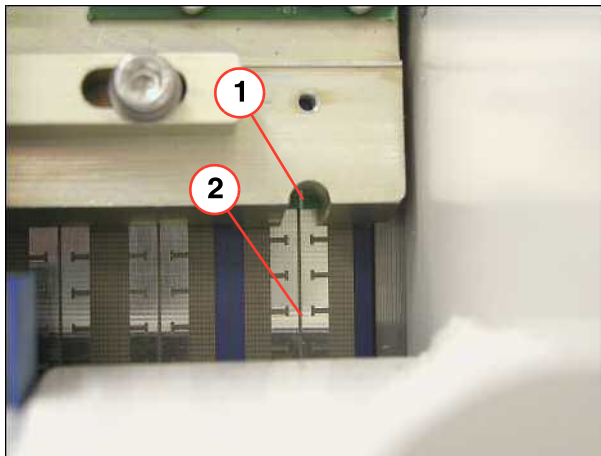
Fig. 90: Alignment jig, large



- Insert the large alignment jig (item 1) between the rotating antenna ring and the triple receiver carrier (item 2)
- Adjust the distance of the triple receiver carrier to the rotating data ring. Refer to (→ Mechanical / Page 185) adjustment of stationary receiver carrier.

4. Check the adjustment of the triple receiver assembly in the Z-direction

Fig. 91: Adjustment mark (Z-position)



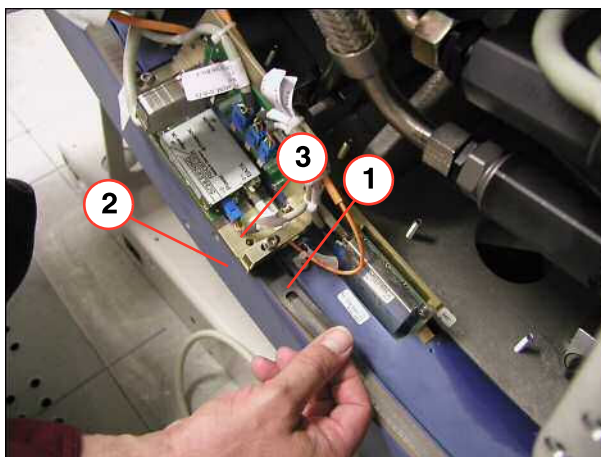
- The 2 marks at the bottom and top (e.g. lower mark, item 1) of the triple receiver assembly have to match the line in the structure of the rotating antenna ring (item 2).
- If the marks do not match, refer to the section on the "adjustment of the stationary receiver carrier" (→ Mechanical / Page 185)

5. Check the adjustment of the rotating receiver assembly (distance to ring)



The adjustment is correct, if the alignment jig can be removed without using force. During the adjustment procedure, rotate the gantry by hand and check at various gantry positions. The alignment jig has to fit all gantry positions.

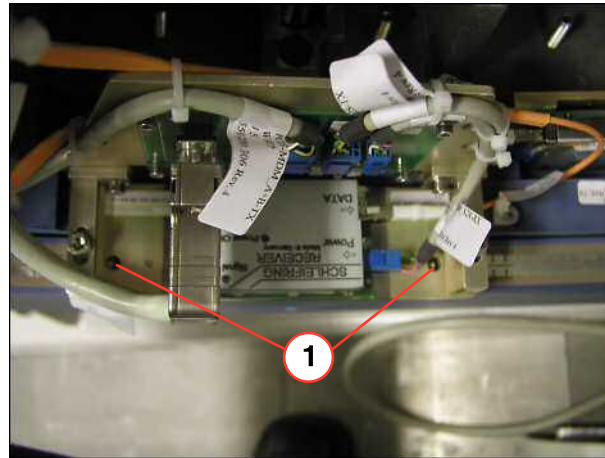
Fig. 92: Alignment jig (small)



- Insert the small alignment jig (item 1) between the stationary antenna ring (item 2) and the receiver carrier (item 3)
- If the adjustment (distance to ring) is too wide or too narrow, refer to (→ Mechanical / Page 191) adjustment of rotating receiver carrier

6. Check the adjustment of the rotating receiver assembly in the Z-direction

Fig. 93: Adjustment marks (Z-position)



- The 2 marks (item 1) of the rotating receiver assembly have to match the structure of the stationary antenna ring.
- If the marks do not match, refer to (→ Mechanical / Page 191) adjustment of the rotating receiver carrier.

7. Rotate the Gantry by hand and check for unusual noise

8. Switch on breakers F2, F3, F5, F6, F7 and F11 in the PDC A

- » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.

9. Switch on the system (**SYSTEM/ON**) at the control box.

10. Check the adjustment of the light markers.

- Move the table in using the panel at the table and check if the light markers match the lines on the tapes. **Table positions must be the same as that used before replacing the light markers.**

If the lines do not match the light markers, adjust the light markers to the lines. Refer to , (→ Adjustment of the lateral Light Markers / Page 118), (→ Adjustment of the sagittal Light Marker / Page 122), (→ Adjustment of the transaxial Light Marker / Page 126), for detailed adjustment instructions.

11. After finishing the adjustment, switch off the gantry (**SYSTEM/OFF**) at the Control Box

12. Enable rotation using the rotation enable switch S301

13. Install all covers

14. Switch on the system (**SYSTEM/ON**) at the control box

3.11.5.2 Tune-up

n.a

3.11.5.3 Testing

Run the Fastlink Test and the Disk Test in Local Service->Test Tools

3.11.6 Final steps

n.a.

3.12 Stationary Data Ring (with liftable top front ring segment)

3.12.1 Prerequisites

3.12.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".



Follow ESD - rules when touching data transmission parts

3.12.1.2 Tools and auxiliary Equipment

- Standard tool kit with extensions.
- Alignment pins (delivered with spare part)
- Alignment jigs (delivered with the system)
- Torque wrench
- Loctite 243

3.12.1.3 Time and Manpower

- 2 persons for removing/installing the data ring and removing the front top ring segment.
- Estimated total repair time: 3.0 hours.

3.12.2 Preliminary Steps

1. Position the tube lifting device at the front end of the patient table and switch on the light markers.
2. Mark the positions of the lateral light markers at the lifting device using strips of tape and a pen. **Write down the value for the table height. Read it off the gantry display**

3. Mark the positions of the sagittal and transaxial light markers at the patient table using strips of tape and a pen. **Write down the value for the horizontal table position. Read it off the gantry display**



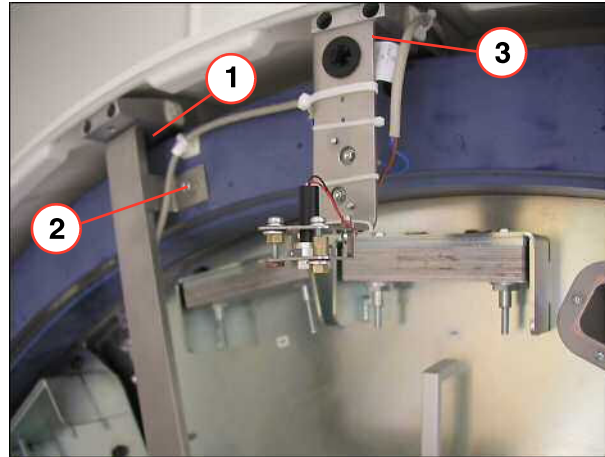
During the procedure, all stationary light markers have to be removed. As a result, it is necessary to mark the light marker positions and write down the table positions for adjusting the light markers later on.

4. Move the table with the installed tube lifting device carefully out of the gantry.
5. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Note that some system components may be very hot , i.e., the Straton tube may reach 130 degrees Celsius
6. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down)
7. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
8. Disable gantry rotation using rotation enable switch S301.
9. Remove the bottom front ring segment (→ Removing / Page 32)
10. Remove the liftable top front ring segment from gantry (→ Removing / Page 37)
11. Turn the gantry by hand until **DMS A** is in the 12 o'clock position
12. Secure against unintended rotation by using the front safety bolt (→ Safety bolts / Page 21)

3.12.3 Removing

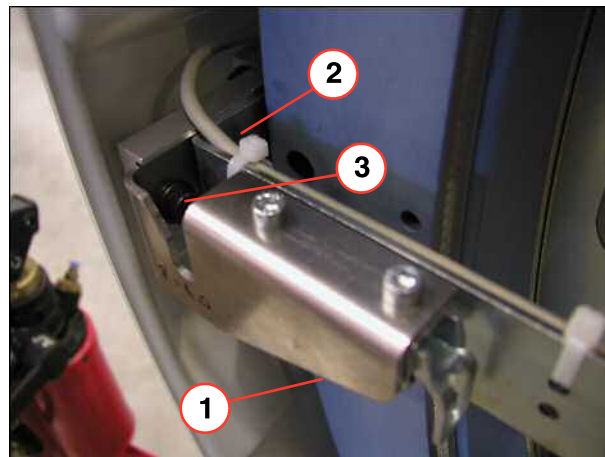
1. Remove the sagittal and transaxial light marker holding brackets including the light markers.

Fig. 94: Light markers, transaxial and sagittal



- Remove all cables on the transaxial light marker. Remove cable ties as necessary. Remove the bracket with the transaxial light marker (item 3)
 - Remove the bracket with the sagittal light marker by removing the screws (item 1 and item 2) at both sides. Remove cable ties as necessary and lift the whole bracket to gantry top. Secure the bracket with some tape to gantry top.
2. Remove the holding bracket of the left lateral light marker as well as the light marker.

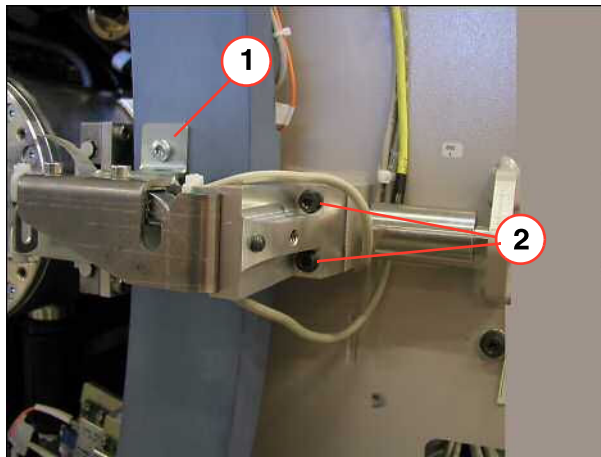
Fig. 95: Removing the left lateral lightmarker bracket



- Remove the M6 screw (item 1) from the stationary data ring.
- Remove the M8 screws (item 2 and item 3). Start with screw item 2.
- Remove the securing lock (for cover of the supporting stands) from gantry.
- Remove cable ties as necessary and carefully secure the bracket with some tape or cable ties at the gantry side.

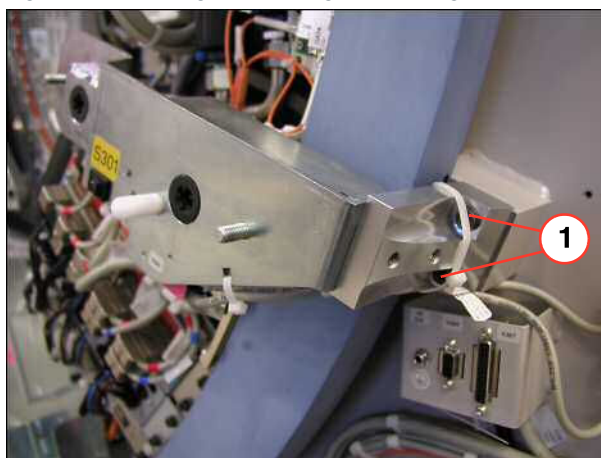
3. Remove the holding bracket of the right lateral light marker including the light marker.

Fig. 96: Right lateral lightmarker bracket



- Remove the M6 screw (item1) from the stationary data ring.
 - Remove the M6 screws (item2).
 - Remove cable ties as necessary and carefully secure the bracket with some tape or cable ties to the gantry frame.
4. Remove the brackets holding the cover. (The image below shows the right bracket.)

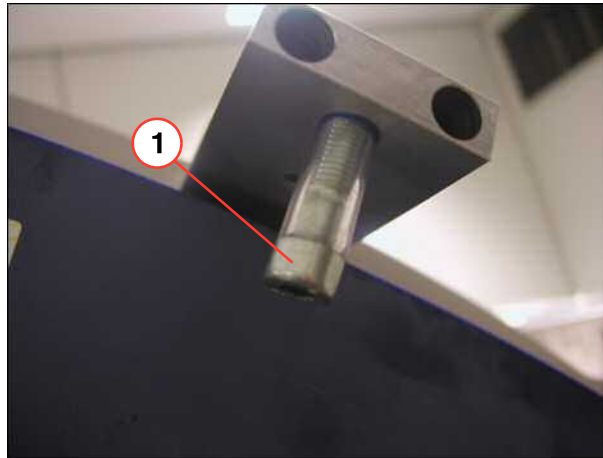
Fig. 97: Removing the holding bracket, right cover



- Remove the 2 (M6) screws from the right bracket (item 1) using a Torx key T30
- Remove cable ties as necessary and carefully secure the bracket with some tape or cable ties to the gantry frame.
- Remove the 2 (M8) screws from the left bracket using a Torx key T40 and an extension.

5. Remove the bolts securing the cover

Fig. 98: Bolt



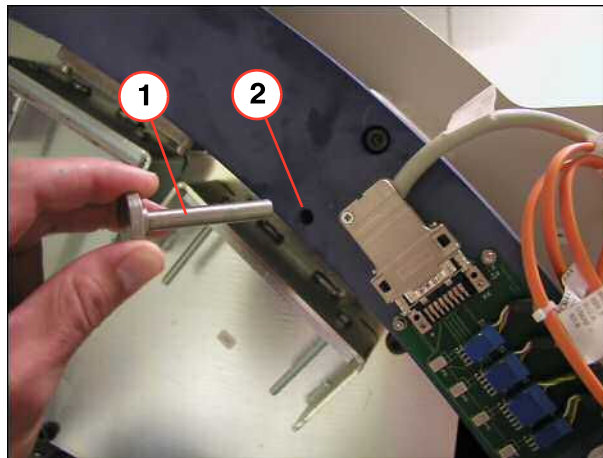
- Remove the 4 bolts (e.g.item1) from the gantry using a 6 mm Allen key

6. Inserting the guide pins



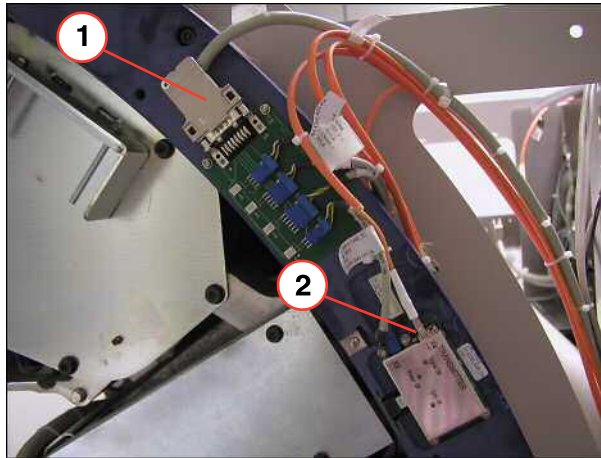
DMS A has to be in the 12 o'clock position during the entire procedure. The guide pins have to be installed through the holes of the stationary data ring in the rotating data ring. If the pins do not fit in the rotating ring, wiggle the rotating part slightly.

Fig. 99: Guide pins



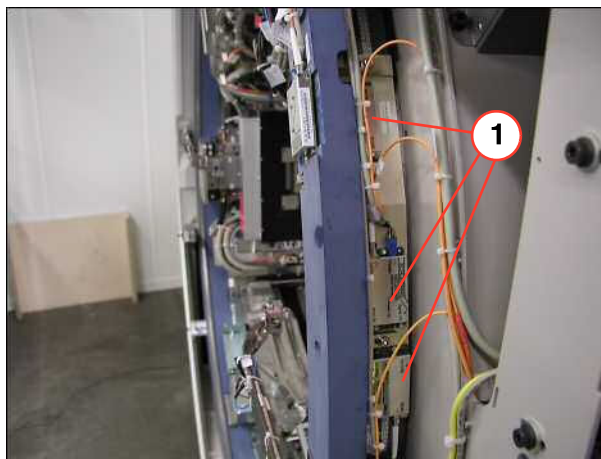
- Insert the 4 guide pins (e.g. item1) in the 4 holes (e.g. item 2) of the stationary data ring.

7. Remove the connectors

Fig. 100: Connectors, wiring interface, and transmitter

- Remove all connectors from the stationary wiring interface (item 1)
- Remove all connectors from the stationary transmitter (item 2)

8. Remove the connectors at the triple receiver assembly

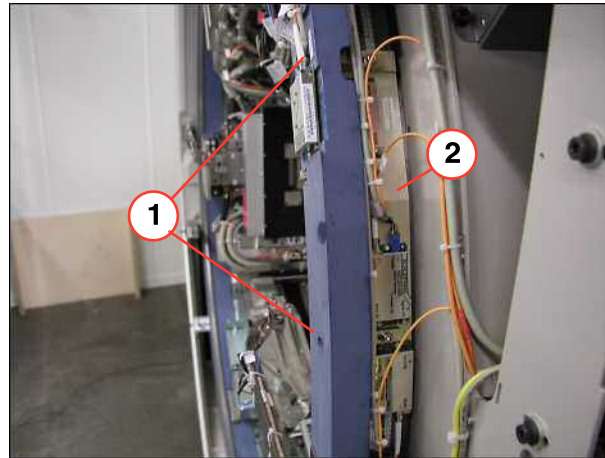
Fig. 101: Connectors at receivers

- Remove all connectors from the stationary receivers (item 1)

9. Remove the cable ties as necessary

10. Remove the triple receiver assembly (do not yet install it at the new ring)

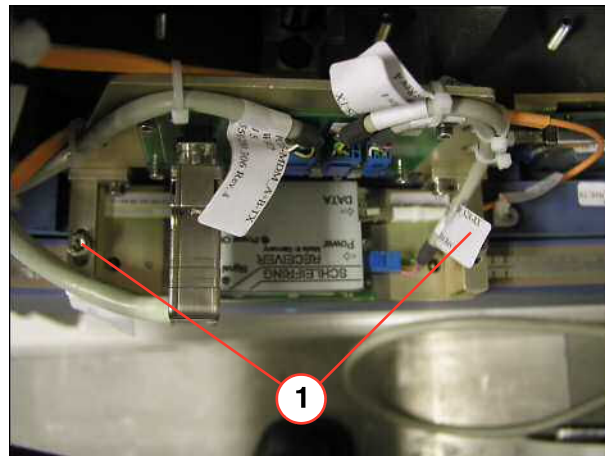
Fig. 102: Removing the triple receiver assembly



- Loosen the 2 (M5) screws (adjustment screws for triple receiver Z-position) on top of the tripple receiver assembly (item 2). (Necessary to get enough space for removal)
- Remove the triple receiver assembly (item 2) by removing the 2 (M4x20) screws (item 1)

11. Remove the rotating receiver assembly (to prevent damage)
(if not already performed during the instructions for "Rotating Data Ring")

Fig. 103: Removing the rotating receiver assembly

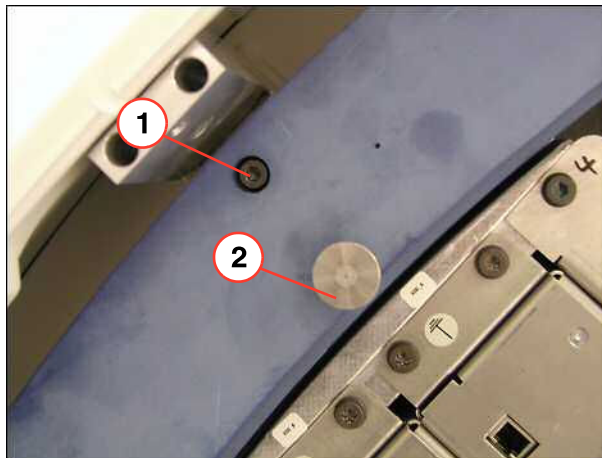


- Disconnect cables and cable ties as necessary.
- Remove the receiver assembly by removing the 2 screws (item 1)

12. Remove the stationary transmitter and install it at the new data ring. Refer to the sections on "Removing/Installing the Stationary Transmitter" (→ Removing / Page 177) (not necessary if the rotating data ring has to be replaced)
13. Remove the stationary wiring interface and install it at the new data ring. Refer to sections on "Removing/Installing the wiring interface" (→ Removing / Page 195) (not necessary if the rotating data ring has to be replaced)

14. Remove the stationary data ring

Fig. 104: Screw and guide pin



- Remove the 8 (M6x35) holding screws (e.g. item 1) using a Torx key T30
- Hold the whole ring with 2 persons and remove the guide pins (e.g. item 2)
- Move the whole ring out slightly. First move the ring down and then out.

3.12.4 Installation



Position the data ring with the holes for the triple receiver assembly at approx. 2 o'clock. Refer to the holes in the stationary data ring for orientation

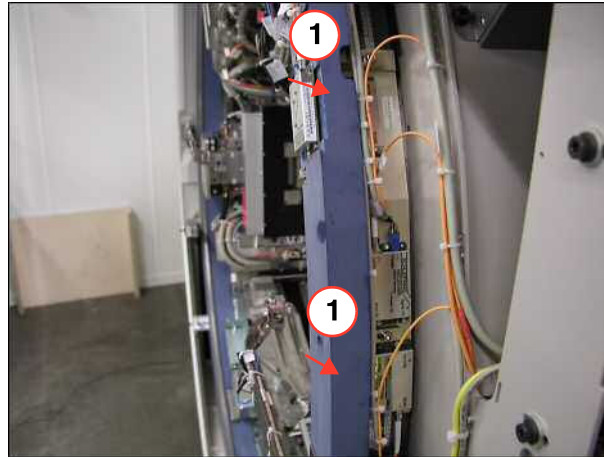
1. Lift the data ring into the gantry and insert at least the 2 upper guide pins in the ring. Be careful not to damage the surface of the antenna.
2. Use Loctite 243 and install the 8 (M6x35) screws while holding the ring. Tighten the screws with a torque of **10Nm**
3. Remove the guide pins.



Do not forget to remove the guide pins!

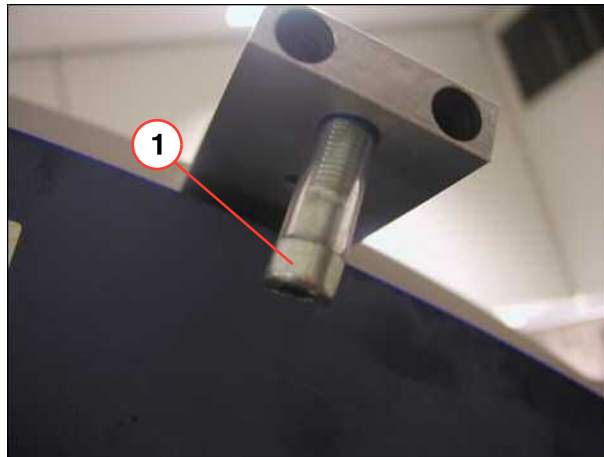
4. Install the triple receiver assembly

Fig. 105: Installing the triple receiver assembly



- Carefully mount the triple receiver assembly to the stationary antenna ring.
 - Move the triple receiver assembly out (item 1) as much as possible (long holes in data ring) and tighten the 2 screws by hand. (Adjustment later on)
5. Connect the cables to the wiring interface. Connect the transmitter and receivers according to the original configuration.
 6. Install the rotating receiver assembly and tighten the 2 screws by hand. (Adjust later on). Connect the cables according to the original configuration.
 7. Install cable ties where you removed them before
 8. Install the bolts securing the cover

Fig. 106: Bolt



- Install the 4 bolts (e.g.item 1) using a 6 mm Allen key and Loctite 243
 - Tighten the bolts unless the plastic part contacts the gantry
9. Install the 2 cover holding brackets (mounting for ring cover left and right).
 - Tighten the M8 screws at the left bracket with a torque of **31Nm**
 - Tighten the M6 screws at the right bracket with a torque of **14Nm**

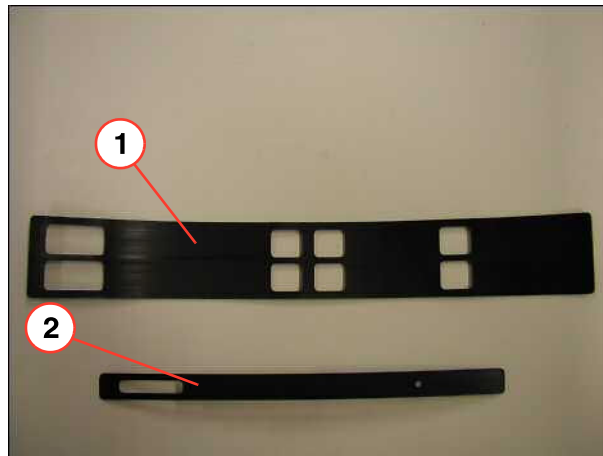
10. Install the 4 stationary light marker holding brackets including the light markers.
 - Tighten the M8 screws attaching the left bracket to the gantry frame with a torque of **31Nm**
 - Tighten the M6 screws attaching the right bracket to the gantry frame with a torque of **14Nm**
 - Tighten the M6 screws attaching the brackets to the stationary data ring with a torque of **10Nm**
11. Reconnect the cables to the transaxial light marker

3.12.5 Start-up

3.12.5.1 Mechanical

1. Unlock the safety bolt for gantry rotation
2. Use alignment jigs to verify and adjust the distance of the transmitter carriers to the data ring

Fig. 107: Alignment jigs

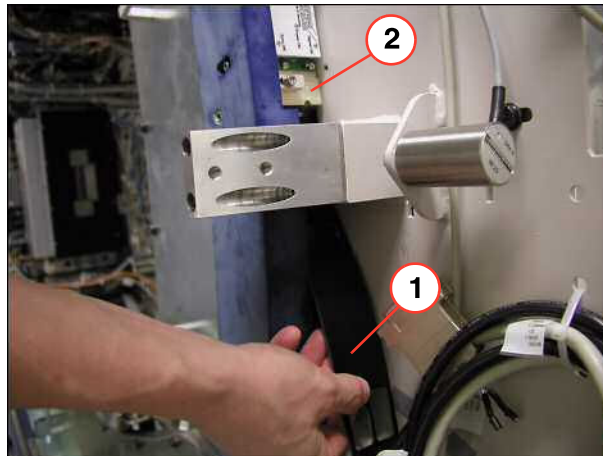


- The large jig (item 1) is used to verify and adjust the distance between the stationary receiver carrier and the rotating data ring
 - The small jig (item 2) is used to verify and adjust the distance between the rotating receiver carrier and the stationary data ring
3. Adjust the triple receiver assembly (distance to ring)



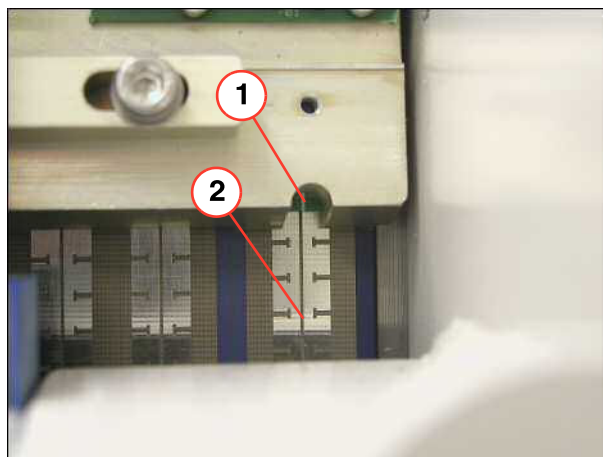
The adjustment is correct if the alignment jig can be removed without using force. During the adjustment procedure, rotate the gantry by hand and check at various gantry positions. The alignment jig has to fit all gantry positions.

Fig. 108: Alignment jig, large



- Insert the large alignment jig (item 1) between the rotating antenna ring and the triple receiver carrier (item 2)
 - Adjust the distance of the triple receiver carrier to the rotating data ring. Refer to (→ Mechanical / Page 185) adjustment of stationary receiver carrier.
4. Check the adjustment of the triple receiver assembly in the Z-direction

Fig. 109: Adjustment mark (Z-position)

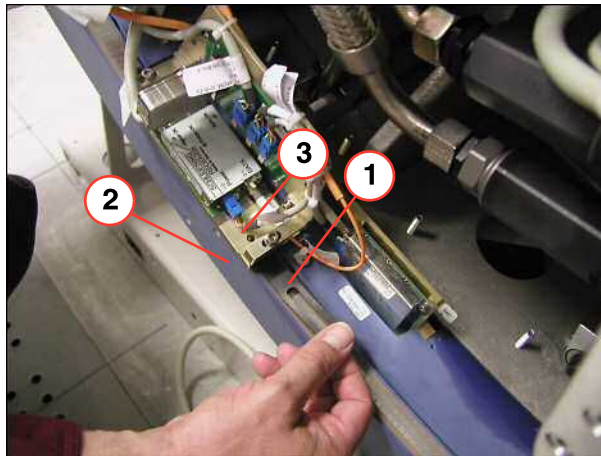


- The 2 marks at the bottom and top (e.g. lower mark, item 1) of the triple receiver assembly have to match the line in the structure of the rotating antenna ring (item 2).
 - If the marks do not match, refer to the section on the "adjustment of the stationary receiver carrier" (→ Mechanical / Page 185)
5. Check the adjustment of the rotating receiver assembly (distance to ring)



The adjustment is correct, if the alignment jig can be removed without using force. During the adjustment procedure, rotate the gantry by hand and check at various gantry positions. The alignment jig has to fit all gantry positions.

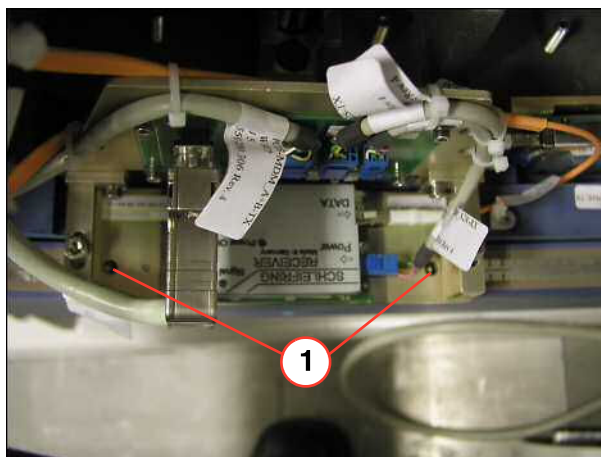
Fig. 110: Alignment jig (small)



- Insert the small alignment jig (item 1) between the stationary antenna ring (item 2) and the receiver carrier (item 3)
- If the adjustment (distance to ring) is too wide or too narrow, refer to (→ Mechanical / Page 191) adjustment of rotating receiver carrier

6. Check the adjustment of the rotating receiver assembly in the Z-direction

Fig. 111: Adjustment marks (Z-position)



- The 2 marks (item 1) of the rotating receiver assembly have to match the structure of the stationary antenna ring.
- If the marks do not match, refer to (→ Mechanical / Page 191) adjustment of the rotating receiver carrier.

7. Rotate the Gantry by hand and check for unusual noise

8. Switch on breakers F2, F3, F5, F6, F7 and F11 in the PDC A

- » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.

9. Switch on the system (**SYSTEM/ON**) at the control box.

10. Check the adjustment of the light markers.

- Move the table in using the panel at the table and check if the light markers match the lines on the tapes. **Table positions must be the same as that used before replacing the light markers.**

If the lines do not match the light markers, adjust the light markers to the lines. Refer to, (→ Adjustment of the lateral Light Markers / Page 118), (→ Adjustment of the sagittal Light Marker / Page 122), (→ Adjustment of the transaxial Light Marker / Page 126), for detailed adjustment instructions.

11. After finishing the adjustment, switch off the gantry (**SYSTEM/OFF**) at the Control Box

12. Install the liftable top front ring segment (→ Installation / Page 40)

13. Enable rotation using the rotation enable switch S301

14. Close all covers

15. Switch on the system (**SYSTEM/ON**) at the control box

3.12.5.2 Tune-up

n.a

3.12.5.3 Testing

Run the Fastlink Test and the Disk Test in Local Service->Test Tools

3.12.6 Final Steps

n.a.

3.13 Rotating data ring

3.13.1 Prerequisites

3.13.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".



Follow ESD - rules when touching data transmission parts

3.13.1.2 Tools and auxiliary Equipment

- Standard tool kit.
- Alignment pins (delivered with spare part)
- Alignment jigs
- Torque wrench
- Loctite 243

3.13.1.3 Time and Man Power

- 2 persons for removing and installing the data ring
- Estimated total repair time: 3.5 hours

3.13.2 Preliminary Steps

1. Position the tube lifting device at the front end of the patient table and switch on the light markers.
2. Mark the positions of the lateral light markers at the lifting device using strips of tape and a pen. **Write down the value for the table height. Read it off the gantry display**

3. Mark the positions of the sagittal and transaxial light markers at the patient table using strips of tape and a pen. **Write down the value for the horizontal table position. Read it off the gantry display**



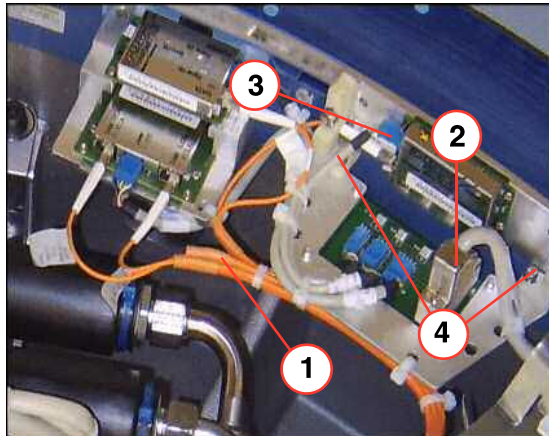
During the procedure, all stationary light markers have to be removed. As a result, it is necessary to mark the light marker positions and write down the table positions for adjusting the light markers later on.

4. Move the table out with the installed tube lifting device.
5. Allow at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Note that some system components may be very hot , i.e., the Straton tube may reach 130 degree Celsius
6. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down)
7. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
8. Remove the bottom front ring segment (→ Removing / Page 32)
9. Disable rotation using the rotation enable switch S301.
10. Remove the top front ring segment, refer to (→ Removing / Page 34). If a liftable front top ring segment is installed, refer to (→ Removing / Page 37)
11. Turn the gantry by hand until **DMS A** is in the 6 o'clock position. If the liftable front top ring segment is installed, move the **DMS A** to the 12 o'clock position.
12. Secure against unintended rotation by using the front safety bolt (→ Safety bolts / Page 21)

3.13.3 Removing

1. Remove the connectors

Fig. 112: Rotating transmitters and receiver



- Remove all 4 fiber optic connectors at the rotating receiver (fastlink data) and at the rotating transmitters (e.g. item 1)
- Remove the Canon plug (item2) at the wiring interface
- Remove the power connector at rotating receiver (item3)
- Remove the cable ties which are securing the fiber optics cables to the bracket.
- Remove the rotating receiver including the bracket from the old data ring by removing the 2 (M4x12) screws using a Torx key T20 (item4).

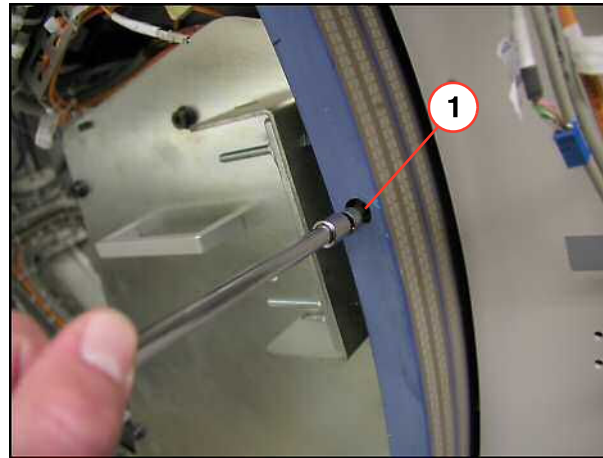
2. Remove the stationary data ring.



Two different replacement instructions for the stationary data ring are available. Depending on the cover type installed for the gantry front, the procedure is different! Refer to the corresponding section (→ Stationary Data Ring (with removable top front ring segment) / Page 132) or (→ Stationary Data Ring (with liftable top front ring segment) / Page 145), depending on your type of cover.

3. Remove the rotating data ring

Fig. 113: Screws of rotating data ring



- Remove the 12 (M6x60) holding screws using a Torx key T30. Hold the ring when removing the last screws.
- Slightly move out the whole ring with 2 people



2 people are required to remove the ring.

3.13.4 Installation

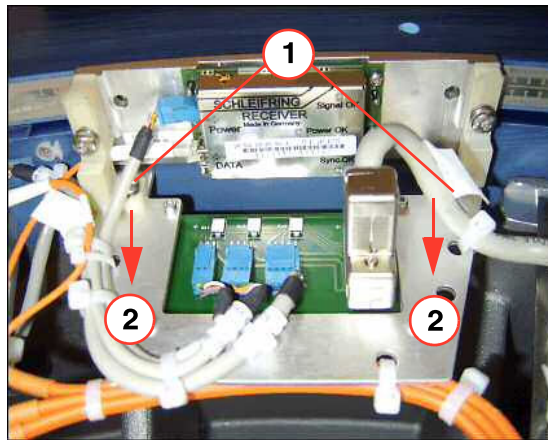
1. Install the new rotating data ring to the gantry
 - Lift the data ring into the gantry with 2 people
 - For correct positioning, refer to the holes in the ring and the plugs in the gantry.
 - Use Loctite 243 and insert the 12 (M6x60) screws while holding the data ring
 - Tighten the 12 screws with a torque of **10Nm** using a Torx key T30
2. Remove all cables from the old wiring interface and install them on the new data ring (if not already installed)
3. Remove the rotating transmitter (DMS data) from the defective data ring and install it on the new data ring. Refer to the section removing/installing the rotating transmitter (DMS data) (→ Removing / Page 168)
4. Remove the rotating transmitter (fastlink data) from the defective data ring and install it on the new data ring. Refer to the section removing/installing the rotating transmitter (fastlink data) (→ Removing / Page 171)
5. Installing the stationary data ring. Refer to the section (→ Installation / Page 139) the stationary data ring



There is no need to do the "Start-up" for the stationary data ring. Just refer to the section on "installing". Start-up for both data rings is included in this section!

6. Install the rotating receiver assembly on the new data ring and connect all cables

Fig. 114: Installing the rotating receiver assembly



- Carefully mount the rotating receiver assembly on the bracket of the new rotating data ring using 2 screws (M4x12) (item 1)
 - Move the receiver assembly out (item 2) as much as possible (long holes) and hand tighten the 2 screws (adjustment later on)
 - Connect all cables according to their original configuration
7. Connect cables to rotating transmitters (if not already done), and the wiring interface according to their original configuration
 8. Install cable ties where removed

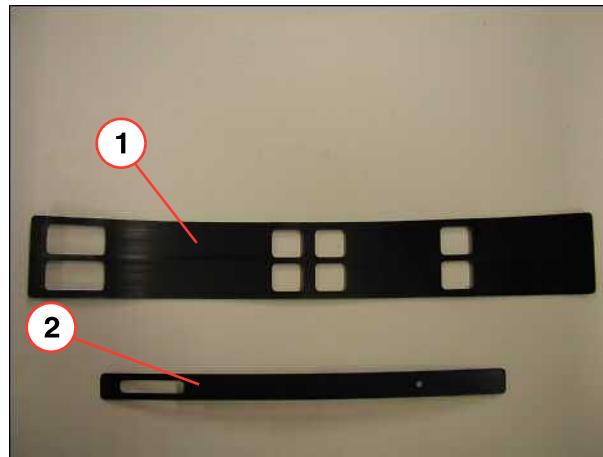
3.13.5 Start-up

3.13.5.1 Mechanical

1. Unlock the safety bolt for gantry rotation

2. Use alignment jigs to verify and adjust the distance of the transmitter carriers to the data ring

Fig. 115: Alignment jigs



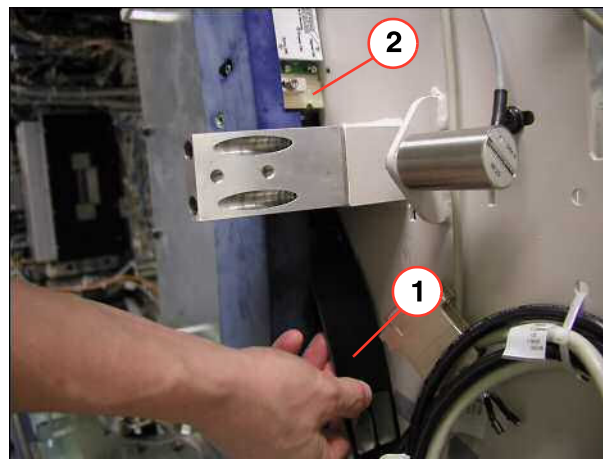
- The large jig (item 1) is used to verify and adjust the distance between the stationary receiver carrier and the rotating data ring
- The small jig (item 2) is used to verify and adjust the distance between the rotating receiver carrier and the stationary data ring

3. Adjust the triple receiver assembly (distance to ring)



The adjustment is correct, if the alignment jig can be removed without using force. During the adjustment procedure, rotate the gantry by hand and check at various gantry positions. The alignment jig has to fit all gantry positions.

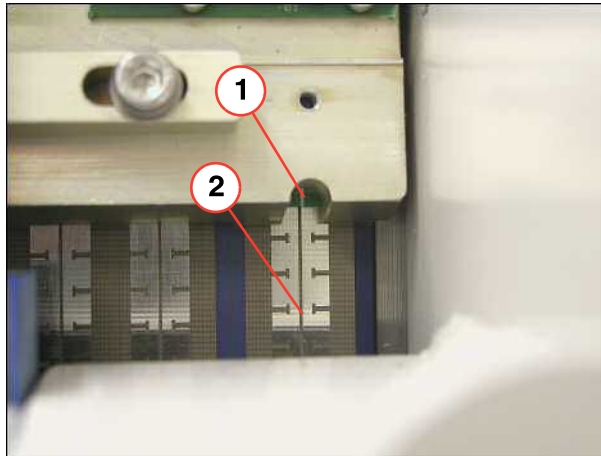
Fig. 116: Alignment jig, large



- Insert the large alignment jig (item 1) between the rotating antenna ring and the triple receiver carrier (item 2)
- Adjust the distance of the triple receiver carrier to the rotating data ring. Refer to (→ Mechanical / Page 185) adjustment of stationary receiver carrier.

4. Check the adjustment of the triple receiver assembly in the Z-direction

Fig. 117: Adjustment mark (Z-position)



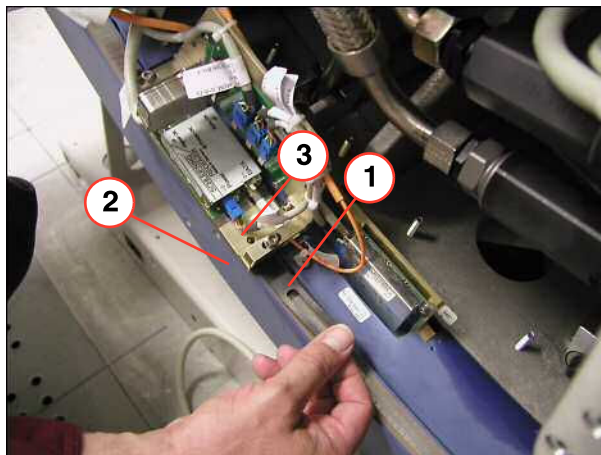
- The 2 marks at the bottom and top (e.g. lower mark, item 1) of the triple receiver assembly have to match the line in the structure of the rotating antenna ring (item 2).
- If the marks do not match refer to (→ Mechanical / Page 185) adjustment of stationary receiver carrier.

5. Adjust the rotating receiver assembly (distance to ring)



The adjustment is correct, if the alignment jig can be removed without using force. During the adjustment procedure, rotate the gantry by hand and check at various gantry positions. The alignment jig has to fit all gantry positions.

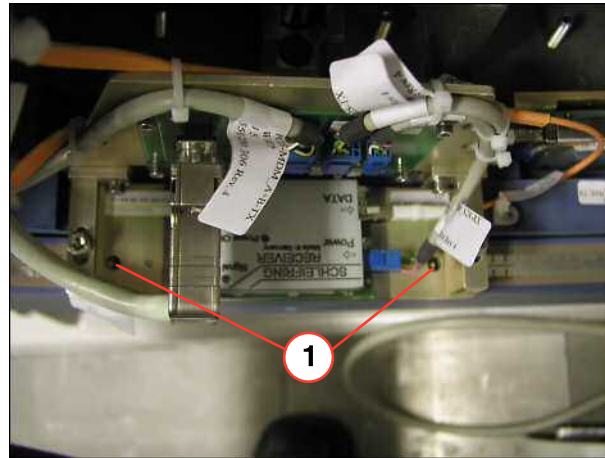
Fig. 118: Alignment jig (small)



- Insert the small alignment jig (item 1) between the stationary antenna ring (item 2) and the receiver carrier (item 3)
- Adjust the distance of the rotating receiver assembly to the stationary data ring. Refer to (→ Mechanical / Page 191) adjustment of rotating receiver carrier

6. Check the adjustment of the rotating receiver assembly in the Z-direction

Fig. 119: Adjustment marks (Z-position)



- The 2 marks (item 1) of the rotating receiver assembly have to match the structure of the stationary antenna ring.
- If the marks do not match, refer to (→ Mechanical / Page 191) adjustment of rotating receiver carrier.

7. Rotate the Gantry by hand and check for unusual noise

8. Switch on breakers F2, F3, F5, F6, F7 and F11 in the PDC A

- » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.

9. Switch on the system (**SYSTEM/ON**) at the control box.

10. Check the adjustment of the light markers.

- Move the table in using the panel at the table and check if the light markers match the lines on the tapes. **Table positions must be the same as that used before replacing the light markers.**

If the lines do not match the light markers, adjust the light markers to the lines. Refer to, (→ Adjustment of the lateral Light Markers / Page 118), (→ Adjustment of the sagittal Light Marker / Page 122), (→ Adjustment of the transaxial Light Marker / Page 126), for detailed adjustment instructions.

11. After finishing the light marker adjustment, switch off the gantry (**SYSTEM/OFF**) at the Control Box

12. Install the top front ring segment. Refer to Section "Covers" for detailed instructions.

13. Enable rotation using the rotation enable switch S301

14. Install all covers

15. Switch on the system (**SYSTEM/ON**) at the control box

3.13.5.2 Tune-up

n.a

3.13.5.3 Testing

Run the Fastlink Test and the Disk Test in Local Service->Test Tools

3.13.6 Final steps

n.a.

3.14 Rotating Transmitter (DMS Data)

3.14.1 Prerequisites

3.14.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

3.14.1.2 Tools and auxiliary Equipment

- Standard tool kit.
- Torque wrench

3.14.1.3 Time and Man Power

- 1 person
- Estimated total repair time: 0.5 hours.

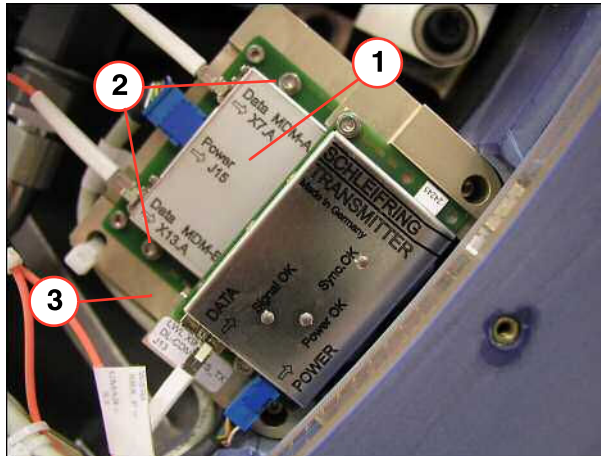
3.14.2 Preliminary Steps

1. Allow at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may be up to 130 degree Celsius.
2. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Turn the gantry by hand until the part to be replaced is in the 6 o'clock position.
6. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

3.14.3 Removing

1. Remove the transmitter

Fig. 120: Transmitter, rotating (DMS data)



- Remove the 3 connectors (2 x data and 1x power) from the DMS data transmitter (item 1)
- Remove the 2 screws (item 2) using a 2.5mm Allen key
- Slightly move the transmitter out of the metal holding bracket (item 3)

3.14.4 Installation

1. Install the new transmitter in reverse order and tighten the 2 screws with a torque of **0.8Nm**.
2. Connect the cables according to the original configuration.

3.14.5 Start-up

3.14.5.1 Mechanical

1. Remove the data ring protections
2. Rotate the Gantry by hand and check for unusual noise
3. Install all covers.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.

3.14.5.2 Tune-up
n.a

3.14.5.3 Testing

1. Run the Disk Test in Local Service -> Test Tools

3.14.6 Final steps
n.a.

3.15 Rotating Transmitter (fastlink data)

3.15.1 Prerequisites

3.15.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the “General” section of this document and/or the “Product specific safety notes”.

3.15.1.2 Tools and auxiliary Equipment

- Standard tool kit.
- Torque wrench

3.15.1.3 Time and Man Power

- 1 person
- Estimated total repair time: 0.5 hours.

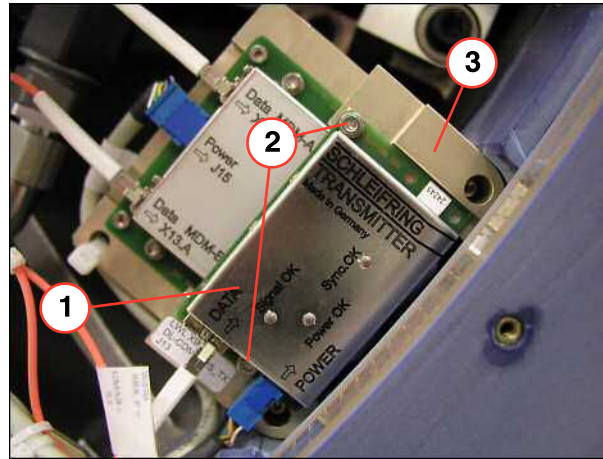
3.15.2 Preliminary Steps

1. Allow at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may be up to 130 degree Celsius.
2. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Turn the gantry by hand until the part to be replaced is in the 6 o'clock position.
6. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

3.15.3 Removing

1. Remove the transmitter

Fig. 121: Transmitter, rotating (fastlink data)



- Remove the connectors (data and power) from the rotating transmitter (item 1)
- Remove the 2 (M3x6) screws (item 2) using a 2.5mm Allen key
- Slightly move the transmitter out of the metal holding bracket (item 3)

3.15.4 Installation

1. Install the new transmitter in the reverse order and tighten the 2 screws with a torque of **0.8 Nm**.
2. Connect the cables according to the original configuration.

3.15.5 Start-up

3.15.5.1 Mechanical

1. Remove the data ring protections
2. Rotate the Gantry by hand and check for unusual noise
3. Install all covers.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.

3.15.5.2 Tune-up

n.a

3.15.5.3 Testing

1. Run the fastlink test in Local Service -> Test Tools

3.15.6 Final steps

n.a.

3.16 Stationary Receivers (DMS data and fastlink data)

3.16.1 Prerequisites

3.16.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

3.16.1.2 Tools and auxiliary Equipment

- Standard tool kit.
- Torque wrench

3.16.1.3 Time and Man Power

- 1 person
- Estimated total repair time: 0.5 hours.

3.16.1.4 Information

The 3 stationary receivers for DMS A data, DMS B data, and fastlink data, are similar.
(same part number)

The replacement procedure is the same for all 3 stationary receivers

3.16.2 Preliminary Steps

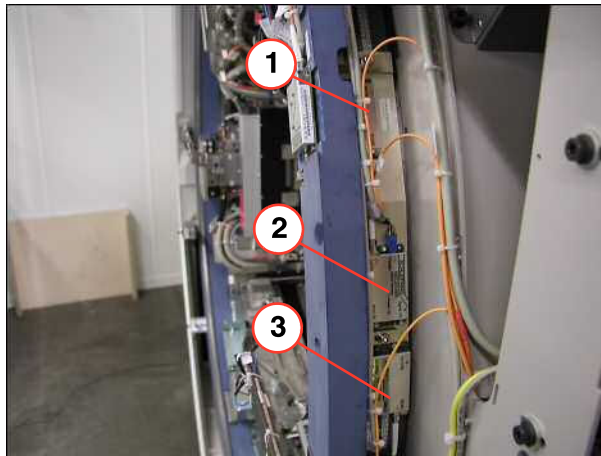
1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may be up to 130 degree Celsius.
2. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).

6. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

3.16.3 Removing

1. Locate the defective receiver

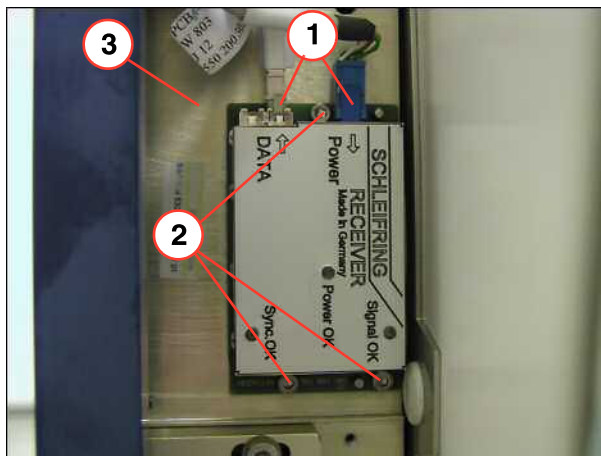
Fig. 122: Stationary receivers (triple assembly)



- Receiver for fastlink data from rotating part (item 1).
- Receiver for data from DMS A (item 2).
- Receiver for data from DMS B (item 3).

2. Remove the receiver

Fig. 123: Removing one stationary receiver



- Remove the 2 connectors (data and power) (item 1) from the corresponding receiver.
- Remove the 3 screws (M3x6) (item 2) using a 2.5mm Allen key.
- Remove the receiver from the carrier (item 3).

3.16.4 Installation

1. Install the new receiver in reverse order and tighten the 2 screws with a torque of **0.8Nm**.
2. Connect the cables according to the original configuration.

3.16.5 Start-up

3.16.5.1 Mechanical

1. Remove the data ring protections
2. Install all covers.
3. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
4. Switch on the system (**SYSTEM/ON**) at the control box.

3.16.5.2 Tune-up

n.a

3.16.5.3 Testing

1. Run the Fastlink Test and Disk Test in Local Service -> Test Tools

3.16.6 Final steps

n.a.

3.17 Stationary Transmitter (fastlink data)

3.17.1 Prerequisites

3.17.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

3.17.1.2 Tools and auxiliary Equipment

- Standard tool kit.
- Torque wrench

3.17.1.3 Time and Man Power

- 1 person
- Estimated total repair time: 0.5 hours.

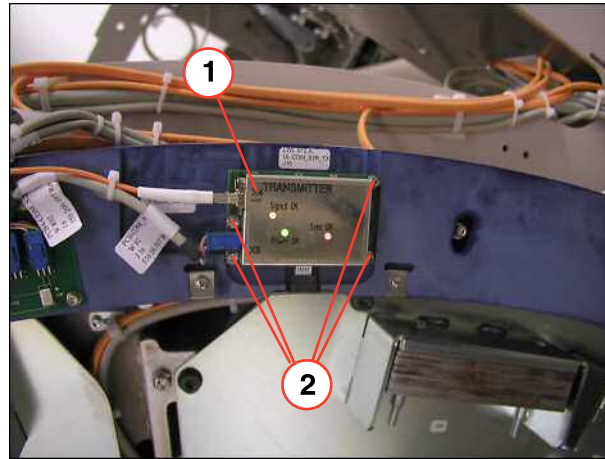
3.17.2 Preliminary Steps

1. Allow at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Note that some system components may be very hot , i.e., the Straton tube may reach 130 degree Celsius
2. Lower the patient table using the table control to open the gantry front segment.
3. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down)
4. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
5. Remove the bottom front ring segment. (→ Removing / Page 32)
6. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
7. Install the data ring protection. (→ Data ring protection / Page 23)

3.17.3 Removing

1. Remove the transmitter

Fig. 124: Transmitter, stationary (fastlink data)



- Remove the connectors (data and power) from the stationary receiver (item 1)
- Remove the 4 (M3x6) screws (item 2) using a 2.5mm Allen key
- Remove the transmitter from the stationary receiver ring.

3.17.4 Installation

1. Install the transmitter in the reverse order and tighten the 4 screws with a torque of **0.8Nm**.
2. Connect the cables according to the original configuration.

3.17.5 Start-up

3.17.5.1 Mechanical

1. Remove the data ring protections
2. Install all covers.
3. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
4. Switch on the system (**SYSTEM/ON**) at the control box.

3.17.5.2 Tune-up

n.a

3.17.5.3 Testing

1. Run the Fastlink Test in Local Service -> Test Tools

3.17.6 Final steps

n.a

3.18 Rotating Receiver (fastlink data)

3.18.1 Prerequisites

3.18.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

3.18.1.2 Tools and auxiliary Equipment

- Standard tool kit.
- Torque wrench

3.18.1.3 Time and Man Power

- 1 person
- Estimated total repair time: 0.5 hours.

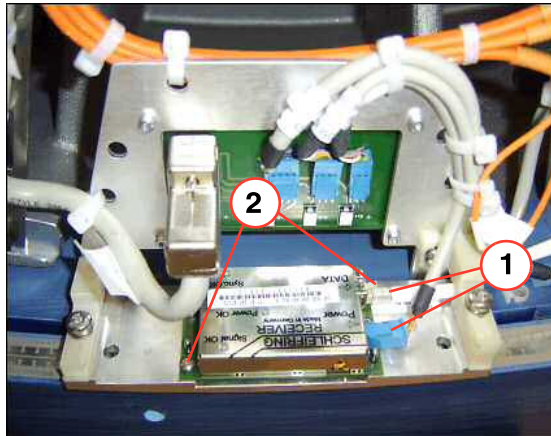
3.18.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may be up to 130 degree Celsius.
2. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Turn the gantry by hand until the part to be replaced is in the 6 o'clock position.

3.18.3 Removing

1. Remove the receiver

Fig. 125: Rotating receiver (fastlink data)



- Remove the connectors (data and power, item 1) from the rotating receiver
- Remove the 2 (M3x6) screws (item 2) using a 2.5mm Allen key
- Remove the receiver

3.18.4 Installation

1. Install the receiver in reverse order and tighten the 2 screws with a torque of **0.8Nm**.
2. Connect the cables according to the original configuration.

3.18.5 Start-up

3.18.5.1 Mechanical

1. Remove the data ring protections
2. Rotate the Gantry by hand and check for unusual noise
3. Install all covers.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.

3.18.5.2 Tune-up

n.a

3.18.5.3 Testing

1. Run the Fastlink Test in Local Service -> Test Tools

3.18.6 Final steps

n.a

3.19 Receiver Carrier (stationary)

3.19.1 Prerequisites

3.19.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product-specific safety notes."

3.19.1.2 Tools and auxiliary Equipment

- Standard tool kit.
- Torque wrench

3.19.1.3 Time and Man Power

- 1 person
- Estimated total repair time: 1 hour.

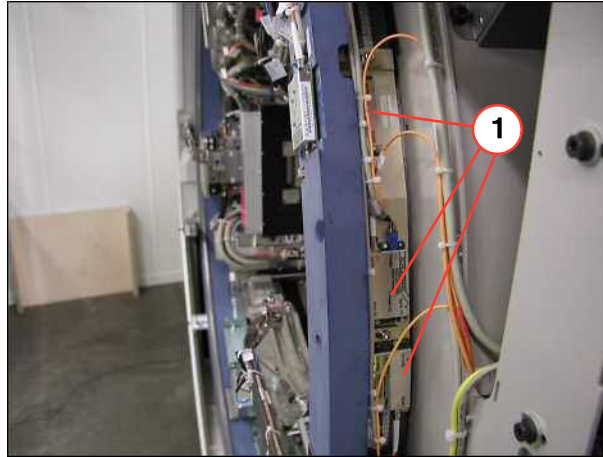
3.19.2 Preliminary Steps

1. Allow at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Note that some system components may be very hot , i.e., the Straton tube may reach 130 degree Celsius
2. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down)
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Install the data ring protections. (→ Data ring protection / Page 23)

3.19.3 Removing

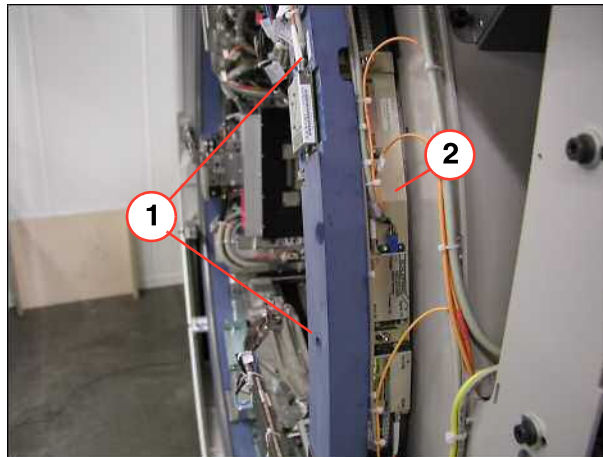
1. Remove the connectors at the receivers

Fig. 126: Connectors at receivers



- Remove all connectors at the stationary receivers (item 1)
2. Remove the cable ties as necessary
 3. Remove the triple receiver assembly

Fig. 127: Removing the triple receiver assembly

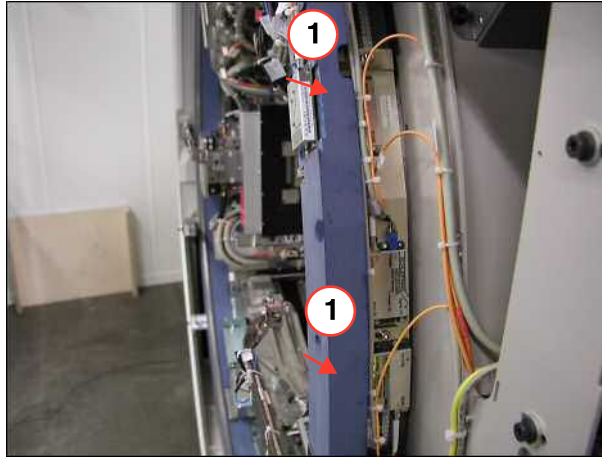


- Remove the 2 (M4x20) screws (item 1) using a 3 mm Allen key.
Be careful not to damage the rotating antenna ring with the triple receiver carrier (item 2).
 - Remove the triple receiver assembly.
4. Remove the 3 receivers from the receiver carrier and install them at the new receiver carrier. Refer to the section "removing/installing stationary receivers (→ Removing / Page 174)

3.19.4 Installation

1. Install the triple receiver assembly

Fig. 128: Installing the triple receiver assembly



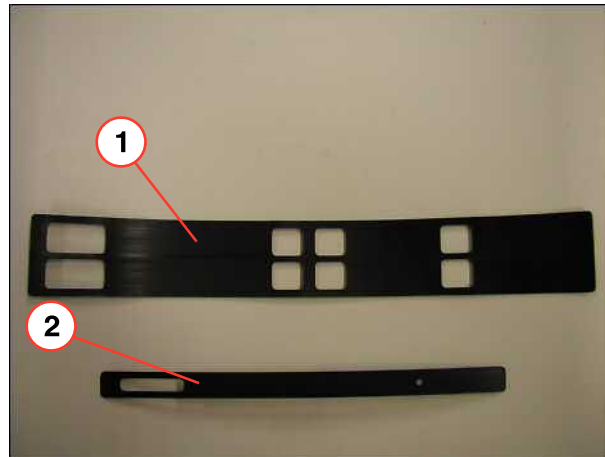
- Carefully mount the triple receiver assembly to the stationary antenna ring.
 - Move the triple receiver assembly out (item 1) as much as possible (long holes) and hand tighten the 2 screws.
2. Connect cables to receivers according to their original configuration
 3. Install cable ties where removed

3.19.5 Start-up

3.19.5.1 Mechanical

1. Use alignment jigs to verify and adjust the distance of the transmitter carriers to the data ring

Fig. 129: Alignment jigs



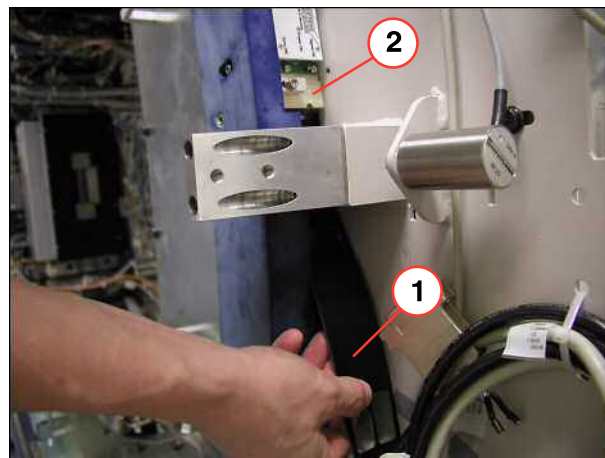
- The large jig (item 1) is used to verify and adjust the distance between the stationary receiver carrier and the rotating data ring
- The small jig (item 2) is used to verify and adjust the distance between the rotating receiver carrier and the stationary data ring

2. Adjust the triple receiver assembly (distance to ring)



The adjustment is correct, if the alignment jig can be removed without using force. During the adjustment procedure, rotate the gantry by hand and check at various gantry positions. The alignment jig has to fit all gantry positions.

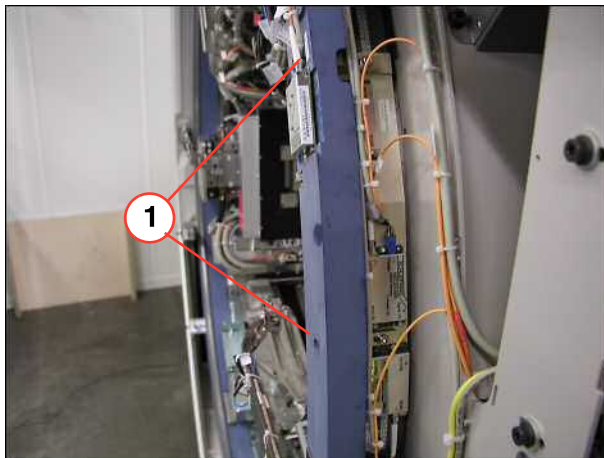
Fig. 130: Alignment jig, large



- Insert the large alignment jig (item 1) between the rotating antenna ring and the triple receiver carrier (item 2)

3. Adjust the triple receiver assembly via the long holes in the data ring (distance to ring)

Fig. 131: Adjustment of triple receiver assembly

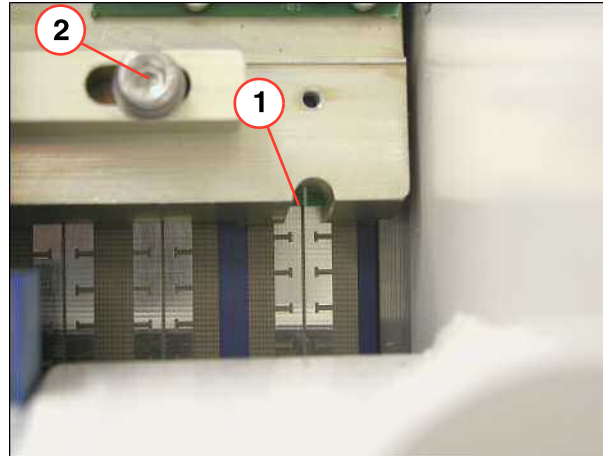


- Open the 2 screws (item 1) of the triple receiver assembly.
- Move the whole triple receiver assembly toward the alignment jig and hand tighten the 2 screws
- Tighten the 2 screws (M4x20) with a torque of **1.8Nm** and check the distance again using the jig.
- If necessary, repeat the procedure

4. Adjust the triple receiver assembly (Z-position)

This example shows the lower adjustment mark only. Another adjustment mark is located at the top of the triple receiver assembly behind the fiber optic cable of the DMS receiver. Use both marks for proper adjustment.

Fig. 132: Lower adjustment mark/screw (Z-direction)



- The 2 marks (e.g. item 1) of the triple receiver assembly have to match the structure of the rotating antenna ring.
 - If the marks do not match, open the 2 screws (M5x20) (e.g. item 2) of the triple receiver assembly using a Torx key T25
 - Adjust the Z-position so that the marks on the triple receiver assembly match the line in the structure of the rotating antenna ring (item 1)
 - Tighten the 2 screws of the triple receiver assembly with a torque of **3.5Nm**.
5. Remove the data ring protections
 6. Rotate the Gantry by hand and check for unusual noise
 7. Install all covers.
 8. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
 9. Switch on the system (**SYSTEM/ON**) at the control box.

3.19.5.2 Tune-up

n.a

3.19.5.3 Testing

1. Run the Fastlink Test and Disk Test in Local Service -> Test Tools

3.19.6 Final steps

n.a

3.20 Receiver Carrier (rotating)

3.20.1 Prerequisites

3.20.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product-specific safety notes."

3.20.1.2 Tools and auxiliary Equipment

- Standard tool kit.
- Torque wrench

3.20.1.3 Time and Man Power

- 1 person
- Estimated total repair time: 1 hour.

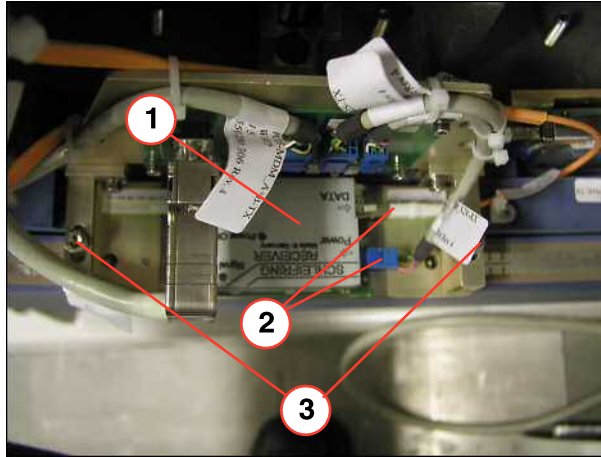
3.20.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may be up to 130 degree Celsius.
2. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down).
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Turn the gantry by hand until the part to be replaced is in the 6 o'clock position.

3.20.3 Removing

1. Remove the receiver carrier including the receiver

Fig. 133: Removing the receiver carrier



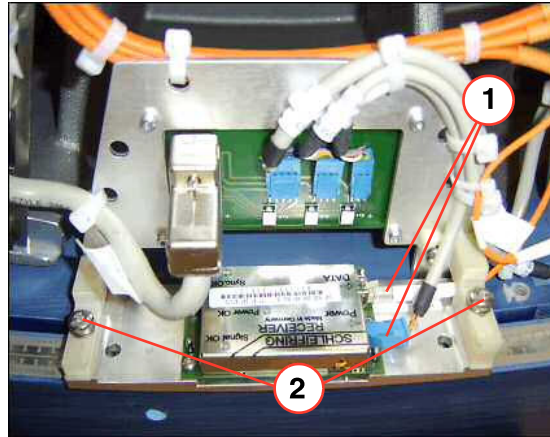
- Remove the connectors (data and power, item 2) at the receiver (item 1)
- Remove the 2 screws (item 3) using an Allen key.
Be careful not to damage the antenna ring with the receiver carrier.
- Remove the receiver carrier including the receiver

3.20.4 Installation

1. Remove the receiver from the receiver carrier and install it at the new carrier. Refer to (→ Removing / Page 180) rotating receiver

2. Install the new receiver carrier

Fig. 134: Rotating receiver carrier



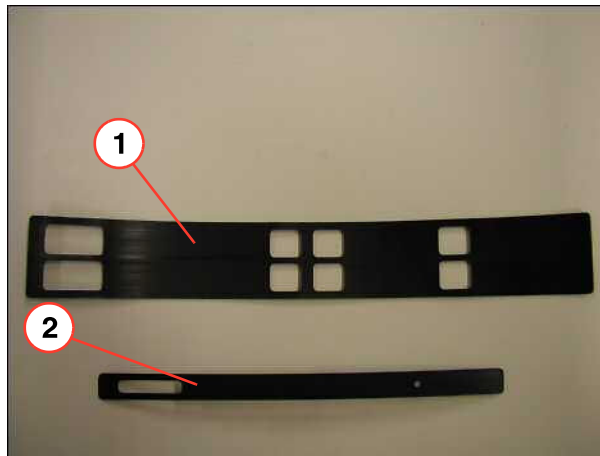
- Carefully mount the new receiver carrier to the holding brackets and hand tighten the screws.
- Reconnect the connectors (data and power) to the receiver

3.20.5 Start-up

3.20.5.1 Mechanical

1. Remove the data ring protections
2. Use alignment jigs to verify and adjust the distance of the transmitter carriers to the data ring

Fig. 135: Alignment jigs



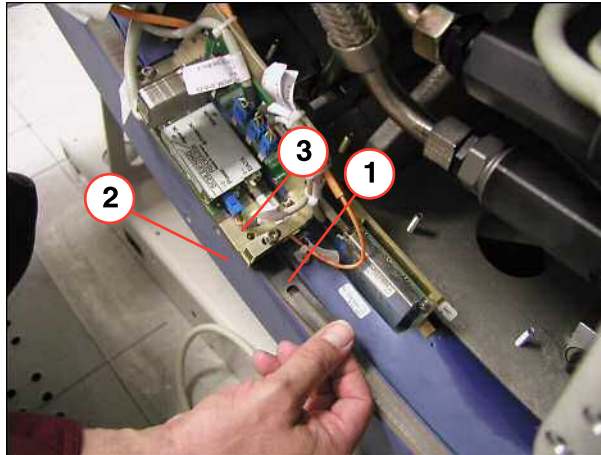
- The large jig (item 1) is used to verify and adjust the distance between the stationary receiver carrier and the rotating data ring
- The small jig (item 2) is used to verify and adjust the distance between the rotating receiver carrier and the stationary data ring

3. Adjust the rotating receiver assembly (distance to ring)



The adjustment is correct, if the alignment jig can be removed without using force. During the adjustment procedure, rotate the gantry by hand and check at various gantry positions. The alignment jig has to fit all gantry positions.

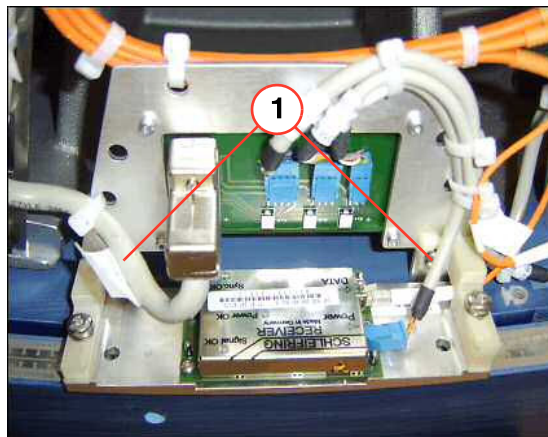
Fig. 136: Alignment jig (small)



- Insert the small alignment jig (item 1) between the stationary antenna ring (item 2) and the receiver carrier (item 3)
- If the distance to the ring is too narrow or too wide, proceed with the next step

4. Adjust the rotating receiver assembly (distance to ring)

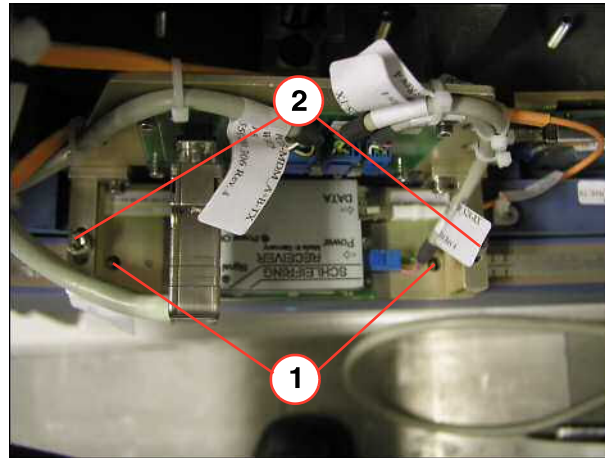
Fig. 137: Adjustment screws (distance to ring)



- Untighten the 2 screws (M4x12) (item 1) of the receiver assembly.
- Move the receiver assembly toward the alignment jig and hand tighten the 2 screws
- Remove the alignment jig
- Tighten the 2 screws with a torque of **1.8Nm** and check the distance again using the jig.

5. Adjust the rotating receiver assembly (Z-position)

Fig. 138: Adjustment marks (Z-position)



- The marks (item 1) of the rotating receiver assembly have to match the structure of the stationary antenna.
 - If the marks do not match, open the 2 screws (M5x20) and (M5x10) (item2) of the receiver assembly using a Torx key T25.
 - Adjust the Z-position until the marks on the receiver assembly match the line in the antenna structure (item 1).
 - Tighten the 2 screws of the receiver assembly with a torque of 3.5 Nm.
6. Remove the data ring protections
 7. Rotate the Gantry by hand and check for unusual noise
 8. Install all covers.
 9. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
 10. Switch on the system (**SYSTEM/ON**) at the control box.

3.20.5.2 Tune-up

n.a

3.20.5.3 Testing

1. Run the fastlink test in Local Service -> Test Tools

3.20.6 Final steps

n.a

3.21 Wiring Interfaces - Data Transmission

3.21.1 Prerequisites

3.21.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

3.21.1.2 Tools and auxiliary Equipment

- Standard tool kit.
- Torque wrench

3.21.1.3 Time and Man Power

- 1 person
- Estimated total repair time: 0.5 hours

3.21.1.4 Information

This section describes the replacement of the rotating wiring interface. Replacement of the stationary one is similar

3.21.2 Preliminary Steps

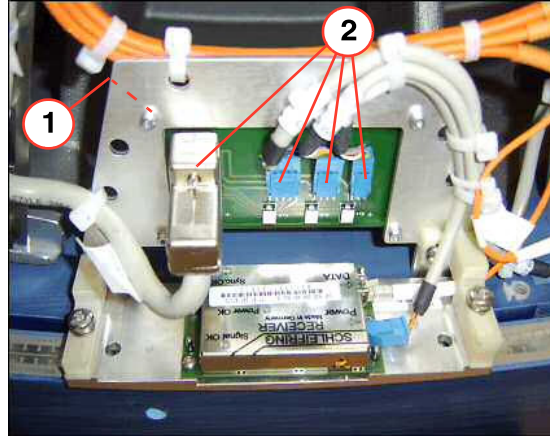
1. Allow at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Note that some system components may be very hot , i.e., the Straton tube may reach 130 degree Celsius
2. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down)
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Remove or open (depending on cover type) the gantry top front ring segment. (For stationary wiring interface only). Refer to (→ Opening/closing the top front ring segment cover / Page 25).

6. Turn the gantry by hand until the part to be replaced is in the 6 o'clock position

3.21.3 Removing

1. Removing the wiring interface

Fig. 139: Wiring interface, rotating



- Remove all connectors from the wiring interface (item 1)
- Remove the 4 nuts (M3) (e.g. item 1, rear side of the mounting plate) use an open end wrench 5.5mm.
- Removing the wiring interface

3.21.4 Installation

1. Install the new wiring interface in reverse order and tighten the 4 nuts with a torque of **1.3 Nm**.
2. Reconnect all cables according to the original configuration.

3.21.5 Start-up

3.21.5.1 Mechanical

1. Remove the data ring protections
2. Rotate the Gantry by hand and check for unusual noise
3. Install all covers.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.

3.21.5.2 Tune-up

n.a

3.21.5.3 Testing

1. Run the fastlink test in Local Service -> Test Tools

3.21.6 Final steps

n.a

3.22 Balancing Sensor at Gantry Front

3.22.1 Prerequisites

3.22.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

3.22.1.2 Tools and auxiliary Equipment

- Standard tool kit
- Torque wrench

3.22.1.3 Time and Man Power

- 1 person
- Estimated total repair time: 0.5 hours

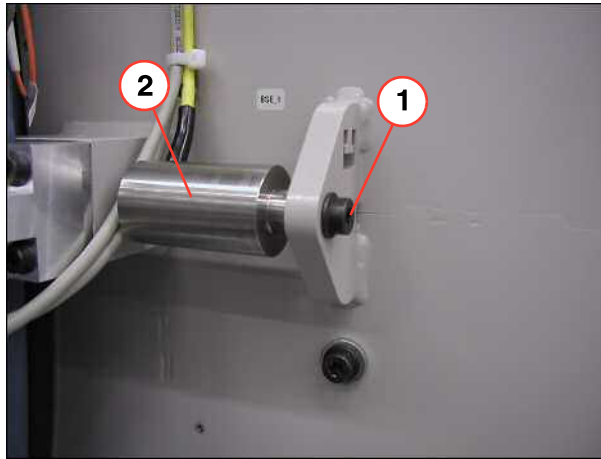
3.22.2 Preliminary Steps

1. Allow at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Note that some system components may be very hot , i.e., the Straton tube may reach 130 degree Celsius
2. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut down)
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
4. Remove the right side cover. (→ Removing / Page 51)
5. Remove the right front cover. (→ Removing / Page 54)

3.22.3 Removing

1. Removing the balancing sensor

Fig. 140: Balancing Sensor, front side



- Remove connector E2 on UMAS (yellow cable)
- Remove the cable ties as necessary
- Remove the screw (M8x16) (item 1) using a Torx key T40
- Remove the Balancing Sensor (item 2)

3.22.4 Installation

1. Install the new Balancing Sensor in the reverse order and tighten the screw with a torque of **10Nm**.
2. Reconnect E2 to UMAS and install cable ties where removed

3.22.5 Start-up

3.22.5.1 Mechanical

1. Rotate the gantry by hand and check for unusual noise.
2. Install all covers.
3. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
4. Switch on the system (**SYSTEM/ON**) at the control box.

3.22.5.2 Tune-up

1. Run the tune up sequence => Balancing, including calibration

3.22.5.3 Testing

n.a.

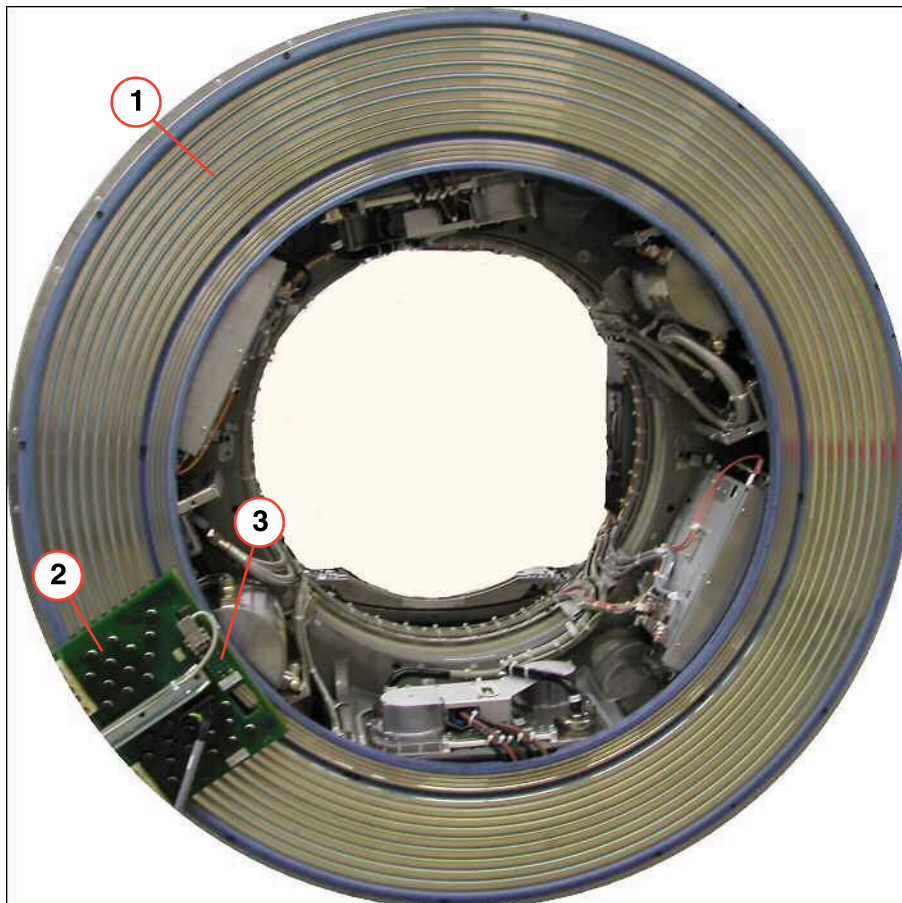
3.22.6 Final steps

n.a.

4.1 Overview

The figure below shows the location of the majority of parts described in this chapter. For replacement of XRS components located at the rear of the gantry, refer to chapter XRS.

Fig. 141: Back view without covers



Parts at the rear of the gantry

- Power slip ring (item 1)
- Brush block, high power (item 2), (color may be different)
- Signal Brush Tip unit (item 3)
- Balancing sensor, rear (not visible in image, located at the left-hand side of the power slip ring)

4.2 Slip Ring

4.2.1 Prerequisites

4.2.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

Tools and auxiliary Equipment

- Standard tool kit.
- Torque wrench
- Loctite 243

Time and Manpower

- 1 person
- Estimated total repair time: 2.0 hours

4.2.2 Preliminary Steps

**WARNING**

[hz_serdoc_F13G01U05M03]

Dangerous voltages are present when the system is switched on. Dangerous voltages may be present even when the system is switched off due to capacity power.

Risk of electric shock!

- » Observe power-off instructions and discharge wait times (allow at least five minutes discharge time after the last scan for all involved HV and UDC/UPS parts).
- » Secure the system against unintended switch-on (e.g. block breaker and/or mark breaker against switch on).
- » Ensure via measurements that all voltages are switched off.
- » Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry.

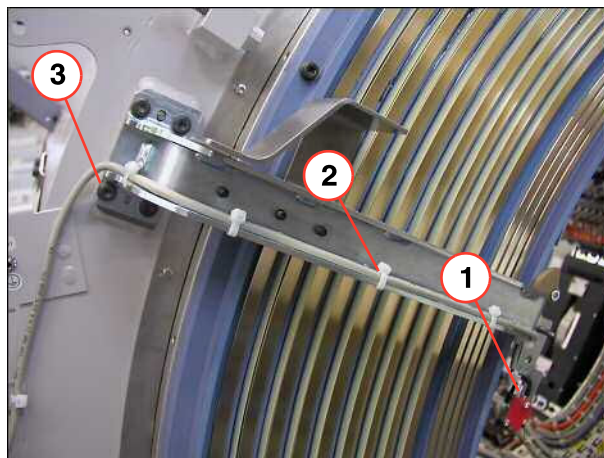
1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.
2. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
3. Switch off the gantry power using service push-button S1 in the PDC A to secure against unintended switch-on. (→ Service Push-Button S1 in PDC-A / Page 20)
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Remove the funnel and the Plexiglas ring. (→ Removing / Page 43)
7. Remove the rear ring cover. (→ Removing / Page 48)
8. Secure against unintended rotation with the safety bolt. (→ Safety bolts / Page 21)

4.2.3 Removing

1. Ensure via measurement at the power brush block that all voltages at the slip ring are switched off before touching any part of the slip ring system.

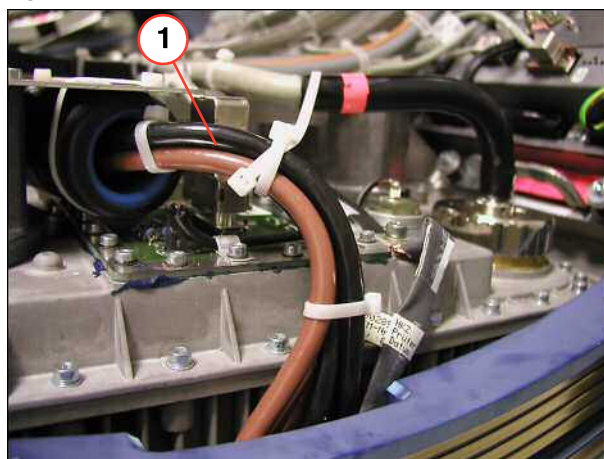
2. Remove the 3 funnel arms. The image below shows how to remove the left funnel arm. Remove the other funnel arms in the same way.

Fig. 142: Funnel holder with switch



- Remove the cables at the switch (item 1), mark the cables for later reconnection. (Only necessary at left funnel holder)
 - Remove the cable ties (e.g. item 2) at the funnel arms. (only necessary at left funnel arm)
 - Remove the 4 screws (M6x16) (e.g. item 3) at each of the three funnel arms using a Torx key T30
3. Remove brush block, high power. Refer to (→ Removing / Page 208) brush block, high power
 4. Remove MVT cables on both HV tanks

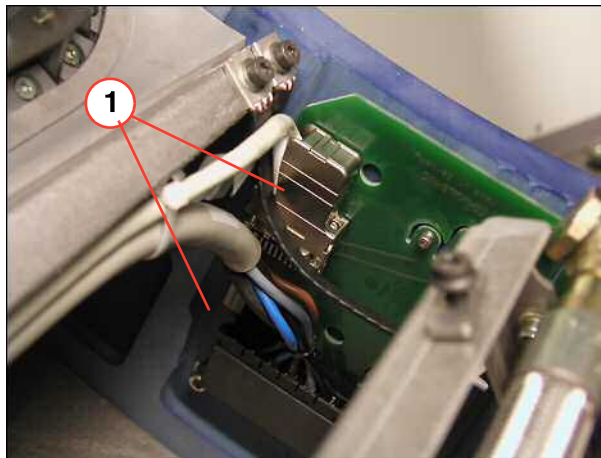
Fig. 143: Cables to HV tank



- Remove the MVT cables (e.g. item 1) on HV tank A and HV tank B
 - Remove the cable ties as necessary
5. Remove the **cover** of the XTC A tube cooling unit to access the connectors at the inner side of the slip ring. Refer to the section on the (→ Removing / Page 332) XTC tube cooling unit.

6. Remove connectors on inner side of slip ring (only visible when XTC cover is removed)

Fig. 144: Cable connections on slip ring (inner side)



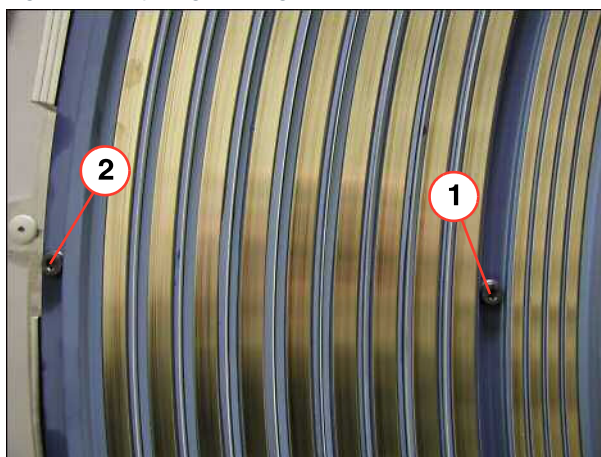
- Remove the power cable and the control cable (item 1) on inner side of slip ring
7. Remove all **10** grounding straps from the rotating part. Some of the grounding straps are connected to rotating components and others are connected to the drum.



2 people are required to remove the slip ring from the gantry.

Remove the slip ring

Fig. 145: Slip ring holding screws



- Remove the 6 inner screws (M6x20) (e.g. item1) using a Torx key T30.
- Remove the 12 outer screws (M6x35) (e.g. item2) using a Torx key T30 while holding the slip ring
- Remove the slip ring from the gantry using 2 people

4.2.4 Installation

1. Check the cable connections on the new slip ring, making sure the cables are connected properly. Refer to the old slip ring for reference
2. Lift the new slip ring into the gantry (2 people are required) and find the correct position for installation. **Use the control and power cable as reference for correct positioning**
 - Additionally, refer to the hole in the slip ring and gantry drum for orientation
3. Install the 6 inner screws (M6x20 with lacquer coated thread) and tighten all screws by hand while holding the ring.
4. Use Loctite 243 for the 12 outer screws (M6x35) and install the screws.
5. Tighten all 18 screws with a torque of **10Nm** in a cross-wise pattern.
6. Reconnect cables to HV tank A and HV tank B
 - Guide the MVT cables through the coils
 - Tighten the screws of the MVT cables on the HV tanks with a torque of **4.8Nm**
7. Reconnect cables to slip ring (power and control cable at inner side of slip ring)
8. Install all **10** grounding straps and tighten the screws with a torque of **10Nm**
9. Install all cable ties where removed
10. Install the covers of the XTC tube cooling units. Refer to (→ Installation / Page 337) XTC tube cooling unit
11. Install the brush block, high power. Refer to (→ Installation / Page 209) brush block, high power
12. Reinstall the funnel arms and tighten the screws with a torque of **14Nm**
13. Reconnect the switch and install cable ties at the left funnel arm

4.2.5 Start-up

4.2.5.1 Mechanical

1. Unlock the safety bolt for gantry rotation.
2. Rotate gantry by hand and check for unusual noise.
3. Install all covers
4. Switch on breakers F2, F3, F5, F6, F7 and F11 in the PDC A
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the Control Box.

4.2.5.2 Tune-up

n.a.

4.2.5.3 Testing

1. Run some spiral and sequence scans

4.2.6 Final steps

n.a.

4.3 Brush Block, High Power

4.3.1 Prerequisites

4.3.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

Tools and auxiliary equipment

- Standard tool kit.
- Alignment pins (delivered with the spare part)
- Torque wrench

Time and Manpower

- 1 person
- Estimated total repair time: 2.0 hours

4.3.2 Preliminary Steps

**WARNING**

[hz_serdoc_F13G01U05M03]

Dangerous voltages are present when the system is switched on. Dangerous voltages may be present even when the system is switched off due to capacity power.

Risk of electric shock!

- » Observe power-off instructions and discharge wait times (allow at least five minutes discharge time after the last scan for all involved HV and UDC/UPS parts).
- » Secure the system against unintended switch-on (e.g. block breaker and/or mark breaker against switch on).
- » Ensure via measurements that all voltages are switched off.
- » Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry.

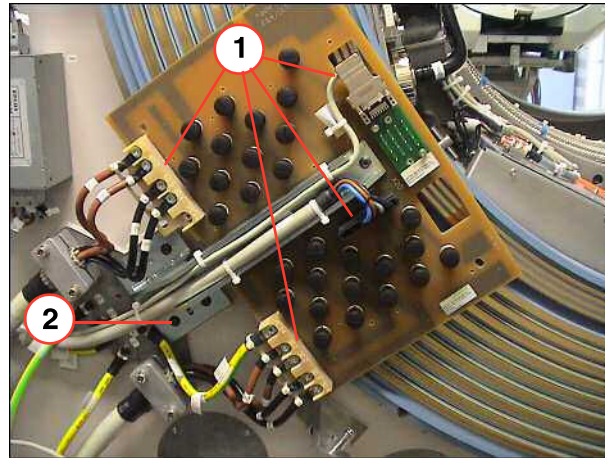
1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.
2. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
3. Switch off the gantry power using service push-button S1 in PDC A to secure against unintended switch-on. (→ Service Push-Button S1 in PDC-A / Page 20)
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Remove the funnel and the Plexiglas ring. (→ Removing / Page 43)
7. Remove the rear ring cover. (→ Removing / Page 48)
8. Secure against unintended rotation with the safety bolt. (→ Safety bolts / Page 21)

4.3.3 Removing

1. Ensure via measurement at the power brush block that all voltages at the slip ring are switched off before touching any part of the slip ring system.

2. Remove the brush block

Fig. 146: Brush block on holding bracket



- Remove all cables and connectors (item 1)
- Remove cable ties on metal bracket
- Remove the 4 screws (M6x12) (item 2) using a Torx key T30.
- Carefully remove the metal holding bracket including the brush block from the gantry.

Do not damage the slip ring tracks and slip ring track separators when removing the brush block

4.3.4 Installation

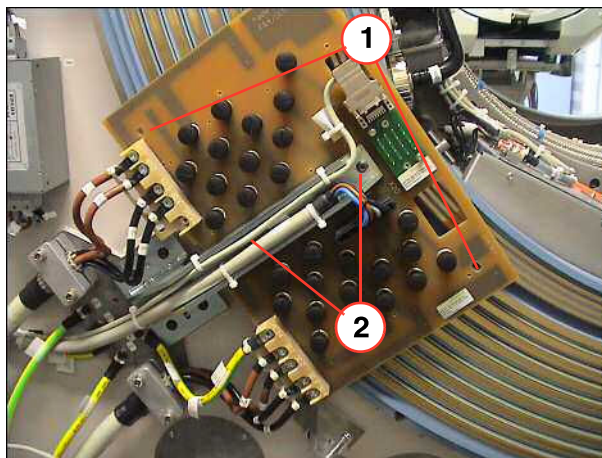
1. Remove the brush block from the metal holding bracket by removing the 2 screws (M6x20) using a Torx key T30
2. Install the new brush block to the metal holding bracket and tighten the 2 screws (M6x20) by hand.
3. Install the metal holding bracket including the brush block in reverse order and tighten the 4 screws (M6x12) with a torque of **14Nm**.

Do not damage the slip ring tracks and slip ring track separators when removing the brush block

4. Unlock the safety bolt for gantry rotation

5. Adjust the brush block and connect all cables.

Fig. 147: Adjusting the power brush block



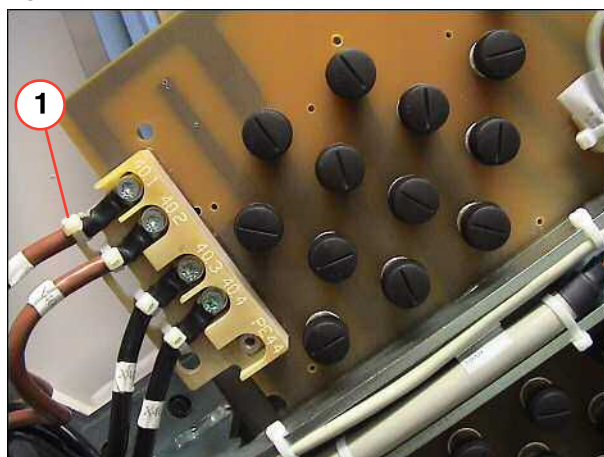
- Rotate the gantry by hand until the 2 holes in the slip ring are underneath the holes of the power brush block (item 1)
- Loosen the 2 screws (M6x20) (item 2)
- Insert the 2 guide pins (delivered with spare part) through the holes in the brush block (item 1) into the holes in the slip ring. If necessary, slightly move the brush block and slightly rotate the gantry.
- Adjust the power brush block so that the brushes match with the slip ring tracks.
- Tighten the 2 screws (item 2) with a torque of **8Nm** and remove the pins
- After tightening the screws, it should be easy to insert and remove the adjustment pins from the centering plate. The gap can be verified by rotating the pins in the holes. If necessary, readjust the brush block.
- Reconnect the connectors to the power brush block and the Signal Brush Tip unit. Install cable ties at the metal bracket.
- Reconnect the MVT cables and tighten all screws of the MVT cables at the connection terminals with a torque of **10Nm**. Ensure that every cable is connected to the correct (labeled) position!

6. Install a cable tie on each cable



It is important that the MVT cables are connected to the correct terminals (labeled) at the brush block to prevent XRS parts from being damaged. Do not mix up the A with the B cable!

Fig. 148: Cables secured with cable ties



- Install a cable tie on every cable at both terminals (e.g.item 1).

4.3.5 Start-up

4.3.5.1 Mechanical

1. Rotate gantry by hand and check for unusual noise.
2. Install all covers
3. Switch on breakers F2, F3, F5, F6, F7 and F11 in the PDC A
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
4. Switch on the system (**SYSTEM/ON**) at the control box.

4.3.5.2 Tune-up

n.a.

4.3.5.3 Testing

1. Run the **cable check** test.

This test checks if the 2 MVT cables are connected to the correct terminals at the power brush block.

Options->Service->Local Service->Test Tools->Generator

In the Generator Tests platform unset all tests and set only:

- **Cable check PDC A to HV A**
- **Cable check PDC B to HV B**

If the test fails, an error message is generated in the Event Log. In this case, follow the instructions in the Event Log.

4.3.6 Final steps

n.a.

4.4 Signal Brush Tip unit

4.4.1 Prerequisites

4.4.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

4.4.1.2 Tools and auxiliary Equipment

- Standard tool kit

4.4.1.3 Time and Manpower

- 1 person
- Estimated total repair time: 1.0 hours

4.4.2 Preliminary Steps



WARNING

[hz_serdoc_F13G01U05M03]

Dangerous voltages are present when the system is switched on. Dangerous voltages may be present even when the system is switched off due to capacity power.

Risk of electric shock!

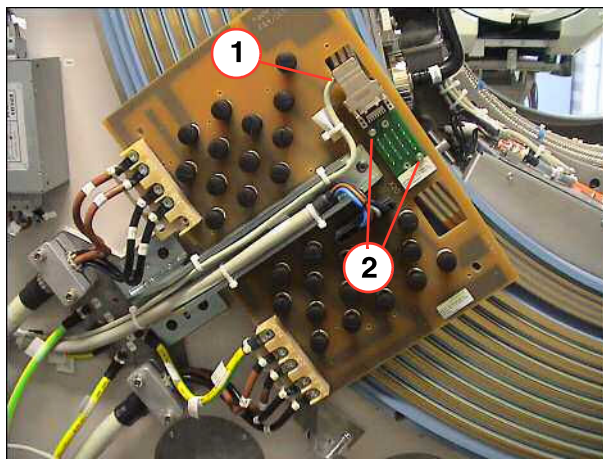
- » Observe power-off instructions and discharge wait times (allow at least five minutes discharge time after the last scan for all involved HV and UDC/UPS parts).
- » Secure the system against unintended switch-on (e.g. block breaker and/or mark breaker against switch on).
- » Ensure via measurements that all voltages are switched off.
- » Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry.

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.
2. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
3. Switch off the gantry power using service push-button S1 in the PDC A to secure against unintended switch-on. (→ Service Push-Button S1 in PDC-A / Page 20)
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Remove the funnel and the Plexiglas ring. (→ Removing / Page 43)
7. Remove the rear ring cover. (→ Removing / Page 48)
8. Secure against unintended rotation with the safety bolt. (→ Safety bolts / Page 21)

4.4.3 Removing

1. **Ensure via measurement at the power brush block that all voltages at the slip ring are switched off before touching any part of the slip ring system.**
2. Remove the Signal Brush Tip unit

Fig. 149: Removing the Signal Brush Tip unit



- Remove connector X2 (item 1) on the Signal Brush Tip unit (item 2)
- Remove the 2 (M5x20)screws (item3) using a Torx key T25
- Pull out the Signal Brush Tip unit

4.4.4 Installation

1. Install the new Signal Brush Tip unit in reverse order and tighten the 2 screws with a torque of **3.5Nm**



Do not mix up the orientation of Signal Brush Tip unit and do not damage the new brushes while installing them.

2. Reconnect X2 to Signal Brush Tip unit

4.4.5 Start-up

4.4.5.1 Mechanical

1. Unlock the safety bolt for gantry rotation.
2. Rotate gantry by hand and check for unusual noise.
3. Install all covers.
4. Switch on breakers F2, F3, F5, F6, F7 and F11 in the PDC A
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.

4.4.5.2 Tune-up

n.a.

4.4.5.3 Testing

1. Run some spiral and sequence scans

4.4.6 Final steps

n.a.

4.5 Balancing sensor, rear

4.5.1 Prerequisites

4.5.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product-specific safety notes."

4.5.1.2 Tools and auxiliary Equipment

- Standard tool kit

4.5.1.3 Time and Manpower

- 1 person
- Estimated total repair time: 1.0 hour

4.5.2 Preliminary Steps



WARNING

[hz_serdoc_F13G01U05M03]

Dangerous voltages are present when the system is switched on. Dangerous voltages may be present even when the system is switched off due to capacity power.

Risk of electric shock!

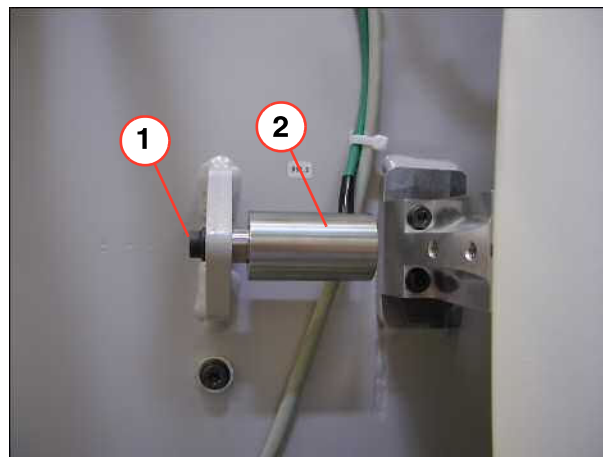
- » Observe power-off instructions and discharge wait times (allow at least five minutes discharge time after the last scan for all involved HV and UDC/UPS parts).
- » Secure the system against unintended switch-on (e.g. block breaker and/or mark breaker against switch on).
- » Ensure via measurements that all voltages are switched off.
- » Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry.

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.
2. Switch off the system (Click <SYSTEM> and select <END> in the user main menu to switch off the system automatically after shut-down).
3. Switch off the gantry power using service push-button S1 in the PDC A to secure against unintended switch-on. (→ Service Push-Button S1 in PDC-A / Page 20)
4. Remove the right side cover. (→ Removing / Page 51)
5. Remove the right rear cover. (→ Removing / Page 58)

4.5.3 Removing

1. **Ensure via measurements at the power brush block that all voltages to the slip ring system are switched off before working at the rear of the gantry.**
2. Removing the balancing sensor

Fig. 150: Balancing Sensor, back



- Remove connector E1 on UMAS
- Remove the cable ties as necessary
- Remove the screw (item 1) (M8x16) on balancing sensor (item 2) using a Torx key T40
- Remove the Balancing Sensor

4.5.4 Installation

1. Install the new Balancing Sensor in the reverse order and tighten the screw with a torque of **10Nm**.
2. Reconnect E1 to UMAS and install cable ties where removed

4.5.5 Start-up

4.5.5.1 Mechanical

1. Rotate gantry by hand and check for unusual noise.
2. Install all covers.
3. Switch on breakers F2, F3, F5, F6, F7 and F11 in the PDC A
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
4. Switch on the system (**SYSTEM/ON**) at the control box.

4.5.5.2 Tune-up

Run tune up sequence => Balancing, including calibration

4.5.5.3 Testing

n.a.

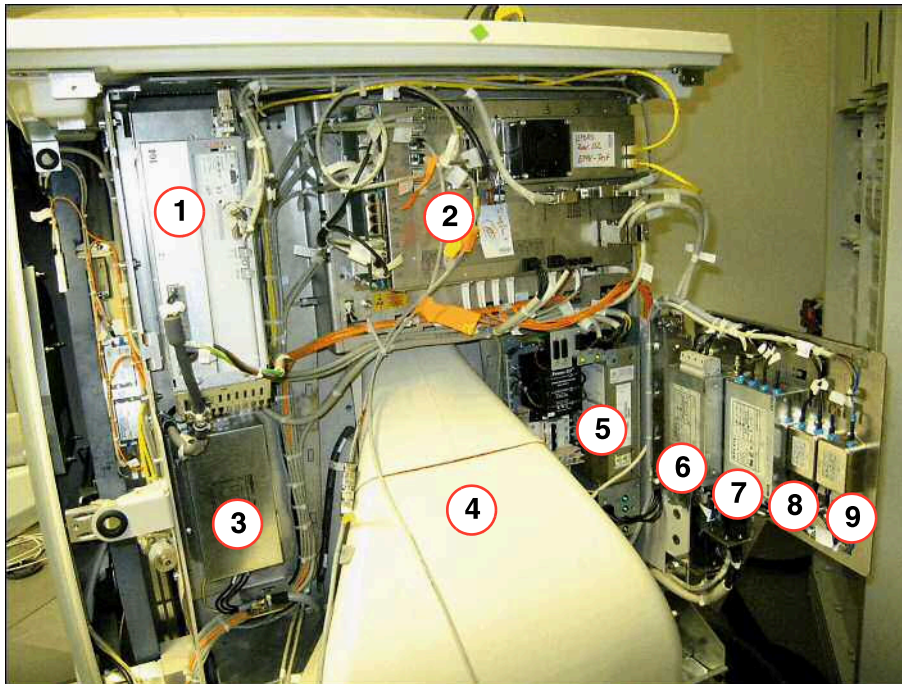
4.5.6 Final Steps

n.a.

5.1 Overview

The figure below shows the location of the parts at the gantry stand.

Fig. 151: Gantry right stand



- (1) Servo Controller Rotation (SCR)
- (2) Universal Master Stationary (UMAS)
- (3) Line Filter Z301
- (4) Air Tube
- (5) Power distribution stationary (PDS)
- (6) Line Filter Z304
- (7) Line Filter Z303
- (8) Line Filter Z305
- (9) Line Filter Z306

Fig. 152: Braking resistor (behind the servo controller)



5.2 Universal Master Stationary (UMAS and UMASplus)

5.2.1 Prerequisites

5.2.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

5.2.1.2 Time and Manpower

- 1 person
- Estimated total repair time: 1 hour.

5.2.1.3 Tools and auxiliary Equipment

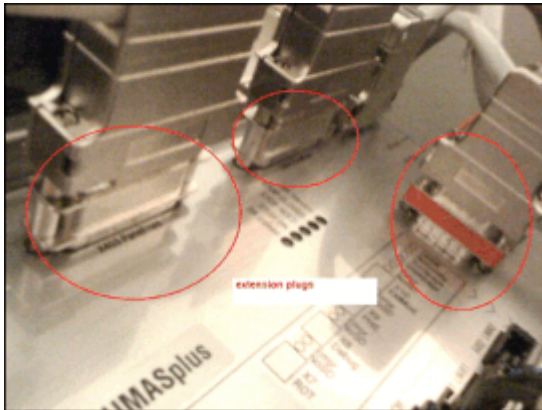
- Standard service kit

5.2.2 Preliminary Steps

1. Perform a backup of site-specific tables stored in UMAS -> Select in **Local Service** -> **Backup/Restore** -> **Command:** Backup -> **Drive:** Select drive to store tune-up tables -> **from Packages:** select "CT- Settings"
 - » Site-specific tables are stored in UMAS. Before replacing the UMAS, these tables have to be saved (if possible).
2. Remove the right stand cover.
3. Switch off the system (Click <SYSTEM> and select <END>).
4. Switch off the gantry power with the service push-button S1 in the PDC A cabinet. Secure against unintended switch-on.

5.2.3 Removing

Fig. 153: Filter capacitors of the UMAS



When replacing a UMAS with a UMASplus, do not remove the filter capacitors from the UMAS. The UMAS with the filter capacitors must be sent back.

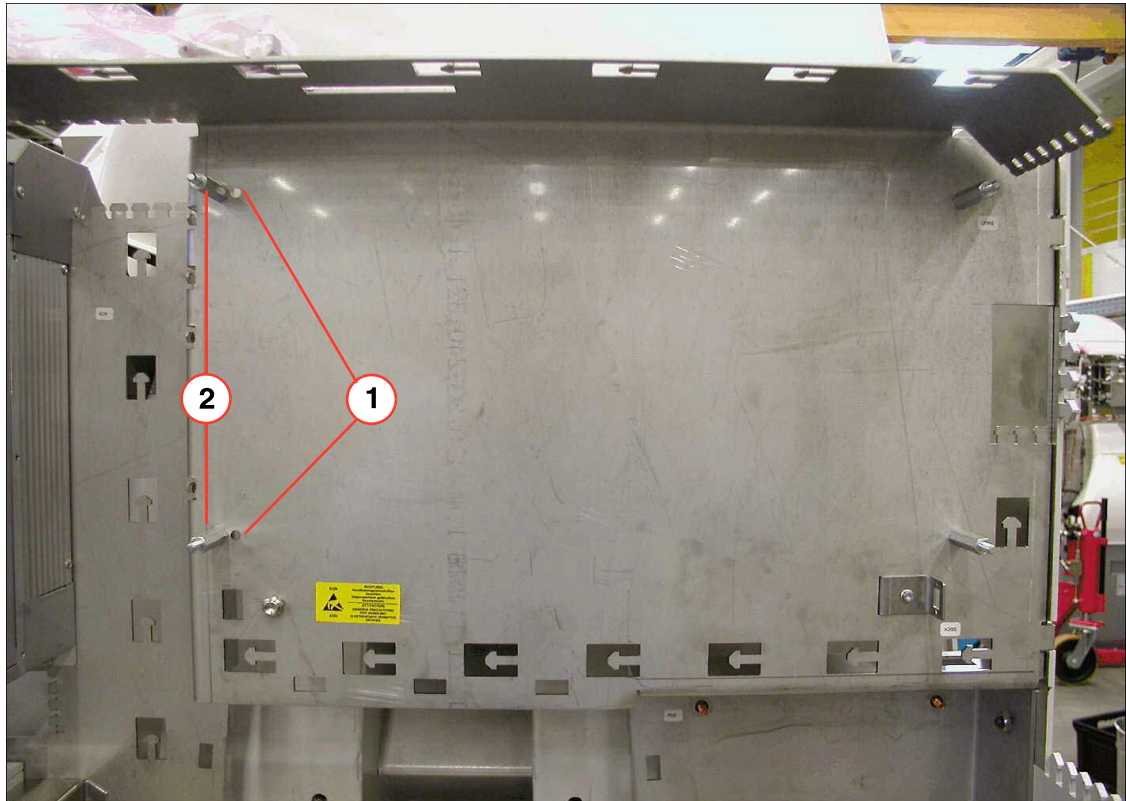
The UMASplus has the filter capacitors integrated on the board.



1. Disconnect all UMAS/UMASplus cables. Note that while all plugs are already labeled, some sockets on the UMAS/UMASplus may not be. Mark the sockets as necessary to ensure that you can reconnect all cables to the correct original locations.
 - » **Observe ESD precautions**
2. Remove the four attachment nuts at the four corners of the UMAS/UMASplus and remove the UMAS/UMASplus assembly.

5.2.4 Installation

Fig. 154: New position of both left bolts (UMAS)



- (1) old bolt position for fixing the UMAS
- (2) new bolt position for fixing the UMAS

1. The new UMAS may have the attaching holes in a slightly different position. If necessary, move the bolts from their old position 1 to the new position 2; refer to fig. (→ Fig. 154 Page 223).
2. Tighten the 2 bolts with a torque of 14 Nm.
3. Install the new UMAS assembly.
4. Tighten the 4 nuts with a torque of **10 Nm**.
5. Connect all UMAS cables.

5.2.5 Start-up

1. Close the right stand cover.
2. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
3. Switch on the system (**SYSTEM/ON**) at the control box.

5.2.5.1 Software

1. Run the tune-up sequence as described under **Local Service > Tune-up > FRU Replace**.

5.2.5.2 Tune-up

n.a

5.2.5.3 Testing

1. Perform some test scans.

5.2.6 Final Steps

n.a

5.3 Power Distribution Stationary (PDS)

5.3.1 Prerequisites

5.3.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

5.3.1.2 Time and Manpower

- 1 person
- Estimated total repair time: 1 hour.

5.3.1.3 Tools and auxiliary Equipment

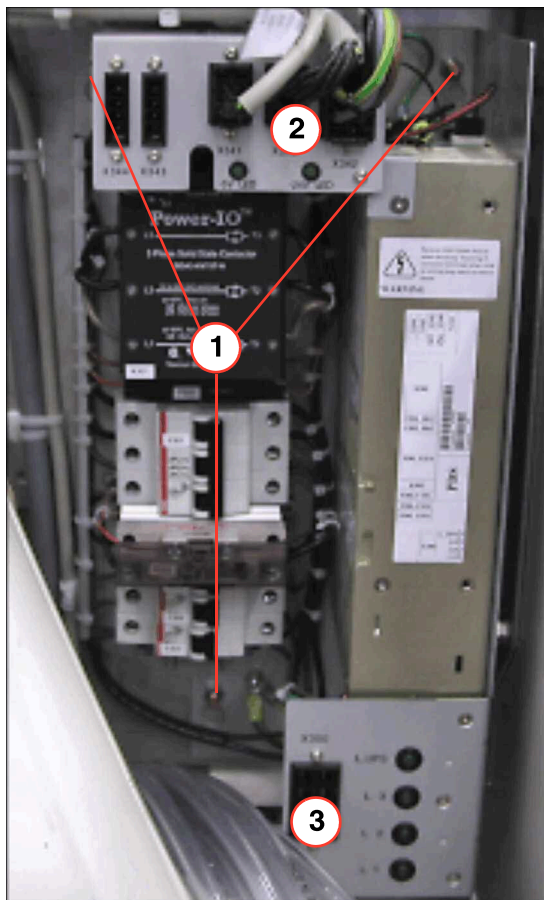
- Standard service kit

5.3.2 Preliminary Steps

1. Remove the right stand cover.
2. Switch off the system (Click <SYSTEM> and select <END>).
3. Switch off the gantry power with the service push-button S1 in the PDC A cabinet. Secure against unintended switch-on.
4. Unlock the line filter assembly in the right gantry stand and open it.

5.3.3 Removing

Fig. 155: Power distribution stationary (PDS)



- (1) 3 securing nuts
- (2) Output cables
- (3) Input cable

1. Disconnect the PDS input and output cables (Pos. 2 and 3)
2. Loosen the three holding nuts of the PDS assembly (Pos. 1).
3. Lift the PDS assembly and remove it.

5.3.4 Installation

1. Install the new PDS assembly.
2. Tighten the 3 nuts with a torque of **10 Nm**.
3. Connect and secure the PDS cables.
4. Close the line filter assembly in the right gantry stand and lock it.

5.3.5 Start-up

1. Close the right stand cover.

2. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
3. Switch on the system (**SYSTEM/ON**) at the control box.

5.3.5.1 Software

n.a

5.3.5.2 Tune-up

n.a

5.3.5.3 Testing

1. Perform some test scans.

5.3.6 Final Steps

n.a

5.4 Z301 Line Filter

5.4.1 Prerequisites

5.4.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

5.4.1.2 Time and Manpower

- 1 person
- Estimated total repair time: 1.0 hours.

5.4.1.3 Tools and auxiliary Equipment

- Standard service kit

5.4.2 Preliminary Steps

1. Remove the right stand cover.
2. Switch off the system (Click <SYSTEM> and select <END>).
3. Switch off the gantry power with the service push-button S1 in the PDC A cabinet. Secure against unintended switch-on.

5.4.3 Removing

1. Disconnect Z301 cables
2. Remove the four holding nuts and remove the Z301.

5.4.4 Installation

1. Install the new Z301.
2. Tighten the 4 nuts with a torque of **10 Nm**.
3. Connect the Z301 cables.

4. Secure the single wires close to the filter with cable ties.

5.4.5 Start-up

1. Close the right stand cover.
2. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
3. Switch on the system (**SYSTEM/ON**) at the control box.

5.4.5.1 Software

n.a

5.4.5.2 Tune-up

n.a

5.4.5.3 Testing

1. Perform some test scans.

5.4.6 Final Steps

n.a

5.5 Z302 Line Filter

5.5.1 Prerequisites

5.5.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

5.5.1.2 Time and Manpower

- 1 person
- Estimated total repair time: 1.0 hours.

5.5.1.3 Tools and auxiliary Equipment

- Standard service kit

5.5.2 Preliminary Steps

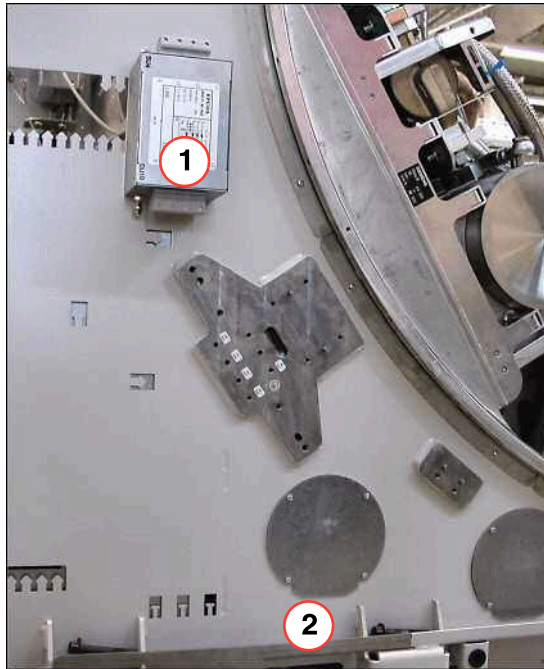
1. Remove the right stand cover and the right rear cover (→ Right rear cover / Page 57).
2. Switch off the system (Click <SYSTEM> and select <END>).
3. Switch off the gantry power with the service push-button S1 in the PDC A cabinet. Secure against unintended switch-on.

5.5.3 Removing

1. Disconnect Z302 cables

2. Remove the M5 holding screws and remove the Z302 filter.

Fig. 156: Gantry back (left side)



- (1) Filter Z302
- (2) Service opening

5.5.4 Installation

1. Install the new Z302 filter.
2. Tighten the screws with a torque of **6 Nm**.
3. Connect the Z302 cables.
4. Secure the single wires close to the filter with cable ties.

5.5.5 Start-up

1. Close the right stand cover and the right rear cover.
2. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
3. Switch on the system (**SYSTEM/ON**) at the control box.

5.5.5.1 Software

n.a

5.5.5.2 Tune-up

n.a

5.5.5.3 Testing

1. Perform some test scans.

5.5.6 Final Steps

n.a

5.6 Z303, Z304, Z305, Z306 Line Filters

5.6.1 Prerequisites

5.6.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

5.6.1.2 Time and Manpower

- 1 person
- Estimated total repair time: 1.0 hours.

5.6.1.3 Tools and auxiliary Equipment

- Standard service kit

5.6.2 Preliminary Steps

1. Remove the right stand cover.
2. Switch off the system (Click <SYSTEM> and select <END>).
3. Switch off the gantry power with the service push-button S1 in the PDC A cabinet. Secure against unintended switch-on.
4. Unlock the line filter assembly in the right gantry stand and open it.

5.6.3 Removing

1. Disconnect the cables of the defective line filter
2. Remove the holding nuts.

5.6.4 Installation

1. Install the new line filter.
2. Tighten the nuts or screws.
3. Connect the cables to the line filter.

4. Secure the single wires close to the filter with cable ties.
5. Close the line filter assembly in the right gantry stand and lock it.

5.6.5 Start-up

1. Close the right stand cover.
2. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
3. Switch on the system (**SYSTEM/ON**) at the control box.

5.6.5.1 Software

n.a

5.6.5.2 Tune-up

n.a

5.6.5.3 Testing

1. Perform some test scans.

5.6.6 Final Steps

n.a

5.7 Servo Controller Rotating (SCR)

5.7.1 Prerequisites

5.7.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".



There are two types of controllers !!

Mat# 77 41 486 (old controller).

Mat# 101 61 844 (new controller).

Take note when replacing.

5.7.1.2 Time and Manpower

- 1 person
- Estimated total repair time: 1.0 hours.

5.7.1.3 Tools and auxiliary Equipment

- Standard service kit

5.7.2 Preliminary Steps

1. Remove the right stand cover.
2. Switch off the system (Click <SYSTEM> and select <END>).
3. Switch off the gantry power with the service push-button S1 in the PDC A cabinet. Secure against unintended switch-on.

5.7.3 Removing

1. Disconnect all SCR cables
 - » Don't disconnect any single wires on X1 or X2. Remove the complete plugs.
2. Remove the service cover from the air duct above the SCR (only by the old controller).
3. Loosen the 4 nuts of the SCR.

4. Lift the SCR and remove it.

5.7.4 Installation

1. Install the new SCR.
2. Tighten the 4 nuts with a torque of **3 Nm**.
3. Install the air duct (only for the old controller).
4. Connect and secure the SCR cables.

5.7.5 Start-up

1. Close the right stand cover.
2. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
3. Switch on the system (**SYSTEM/ON**) at the control box.

5.7.5.1 Software

- Go to the Service home menu, select Options -> Service -> Local Service
- In the service software menu select **TuneUp -> FRU Replace**
- Select the corresponding part replaced and click GO
 - Note:** If the part you replaced is not in the list, press go and continue. The firmware check will be performed anyhow
- Follow the output instructions



If the firmware update aborts with an error and the firmware status is NOT ok, perform the firmware update again.

5.7.5.2 Tune-up

n.a.

5.7.5.3 Testing

1. Perform some test scans.

5.7.6 Final Steps

n.a.

5.8 Braking resistor (for SCR)

5.8.1 New type of braking resistor



The original braking resistor for the Definition, part no. 10161826, is no longer available. From now on, if the Definition's braking resistor needs to be replaced, the one for the Definition Flash (part no. 10430657) should be used.

Before the Definition Flash braking resistor can be installed in the Definition, a mechanical modification of the gantry must be carried out. The electrical connections remain the same.

For serial numbers 60142 and lower, the braking resistor kit with part no. 10161796 is required. The kit contains a mounting plate and the braking resistor.

For serial numbers 60143 and higher, only the braking resistor (part no. 10430657) is required. But the existing mounting plate must first be removed from the gantry, and then the braking resistor can be installed directly in the gantry.

5.8.2 Prerequisites

5.8.2.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

5.8.2.2 Time and Manpower

- 1 person
- Estimated total repair time: 1.0 hours.

5.8.2.3 Tools and auxiliary Equipment

- Standard service kit

5.8.3 Preliminary Steps

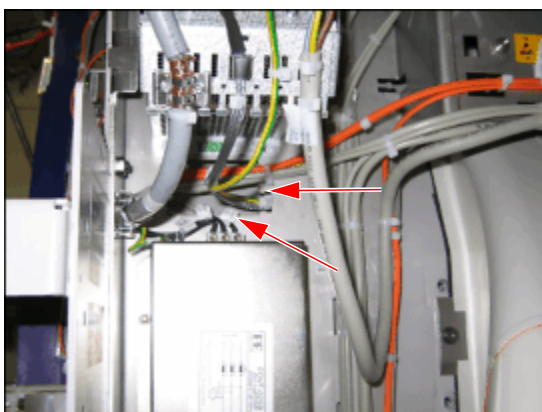
1. Switch off the system (Click <SYSTEM> and select <END>).
2. Switch off the gantry power with the service push-button S1 in the PDC A cabinet. Secure against unintended switch-on.

3. Remove the right stand cover.
4. Remove the right front and the rear cover.

5.8.4 Removing

1. Disconnect all cables coming from the braking resistor.

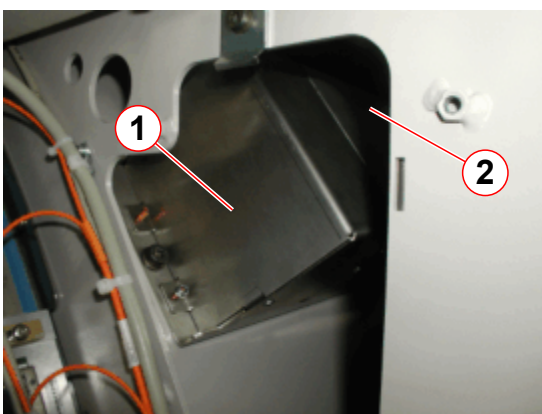
Fig. 157: Cables to the braking resistor



Before removing the cables of the braking resistor, check the cable routing as well as the distances between each cable to be removed and the neighboring cables. The new cables must be routed the same way as the old ones.

2. Cut cable ties, as necessary.
3. Remove the cover from the air duct.

Fig. 158: Overview of braking resistor



- (1) Cover of the air duct
(2) Braking resistor

4. Loosen the 4 nuts of the braking resistor.
5. Pull out the braking resistor.

6. Remove the four screws (with serial numbers 60142 and lower, there are only two screws) of the braking resistor mounting plate. Remove the mounting plate.



Keep the screws because they will be used for the installation later on.

5.8.5 Installation

1. Serial numbers 60142 and lower:

Install the mounting plate with the two screws inside the gantry, and then tighten the screws with a torque of **31 Nm**.

Fig. 159: New mounting plate for serial numbers 60142 and lower



Serial numbers 60143 and higher:

No mounting plate is used, so the braking resistor will be installed directly in the gantry with the four screws.

2. Serial numbers 60142 and lower:

Install the new braking resistor, and tighten the four nuts (delivered with the kit) with a torque of **12 Nm**.

Serial numbers 60143 and higher:

Install the new braking resistor and tighten the four screws with a torque of **31 Nm**.

3. Connect and secure the braking resistor cables.

- » Fasten the shielded resistor cable as far away from the neighboring cables as possible.

5.8.6 Start-up

1. Close the right stand cover.
2. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
3. Switch on the system (**SYSTEM/ON**) at the control box.

5.8.6.1 Software

The parameters will be loaded automatically after system switch-on.

5.8.6.2 Tune-up

n.a.

5.8.6.3 Testing

1. Perform some test scans.

5.8.7 Final Steps

n.a.

5.9 Water Cooling System (WCS) unit



WCS in systems up to serial number 60181:

There is not enough space in the gantry to lift the WCS to the correct height with the lifting device. The WCS must be pulled out by hand to get the WCS on the top cover of the spare part box.

Due to the heavy weight of the WCS, a carrier has to be requested to remove the WCS from the gantry.

For further details on that subject, contact CS HSC.

5.9.1 Prerequisites

5.9.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

5.9.1.2 Time and Manpower

- Persons: 2
- Estimated total repair time: 5 hours

5.9.1.3 Tools and auxiliary Equipment

- Wrench size 41
- Adjustable wrench
- Standard service kit
- To drain the WCS: a bucket and cleaning rags.

5.9.2 Preliminary Steps

Since you will be opening the facility cooling lines, inform the responsible facility personnel in advance.

Prior to starting with the work, check the contents of the spare part box.

1. Open the spare part box

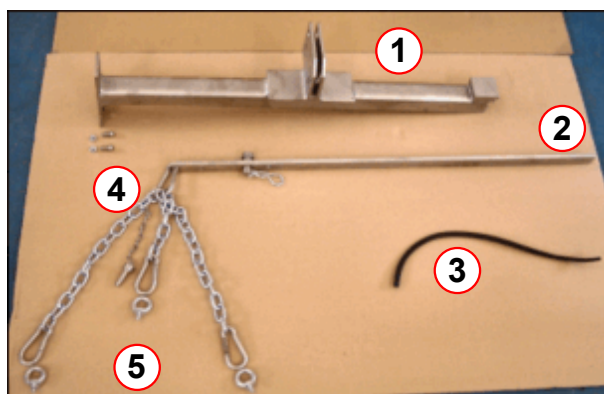
Fig. 160: spare part box



- (1) Top cover
- (2) Horizontal bar
- (3) Wooden bars

- Remove the top cover
- If you remove the insert plate with the tools and the side covers, you can turn the roller mounting bolts clockwise in the bottom of the box. To raise the box up on its rollers, roll the crate to a location near the gantry.
- Remove the parts of the lifting device

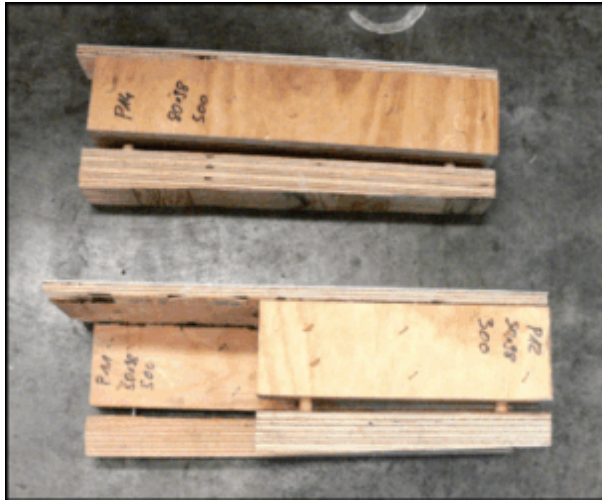
Fig. 161: Lifting assembly



- (1) Horizontal bar
- (2) Lever
- (3) Edge protection
- (4) Chain assembly
- (5) 3 eyebolts

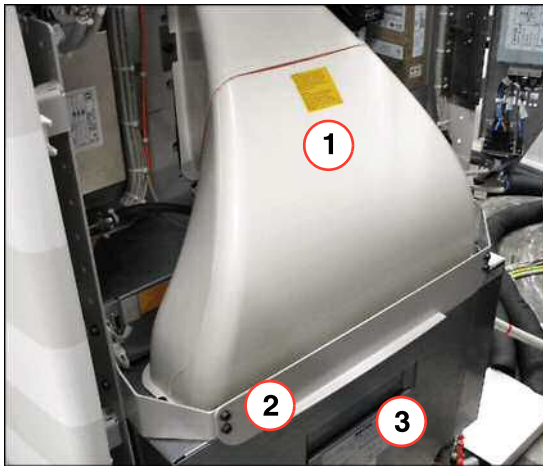
- Remove the wooden bars

Fig. 162: Wooden bars



2. Switch off the system (Click <SYSTEM> and select <END>).
3. Switch off the gantry power with the service push-button S1 in the PDC A cabinet. Secure against unintended switch-on.
4. Remove the 3 right gantry stand covers (side, front and back).

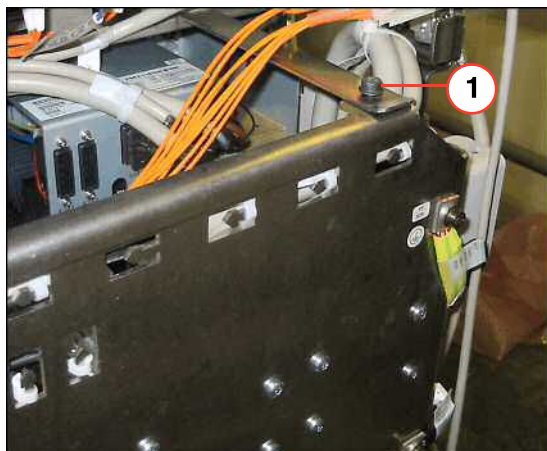
Fig. 163: Right gantry stand



- (1) Air duct
 - (2) Holding bracket
 - (3) WCS assembly
5. Remove the holding bracket (pos.2) of the WCS (4 outer screws).

6. Unlock the line filter assembly in the right gantry stand and open it.

Fig. 164: Filter assembly



(1) Locking screw

7. Remove the air duct.

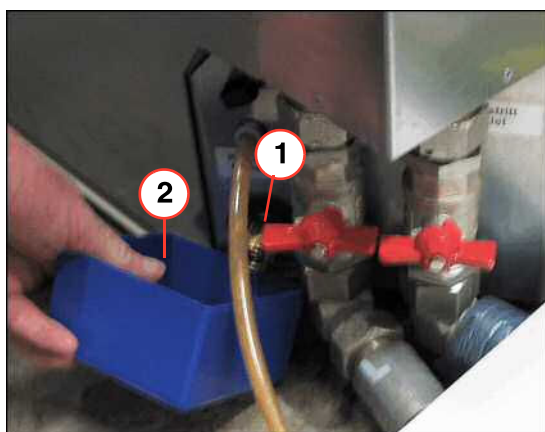
- Open the 7 holding screws. (Top screws are accessible from the back.)
- Remove the air duct.

8. Remove the WCS electronic unit, refer to: (→ WCS Electronic unit / Page 254).

9. Drain the WCS:

- Close the water in/out valves (the red winged screws must be in horizontal position)
- Place the flat collecting vessel under the drain valve (pos.2).
- Open the drain valve (pos.1)
- To speed up the drain procedure, remove the drain water tube (next pos. 1) from the WCS (push back the grey plastic ring to pull out the tube).

Fig. 165: WCS right lower part



(1) Drain valve

(2) Collect vessel

- Open the venting valve

Fig. 166: WCS right upper part



(1) Venting valve

- When the collect vessel is full of water, close the drain valve and empty the collect vessel.
- Repeat this procedure until the WCS is empty. **Note:** The WCS contains approx. **8 liters** of water.
- Unscrew the water in/out pipes from the WCS (above the red wing nuts, wrench size 41mm).

5.9.3 Removing the WCS from serial number 60182 onwards



WCS in systems up to serial number 60181:

There is not enough space in the gantry to lift the WCS to the correct height with the lifting device. The WCS must be pulled out by hand to get the WCS on the top cover of the spare part box.

Due to the heavy weight of the WCS, a carrier has to be requested to remove the WCS from the gantry.

For further details on that subject, contact CS HSC.

1. Remove the cable ties

Note: The cables must be positioned in such a way to provide enough open space above the WCS to lift it up by about 16 cm.

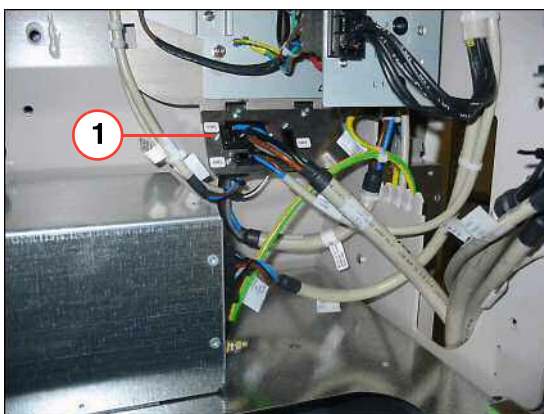
Fig. 167: WCS cable ties



2. Disconnect the WCS cables at X301 and X345.

- Remove the connector plate (pos. 1).

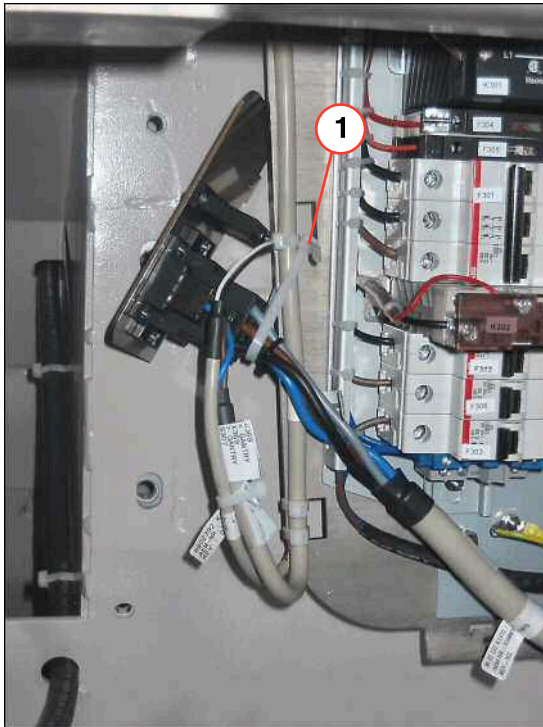
Fig. 168: WCS controller (right-hand side)



(1) Connector plate

3. Attach it with a cable tie to the mounting plate (pos. 1).

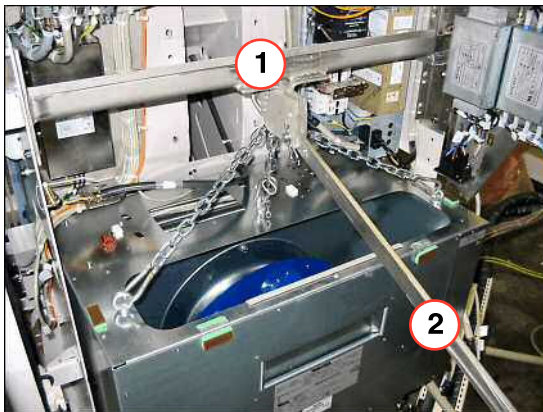
Fig. 169: Cable plate



(1) Holding point

4. Installation of the lifting device.

Fig. 170: lifting device

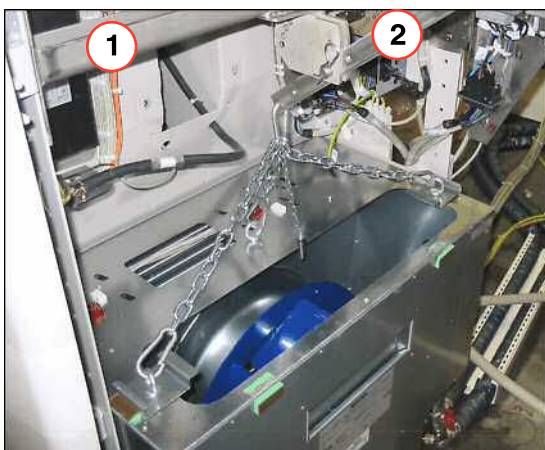


(1) Horizontal bar

(2) Lever

- Put the right side of the horizontal bar on to the support plate above the filter assembly.
- Attach the left side of the horizontal bar with 2 screws and nuts to the gantry stand plate.
- Attach the lever to the horizontal bar.

Fig. 171: lifting device



(1) Horizontal bar

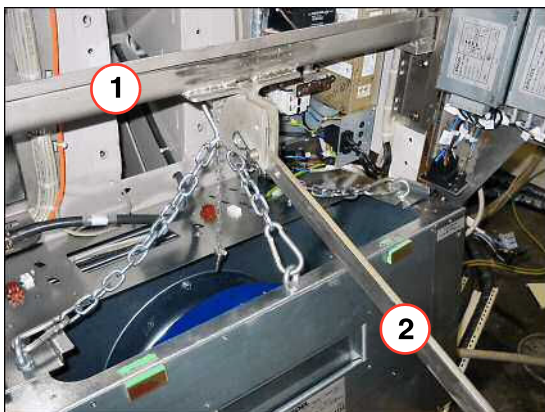
(2) Lever

- Attach the 3 eyebolts to the WCS
- Attach the middle part of the chain assembly to the lever.
- Attach the 3 ends of the chain to the 3 eyebolts. The shorter one must be attached at in back of the WCS.

5. To lift up the WCS to the correct height, two steps are required.

Lift up the WCS to the first step: Pull down the lever and lock the lever with the locking bolt.

Fig. 172: lifting device



(1) Horizontal bar

(2) Lever

6. Put the right and left wooden bars (13 cm height) under the WCS.

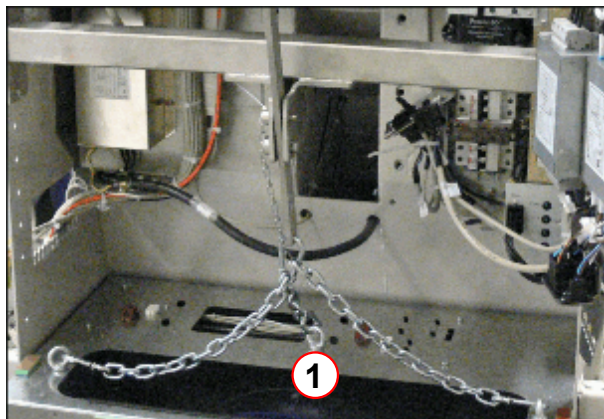
Hinweis: The aluminium side must be on top. The narrow wooden bar must be positioned on the left side, the wider wooden bar must be positioned on the right side.

Fig. 173: WCS height after first step



7. Unlock the lever and lower the WCS to the wooden bars.
8. To be able to raise the WCS once more, move the middle chain link (that can be opened) up by one link to shorten the chain length by one link.

Fig. 174: Chain assembly

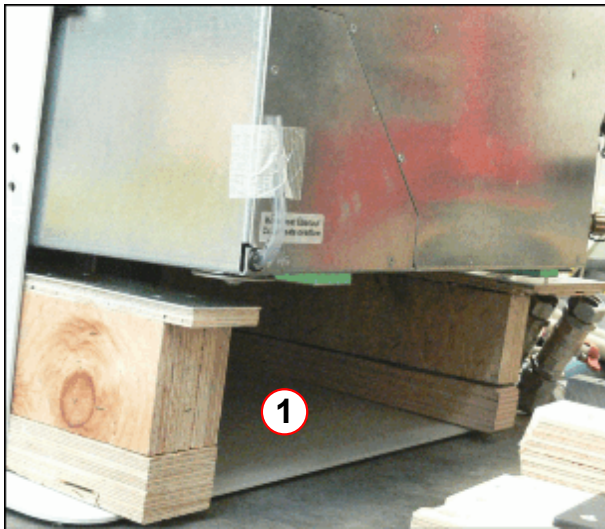


(1) Middle chain

9. **Lift up the WCS second step:** Pull down the lever and lock it with the locking bolt.

10. Remove the two bars and attach the smaller bars underneath them to form a higher platform. Put the extended right and left wooden bars back under the WCS again.

Fig. 175: WCS lifted



(1) Wooden bars with extension

11. Unlock the lever and lower the WCS to the extended wooden bars.
12. Moving the WCS onto the top cover of the spare part box:
 - Place the top cover of spare part box in front of the WCS.
 - Adjust the height of the top cover with the roller mounting bolts to the same level of the wooden bars and secure it.

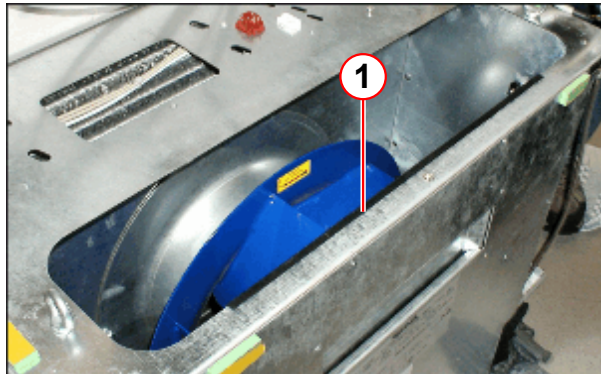
Fig. 176: Adjustment of the top cover



- Remove the 3 chain connections from the WCS

- Attach the edge protection at the upper part of the WCS to avoid injury when pulling out the WCS.

Fig. 177: WCS



(1) Edge protection

- Pull the WCS out of the gantry stand until the WCS is positioned on the top cover of the spare part box.
- Turn the four roller bolts CW to raise the 4 transportation wheels until the WCS can be moved away. Move the WCS away.

5.9.4 Installation of the new WCS

1. Remove the 4 side parts of the spare part box.
2. Take off the cardboard that protects the radiator. There will be no airflow if it is not removed.

Fig. 178: Cardboard



3. Turn the roller mounting bolts clockwise to lower the 4 transportation wheels until the new WCS can be moved.
4. Position the new WCS in front of the right gantry stand.
5. Adjust the height of the cover with the roller mounting bolts to the same level as that of the wooden bars and secure it.
6. Remove the 3 hooks from the "old" WCS and install them to the "new" WCS.
7. Attach the edge protection at the upper part of the new WCS to avoid injury.
8. Push the WCS into the right gantry stand.
9. Attach the chain to the 3 hooks.
10. Remove the transportation device.
11. Lift the WCS with the lever and lock it with the safety bolt.
12. Remove the two wooden bars.
13. To lower the WCS, two steps are required.
14. Unlock the lever and lower the WCS to the bottom of the right gantry stand.
15. Remove the lifting device from the right gantry stand.
16. Check the drain valve and the venting valve. They must be closed
17. Install the WCS electronic(→ WCS Electronic unit / Page 254)
18. Install the air duct with its holding bolts using a Tx 40 bit.
 - Attach the upper 4 screws (including the T plates) with a torque of **5 Nm**, from inside the frame of the gantry stand.
 - Attach the lower 3 screws to the WCS so that the foam at the lower side of the air duct is compressed.
19. Install the holding bracket(→ Fig. 163 Page 243).
20. Close the line filter assembly in the right gantry stand and lock it.

5.9.4.1 Filling the cooling system with water:

1. Reconnect water lines.
2. Ensure that the hose for condensed water is connected to the on-site outlet.
3. Open the two on-site water supply valves.
4. Open the bleeding valve for the bleeding air outflow. Use the four-square Allen key attached at the side.
5. Look to determine whether the clear hose for condensed water or bleeding air still contains air bubbles.
6. Close the bleeding valve for the outflow of condensed water or for bleeding air.
7. Wait 15 minutes and repeat the process beginning at step 4.



After the WCS is completely filled, the on-site connections as well as the connections of the gantry have to be checked at approx. 30 min. intervals for possible leaks.

If the connections are free of leaks, the connections have to be insulated as well with the insulation material .

5.9.5 Start-up

1. Close the right stand cover.
2. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
3. Switch on the system (**SYSTEM/ON**) at the control box.

5.9.5.1 Software

If the application was not restarted after replacing the WCS, restart the application now to ensure that the firmware is actualized.

5.9.5.2 Tune-up (from software version VA30 upward)

The **WCS** procedure is used to adjust the water flow sensor of the water cooling system:

- Select **Tune-up**.
- Select **Expert Mode**.
- Select **WCS**.
- Press **<Go>** to start the water flow sensor adjustment procedure.
 - The service user is asked to turn off the water supply manually to reset the water flow sensor. Afterwards click **OK** for confirmation.
 - The service user is asked to turn on the water supply and confirm it with **OK**.
 - The message "**Water Cooling System Adjustment**" **completed** appears and the adjustment results are saved automatically.

5.9.5.3 Testing

1. Perform some test scans.

5.9.6 Final Steps

n.a.

5.10 WCS Electronic unit

5.10.1 Prerequisites

5.10.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

5.10.1.2 Time and Manpower

- 1 person
- Estimated total repair time: 1.5 hours.

5.10.1.3 Tools and auxiliary Equipment

- Standard service kit

5.10.2 Preliminary Steps

1. Remove the right stand cover.
2. Switch off the system (Click <SYSTEM> and select <END>).
3. Switch off the gantry power with the service push-button S1 in the PDC A cabinet. Secure against unintended switch-on.
4. Unlock the line filter assembly in the right gantry stand and open it.

5.10.3 Removing

1. Remove the air duct, using a Tx 40 Torx bit.
 - Open the 7 holding screws
 - Remove the air duct.
2. Disconnect the cables from the WCS electronic unit.
 - Remove all cables from the WCS electronics unit (left end and right end).
 - Unlock the WCS electronics unit: Using a 10mm Allen key, turn the lock 90 degrees CCW.

3. Pull the WCS electronic unit toward you, lift it off its mounting studs, tip it 90 degrees (bottom of unit goes toward gantry, as shown below). Take care not to damage the cables that feed into the WCS Electronics Box from below as you tip the unit up!

Fig. 179: WCS electronics box during removal



4. Reach over the Electronics Box and disconnect the 4 cables that come from the WCS.
5. Remove the defective WCS electronic unit.

» **Observe ESD precautions**

5.10.4 Installation

1. Install the new WCS electronic unit by reversing the above procedure.



Before you try and lock the box down, make sure all mounting studs are firmly engaged in the slotted holes in the bottom of the electronics box.

2. Ensure that all connectors are connected properly and are secure.
Tighten the PE-cable with a torque of **5 Nm**
3. Install the air duct with its holding bolts using a Tx 40 bit.
 - Attach the upper 4 screws (including the T plates) with a torque of **5 Nm**, from inside the frame of the gantry stand.
 - Attach the lower 3 screws to the WCS so that the foam at the lower side of the air duct is compressed.

5.10.5 Start-up

1. Close the right stand cover.
2. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the “0” position and then to the “1” position.
3. Switch on the system (**SYSTEM/ON**) at the control box.

5.10.5.1 Software

- Go to the Service home menu, select Options -> Service -> Local Service

- In the service software menu select **TuneUp** -> **FRU Replace**
- Select the corresponding part replaced and click GO

Note: If the part you replaced is not in the list, press go and continue. The firmware check will be performed anyhow

- Follow the output instructions



If the firmware update aborts with an error and the firmware status is NOT ok, perform the firmware update again.

5.10.5.2 Tune-up (from software version VA30 upward)

The **WCS** procedure is used to adjust the water flow sensor of the water cooling system:

- Select **Tune-up**.
- Select **Expert Mode**.
- Select **WCS**.
- Press **<Go>** to start the water flow sensor adjustment procedure.
 - The service user is asked to turn off the water supply manually to reset the water flow sensor. Afterwards click **OK** for confirmation.
 - The service user is asked to turn on the water supply and confirm it with **OK**.
 - The message "**Water Cooling System Adjustment**" **completed** appears and the adjustment results are saved automatically.

5.10.5.3 Testing

1. Perform some test scans.

5.10.6 Final Steps

n.a.

5.11 WCS Air Temperature Sensor (gantry out)

5.11.1 Prerequisites

5.11.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

5.11.1.2 Time and Manpower

- 1 person
- Estimated total repair time: n.a.

5.11.1.3 Tools and auxiliary Equipment

- Wrench M41
- Adjustable wrench
- Standard service kit

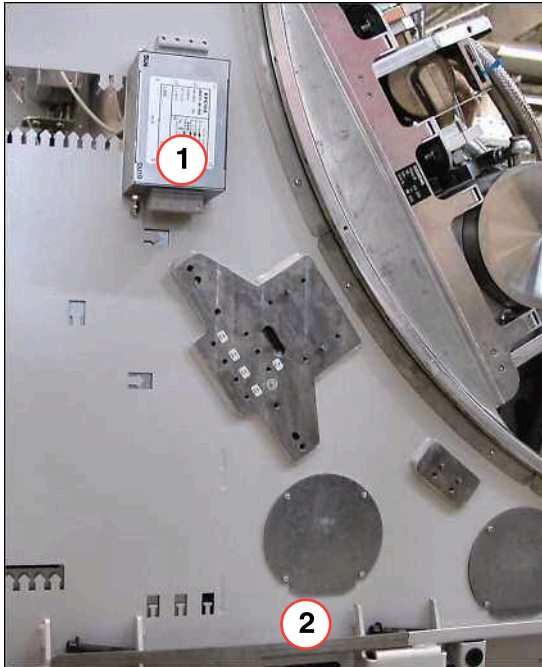
5.11.2 Preliminary Steps

1. Remove the right rear cover (→ Right rear cover / Page 57).
2. Switch off the system (Click <SYSTEM> and select <END>).
3. Switch off the gantry power with the service push-button S1 in the PDC A cabinet. Secure against unintended switch-on.

5.11.3 Removing

1. Remove the service opening

Fig. 180: Gantry back (left side)

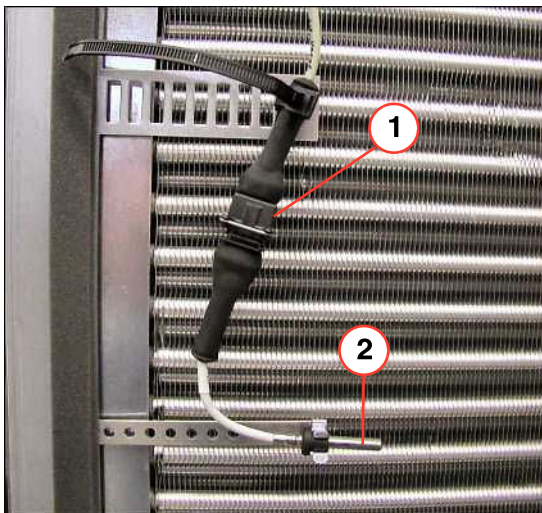


(1) Filter Z302

(2) Service opening

2. Removing the temperature sensor

Fig. 181: WCS air temperature sensor (gantry out)



(1) Connector

(2) Temperature sensor

The air temperature sensor is attached to the WCS. The air temperature sensor must be replaced with one hand through the service opening.



The thin blades of the WCS behind the temperature sensor are sharp. Be careful to avoid injuries.

- Disconnect the cable first.
- Then pull the sensor out of its holding device and remove it.

5.11.4 Installation

1. Install the new air temperature sensor.

5.11.5 Start-up

1. Close the right rear cover.
2. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
3. Switch on the system (**SYSTEM/ON**) at the control box.

5.11.5.1 Software

n.a.

5.11.5.2 Tune-up

n.a.

5.11.5.3 Testing

1. Perform some test scans.

5.11.6 Final Steps

n.a.

5.12 WCS Air Temperature Sensor (gantry in)

5.12.1 Prerequisites

5.12.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the “General” section of this document and/or the “Product specific safety notes”.

5.12.1.2 Time and Manpower

- 1 person
- Estimated total repair time: n.a.

5.12.1.3 Tools and auxiliary Equipment

- Standard service kit

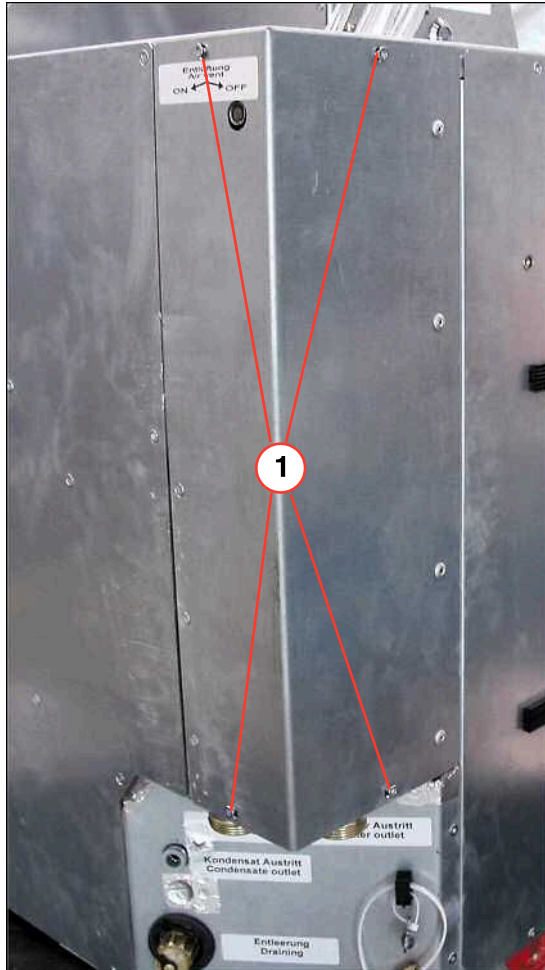
5.12.2 Preliminary Steps

1. Remove the right stand cover.
2. Switch off the system (Click <SYSTEM> and select <END>).
3. Switch off the gantry power with the service push-button S1 in the PDC A cabinet. Secure against unintended switch-on.

5.12.3 Removing

1. Remove the 4 screws attaching the cover and remove it.

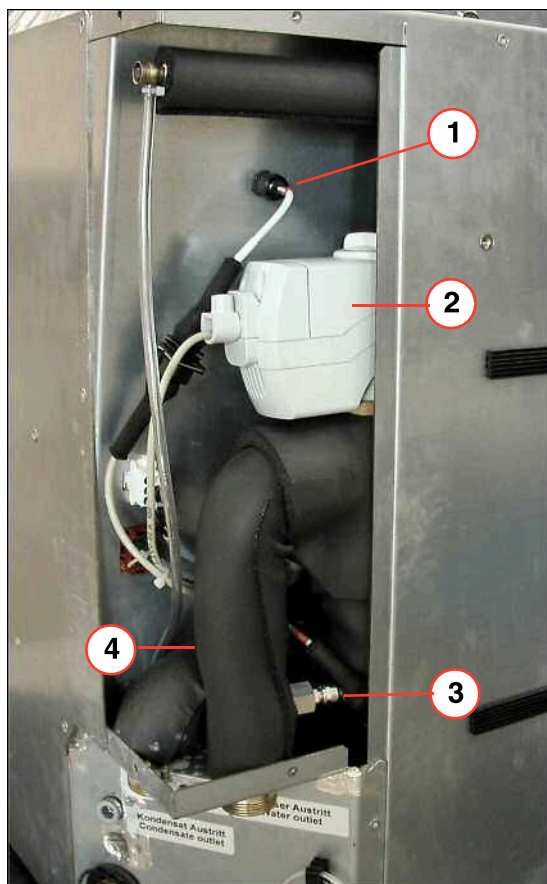
Fig. 182: WCS



(1) 4 screws securing the cover

2. Loosen the plastic screw at the air temperature sensor and pull the sensor out.

Fig. 183: WCS right side



- (1) Air temperature sensor (gantry in)
- (2) Adjustment drive
- (3) Water flow rate measurement device
(not replaceable in the field)
- (4) Location of the water temperature sensor

3. Disconnect the sensor

5.12.4 Installation

1. Insert the new air temperature sensor and connect it.
2. Install the service cover of the WCS.

5.12.5 Start-up

1. Close the right cover.
2. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.

3. Switch on the system (**SYSTEM/ON**) at the control box.

5.12.5.1 Software

n.a.

5.12.5.2 Tune-up

n.a

5.12.5.3 Testing

1. Perform some test scans.

5.12.6 Final Steps

n.a.

5.13 WCS Water Temperature Sensor

5.13.1 Prerequisites

5.13.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the “General” section of this document and/or the “Product specific safety notes”.

5.13.1.2 Time and Manpower

- 1 person
- Estimated total repair time: 1.5 hours.

5.13.1.3 Tools and auxiliary Equipment

- Standard service kit

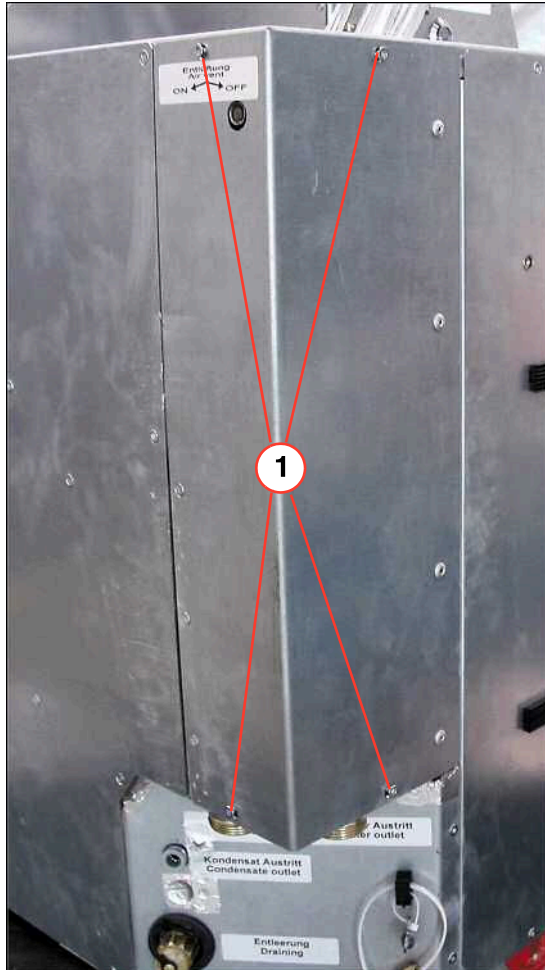
5.13.2 Preliminary Steps

1. Remove the right stand cover.
2. Switch off the system (Click <SYSTEM> and select <END>).
3. Switch off the gantry power with the service push-button S1 in the PDC A cabinet. Secure against unintended switch-on.

5.13.3 Removing

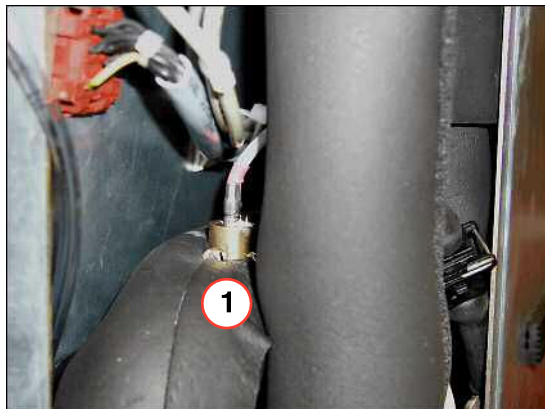
1. Remove the 4 screws attaching the cover and remove it.

Fig. 184: WCS



- (1) 4 screws securing the cover
2. Loosen the attachment screw of the water temperature sensor.

Fig. 185: WCS water temperature sensor



- (1) Sensor

3. Disconnect the water temperature sensor and remove it.

5.13.4 Installation

1. Install the new water temperature sensor.
2. Install the service cover of the WCS.

5.13.5 Start-up

1. Close the right stand cover.
2. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
3. Switch on the system (**SYSTEM/ON**) at the control box.

5.13.5.1 Software

n.a.

5.13.5.2 Tune-up

n.a

5.13.5.3 Testing

1. Perform some test scans.

5.13.6 Final Steps

n.a.

5.14 WCS Adjustment Drive

5.14.1 Prerequisites

5.14.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

5.14.1.2 Time and Manpower

- 1 person
- Estimated total repair time: 1.5 hours.

5.14.1.3 Tools and auxiliary Equipment

- Standard service kit

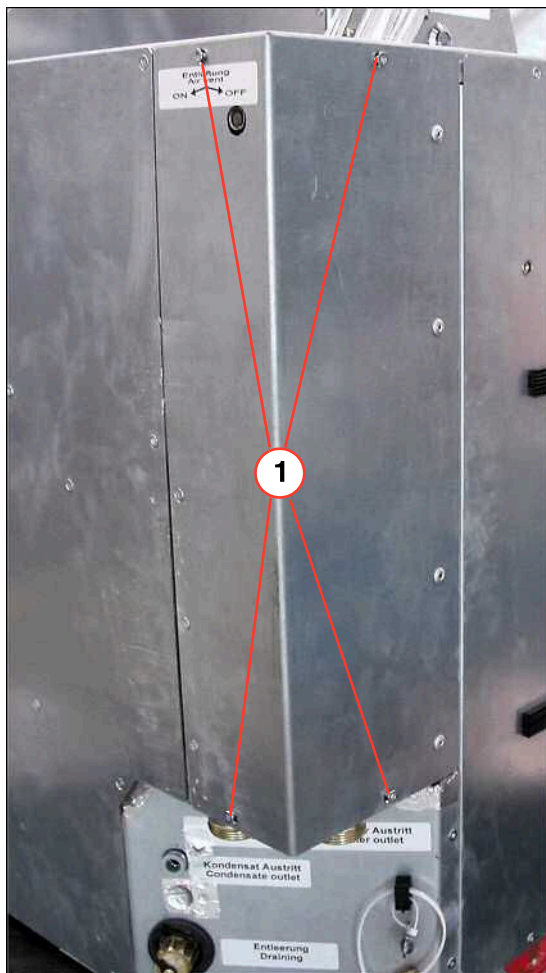
5.14.2 Preliminary Steps

1. Remove the right stand cover.
2. Switch off the system (Click <SYSTEM> and select <END>).
3. Switch off the gantry power with the service push-button S1 in the PDC A cabinet. Secure against unintended switch-on.

5.14.3 Removing

1. Remove the 4 screws attaching the cover and remove it.

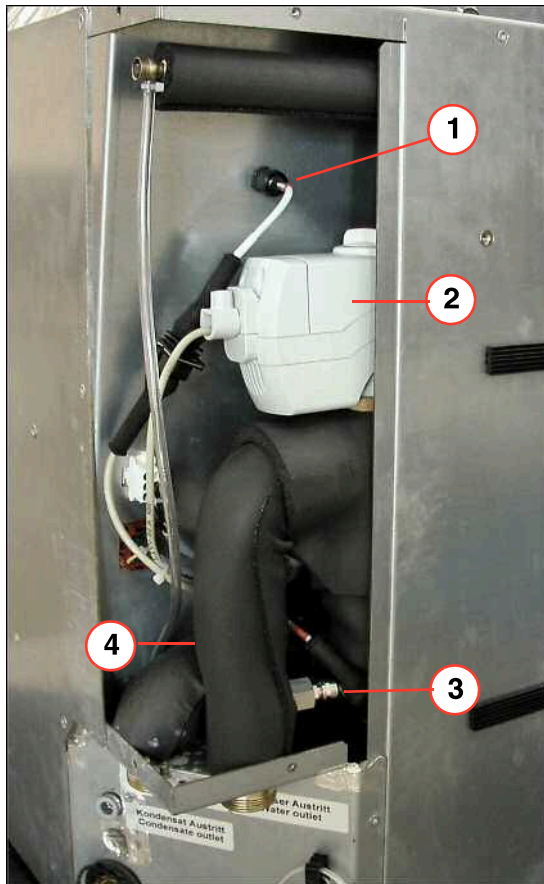
Fig. 186: WCS



(1) 4 screws securing the cover

2. Disconnect the cable of the adjustment drive.

Fig. 187: WCS right side



- (1) Air temperature sensor (gantry in)
 - (2) Adjustment drive
 - (3) Water flow rate measurement device
(not replaceable in the field)
 - (4) Location of the water temperature sensor
3. Hold the attachment screw of the adjustment drive and move the housing of the adjustment drive until the attachment screw can be turned by hand.
Remove the adjustment drive.

5.14.4 Installation

1. Install the new adjustment drive. Tighten the attachment screw by hand.
2. Connect the cable to the new adjustment drive.

5.14.5 Start-up

1. Close the right stand cover.

2. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
3. Switch on the system (**SYSTEM/ON**) at the control box.

5.14.5.1 Software

n.a.

5.14.5.2 Tune-up

n.a

5.14.5.3 Testing

1. Perform some test scans.

5.14.6 Final Steps

n.a.

5.15 WCS Condensate Water Pump Kit

5.15.1 Prerequisites

5.15.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

5.15.1.2 Time and Manpower

- 1 person
- Estimated total repair time: 1.5 hours.

5.15.1.3 Tools and auxiliary Equipment

- Standard service kit

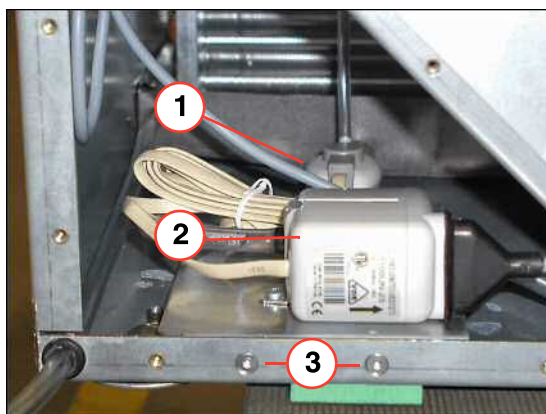
5.15.2 Preliminary Steps

1. Remove the right stand cover.
2. Switch off the system (Click <SYSTEM> and select <END>).
3. Switch off the gantry power with the service push-button S1 in the PDC A cabinet. Secure against unintended switch-on.

5.15.3 Removing

1. Remove the service cover at the lower left corner of the WCS.
Remove the 2 attachment screws (Pos. 3) of the condensate water pump kit.

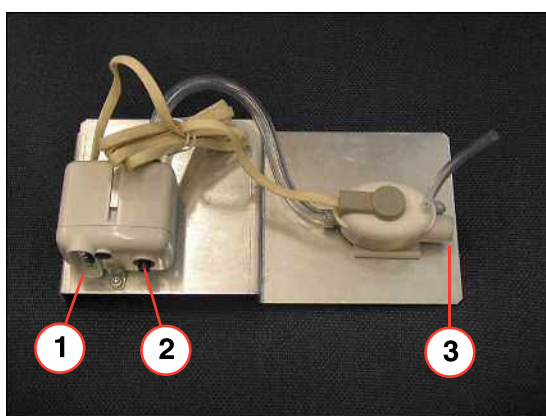
Fig. 188: WCS condensate water assembly



- (1) Float switch
- (2) Water pump
- (3) 2 attachment screws for the water pump kit

2. Remove the electrical connector (Pos. 1) and the water connection (Pos.2) from the water pump and the water connection (Pos. 3) from the float switch.

Fig. 189: WCS condensate water pump kit



- (1) Water pump connector
- (2) Water connection at the pump
- (3) Water connection at the float switch

3. Remove the condensate water pump kit from the WCS.

5.15.4 Installation

1. Install the new condensate water pump kit to the WCS.
2. Connect the water hoses to the float switch and to the pump.
Ensure that the water connectors are connected properly.
3. Connect the electrical connector to the pump.

4. Install the service cover at the lower left corner of the WCS.

5.15.5 Start-up

1. Close the right stand cover.
2. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
3. Switch on the system (**SYSTEM/ON**) at the control box.

5.15.5.1 Software

n.a.

5.15.5.2 Tune-up

n.a

5.15.5.3 Testing

1. Perform some test scans.

5.15.6 Final Steps

n.a.

5.16 WCS Condensate Float Switch

5.16.1 Prerequisites

5.16.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

5.16.1.2 Time and Manpower

- 1 person
- Estimated total repair time: 1.5 hours.

5.16.1.3 Tools and auxiliary Equipment

- Standard service kit

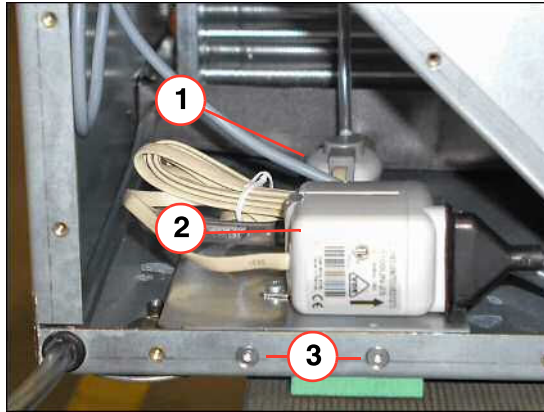
5.16.2 Preliminary Steps

1. Remove the right stand cover.
2. Switch off the system (Click <SYSTEM> and select <END>).
3. Switch off the gantry power with the service push-button S1 in the PDC A cabinet. Secure against unintended switch-on.

5.16.3 Removing

1. Remove the service cover at the lower left corner of the WCS.
Remove the 2 attachment screws (Pos. 3) of the condensate water pump kit.

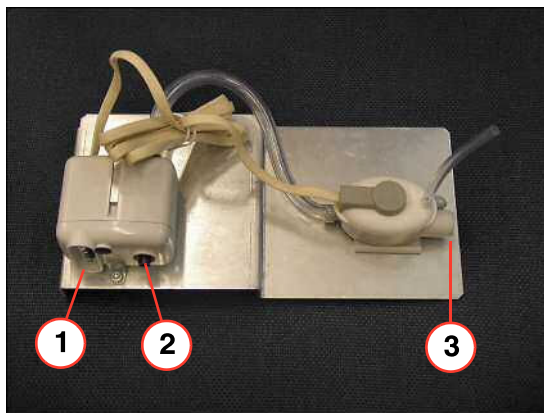
Fig. 190: WCS condensate water assembly



- (1) Float switch
- (2) Water pump
- (3) 2 attachment screws for the water pump kit

2. Remove the electrical connector (Pos. 1) and the water connection (Pos.2) from the water pump and the water connection (Pos. 3) from the float switch.

Fig. 191: WCS condensate water pump kit



- (1) Water pump connector
- (2) Water connection at the pump
- (3) Water connection at the float switch

3. Remove the condensate water pump kit from the WCS.
4. Remove the float switch from the condensate water pump kit.

5.16.4 Installation

1. Install the new float switch to the condensate water pump kit.
2. Install the condensate water pump kit at the WCS.

3. Connect the water hoses to the float switch and to the pump.
Ensure that the water connectors are connected properly.
4. Connect the electrical connector to the pump.
5. Install the service cover at the lower left corner of the WCS.

5.16.5 Start-up

1. Close the right stand cover.
2. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
3. Switch on the system (**SYSTEM/ON**) at the control box.

5.16.5.1 Software

n.a.

5.16.5.2 Tune-up

n.a

5.16.5.3 Testing

1. Perform some test scans.

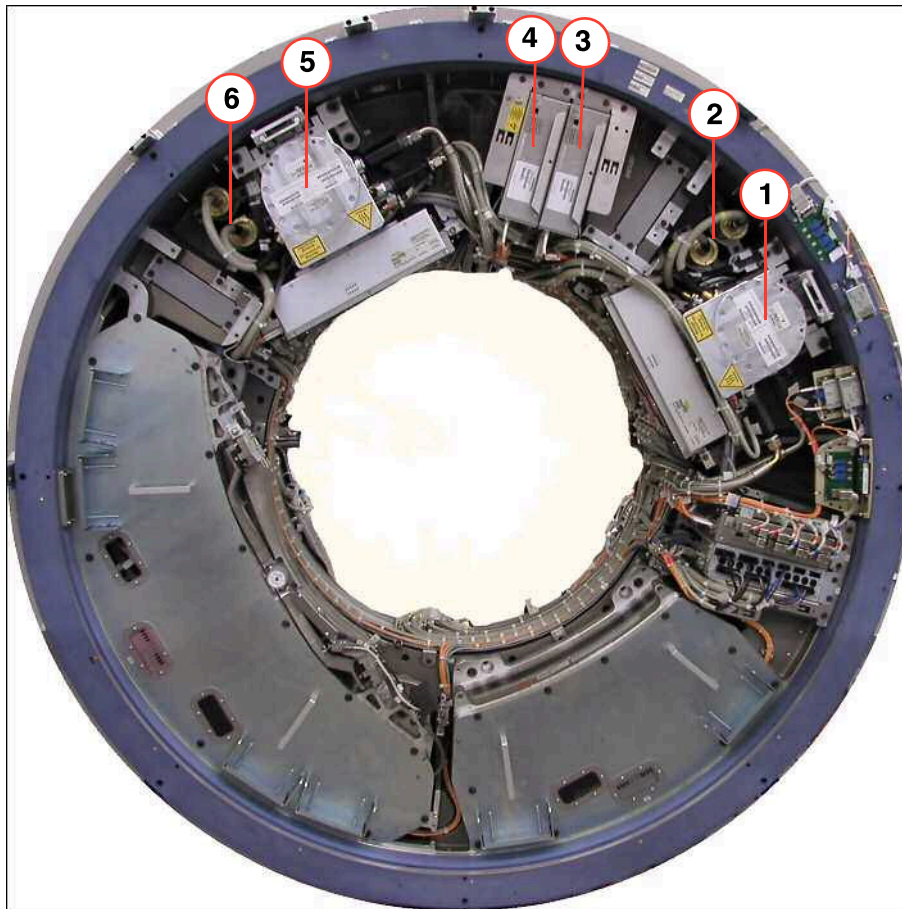
5.16.6 Final Steps

n.a.

6.1 Overview

The figure below shows the location of XRS parts in front of the gantry

Fig. 192: Overview gantry front, XRS parts

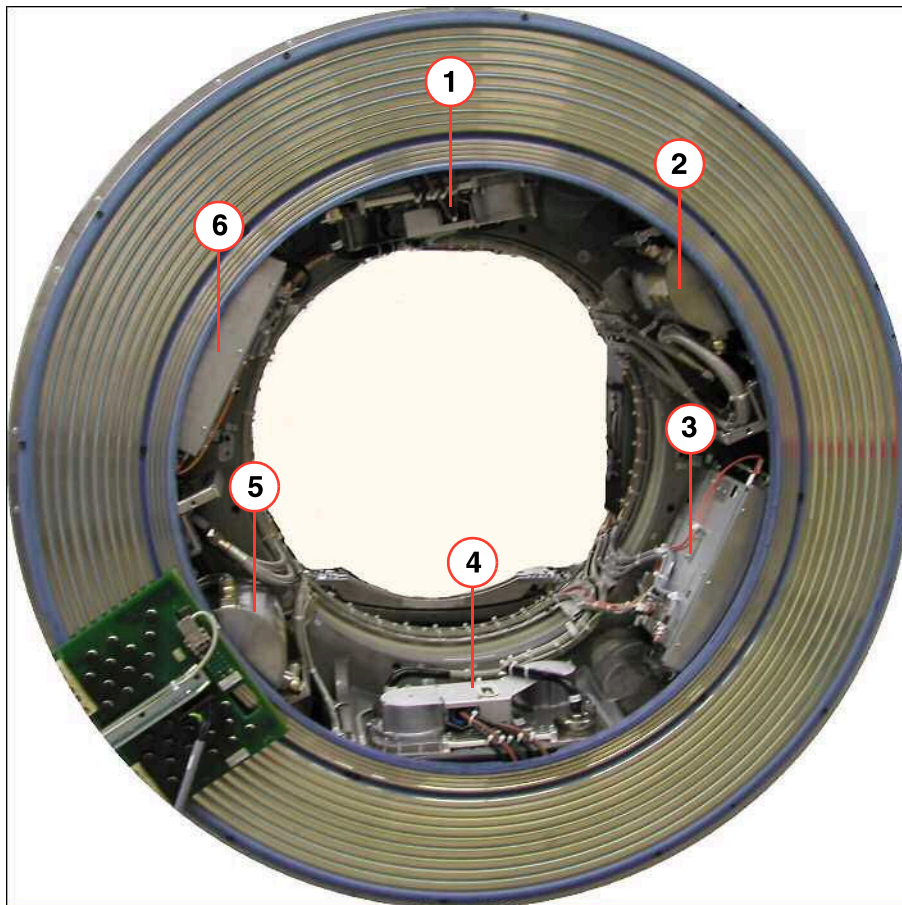


Parts in gantry front:

1. Straton MX A
2. HV plug connectors A
3. XDC A
4. XDC B
5. Straton MX B
6. HV plug connectors B

The figure below shows the location of the XRS parts at the rear of the gantry.

Fig. 193: Overview XRS parts in back



Parts at the rear of the gantry

1. HV-tank A
2. X-ray tube cooling (XTC) B
3. X-ray generator rotating (XGR) B
4. HV-tank B
5. X-ray tube cooling (XTC) A
6. X-ray generator rotating (XGR) A

6.2 Use of lifting strap

- Use of lifting strap.

Fig. 194: Belt inserted through the axle 10...15 cm (app. 4"...6").



- Insert the strap through the slot in the axle. Pull out 10 - 15 cm on the other side.
- Avoid pulling through the entire remaining length.
- Turn the handle several times to roll-up the loose end on the axle.

Fig. 195: The belt is rolled up several turns before it is tight.



- If installed properly, the strap is coiled up several turns with the leading 10 - 15 cm rolled up neatly in the middle, doubling up against the strap.

6.3 Straton MX

The video sequence of tube replacement has temporarily been removed due to changes in the installation that are not (yet) included in the video.

6.3.1 Prerequisites

6.3.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

6.3.1.2 Time and Man Power

- 1 Person
- Estimated total repair time: 7 hrs.

6.3.1.3 Tools and auxiliary Equipment

- Standard tool kit
- Lifting support
- HV plug tightening tool

Fig. 196: HV plug tightening tool



- HV plug tightening tool
- Strap with tensioner
- T50 Torx bit

- The following tools are included with the part:
 - One set of new screws
 - Insulation parts
 - Tube papers
 - Protectors for oil couplings



Note: Tube manuals are no longer sent with the tube. Refer to (→ Final steps / Page 303) for more information.

6.3.2 Preliminary steps



Beginning with software version VA20, it is no longer necessary to run "Tube Replace" and enter the reason for replacement before the "old" tube is deinstalled.

The replacement reason will be inserted after the new tube is installed and "Tube Replace" has started. This information will be sent via SRS so that the tube factory can access the data.



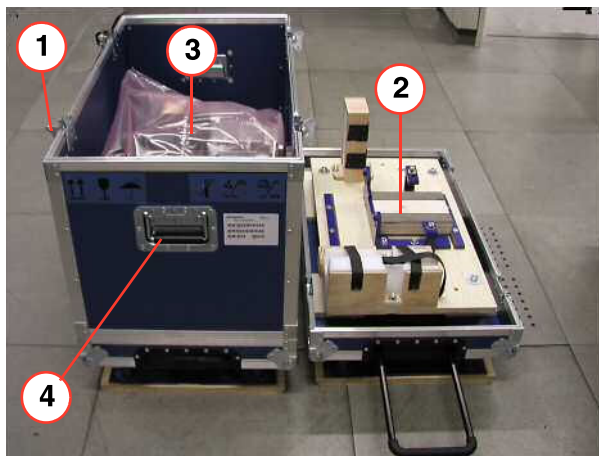
From 11/2009 on, new tubes need a pressure of 1.25 bar (+ altitude correction). These tubes are clearly labelled, and also the expansion unit is labelled with a red sticker.

Set the pressure on "old" tubes without label to 1.05 bar, and on newer tubes to 1.25 bar according to the label.

1. Allow the system to cool down for **at least 10 minutes** after the last scan. During the cooling time, the **system must be switched on**.
 - » Be aware that some system components may be very hot, e.g., the Straton tube, XTC ... may be up to 130 degree Celsius.

2. While the system is cooling down prepare the transport box.

Fig. 197: Tube transport box

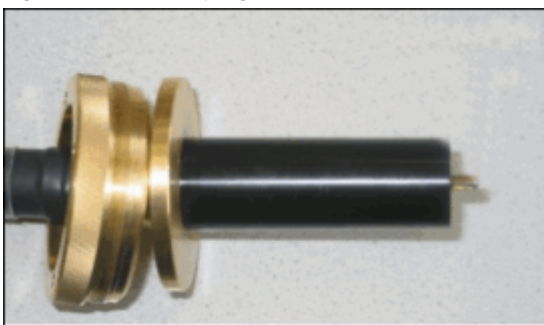


- Open the 4 locks at the top (item 1) and remove the top (item 2) .
- Remove all service tools and papers from the inside (item 3).
- Remove the side cover of the transport box (item 4).

Please check the following items:

- The expansion unit must be connected correctly (if it is not connected correctly, order a new tube).
- The BNC connector (X5) for the X-ray diode must be connected correctly.
- The X-ray window must not have any damage; clean the X-ray window.
- Check the HV plugs for damage:
 - The pin of the mini HV plug may vary depending on the production date of the tube. Do not try to adjust the pin length.

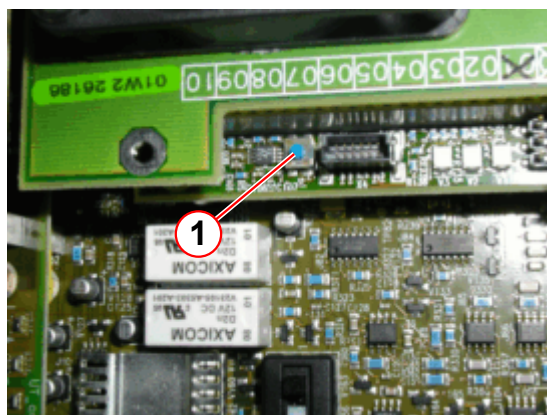
Fig. 198: HV mini plugs



3. Wait until the tube and the cooling system have cooled down, then switch the system to **COMP/ON** status at the control box.
4. Switch off the gantry power using service pushbutton S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on the breakers that S1 switched off (e.g. lock and tag).

5. Press the blue Reset button on the MCB of the PDC XGS Control piggyback board in the PDC A to enable service power to the PHS.

Fig. 199: MCB on D700



(1) Blue reset button

6. Remove the gantry lower front ring segment. Refer to (→ Removing / Page 32) the lower front ring segment..
7. Turn the gantry by hand until the Straton MX tube (A or B) is in the 6 o'clock position.
8. Secure against unintended rotation by using the front safety bolt (→ Safety bolts / Page 21).
9. Install the data ring protection (→ Data ring protection / Page 23).

6.3.3 Removing



CAUTION

[hz_serdoc_F13G02U01M01]

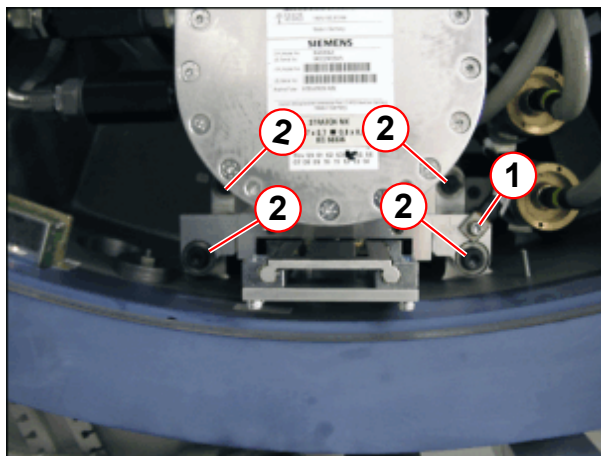
Temperature of tube and/or tube cooling device parts may be above 70C (with Straton up to 130C).

Risk of burns!

- » Avoid contact with oil and exposed parts of X-ray tube and cooling device.
- » Wait until tube and cooling device parts are chilled down.

1. Remove the tube ground connection and the attachment screws

Fig. 200: Straton MX attachment screws



- Disconnect ground straps (item 1) using a T30 Torx bit.
- Remove the 4 M10*40 mm tube attachment Torx bolts using a T50 Torx bit (item 2).
- Remove cable ties as needed.
- Move the tube gently out of the gantry until you gain access to remove the oil hoses and plugs.
 - » **The tube will be guided by the sliding carriage.**

2. Disconnect the oil couplings

Fig. 201: Tube A oil connections

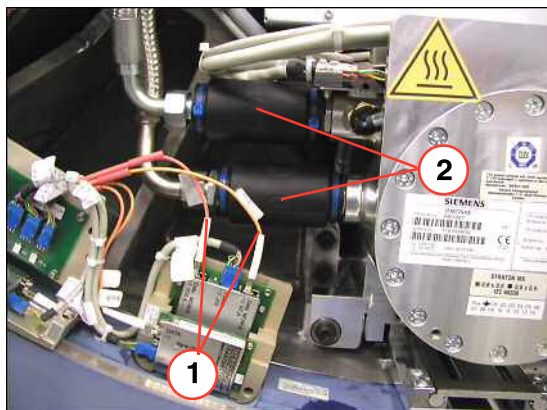
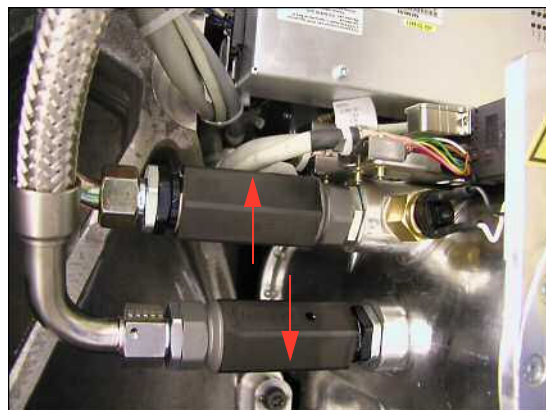


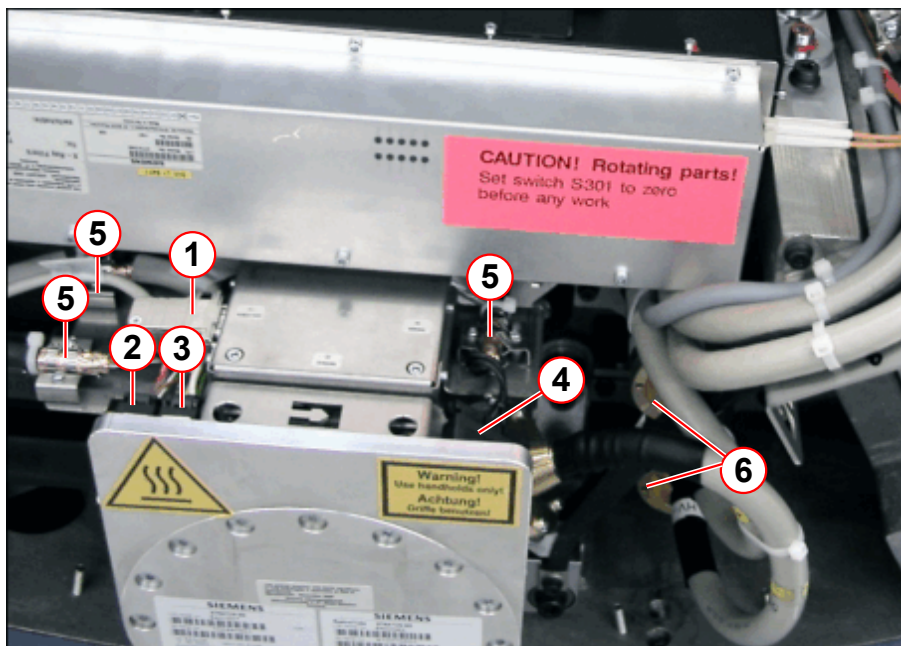
Fig. 202: Oil connections without protection



- Remove 2 fiber optic cables from transmitter (item 1). Not necessary with tube B!
- Open the 4 tie wraps and remove protections from oil couplings (item 2).
- Put some paper towels underneath the couplings before removing them.
 - » **A few drops of oil will drip from the couplings during disassembly.**
- Open the oil couplings at the X-ray tube. To loosen them, the couplings have to be turned in the direction of the red arrows. If necessary use a wrench to open the couplings!
- Dry the couplings to prevent oil from dripping into the gantry.

3. Disconnect tube connections

Fig. 203: Tube cable connections



- Remove cable ties as needed to release cables from the tube.
 - Disconnect the following cable /plugs from the X-ray tube:
 - Plug X1 from X-ray tube (Item 1).
 - Plug X8 from X-ray tube (Item 2).
 - Plug X4 from X-ray tube (Item 3).
 - Plug X5 from X-ray tube (Item 4).
 - Remove all cable ties as necessary to release cables from the tube.
 - Remove the 3 cable clamps (item 5)
 - Remove high voltage cables from HV connector (item 6) by unscrewing the threaded rings using the special tool. Remove cable ties as needed.
4. Check the HV connectors for any obvious damage caused by high-voltage flashovers. If the plug is damaged, check the receptacles and clean them if necessary. If the receptacles of the sockets are severely damaged, replace the cable/sockets to the HV tank.

5. Remove the anode motor connections

Fig. 204: Rotating anode connection (old type)

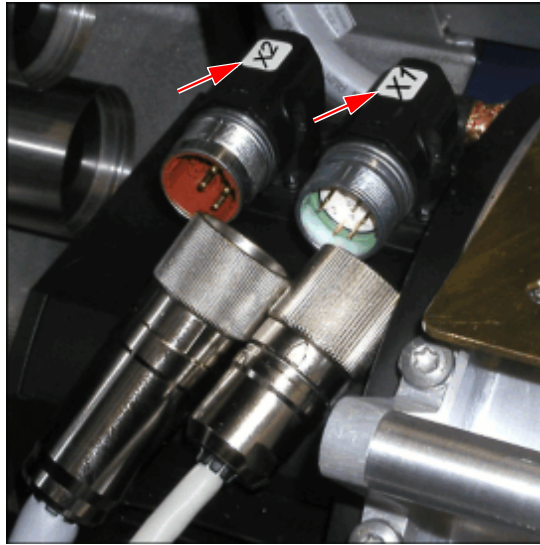
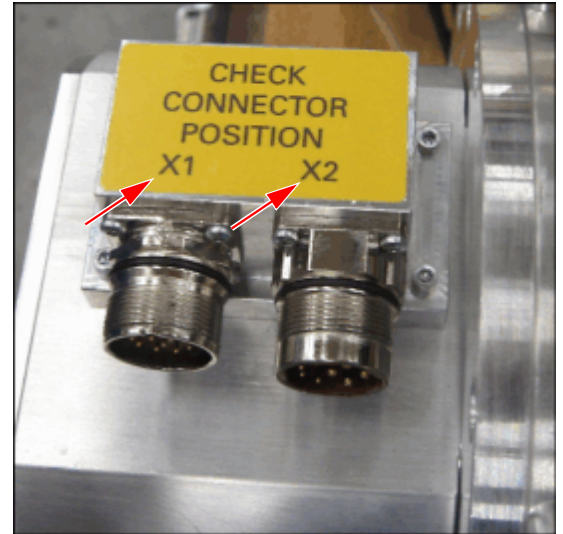
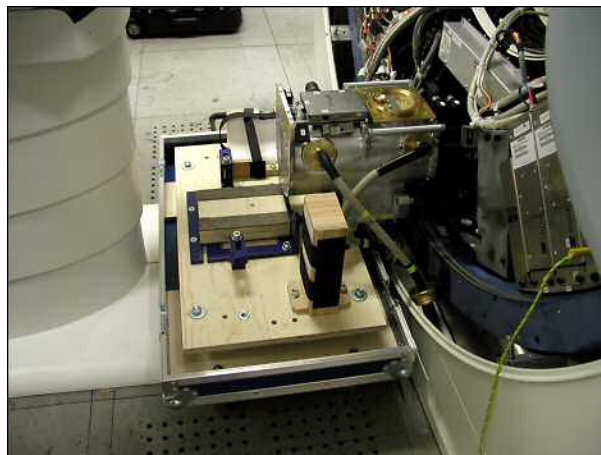


Fig. 205: Rotating anode connection (new type)



- Move the tube to its outermost position.
 - Disconnect connector X1 and X2 plugs from the anode motor
 - » **Note:** With tube revision 11 a new type of rotating anode connector is installed. In particular, note that the X1 connector is now located on the left and the X2 connector on the right side installed, which is the opposite of what it was previously (see arrows left/right image). Please pay attention to this difference, especially when reconnecting the plugs at the new tube.
 - Move the tube into the gantry again to access the transport device.
6. Position the transport device and the old tube for replacement.

Fig. 206: Old tube over the transport device



- Position the transport device between table and gantry.
- Move the tube out of the gantry again until the end position of the sliding carriage is reached.

7. Switch on F7 in PDC A to switch on power to the patient table.

» **After a few minutes the table control at the left table side is ready for use .**

8. Place the lifting support at the end of the tabletop plate.

9. Remove the Straton MX tube

Fig. 207: Straton MX tube secured to the lifting support

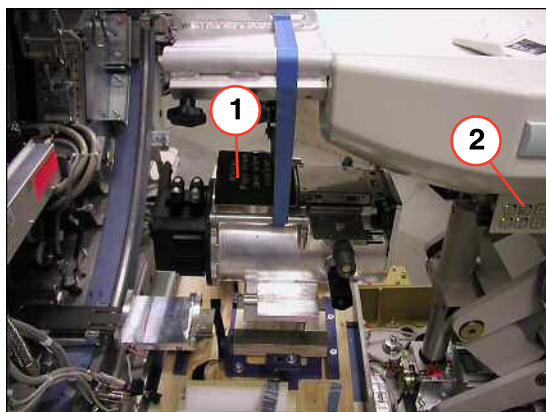
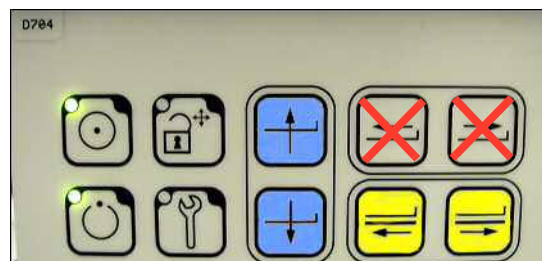


Fig. 208: PHS control panel



- Secure the tube to the lifting strap (left image). Refer to (→ Use of lifting strap / Page 279).
- Lift the tube carefully using the "up" button of the table control (right image, blue marked buttons).
- Move the empty sliding carriage back into the gantry for better access.
- Move the tube gently out of the gantry until the tube is positioned correctly over the transport device. Use the tabletop control buttons (right image, yellow marked buttons) to control the movement.
- Lower the tube onto the transport device using the down button of the table control.

10. Secure the replaced tube to the transport frame.

Fig. 209: Securing the tube to the transport box

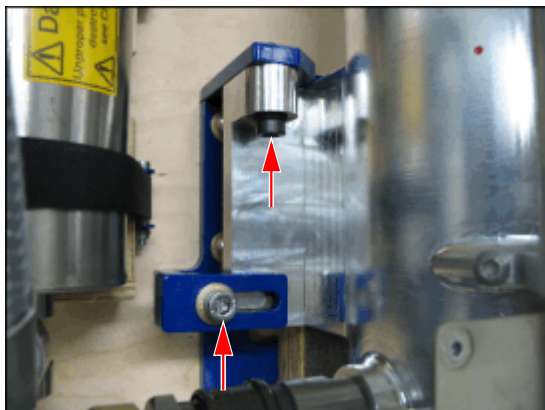
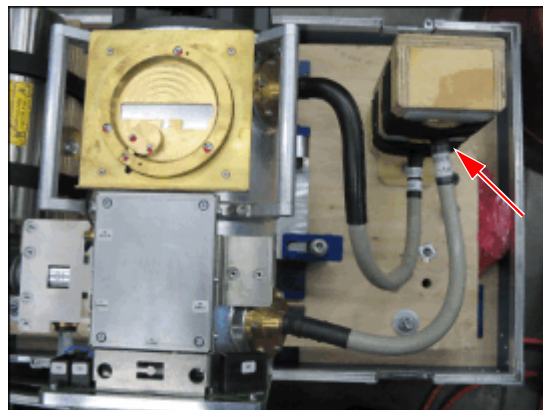


Fig. 210: HV plug transport position



- Secure the old tube to the transport device (left image). Tighten the 4 M10*40 mm Allen screws with a torque of **17 Nm**.
- Secure the HV plugs of the old tube to the transport device using cable ties (right image).
- Remove the old tube from between the table and the gantry.
 - » **Move the transport device with the old tube to the right so it does not hit the electronic table duct.**

6.3.4 Installation

1. Place the new tube between the gantry and the patient table.
2. Loosen and remove the screws on the transport device using a 10 mm Allen key.
3. Remove the 2 plastic protectors from the HV plugs and the protection caps from X1, X2 at the anode motor. Install the removed parts on the old tube .
4. Remove the expansion unit from the transport frame. Newer tubes are delivered with a second expansion unit that will often not be required. Only if the pressure of the

replaced tube was significantly too low, the oil volume inside the first expansion unit may not be sufficient to increase the pressure to 1.25 bar.

Fig. 211: Expansion unit at the new tube

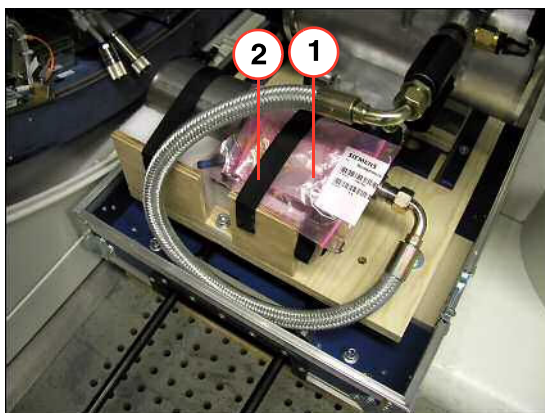
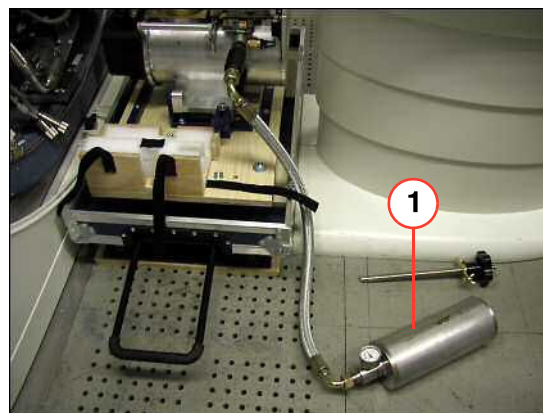


Fig. 212: Removing the expansion unit



- Remove the service material and the plastic bag from the expansion box (left image, item 1)
- Open the velcro tape (item 2) that secures the expansion unit and position the expansion unit next to the transport device (item 1).

5. Install the Straton MX tube

Fig. 213: Straton MX tube secured to the lifting support

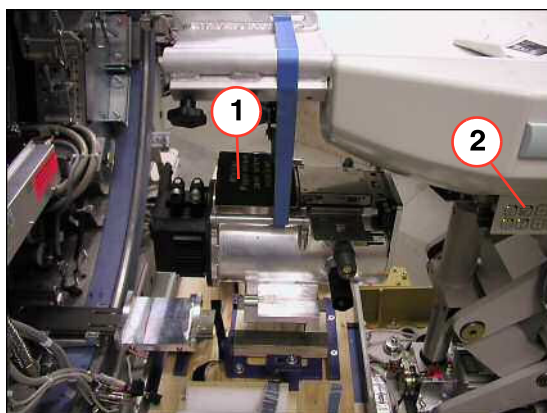
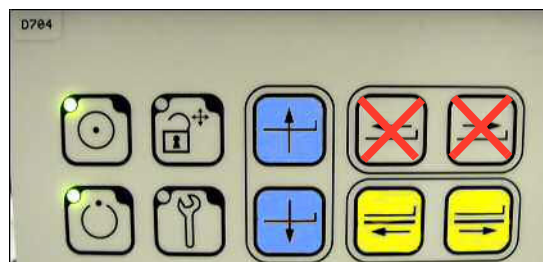


Fig. 214: PHS control panel



- Remove the protective foam from the X-ray exit window (item 1) of the new tube. Install the removed foam on the X-ray exit window of the old tube.
 - Secure the new tube to the lifting strap.
 - Lift the tube carefully using the up button of the table control.
 - Move the sliding carriage out of the gantry to its end position
 - Move the tube gently towards the gantry until it is positioned correctly over the sliding carriage. Use the tabletop support control buttons (right image, yellow marked buttons) to control the movement.
 - Lower the tube onto the sliding carriage using the down button of the table control..
 - Remove the strap and the lifting device.
6. Retract the table fully to its outer end position to gain access for the next work steps
 7. Remove the empty transport device from between the table and the gantry

8. Install anode motor connections

Fig. 215: Rotating anode connection (old type)

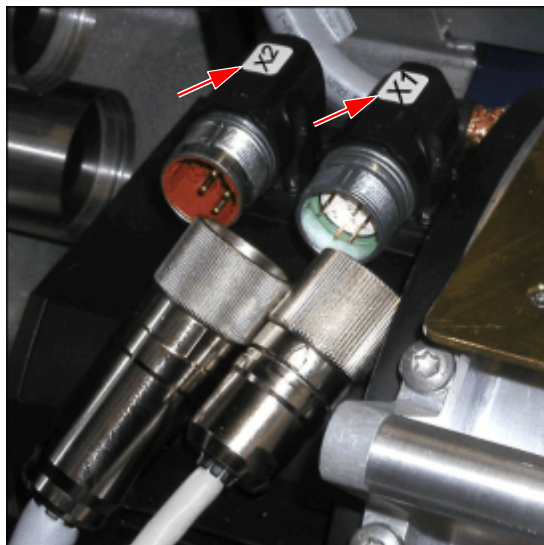
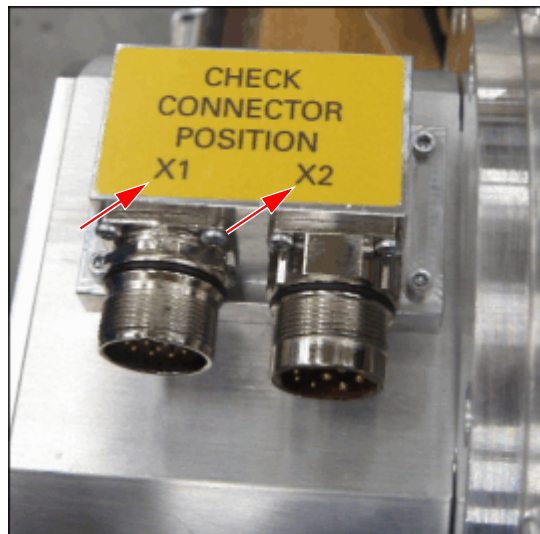


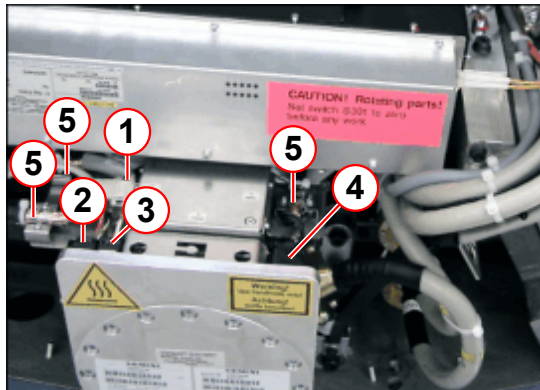
Fig. 216: Rotating anode connection (new type)



- Move the tube into the gantry until the next steps can be performed.
- Connect anode motor cable X1 and X2 to the anode motor connector X1 and X2.
 - » To avoid damage to the pins, do not mix the 2 plugs!
 - » Note: With tube revision 11 a new type of rotating anode connector is installed. In particular, note that the X1 connector is now located on the left and the X2 connector on the right side installed, which is the opposite of what it was previously (see arrows left/right image). Please pay attention to this difference, especially when reconnecting the plugs at the new tube to avoid damage to the pins.

9. Install cable clamps

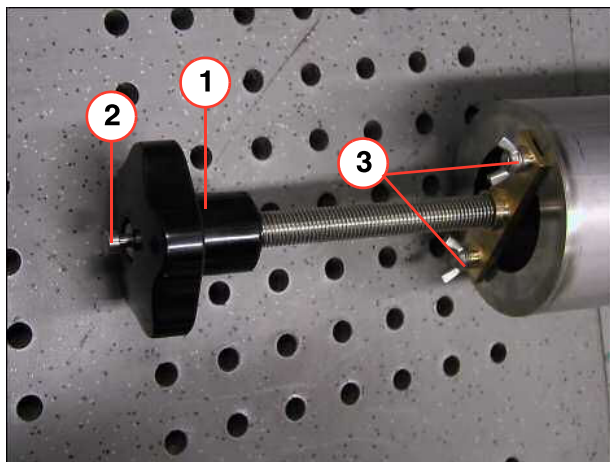
Fig. 217: Tube cable connection



- Move the tube gently into the gantry to gain access for installing the 4 plugs (see Figure, items 1, 2, 3, 4).
- Install the 3 cable clamps (right image, item 5).
 - » Check if the shielding of the cables is ok; if not, the relevant cable has to be replaced.

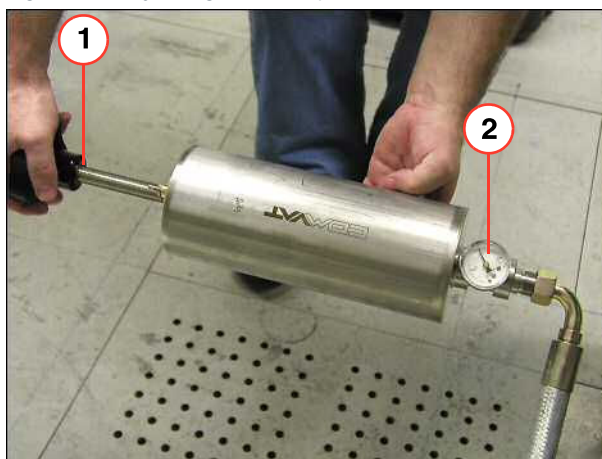
10. Prepare the expansion vessel.

Fig. 218: Adjusting unit for expansion membrane



- Install the adjustment handle (item 1) on the expansion vessel.
 - Press the button (item 2) to lock the thread rod on the membrane.
 - Hand-tighten the 2 screws of the adjustment thread rod (item 3).
- To ensure that the spindle is properly engaged adjust the oil pressure.

Fig. 219: Adjusting the tube pressure



- Preadjust the oil pressure of the new tube to 1.25 bar using the adjustment handle.
 - » **This step ensures that the spindle of the adjustment handle is correctly engaged with the pressure membrane..**

11. Connect the oil hose coming from the XTC to the unused (top) quick coupling at the tube.



The pressure indicator at the expansion vessel shows the actual pressure in the oil circuit after the oil hose from XTC has been connected.

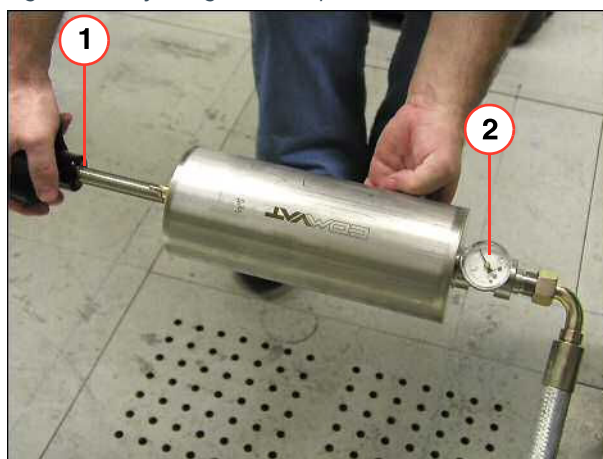
12. Adjust the oil pressure using the connected XTC oil circuit



From 11/2009 on, new tubes need a pressure of 1.25 bar (+ altitude correction). These tubes are clearly labeled, and also the expansion unit is labeled with a red sticker.

Set the pressure on "old" tubes without label to 1.05 bar, and on newer tubes to 1.25 bar according to the label.

Fig. 220: Adjusting the tube pressure



- Using the expansion handle (item 1), check the pressure at the pressure indicator (item 2) attached to the expansion vessel.

The correct pressure for **tubes without a label (see NOTE above)** is **1.05 bar** +/- 0.05 bar at 20 °C at sea level. See the following table for altitude correction of pressure.

Newer tubes are clearly **labelled with a sticker that requests a pressure of 1.25 bar** at sea level. To make sure the pressure can always be reached, even if a 1.05 bar tube is replaced by a 1.25 bar tube, a second oil filled expansion unit is delivered inside the crate. Most times, though, the contents of one expansion unit is enough to increase the pressure to 1.25 bar. See following table for altitude correction of pressure.

- If the system is located at a higher altitude than sea level > add the following correction factor to the sea level pressure (see following table).
 - » For example, if the altitude is 1000 m -> the correction value is 0.12 bar.

Altitude [m]	Altitude correction of pressure[bar]
0	0.00
500	0.06
1000	0.12
1500	0.17
2000	0.22
2500	0.27

Altitude [m]	Altitude correction of pressure[bar]
3000	0.31
3500	0.35
4000	0.39
4500	0.43
5000	0.47

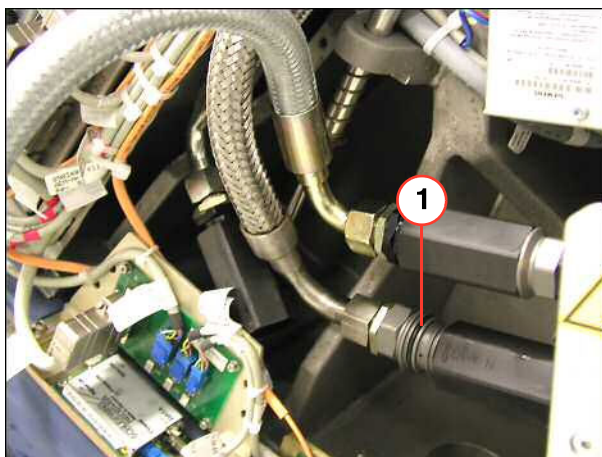
- If the pressure is not correct, adjust the membrane position using the adjustment handle (item 1)

The goal is to obtain the right oil pressure for all operating conditions. If the pressure is too low, arcs may occur in scans with the fastest anode rotation.

- After the pressure adjustment is finished, remove the expansion vessel (remove the oil quick-snap connection to the XRT)

13. Attach the second oil hose to the tube

Fig. 221: Installation of oil couplings at new tube



- Put some paper towels underneath the couplings.
- Clean the inside of the oil coupling.
 - » **Dirt in the coupling may lead to oil leakage!**
- Attach the second oil hose to the tube. Hand-tighten the oil coupling (don't use any tools) and ensure that it is tightened.

- Hand-tighten the oil coupling (don't use any tools!). The coupling has to be tightened until you can no longer see a gap between cover and coupling.

Fig. 222: Insufficiently tightened oil coupling

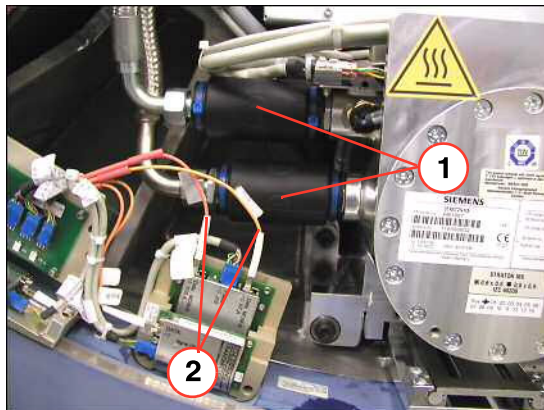


Fig. 223: Correctly tightened oil coupling



14. Install protections to oil couplings at the new tube

Fig. 224: Tube A oil connections



- Install the protections at the oil couplings. Secure them with 4 tie wraps (item 1).
 - » **Make sure that the seam on the protectors is positioned to the front (item 1)**
- Reinstall the 2 fiber optic cables to the transmitter (item 2). Not necessary with tube B!

15. Install the expansion unit at the old tube (**IMPORTANT STEP**)

i

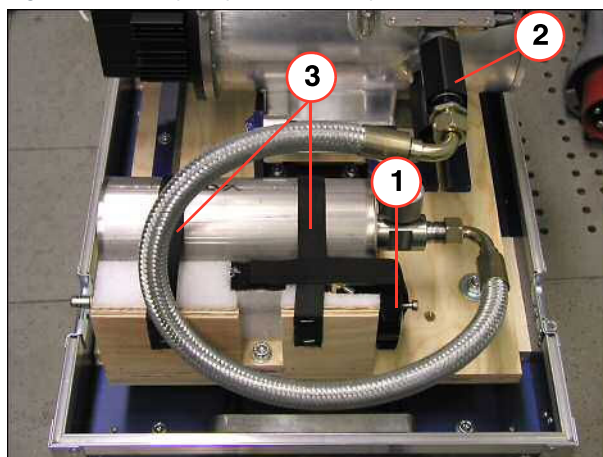
To avoid damaging the old tube, it is very important to connect the expansion unit immediately after adjusting the pressure of the new tube according to the description in the following chapter.

If the tube cools down without the expansion unit, air will be sucked into the tube due to contracting oil. The air will be distributed in the oil circuit and lead to one-sided arcs if the tube is used again during troubleshooting.

i

The expansion unit must be installed at the tube. If the tube is returned without the expansion unit, no failure analysis of the tube will be made, and the tube factory will reject liability.

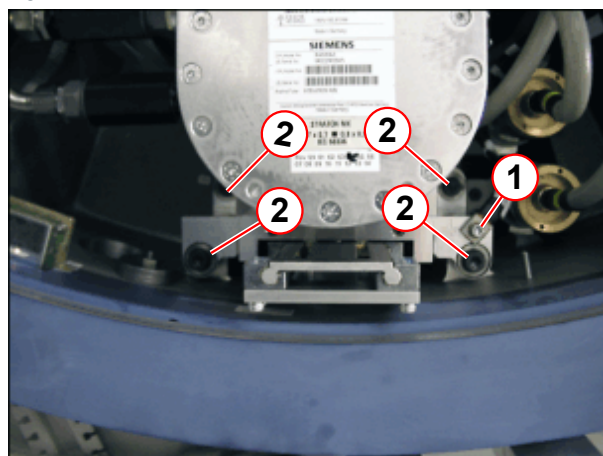
Fig. 225: Transport position of expansion unit



- Remove the adjustment handle from the expansion unit and place it into the foam at the old tube (item 1).
- Connect the oil coupling of the expansion unit to the old tube (item 2). Hand-tighten the connector.
- Secure the expansion unit to the transport frame using Velcro tape (item 3).

16. Install the ground connections and the attachment screws

Fig. 226: Straton MX attachment screws

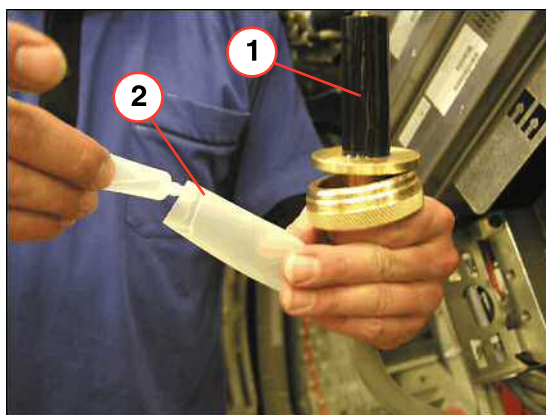
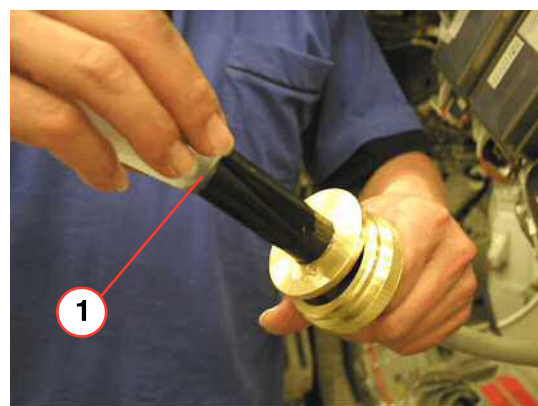


- Move the tube gently to its original position in the gantry.
- Install the 4 **new** M10*40 mm Torx bolts (supplied) and tighten first the 2 back screws and then the 2 front screws using **48 Nm (item 2)**.
- Connect the two X12 ground straps to their original position (item 1). Tighten the screw with a torque of **10 Nm**. Secure the straps with cable ties.

17. Prepare the high voltage connectors



Please do NOT use a marker or ballpoint on the HV cables. If absolutely necessary use appropriate tape or stickers to mark the cable.

Fig. 227: Silicon oil on HV plugs*Fig. 228: Silicon oil on HV plugs*

- Ensure that the connector is clean and inspect it for damage.
- Put oil (supplied with the part) on the whole plug surface(left image, item 1).
 - » **Use about half of the oil from the tube for both plugs.**
- Put oil into the **new** silicon insulator (item 2).
- Install the silicon insulator part over the plugs (left image, item 1)
 - » **Ensure there is no air left in the silicon insulator.**

18. Prepare the high voltage connectors

Fig. 229: Silicon oil on HV plugs

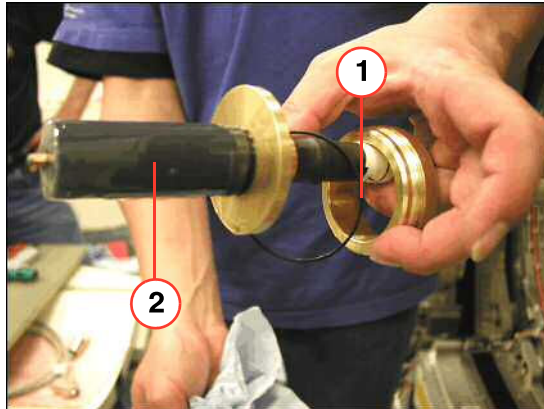


Fig. 230: HV sockets

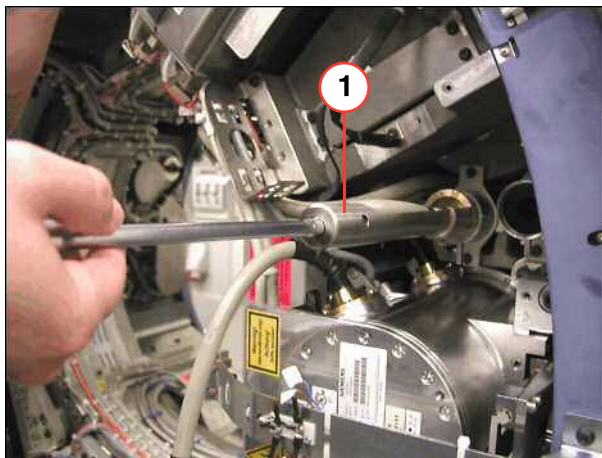


- Check that there is no air left in the silicon insulator.
- Insert the O-ring in the HV plug (item 1).
- Apply oil to the silicon insulator (item 2).
- Insert the high-power plugs into the receptacles.
 - » **Verify the position of the plugs (+ or -).**
 - » **The (-) tube HV plug is marked with a blue shrink hose (if produced after Sept. 2009).**
 - » **The (+) tube HV plug is marked with a black shrink hose (if produced after Sept. 2009).**
- Install the O-ring into the HV socket as shown (right image, arrow)
- Hand-tighten the fastening rings.



19. Tighten the HV plugs.

Fig. 231: Tightening the HV plugs



- Tighten the HV plugs using the special tightening tool (delivered with the tube) with a torque of **25 Nm**.
 - » Hold the cable to avoid turning during tightening.
 - » To ensure the correct connection, it is very important to tighten the HV plugs with the correct torque.

20. Connect all cables according to the original configuration.

Fig. 232: Installed Straton MX tube, A assembly



Fig. 233: Installed Straton MX tube, B assembly



1. Connect all cables according to the original configuration. Secure cables with cable ties.
 - » **Note: Cable routing is different between Straton MX A and Straton MX B.**

6.3.5 Start-up

6.3.5.1 Mechanical

1. After finishing all mechanical work, remove the data ring protection.

2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Close all covers opened before.
5. Switch on circuit breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
6. Switch on the system (**SYSTEM/ON**) at the control box.

6.3.5.2 Software

n.a.

6.3.5.3 Tune-up

1. Run the tune-up sequence as described under **Local Service -> Tune Up -> FRU replacement**.



Disable the gantry rotation using S301 when performing mechanical adjustments inside the gantry during tune up.

6.3.5.4 Testing

n.a.

6.3.6 Final steps

6.3.6.1 Labeling

A second label for the X-ray tube is included in the tube delivery.

Install a second label for the X-ray tube outside the gantry if Regulatory Affairs requires it. To attach them, refer to the location of the labels as described in the Operator Manual (System Owner Manual / location of labels).

6.3.6.2 Filling out Form "Questionnaire for X-Ray Tube Housing Assemblies"

The form "Questionnaire for X-Ray Tube Housing Assemblies" (supplied with the new tube) has to be filled out and sent back with the old tube.

6.3.6.3 Tube manuals

Tube manuals are no longer sent with the tube. You will find the relevant manuals in the Siemens Intranet.

Intranet address: <https://intranet.healthcare.siemens.com/cms/CV/en/Home/Pages/Default.aspx> -> **Products & Solutions -> Products -> Computed Tomography -> select: Domain: Power & Vacuum Technology -> Product group: Straton -> Product: Straton MX**

6.3.6.4 Prepare tube transport box

1. Make sure all screws securing the tube and expansion unit are tightened correctly

Note: If two expansion units were delivered, ensure that both are returned, even if the second one was not used.

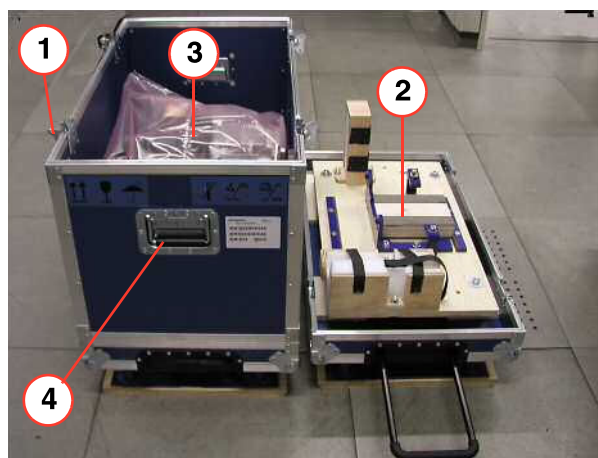
Fig. 234: Tube in transport box



Don't forget to include the old silicone sleeves of the replaced HV connectors!

2. Close transport box.

Fig. 235: Tube transport box



- Reassemble the transport box side cover (item 4) and ensure that all service tools, papers have been placed inside (item 3).
- Install the top (item 2) and secure it using the 4 closing locks (item 1).

6.4 HV sockets / cable

The following instructions apply to both HV sockets/cables

6.4.1 Prerequisites

6.4.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

6.4.1.2 Time and Man Power

- 1 Person
- Estimated total repair time: 2 hours.

6.4.1.3 Tools and auxiliary Equipment

- Standard tool kit

6.4.2 Preliminary steps

1. Allow at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube, XTC ... may be up to 130 degree Celsius.
2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
4. Remove the gantry bottom front ring segment (→ Removing / Page 32).
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Remove the gantry funnel and the Plexiglas ring (→ Removing / Page 43).
7. Turn the gantry by hand until the Straton MX tube (A or B) is in the 6 o'clock position.
8. Secure against unintended rotation by using the front safety bolt (→ Safety bolts / Page 21).

9. Install the data ring protection (→ Data ring protection / Page 23).

6.4.3 Removing



CAUTION

[hz_serdoc_F13G02U01M01]

Temperature of tube and/or tube cooling device parts may be above 70C (with Straton up to 130C).

Risk of burns!

- » Avoid contact with oil and exposed parts of X-ray tube and cooling device.
- » Wait until tube and cooling device parts are chilled down.



In case of problems with HV cables, it is necessary to replace the "+" and "-" cable (only one spare part number).

1. Wait until the X-ray tube and cooling system parts have cooled down.
2. Disconnect the HV cable connections of the tube

Fig. 236: HV connectors



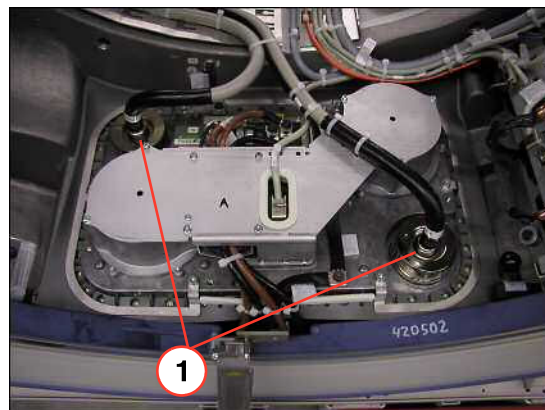
- Remove the two high voltage cables (item 1) from the HV sockets by unscrewing the securing rings using the special tool (→ Fig. 231 Page 302).

3. Disconnect the high voltage cables from the HV tank

Fig. 237: Set screws at securing ring

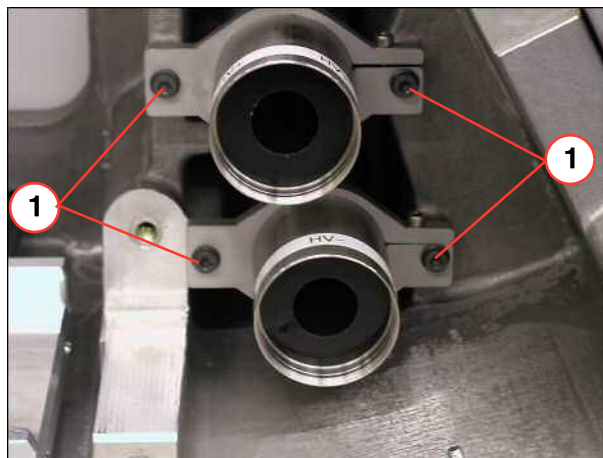


Fig. 238: HV cables at HV tank



- Open the 2 Allen set screws on both securing rings (left image , item 1).
 - Remove all tie wraps as needed to release the HV cables from the HV tank
 - Unscrew the 2 securing rings (right image, item 1) and remove the 2 HV plugs.
4. Remove the HV sockets

Fig. 239: HV_receptacles



- Open the 2 holding screws (item 1) of each HV coupling to remove the HV couplings.
5. Open all tie wraps as needed and remove the HV cables out of the gantry.
- » **The new HV cables must be wired through the gantry similar to the old cables. For this reason, memorize the position of the old cables.**

6.4.4 Installation

1. Install the HV plugs at the HV tank

Fig. 240: HV cables at HV tank

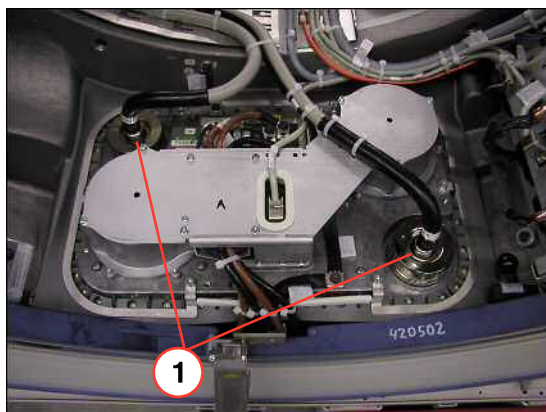


Fig. 241: Set screws at securing ring



- Install the HV connectors at the HV tank (left image, item 1).
 - » **New silicon disks have to be used when connecting the HV cables to the HV tank receptacles.**
 - » If present, refer to the **blue shrink hose** that marks the **(-) connector**
- Hand-tighten the fastening rings and secure them lightly with the two Allen screws (right image, item 1) on each ring.
 - » **If the 2 Allen screws are tightened too much, the HV receptacle may get damaged.**

2. Routing of HV cables plane A

Fig. 242: Routing of HV cables , plane A

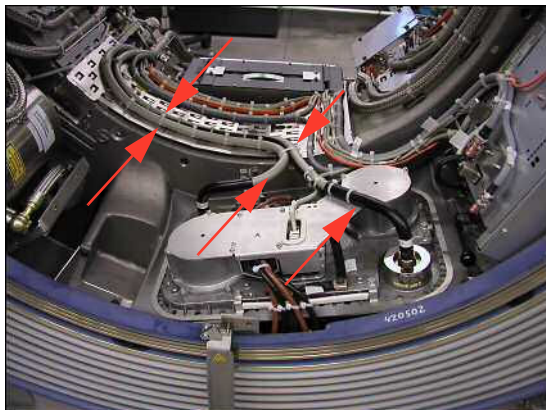
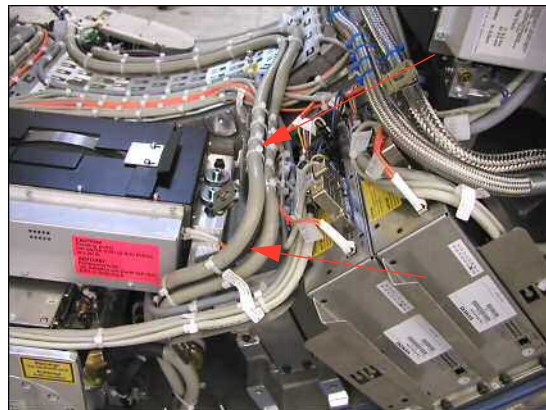
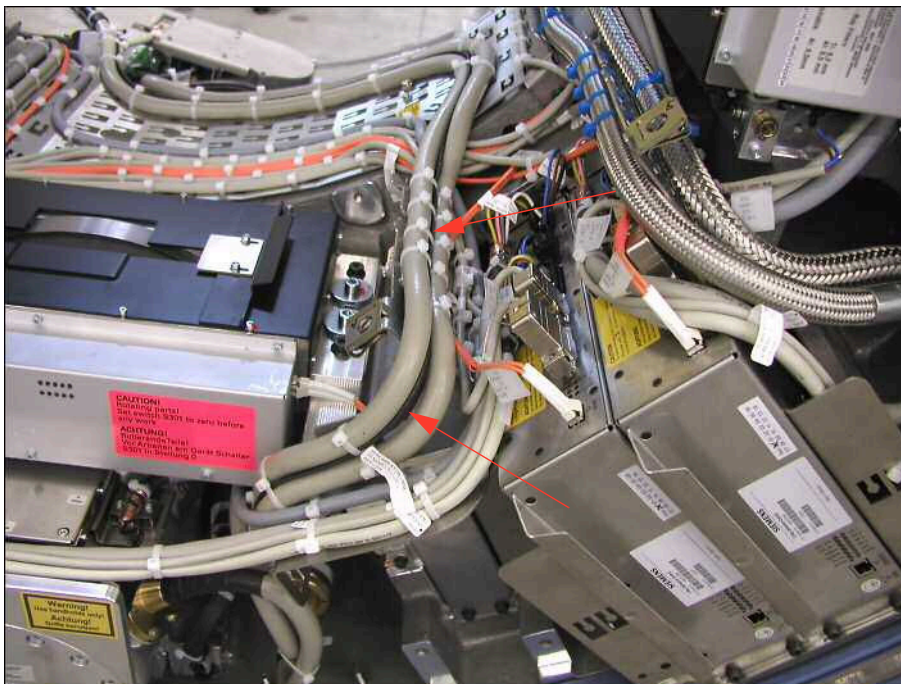


Fig. 243: Routing of HV cables Straton MX A



- Route the two HV cables according to the original configuration (arrows) from the HV tank (left image) to the area of the corresponding HV sockets (right, bottom image).
- Secure cables with cable ties.

Fig. 244: Routing of HV cables Straton MX A



3. Routing of HV cables plane B

Fig. 245: Routing of HV cable Straton MX B

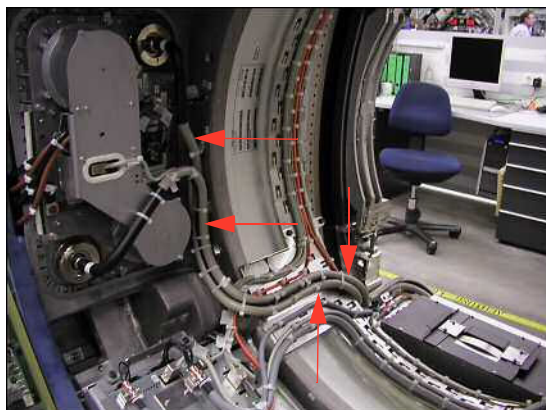
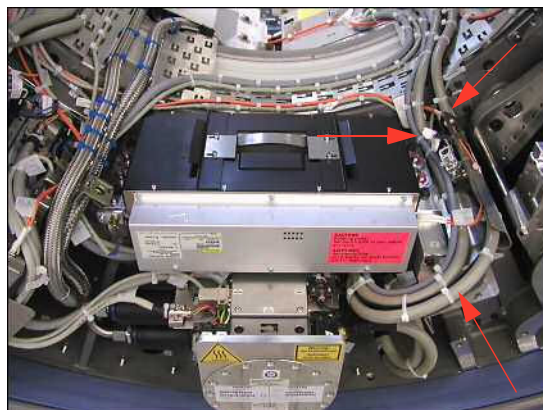
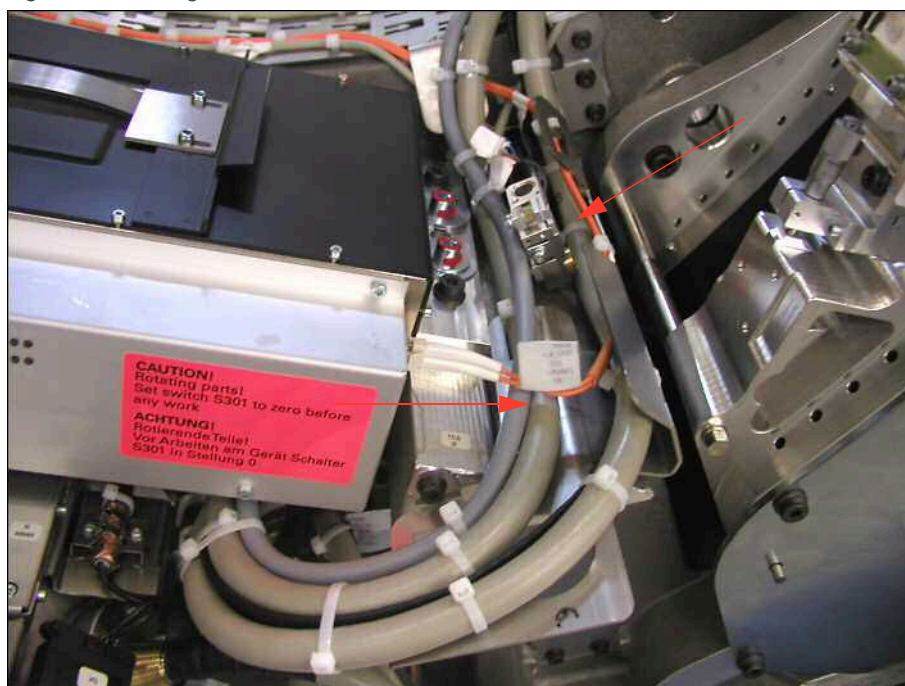


Fig. 246: Routing of HV cables Straton MX B



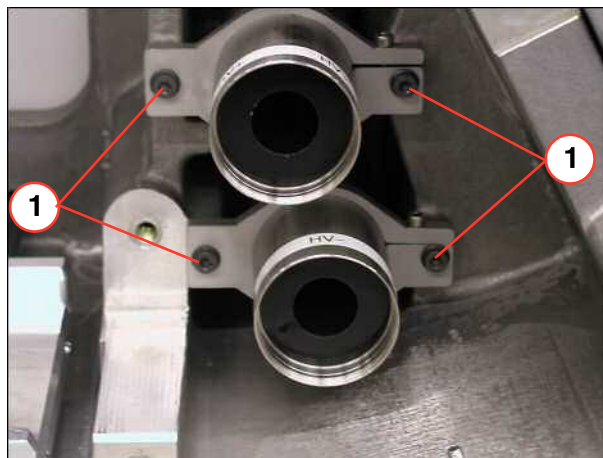
- Route the two HV cables according to their original configuration (arrows) from the HV tank (left image) to the area of the corresponding HV sockets (right, bottom image).
- Secure cables with cable ties.

Fig. 247: Routing of HV cables Straton MX B



4. Install the HV sockets

Fig. 248: HV_receptacles



- Install the HV socket according to the original configuration. Tighten the 4 M6 * 20 mm holding screws (item 1) with a torque of **14 Nm**.
- 5. Reinstall the tube HV cables into the HV receptacles of the HV couplings. Hand-tighten the fastening rings.
 - » **Verify the position of the plugs (+ or -)!**
If present a **blue shrink hose** marks the **(-) plug**.
 - » **New silicon insulators have to be used when connecting the tube HV cables to the receptacles**

6.4.5 Start-up

6.4.5.1 Mechanical

1. After finishing all mechanical work, remove the data ring protection.
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.

6.4.5.2 Software

n.a

6.4.5.3 Tune-up

1. Run the tune-up sequence as described under **Local Service -> Tune Up -> FRU replacement**.

2. Close the gantry covers when instructed within the guided tour or after you performed the guided tour.



Disable the gantry rotation using S301 when performing mechanical adjustments inside the gantry during tune-up!

6.4.5.4 Testing

1. Run some test scans with radiation ON.

6.4.6 Final steps

n.a.

6.5 X-ray generator rotating (XGR)

The following instructions apply to both generator electronic boxes.

6.5.1 Prerequisites

6.5.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

6.5.1.2 Time and Man Power

- 1 Person
- Estimated total repair time: 4 hours.

6.5.1.3 Tools and auxiliary Equipment

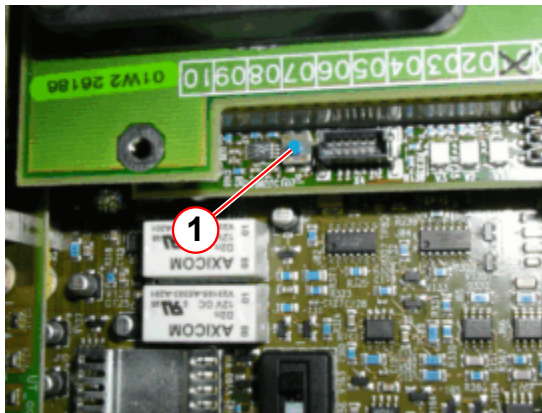
- Standard tool kit
- Lifting device

6.5.2 Preliminary steps

1. Switch system to status **COMP/ON** at the control box.
2. Switch off the gantry power using service push-button S1 in PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.

3. Press the blue Reset button on the MCB of the PDC XGS Control piggyback board in the PDC A to enable service power to the PHS.

Fig. 249: MCB on D700



(1) Blue reset button

4. Remove the gantry bottom front ring segment (→ Removing / Page 32).
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Remove the gantry funnel and the Plexiglas ring (→ Removing / Page 43).
7. Turn the gantry by hand until the part to be replaced is in the 6 o'clock position.
8. Secure against unintended rotation by using the rear safety bolt (→ Safety bolts / Page 21).

6.5.3 Removing

1. Disconnect the XGR assembly connections

Fig. 250: XGR plug/cable connections

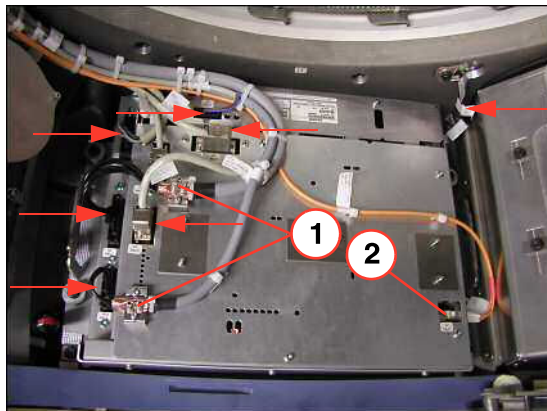
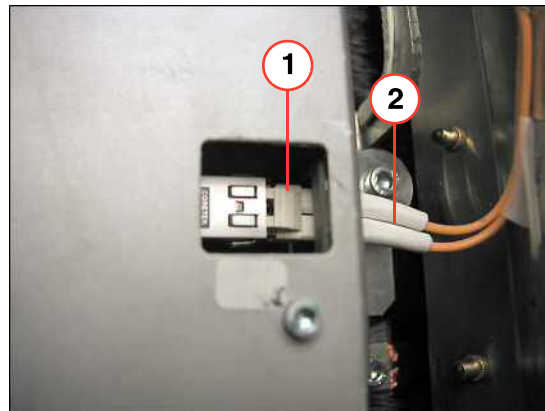


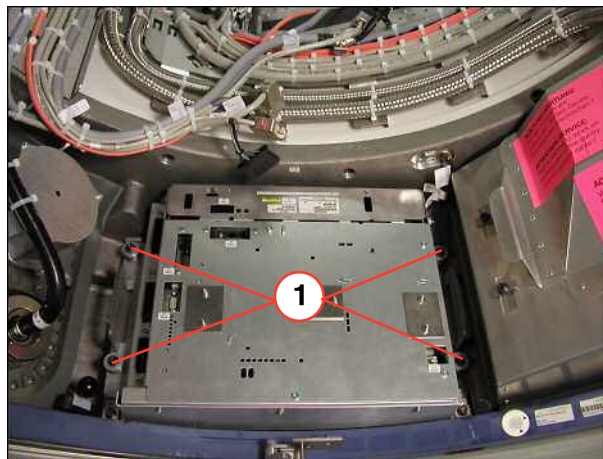
Fig. 251: XGR fiber optic connection



- Remove all tie wraps as needed to release the cables from the XGR assembly.
- Disconnect all cable /ground connections (arrows).
- Remove the 2 cable clamps (left image, item 1).
- Disconnect the fiber optic cable (left image, item 2)
 - » Press on the plug unlocker (right image, item 1) while pulling the fiber optic cable (right image, item 2).

2. Remove the XGR assembly holding screws

Fig. 252: XGS with cables removed



- Remove the 4 M6 *60 mm torx screws (item 1).
- #### 3. For easier access to the lifting handles, also remove the cover of the XTC (tube cooling) next to the XGR box. See (→ Removing / Page 332), step 2 for further details

4. Prepare the XGR box for lifting

Fig. 253: XGS secured to lifting support

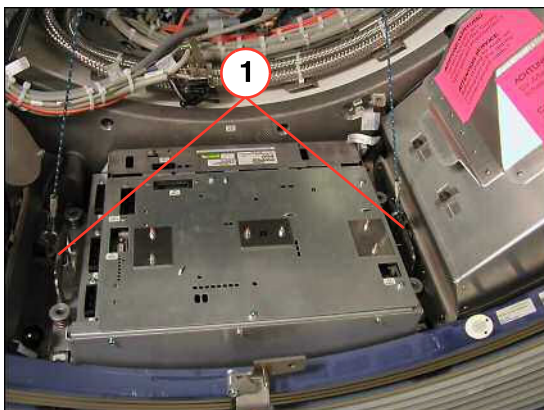
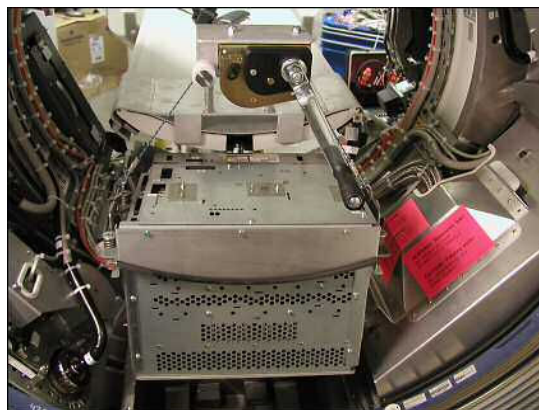


Fig. 254: XGS secured to the lifting support



- Place the lifting support at the end of the tabletop (→ Lifting support / Page 24).
 - Secure the XGR assembly to the cords of the lifting support and tighten the cords using the handle of the lifting support.
 - Lift the XGR assembly carefully using the handle of the lifting support.
5. Move the XGR assembly carefully to the back gantry side using the tabletop.
 6. Lower the XGR assembly to the floor, disconnect the cord and remove the assembly.
 7. Remove the air seals from the old XGR assembly



In newer XGR assemblies the air seal is already installed at the new XGR box. In this case only the front air seal has to be moved from the old to new XGR box.

Fig. 255: Front air seal at XGS assembly

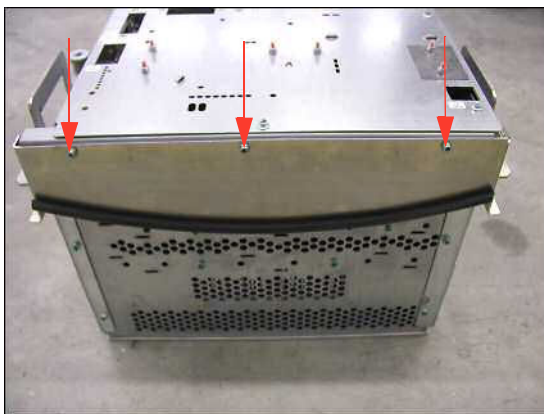
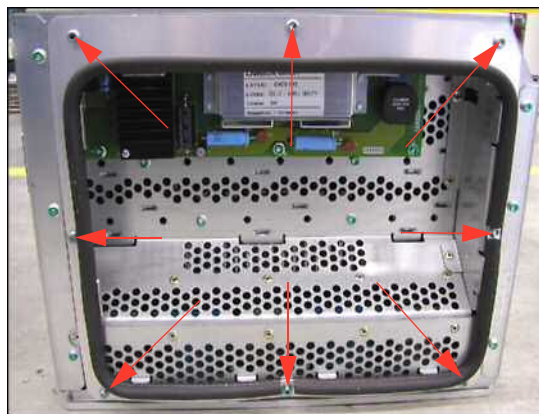


Fig. 256: Back air seal at XGS assembly



- Remove the 3 torx screws (left image, arrows) and remove the air seal.
- Remove the 8 torx screws (right image, arrows) and remove the air seal.
- Remove the ground strip from the XGR assembly (if not done before)

6.5.4 Installation

1. Install the air seals and ground strap on the new XGR assembly



In newer XGR assemblies the air seal is already installed at the new XGR box. In this case only the front air seal has to be moved from the old to new XGR box.

Fig. 257: Front air seal at XGS assembly

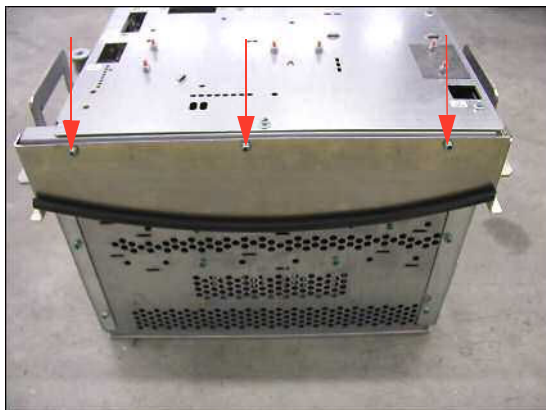
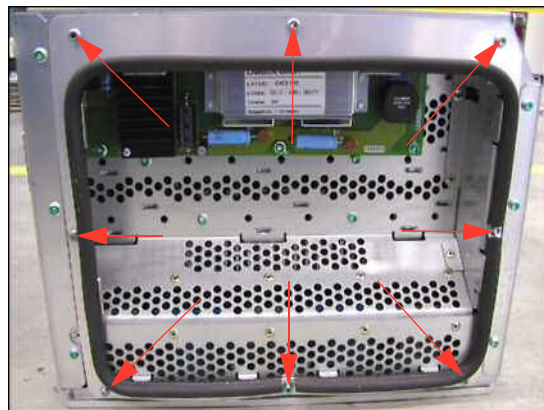


Fig. 258: Back air seal at XGS assembly



- Install the air seal and secure it with 3 torx screws (left image, arrows) . Tighten the screws with **3 Nm**.
 - Install the air seal and secure it with 8 torx screws (right image, arrows). Tighten the screws with **3 Nm**.
 - Connect the ground strap to the new XGR box. Tighten the strap with a torque of **6 Nm**
2. Attach the cords of the lifting support to the new XGR assembly.
 3. Lift the XGR assembly and install the new XGR by reversing the order for removing it.



Make sure that the XGR box is correctly placed in the drum (aligned to the attachment bolt).

4. Remove the lifting support.
5. Tighten the 4 M6 *60 mm torx screws in a crosswise pattern using **14 Nm**.
6. Connect all cables according to the original configuration. Secure cables with cable ties.
7. If you have removed the cover of the XTC (tube cooling) for easier access, reinstall the cover. See (→ Installation / Page 337), step 10 for further details.
8. Connect the ground strap to the original position. Tighten the M6 mm screw with a torque of **10 Nm** at the drum and **6 Nm** at the XGR assembly.

6.5.5 Start-up

6.5.5.1 Mechanical

1. Unlock the safety bolt for gantry rotation.
2. Rotate the gantry by hand and check for unusual noise.
3. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
4. Switch on the system (**SYSTEM/ON**) at the control box.

6.5.5.2 Software

The following section has to be performed to load the actual firmware version.

- Go to the Service home menu, select Options -> Service -> Local Service
 - In the service software menu select **TuneUp -> FRU Replace**
 - Select the corresponding part replaced and click GO
- Note:** If the part you replaced is not in the list, press go and continue. The firmware check will be performed anyhow
- Follow the output instructions



If the firmware update aborts with an error and the firmware status is NOT ok, perform the firmware update again.

6.5.5.3 Tune-up

1. Run the tune-up sequence as described under **Local Service -> Tune Up -> FRU replacement**.
2. Close the gantry covers when instructed within the guided tour or after you performed the guided tour.



Disable the gantry rotation using S301 when performing mechanical adjustments inside the gantry during tune-up!

6.5.5.4 Testing

n.a.

6.5.6 Final steps

6.5.6.1 Prepare the XGR transport box

1. Reassemble the transport box before sending it back.

Note: Return the two carabiners together with the part being replaced!

6.6 HV tank (HVT)

The following instructions apply to both HV tanks

6.6.1 Prerequisites

6.6.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

6.6.1.2 Time and Man Power

- 1 Person
- Estimated total repair time: 3,0 hours.

6.6.1.3 Tools and auxiliary Equipment

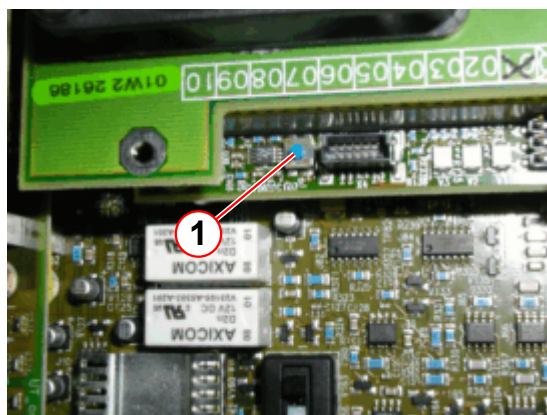
- Standard tool kit
- Wrench extension 40 cm long
- Torx bit TX50
- Lifting support
- 2 new silicon disks # 3810236 for HV plugs
- The following parts are included with the part:
 - 4 new M8*50 mm holding screws

6.6.2 Preliminary steps

1. Switch system to status **COMP/ON** at the control box.
2. Switch off the gantry power using service push-button S1 in PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.

3. Press the blue Reset button on the MCB of the PDC XGS Control piggyback board in the PDC A to enable service power to the PHS.

Fig. 259: MCB on D700



(1) Blue reset button

4. Remove the gantry bottom front ring segment (→ Removing / Page 32).
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Remove the gantry funnel and the Plexiglas ring (→ Removing / Page 43).
7. Turn the gantry by hand until the part to be replaced is in the 6 o'clock position.
8. Secure against unintended rotation by using the back safety bolt (→ Safety bolts / Page 21).

6.6.3 Removing

1. Prepare the HVT transport box

Fig. 260: HVT transport box, closed

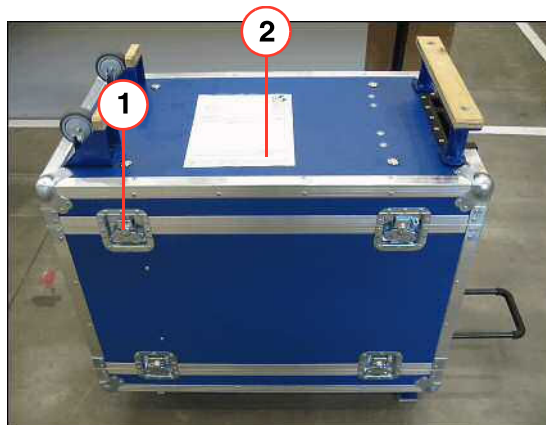
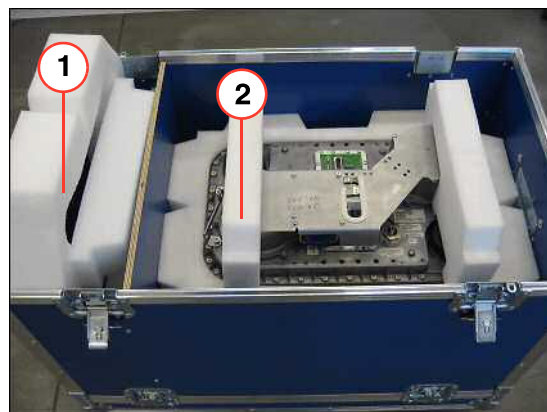


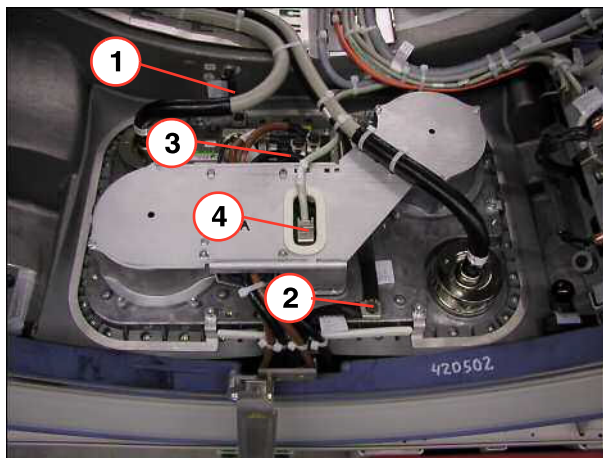
Fig. 261: HVT transport box without top cover



- Open the 4 locks at the top (left image, item 1) and remove the top (left image, item 2).
- Don't remove the foam (right image, item 1/2) at this time.

2. Disconnect the HV tank cable/ground connections

Fig. 262: Air seal securing screws



Disconnect the following cable and connectors from the HV tank.

- Remove all tie wraps as needed to release cables from the HV tank.
- Ground connection (Item 1/2)
- Cable connection X1, X2, X3, X4 (Item 3)
- Plug D732 X9 (Item 4)

3. Disconnect the high voltage cables from the HV tank

Fig. 263: Set screws at securing ring

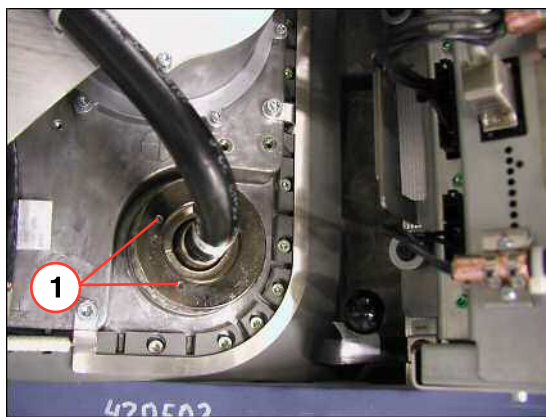
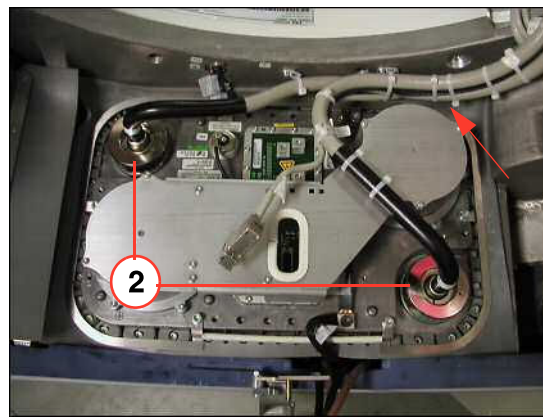


Fig. 264: Cable connection, HV tank



- Open the 2 Allen set screws (2 mm) on both securing rings (left image , item 1).
- Unscrew the 2 securing rings (right image, item 2) and remove the 2 HV plugs.

4. Remove the air seal

Fig. 265: Air seal at HV tank

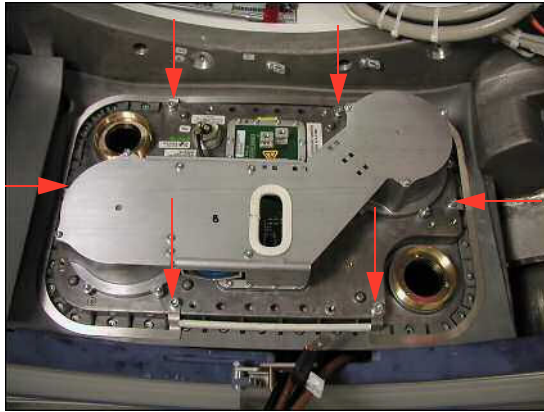
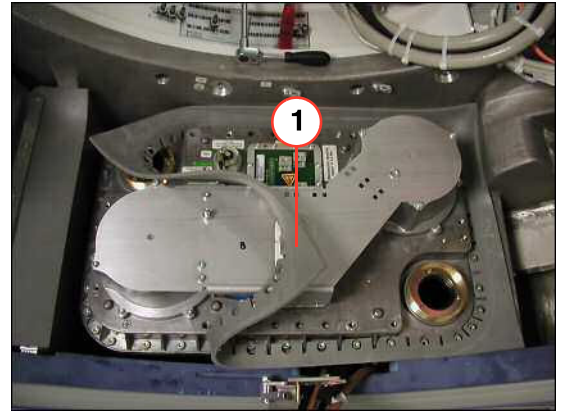


Fig. 266: Removing the air seal



- Remove the air seal frame from the defective HV tank by opening the 6 securing screws (arrows).
 - » Don't lose the washers
- Remove the air seal from the HV tank (right image, item 1)

5. Install lifting brackets from new to old HVT

Fig. 267: HVT transport box

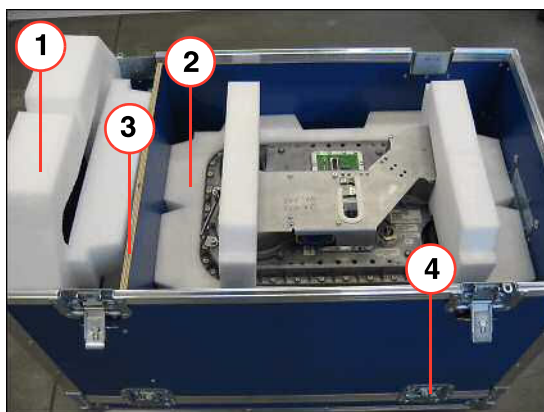
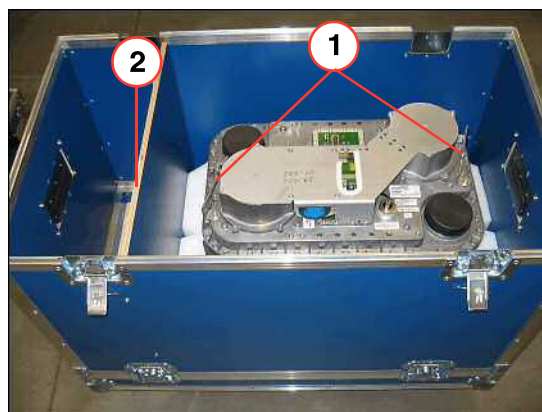


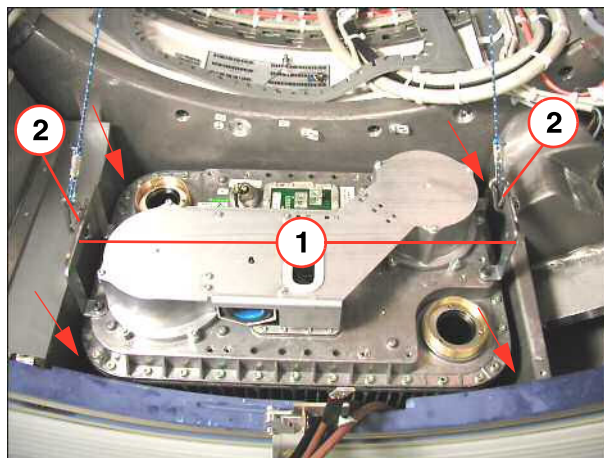
Fig. 268: HVT transport box



- Remove foam (left image, 1/2)
 - » The old HVT will be placed later on the left foam part (left image, item 1).
- Remove wooden attachment bar (left image, item 3) and the side cover by opening the opening locks (left image, item 4).
- Remove the 2 lifting brackets for attaching the cords of the lifting support from the new HVT (right image, item 1).
 - » **This step is only necessary, if the installed (defective) HVT has not already 2 brackets installed (with the first versions of HVT's the brackets were missing)!**
- Install the 2 just removed brackets at the defective HV tank. Use only the supplied holding screws!
 - » **This step is only necessary, if the installed HVT has not already 2 brackets installed (with the first versions of HVT's the brackets were missing)!**
 - » **The immersion depth of the screws of the HV tank cannot exceed a maximum of 8 mm. The use of longer screws will damage the HV tank!**

6. Open the HV tank attachment screws and prepare for lifting.

Fig. 269: Removing the HV tank



- Open the 4 M8*50 mm torx attachment screws (arrows) using a TX50 torx bit.
 - » **After opening the attachment screws, the HV tank is released from the drum!**
- Place the lifting support at the end of the tabletop (→ Lifting support / Page 24).
- Secure the HV tank to the cords of the lifting support and tighten the cords using the handle of the lifting support (item 2).

7. Lift the HV-tank out of the gantry

Fig. 270: HV tank secured to the lifting device

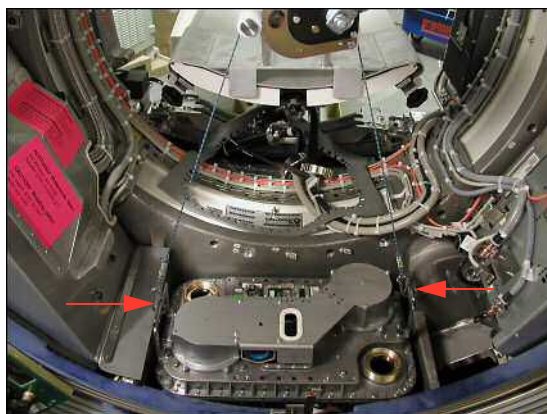
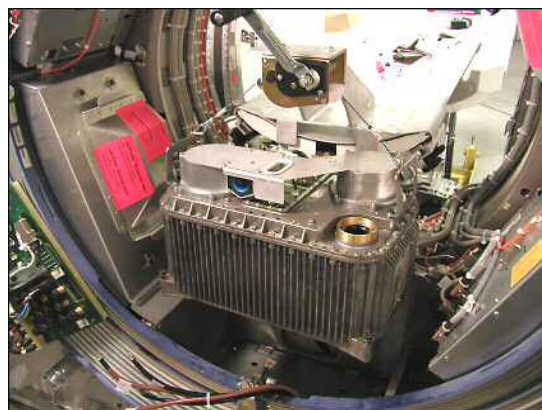


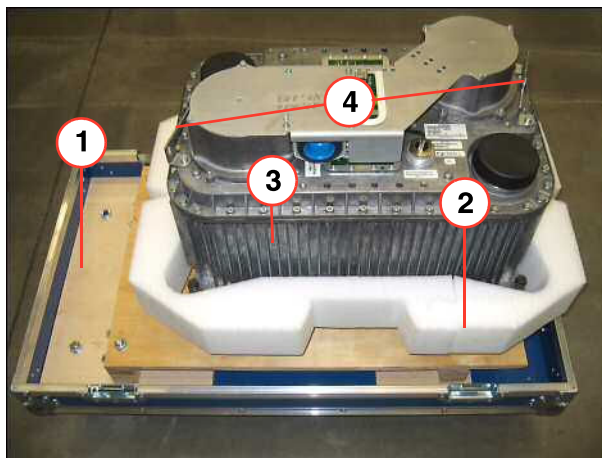
Fig. 271: HV tank attached to the lifting device



- Make sure that the cords of the lifting device are correctly attached to the HV tank brackets.
- Lift the HV tank carefully out of the gantry using the handle of the lifting support (right image).

8. Lower and secure the HV tank assembly to the transport device

Fig. 272: HVT on the transport device



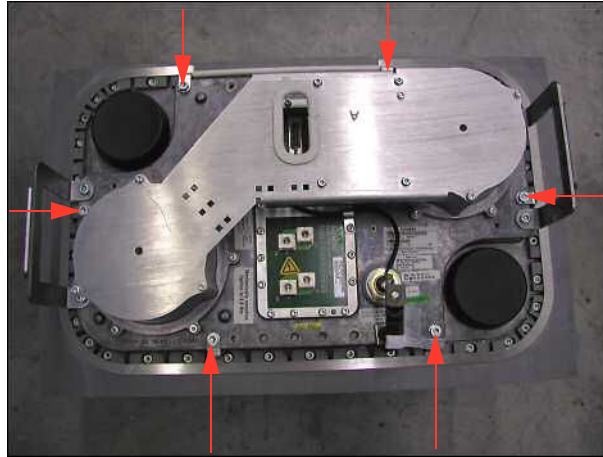
- Roll the empty HV tank transport device (left image, item 1) to the back of the gantry.
- Place the stabilizing foam onto the transport device (left image, item 2).
- Move the HV tank out of the gantry so that it is positioned over the transport device.
- Lower the HV tank assembly onto the transport device (item 3) using the handle of the lifting support.
- Remove the cords of the lifting device from the brackets and secure the HV tank to the transport device.
- Remove the 2 lifting brackets (item 4) from the old HVT and reinstall them at the new HVT.
 - » **This step is only necessary, if the HVT that was just removed do not have brackets installed (with the first versions of HVT's the brackets were missing)!**

9. Remove the transport box with the old HV tank from the back gantry side.

6.6.4 Installation

1. Install the air seal on the new HV tank

Fig. 273: HV tank, air seal



- Install the air seal and tighten the 6 attachment screws holding the frame and seal with **10 Nm**.
2. Place the new HV tank to the back gantry side and attach the cords of the lifting device to the HV tank.
 3. Remove the transportation attachment screws and lift the new HV tank out of the transport box using the handle of the lifting support.
 4. Remove the empty transport box from the back gantry side.
 5. Install the HV tank by reversing the order for removing it.
 6. Tighten the 4 M8*50 mm torx attachment bolts of the HV tank (supplied) in a cross-wise pattern using **31 Nm**.

7. Install the cable connections

Fig. 274: Cable connection, HV tank

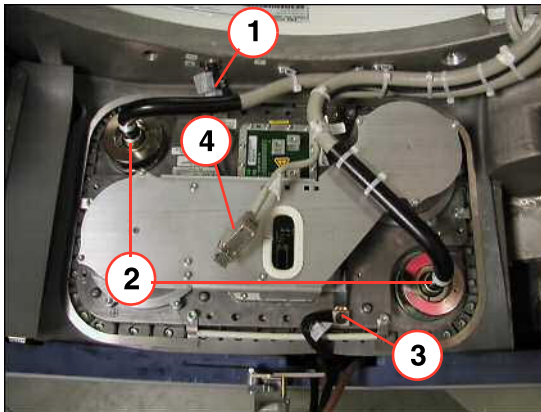


Fig. 275: Set screws at securing ring



- Install the ground strap X528 (item 1) to the HV tank (**HV tank:** block-> contact washer -> ground strap -> washer, **Drum:** ground strap -> washer). Tighten the M6 screws on the HV tank and on the drum with **10 Nm**.
 - » **Maximum one screwing depth of the HV-tank ground screw is 8 mm! The use of longer screws will damage the HV tank!.**
 - Install 2 new silicon disks on the HV plugs.
 - Install the HV cables (item 2) and hand-tighten the fastening rings. Secure them lightly with the two Allen screws (right image, item 1) on each ring.
 - » **Tightening the 2 Allen screws too much may damage the HV receptacles!**
 - Install 2 ground straps (item 3) and tighten them with a torque of **10 Nm..**
 - » **Maximum one screwing depth of the HV-tank ground screw is 8 mm! The use of longer screws will damage the HV tank!.**
 - Connect plug D732 X9 (item 4)
8. Install the 4 MV wires according to their original configuration. Connect the wires to X1, X2, X3, X4. Secure with a M6*10 mm screw (connector block-> cable-> washer) with a torque of **4.8Nm**. Make sure that the wires are not touching the metal holder of the back ring after using the tie wraps.
9. Install all tie wraps in their original position.

6.6.5 Start-up

6.6.5.1 Mechanical

1. Unlock the safety bolt for gantry rotation.
2. Rotate the gantry by hand and check for unusual noise.
3. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.

4. Switch on the system (**SYSTEM/ON**) at the control box.

6.6.5.2 Software

Insert serial number of HV tank under **local Service -> Configuration -> Serial**

6.6.5.3 Tune-up

1. Run the tune-up sequence as described under **Local Service -> Tune Up -> FRU replacement**.
2. Close the gantry covers when instructed within the guided tour or after you performed the guided tour.



Disable the gantry rotation using S301 when performing mechanical adjustments inside the gantry during tune-up!

6.6.5.4 Testing

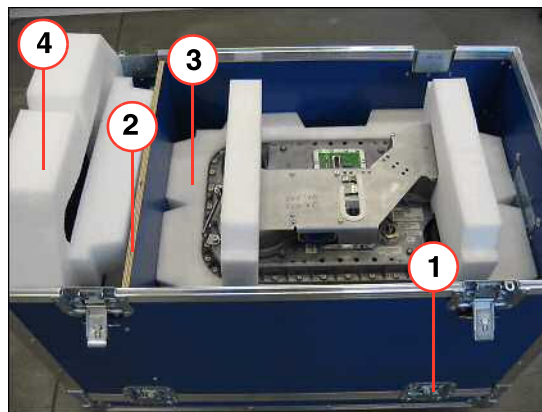
n.a.

6.6.6 Final steps

6.6.6.1 Prepare the HV tank transport box

1. Prepare and close the transport box.

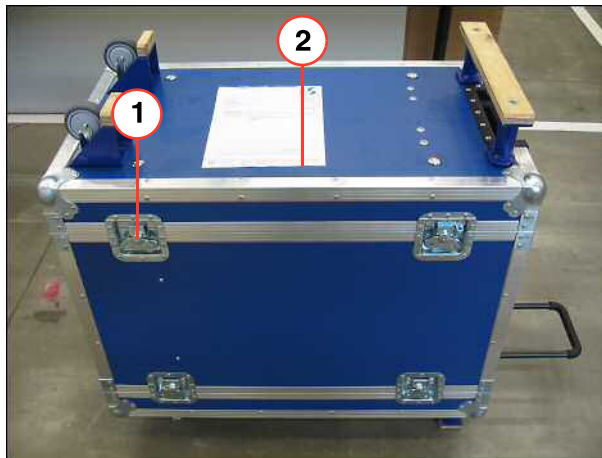
Fig. 276: HVT transport box



- Reinstall the transport box side cover and lock it using the 4 closing locks (item 1) .
- Reassemble the transport box. Ensure that all parts are correctly placed (item 2,3,4).

2. Close the transport box.

Fig. 277: HVT transport box, closed



- Install the top (item 2) and secure it using the 4 closing locks (item 1).

6.7 Tube cooler (XTC)

The following instructions apply to both tube coolers

6.7.1 Prerequisites

6.7.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

6.7.1.2 Time and Man Power

- 1 Person
- Estimated total repair time: 3,5 hours.

6.7.1.3 Tools and auxiliary Equipment

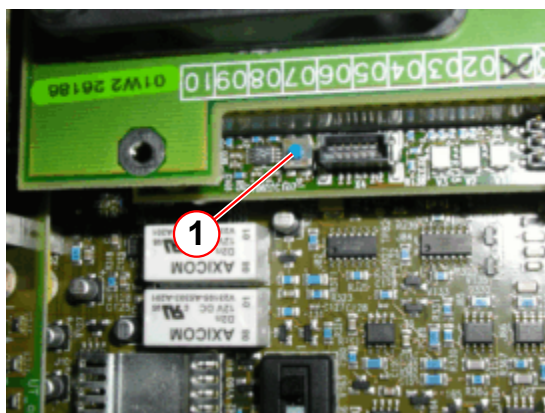
- Standard tool kit
- Lifting support
- The following tools are included with the part:
 - 2 new attachment screws.
 - Heat-proof cable ties (blue colored).

6.7.2 Preliminary steps

1. Allow the system to cool down for at least **10 minutes** after the last scan. During the cooling time, the **system must be switched on**.
 - » Be aware that some system components may be very hot, e.g., the Straton tube, XTC ... may be up to 130 degree Celsius.
2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.

4. Press the blue Reset button on the MCB of the PDC XGS Control piggyback board in the PDC A to enable service power to the PHS.

Fig. 278: MCB on D700



(1) Blue reset button

5. Remove the gantry bottom front ring segment (→ Removing / Page 32).
6. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
7. Remove the gantry funnel and the Plexiglas ring (→ Removing / Page 43).
8. Turn the gantry by hand until the XTC assembly which has to be replaced is in the 6 o'clock position.
9. Secure against unintended rotation by using the backsafety bolt (→ Safety bolts / Page 21).

6.7.3 Removing



CAUTION

[hz_serdoc_F13G02U01M01]

Temperature of tube and/or tube cooling device parts may be above 70C (with Straton up to 130C).

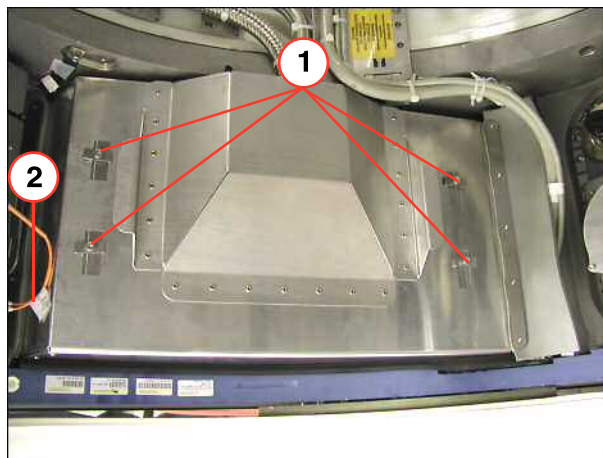
Risk of burns!

- » Avoid contact with oil and exposed parts of X-ray tube and cooling device.
- » Wait until tube and cooling device parts are chilled down.

1. Wait at least 10 minutes for the X-ray tube and cooling device parts to cool down.

2. Remove the tube cooler cover.

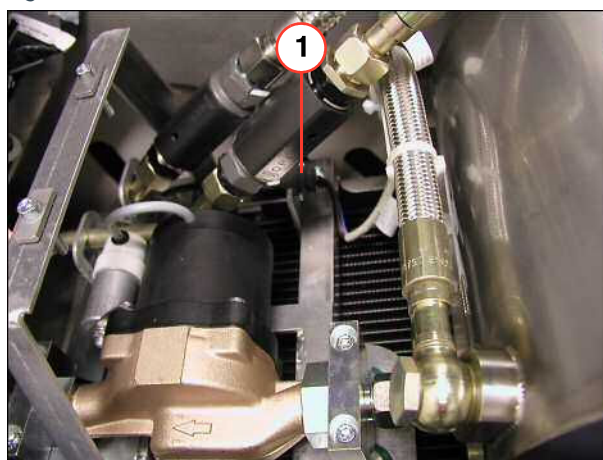
Fig. 279: XTC cover



- Loosen the 4 cover holding screws (item 1) using a TX 30 bit.
- Turn the holding bracket in this way so that the cover can be removed.
- Remove the fiber optic cable and cable ties as necessary from the XGR assembly (item 2).
- Remove the XTC cover carefully.

3. Disconnect the cable connections at the tube cooler.

Fig. 280: XTC cable connection

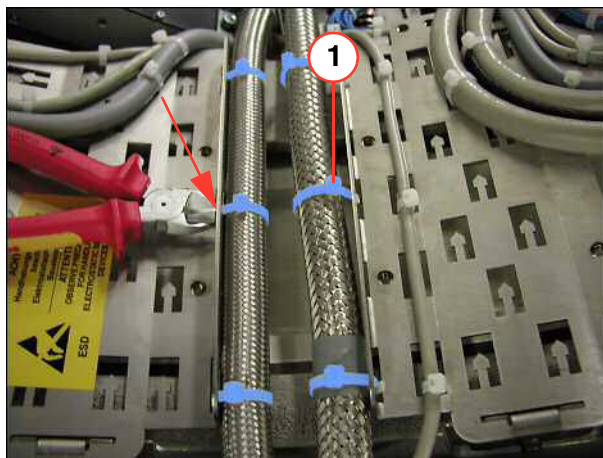


Disconnect the following cable and connectors from the tube cooler.

- Plug X1 (item 1) on the XTC cooling pump.
- Remove all tie wraps necessary to release the cable.

4. Remove cable ties.

Fig. 281: Remove cable ties from oil hoses

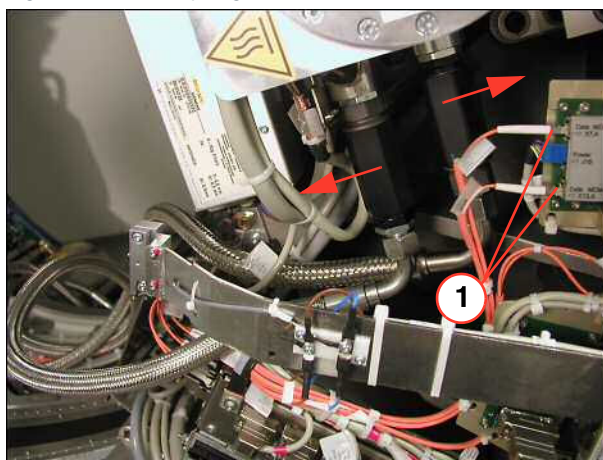


- Remove all cable ties (item 1) necessary to remove the oil hoses.
 - » **Always cut the ties in this way so that damage to the oil hoses shielding is avoided (arrow).**

5. Disconnect the oil couplings at the Straton MX tube.

- » Do not forget to replace the oil hoses, not only the XTC.

Fig. 282: Oil couplings at Straton MX A



- Open the tie wraps from the oil protectors located over the oil couplings.
- Remove the half-shells of the protectors.
- Put some paper towels underneath the couplings before removing them.
 - » **A few drops of oil may drip out of the couplings during disassembly.**
- **Only for Straton MX A:** Disconnect the fiber optic cable from the DMS transmitter to prevent damage (item 1).
- Open the oil couplings on the X-ray tube. To loosen them, the couplings have to be turned in the direction of the red arrows. If necessary, use a wrench to open the couplings.
- Dry the couplings to prevent oil from dripping into the gantry.

6. Move the oil hoses to the back gantry side.

Fig. 283: Oil hoses, removed



- Carefully move the oil hoses to the back gantry side.

7. Remove the securing screws on the XTC assembly

Fig. 284: Left XTC assembly, attachment screws

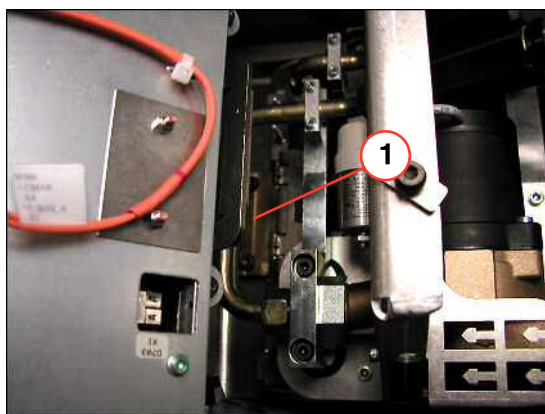
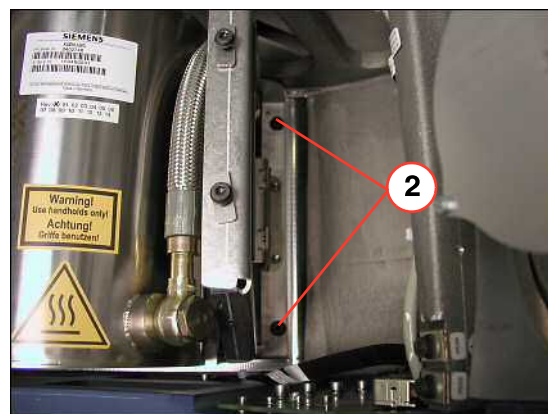


Fig. 285: Right XTC assembly, attachment screws



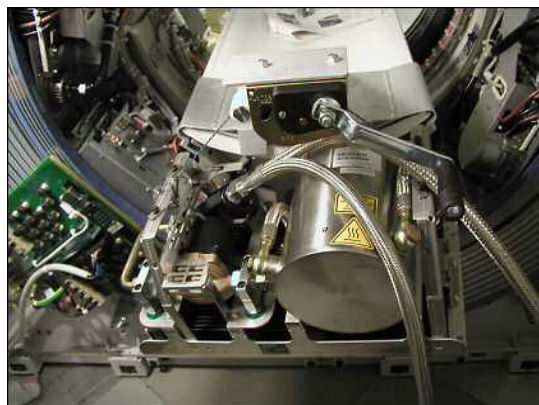
- **Don't loosen the securing screws (item 1)!**
 - » **This retaining bar must not be removed during replacement.**
- Remove the 2 M8*20 mm securing screws (item 2)
- Move the XTC assembly to the right, so that the left side of the XTC is moved out of its holder.

8. Install the lifting support and remove the XTC assembly

Fig. 286: XTC assembly secured to the lifting support



Fig. 287: XTC assembly secured to the lifting support



- Place the lifting support at the end of the tabletop (→ Lifting support / Page 24).
- Secure the cooler assembly to the cords of the lifting support.
- Lift the cooler carefully out of the gantry using the handle of the lifting support.
 - » **Lift the cooler on the right side by hand supported by the lifting support until the cooler is above the gantry frame (left image).**

9. Position the XTC assembly on the transport device

Fig. 288: XTC assembly on transport device

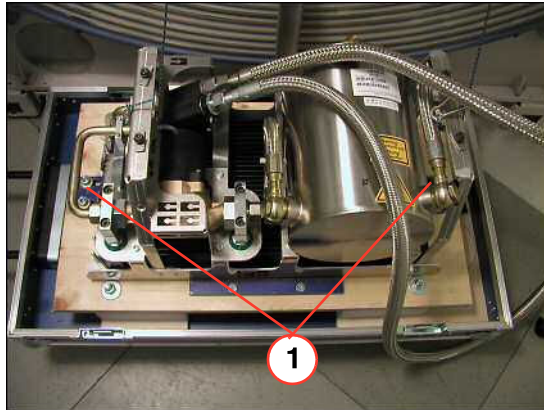
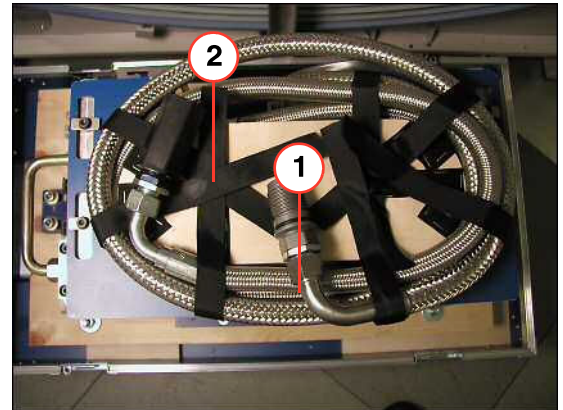


Fig. 289: XTC oil cable, transport position



- Roll the empty cooler transport device to the back gantry side.
- Move the cooling unit out of the gantry using the tabletop so that it is positioned over the transport device.
- Lower the cooler assembly onto the transport device using the handle of the lifting support.
- Remove the cord of the lifting support and secure the cooler to the transport device using a torque of **17 Nm** (left image, item 1).
- Remove the wooden board from the new XTC assembly and install the oil hoses on the board (right image, item 1). Secure the hoses with Velcro tape (right image, item 2) to the board.

10. Remove the old cooler from the back gantry side.

6.7.4 Installation

1. Place the transport device with the new tube cooler in back of the gantry and attach the cords of the lifting support to the cooler.
2. Open the transportation attachment screws and lift the new tube cooler out of the transport box using the handle of the lifting support.
3. Remove the transport device.
4. Move the cooler gently toward the gantry until it is positioned correctly.

5. Install the tube cooler and assemble it to the drum

Fig. 290: Left XTC assembly, attachment screws

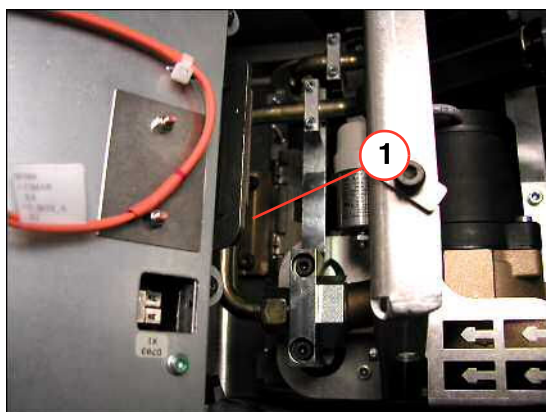
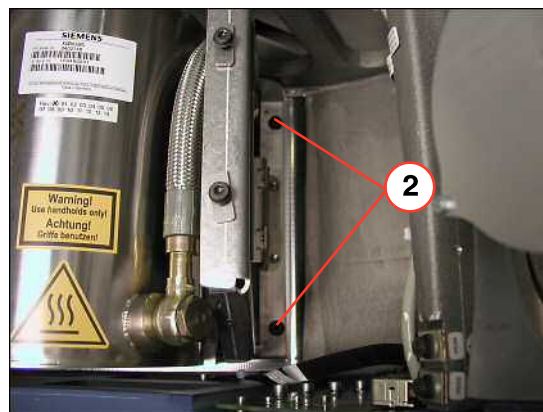


Fig. 291: Right XTC assembly, attachment screws



- Install the new cooler by reversing the order for removing the cooler.
 - » **Lift the cooler on the right side by hand and lower it supported by the lifting support until the cooler is located on the drum.**
 - Remove the lifting support from the table and move tabletop out of the gantry
 - Shift the XTC assembly to the left until it is positioned correctly in the attachment holder (item 1).
 - Install 2 new M8 * 20 mm torx bolts (supplied) at the right XTC side (item 2) and tighten them using **31 Nm**.
6. Install oil hoses between **XTC A** and **Straton MX A** (XTC B and Straton MX B see next step) according to the original configuration. Secure the oil hoses with the **heat-proof cable ties (supplied)**.



Use only heat-proof cable ties for securing the cooling hoses!

Make sure that the oil hoses are not touching any rotating part after installing the funnel/cover

Fig. 292: Oil hoses routing at Straton MX A

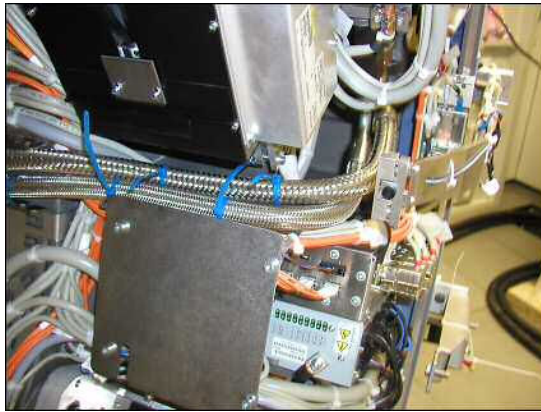
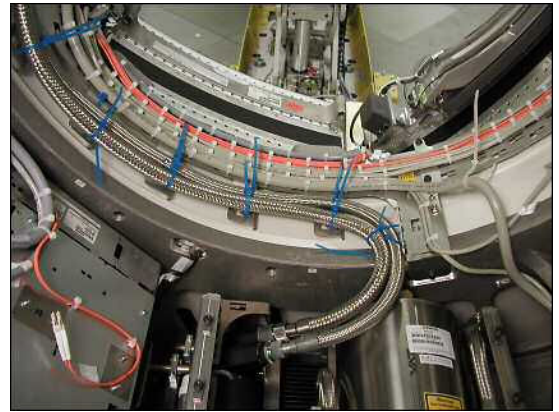


Fig. 293: Oil hoses routing at XTC-A



- Install the oil hoses starting from the Straton MX tube (upper left image) and ensure that the coupling is tightened firmly.
- Hand-tighten the oil coupling (don't use any tools!). The coupling has to be tightened until you can no longer see a gap between cover and coupling.

Fig. 294: Insufficiently tightened oil coupling

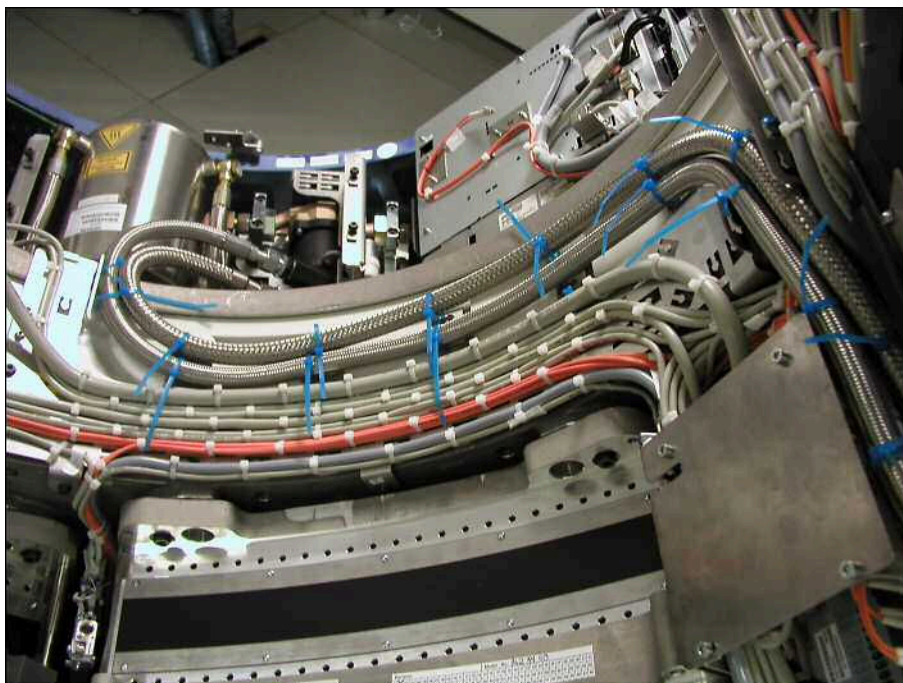


Fig. 295: Correctly tightened oil coupling



- Route the oil hoses between XTC A and Straton MX A as shown in the image below.
- Reinstall the fiber optic cables on the DMS transmitter.
- Finish assembling the oil hoses on the XTC A (see image, upper right)

Fig. 296: Oil hose routing at XTC-A



7. Install the oil hoses between XTC B and Straton MX B according to the original configuration. Secure the oil hoses with the **heat-proof cable ties (supplied)**.



Use only heat-proof cable ties for securing the cooling hoses!

Make sure that the oil hoses are not touching any rotating part after installing the funnel/cover

Fig. 297: Oil hose routing at Straton MX B

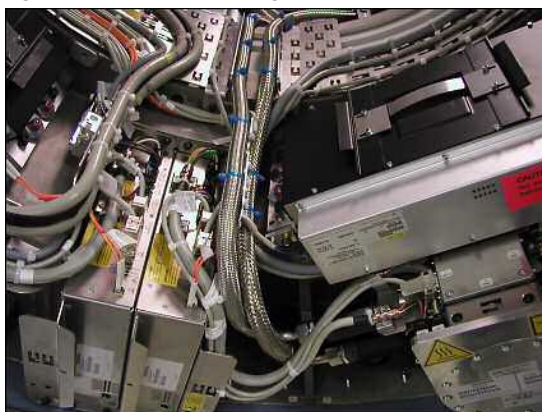
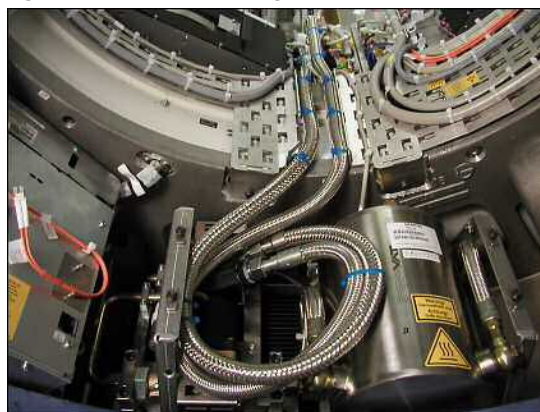
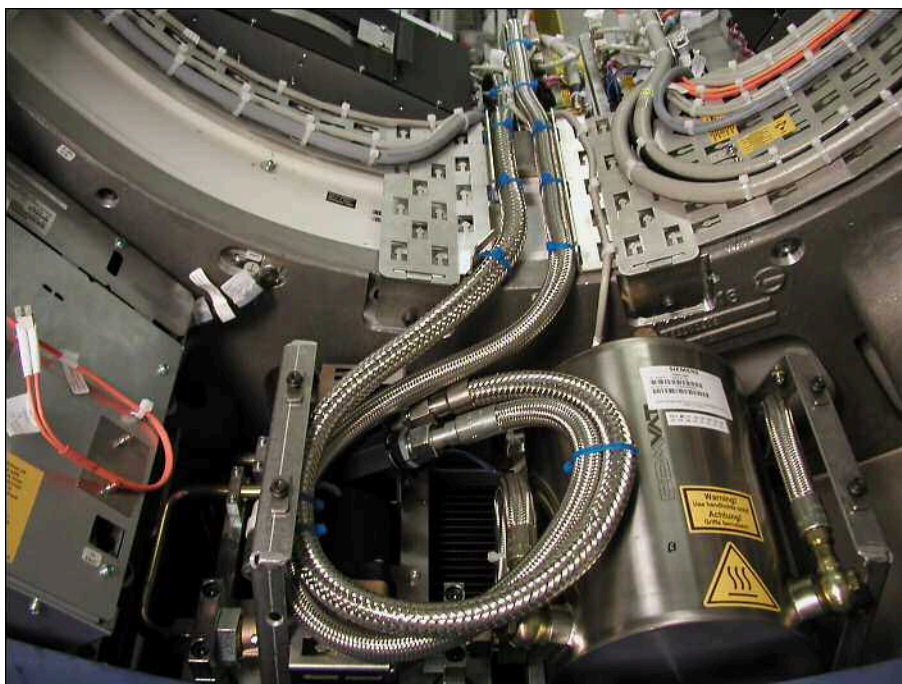


Fig. 298: Oil hose routing at XTC-B



- Install the oil hoses starting from the Straton MX tube (upper left image) and ensure that the coupling is tightened firmly (see installation of XTC A for pictures and further details).
- Route the oil hoses between XTC-B and Straton MX -B.as shown in the image below.
- Finish assembling the oil hoses at the XTC-B assembly (upper right image)

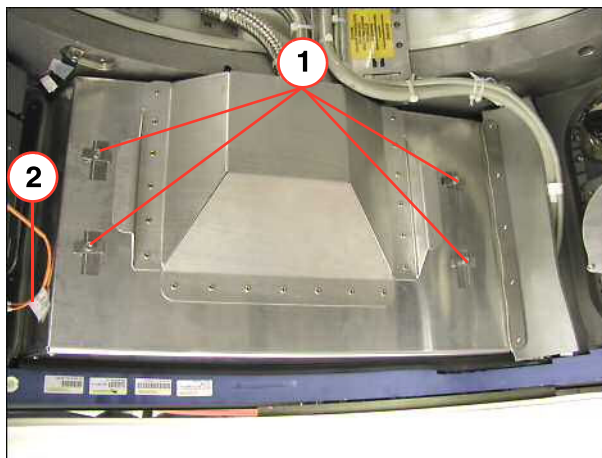
Fig. 299: Oil hose routing at XTC-A



8. Reinstall the oil protectors over both oil couplings and secure them with 2 tie wraps each.
 - » **Make sure that the seam on the protectors is positioned to the front.**
9. Connect Plug X1 (item 1) on the XTC cooling pump. Secure with cable ties.

10. Install the tube cooler cover.

Fig. 300: XTC cover



- Install the XTC cover carefully.
- Turn the holding brackets in this way so that the cover is secure.
 - » **Make sure that the brackets are positioned correctly.**
- Tighten the 4 cover holding screws (item 1) using a TX 30 bit with a torque of **8 Nm**.
- Install the fiber optic cable and cable ties as necessary from the XGR assembly (item 2).

6.7.5 Start-up

6.7.5.1 Mechanical

1. Unlock the safety bolt for gantry rotation.
2. Rotate the gantry by hand and check for unusual noise.
3. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
4. Switch on the system (**SYSTEM/ON**) at the control box.

6.7.5.2 Software

n.a.

6.7.5.3 Tune-up

1. Run the tune-up sequence as described under **Local Service -> Tune Up -> FRU replacement**.

2. Close the gantry covers when instructed within the guided tour or after you performed the guided tour.



Disable the gantry rotation using S301 when performing mechanical adjustments inside the gantry during tune-up!

6.7.5.4 Testing

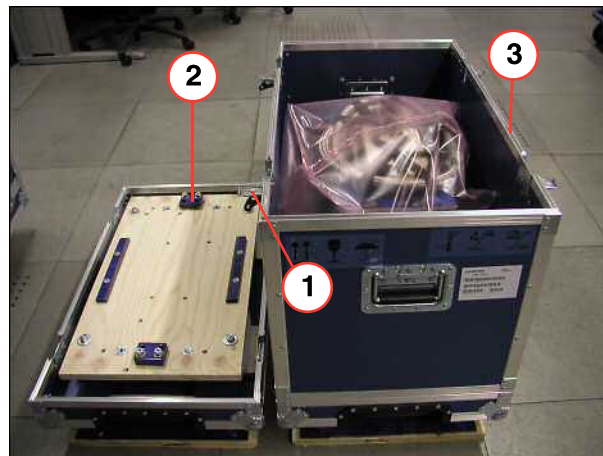
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6.7.6 Final steps

6.7.6.1 Prepare the tube cooler transport box

1. Close transport box.

Fig. 301: XTC transport box



- Reassemble the transport box and ensure that all service tools, papers have been placed inside the box.
- Install the side cover (item 3)
- Install the top cover (item 2) and secure the top using the locks (item 1).

6.8 XDC boards

The following instructions apply to both XDC boards

6.8.1 Prerequisites

6.8.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

6.8.1.2 Time and Man Power

- 1 Person
- Estimated total repair time: 2 hours.

6.8.1.3 Tools and auxiliary Equipment

- Standard tool kit

6.8.2 Preliminary steps

1. Switch system to status **COMP/ON** at the control box.
2. Switch off the gantry power using service push-button S1 in PDC A (→ Service Push-Button S1 in PDC-A / Page 20). Secure against unintended switch-on.
3. Remove the gantry bottom front ring segment (→ Removing / Page 32).
4. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
5. Turn the gantry by hand until the Straton MX tube - A is in the 6 o'clock position (XDC is in an approx. 4 o'clock position).
6. Secure against unintended rotation by using the front safety bolt (→ Safety bolts / Page 21).
7. Install the data ring protection (→ Data ring protection / Page 23).

6.8.3 Removing

1. Remove the metal cover located over the XDC boards.

2. Remove all tie wraps as needed to release the cables and disconnect all XDC cable connections necessary.
3. Remove the XDC board
 - » **Follow ESD rules**
4. Open the 4 holding screws and remove the XDC board

6.8.4 Installation

1. Reinsert the new XDC board and tighten the new 4 M6 *20 mm holding screws (delivered with the part) with a torque of **14 Nm**.
2. Connect all cables according to the original configuration. Secure cables with cable ties.
3. Install the metal cover over the XDC boards.

6.8.5 Start-up

6.8.5.1 Mechanical

1. Remove the data ring protection.
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.

6.8.5.2 Software

The following section has to be performed to load the actual firmware version.

- Go to the service home menu, select Options -> Service -> Local Service
- In the service software menu, select **TuneUp -> FRU Replace**
- Select the corresponding part replaced and click GO
 - Note:** If the part you replaced is not in the list, click Go and continue. The firmware check will be performed anyway.
- Follow the output instructions.



If the firmware update aborts with an error and the firmware status is NOT ok, perform the firmware update again.

6.8.5.3 Tune-up

1. Run the tune-up sequence as described under **Local Service -> Tune Up -> FRU replacement**.

6.8.5.4 Testing

1. Run some test scans with X-radiation on.
2. Close all covers opened before.

6.8.6 Final steps

n.a.

7.1 Overview

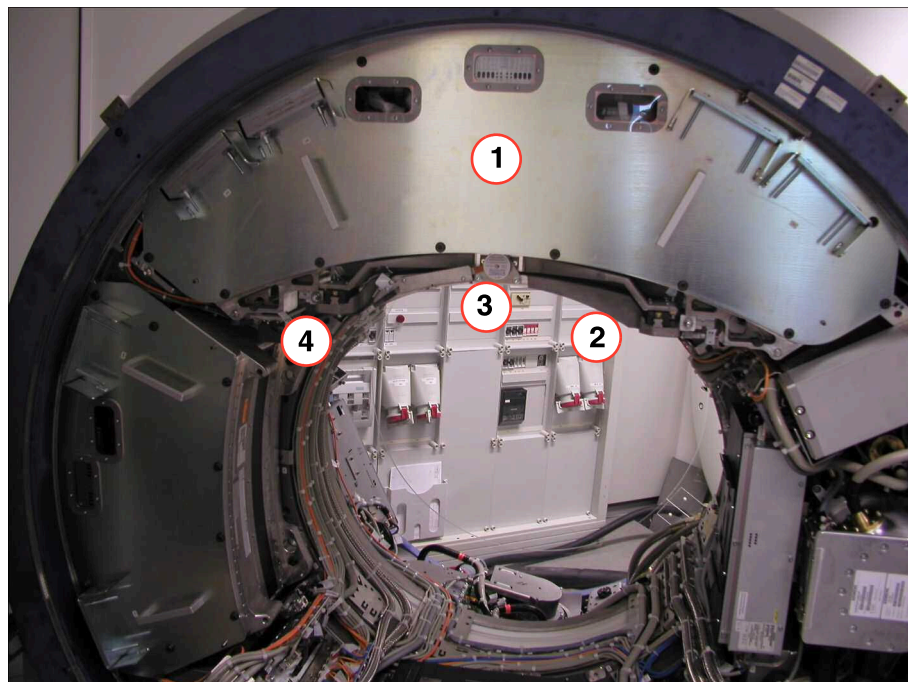
- The first two figures show an overview of all DMS A parts involved in the data measurement and their location inside the gantry (stationary light markers removed).
The next section contains detailed descriptions of how to exchange all major parts of the DMS A.



Follow the ESD guidelines when handling the DMS

- This figure shows DMS A, including all covers and the stationary light markers which have been removed.

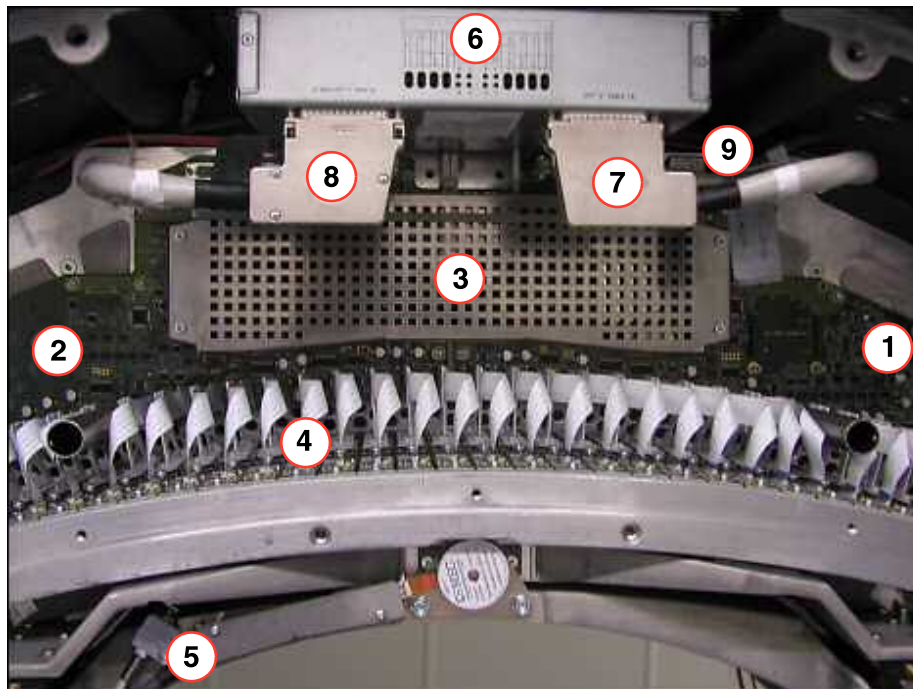
Fig. 302: DMS A with all covers installed



- DMS A with metal front cover (item 1)
- UHR assembly (item 2)
- UHR motor (item 3)
- PHI/Z control board (item 4)

- This figure shows the DMS A without covers.

Fig. 303: DMS A without covers (metal and plastic covers removed)



- Left backplane (item 1)
- Right backplane (item 2)
- DAS Controller with cover (item 3)
- Modules (item 4)
- Home Sensor (item 5)
- Power supply (item 6)
- Power cable left (item 7)
- Power cable right (item 8)
- UHR comb cable (item 9)

7.2 DMS Main Covers

7.2.1 Prerequisites

7.2.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage of parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the “General” section of this document and/or “Product-specific safety notes”.



Follow the ESD guidelines when handling the DMS

7.2.1.2 Tools and auxiliary equipment

- Standard tool kit
- Torque wrench 5 Nm, 10Nm, 14 Nm, 31Nm

7.2.1.3 Time and manpower

Estimated total repair time: 0.5 hours

7.2.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.



If the covers have to be removed in 12 o'clock position, it is necessary to remove the sagittal light marker. Mark the position of the laser as a reference on the tabletop, using a piece of tape and a pen.

2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).

6. Turn the gantry by hand until the part to be replaced is in the 6 o'clock or 12 o'clock position.
7. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)
8. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

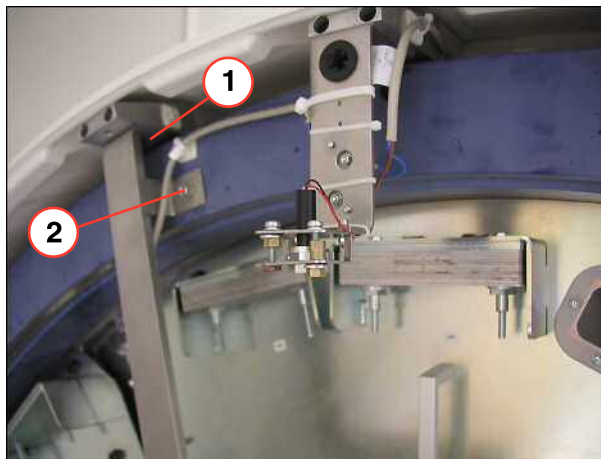
7.2.3 Removing



This chapter describes how to remove the 2 Main covers of DMS A. For some steps, the DMS has to be in 6 o'clock position before removing the covers. For steps in the 12 o'clock position, the sagittal light marker has to be removed before removing the DMS covers.

1. Remove the sagittal light marker, including the holding bracket (if necessary)

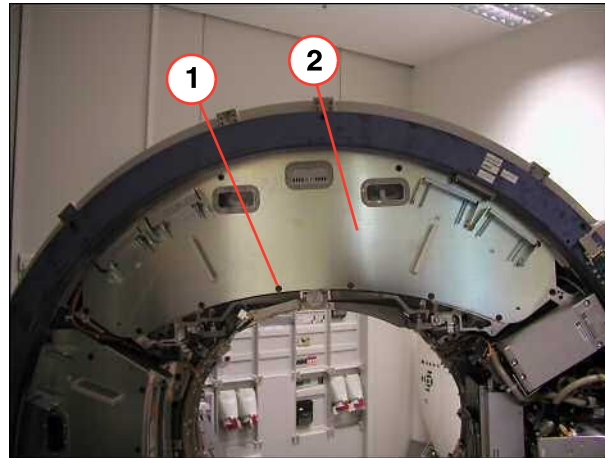
Fig. 304: Transaxial and sagittal light markers



- Remove the connectors on the sagittal light marker
- Remove cable ties as necessary.
- Remove the bracket with the sagittal light marker by removing the screws (items 1 and 2)

2. Remove the metal DMS front cover

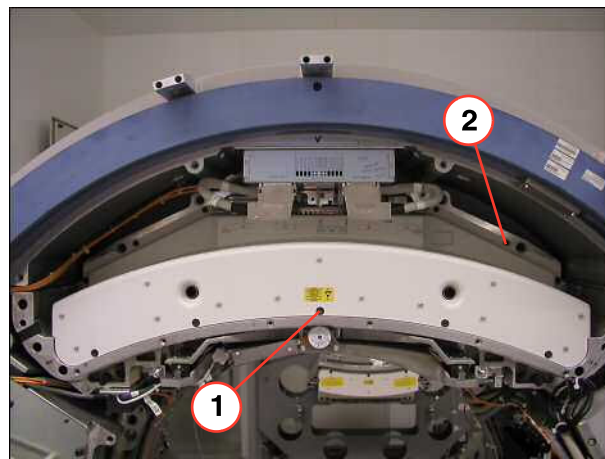
Fig. 305: DMS metal front cover



- Remove the 12 (M8x20) screws (e.g. item 1) using a T40 Torx key
- Remove the metal DMS front cover (item 2) using the 2 handles.

3. Remove the plastic DMS cover.

Fig. 306: DMS plastic front cover



- Remove the 5 (M6x35) screws in front (e.g. item 1) using a T30 Torx key.
- Remove the 6 (M6x35) screws at the rear (e.g. item 2) using a T30 Torx key.
- Remove the plastic cover by pulling outwards and lifting up to clear the power supply.

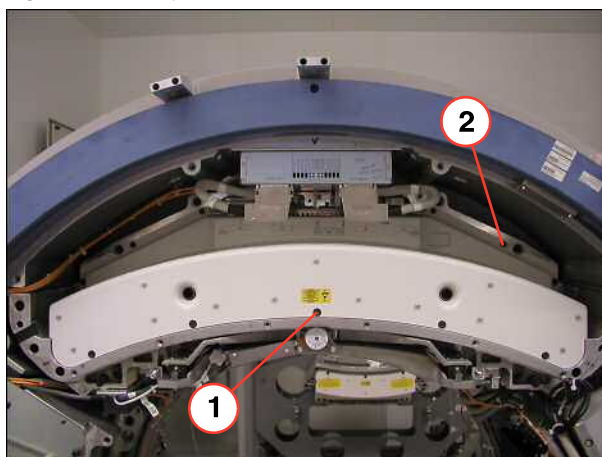
7.2.4 Installation

1. Install the plastic DMS cover.



If screws with lacquer-coated thread were installed, clean the thread prior to installing the screws (e.g. use a wire brush)

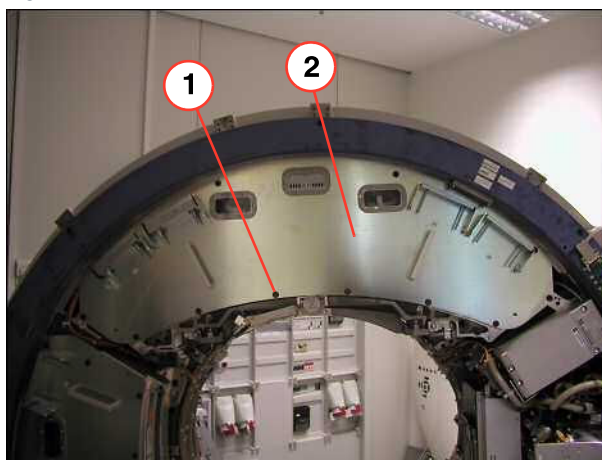
Fig. 307: DMS plastic front cover



- Insert the 11 screws. Use Loctite 243
- Tighten the 5 (M6x35) screws in front (e.g. item 1) with a torque of **14 Nm** using a T30 Torx key.
- Tighten the 6 (M6x35) screws at the rear (e.g. item 2) with a torque of **5 Nm** using a T30 Torx key.

2. Install the metal DMS front cover

Fig. 308: DMS metal front cover



- Install the 12 (M8x20) screws (e.g. item 1) using a T40 Torx key.
- Move the cover (item 2) upwards while tightening the screws to close the slot between the cover and gantry frame (not necessary in 6 o'clock position).
- Tighten the 12 screws with a torque of **31 Nm**.

3. Install the sagittal light marker in the reverse order (if removed previously).

- Tighten the M8 screws securing the bracket to the gantry frame with a torque of **31 Nm**.
- Tighten the M6 screw securing the bracket to the stationary data ring with a torque of **10 Nm**.

7.2.5 Startup

7.2.5.1 Mechanical

1. Remove the data ring protections (delivered with the system)
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.xxxxxxx
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.



Rotation may start during during init for positioning!

6. Disable gantry rotation using rotation enable switch S301.
7. Check the adjustment of light marker and correct if necessary. Refer to (→ Adjustment of the sagittal Light Marker / Page 122)
8. Enable gantry rotation using rotation enable switch S301.
9. Install all covers.

7.2.5.2 Tune-up

n.a.

7.2.5.3 Testing

n.a.

7.2.6 Final steps

n.a.

7.3 Complete DMS

7.3.1 Prerequisites

7.3.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

7.3.1.2 Tools and auxiliary equipment

- Standard toolkit with Torx
- Torque wrench 10 Nm, 14 Nm, 30 Nm, 31 Nm, 62 Nm
- Tools delivered with the spare part:
 - Strap, 2 m with carabineer
 - Torx T50 3/8"
 - Extension 3/8"
 - Reduction 1/2" to 3/8"
 - Pad for tabletop
 - Set of new screws
 - DMS guide pins
 - Transport brackets for old DMS
 - Wooden block
 - Handle

7.3.1.3 Time and manpower

- Estimated total repair time: 7 hours
One person is required to replace the DMS.
A second person is required only for opening and closing the cover of the DMS transport box.

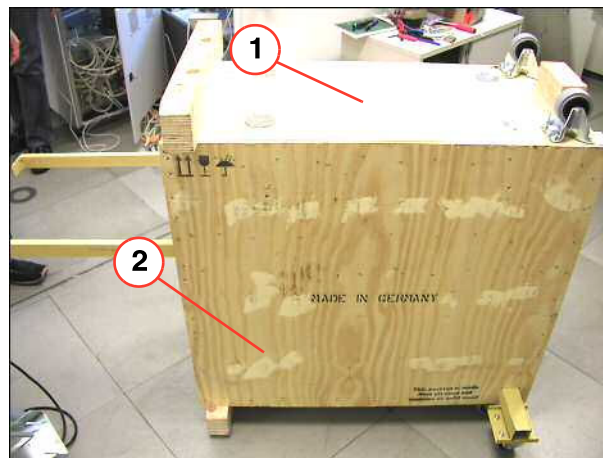
7.3.2 Preliminary Steps

1. Allow at least five minutes cooling time after the last scan. During the cooling time the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may be up to 130 degree Celsius.
2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Press the blue Reset button on the MCB of the PDC_XGS_Control (piggyback board of the D700 main board) in the PDC-A to enable service power to the PHS.
5. Remove the bottom front ring segment. (→ Removing / Page 32)
6. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
7. Turn the gantry by hand until DMS B is in the 6 o'clock position.
8. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)
9. Install the data ring protection. (→ Data ring protection / Page 23)

7.3.2.1 Prepare the transport crate

1. Open the transport crate.

Fig. 309: DMS transport crate



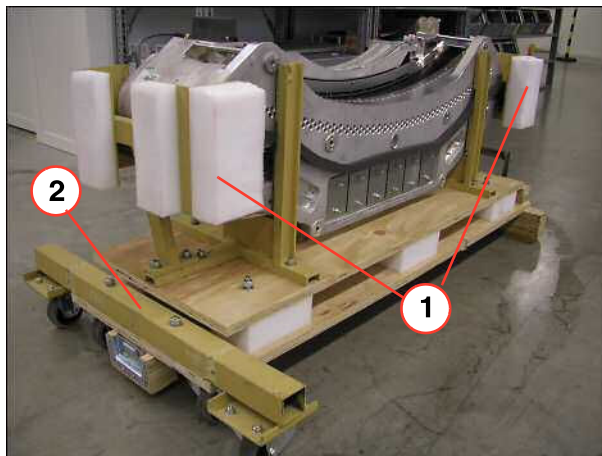
- Open and remove the top cover (item 1) of the transport crate.
 - Remove the service tools.
 - Remove the side covers (item 2) of the transport crate by lifting them up.
2. Install the transport brackets (delivered with the spare part) on the top cover of the transport crate using a 17mm hex socket.



The top of the crate will serve as transport device for the old DMS. For orientation of the transport brackets, refer to the transport device on the new DMS.

4. Tighten the 2 nuts holding the transport brackets with a torque of **30 Nm**.
5. Remove the transport protection from the new DMS. (The figure below is an example. The DMS might look different.)

Fig. 310: Transport protection and stabilizer



- Remove the transport protection (item 1) from the transport device using an 8 mm Allen key.
- Remove the transport stabilizer (item 2) from the transport device using a 17 mm hex socket.
- Mount the wooden block (delivered with the part) to the transport device instead of the transport stabilizer, and tighten the 2 nuts using a 17 mm hex socket.

7.3.3 Removing

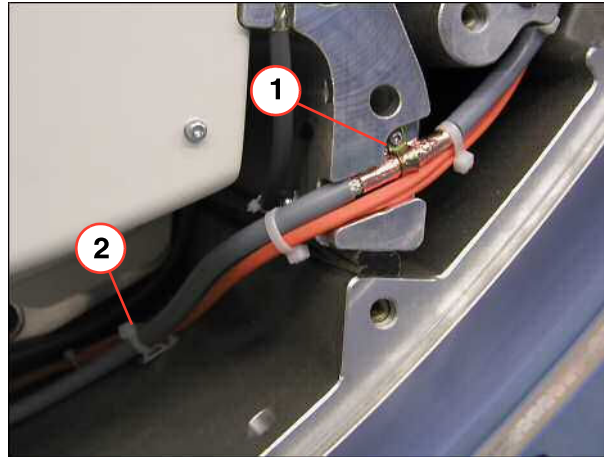


The metal front cover may be heavy because it contains balancing weights. The entire procedure has to be performed in the 6 o'clock position with the safety bolt installed to prevent unintended gantry rotation.

1. Remove the metal DMS front cover. Refer to (→ Removing / Page 350) DMS main covers.
2. Remove DMS power supply. Refer to (→ Removing / Page 379) DMS power supply.
3. Remove the 2 power cables X30 and X31 from DMS backplanes.
4. Disconnect the fiber optic cables at the DAS controller.
5. Remove the grounding strap on the **DMS frame** using a T30 Torx key.

6. Pull out the cables from the DMS.

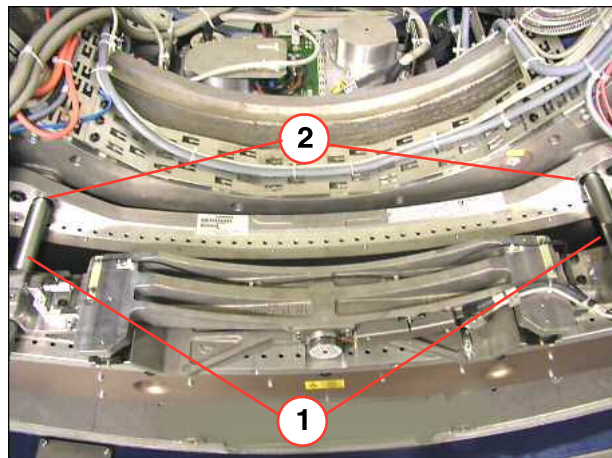
Fig. 311: Cable clamp DMS A



- Remove the cable clamp (item 1) on the right side of the DMS.
- Remove cable ties as necessary (e.g. item 2). Be careful not to damage the cables.
- Pull out the cables on the right side of the DMS (item 1).

7. Install the DMS guide pins.

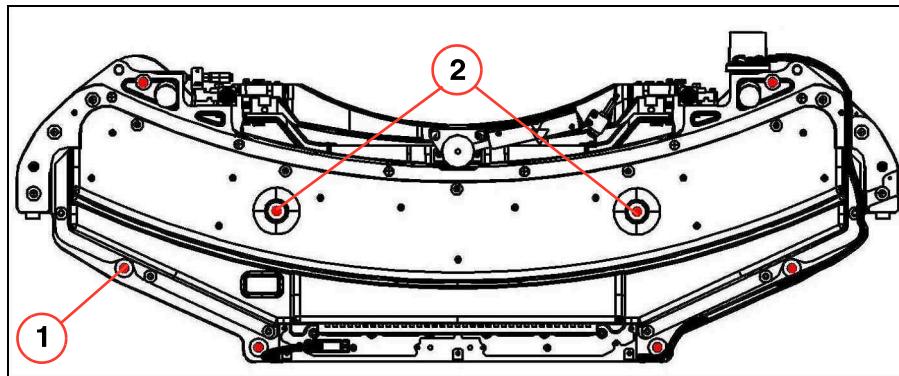
Fig. 312: DMS Guide Pins



- Install the 2 DMS guide pins (item 1) through the holes in the DMS (item 2).
- Tighten the guide pins by hand.

8. Remove the DMS holding screws and position the transport device.

Fig. 313: DMS holding screws



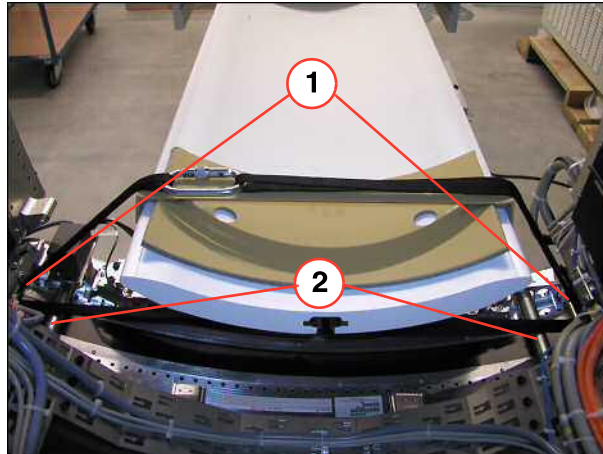
- Remove the 8 Torx holding screws M10x40 (e.g. item 1) using a T50 Torx key.
- Two of the holding screws are inside the protection tubes (item 2) and are therefore not visible. **Use the delivered Torx socket to open these 2 screws.**

9. Remove the DMS.



Be careful not to damage or misadjust the rotating light marker with the strap when removing the DMS.

Fig. 314: Removing the DMS



- Position the tabletop above the DMS using the table control on the left side of PHS.
- Place the pad on the tabletop.
- Attach the strap through the **lifting bolts** (item 1) and around the pad on the tabletop.

Do not mix up lifting bolts (item 1) with guide pins (item 2)!

- Gently move the table up using table control, until the DMS is at the correct height to be moved out via the guide pins.
- Move the DMS out using the table control. **Move only the tabletop support, if possible!**

10. Position the DMS on the transport device.

- Position the transport device underneath the DMS between the gantry and the table. For orientation, refer to the new DMS on its transport device.
- Position the DMS onto the transport device using the table control.
- Secure the DMS with 2 screws, tighten the screws with a torque of **30 Nm**, and remove the strap

11. Remove the transport device. (The figure below is an example. The DMS might look different.)

Fig. 315: Handle on transport device



- Remove pad on tabletop and move the table out for better access.
- Remove the transport device, including the old DMS, from between the gantry and table.
- Use the handle (item 1) to lift and move the transport device

7.3.4 Installation

1. Position the transport device with the new DMS between the gantry and the PHS.
2. Position the pad on the tabletop and move the table above the DMS.
3. Attach the strap through lifting bolts and around pad on the tabletop.
4. Gently move the table up using the table control to add tension to the strap.
5. Remove the 2 holding screws from the transport device and remove the transport device.
6. Gently move the table up using table control, until the DMS is at the correct height to be moved in via the guide pins.



Be careful not to damage or misadjust the rotating light marker with the strap when installing the DMS.

7. Gently move the table inwards using table control until the DMS is completely in.
8. Lower the patient table.
 - Remove the strap.
 - Remove the pad from the tabletop.
 - Move the table out.
9. Insert the 8 (M10x40) holding screws (delivered with new DMS) and tighten with a torque of **62 Nm**.
10. Remove the DMS guide pins

11. Reconnect the connectors according to the original configuration and reinstall all cable ties that were removed.
12. Reattach the grounding strap to the DMS and tighten the screw with a torque of **10 Nm**.
13. Install the DMS Power supply. Refer to (→ Installation / Page 381) DMS power supply.
14. Reconnect X30 and X31 to the backplanes.
15. Install the metal DMS front cover. Refer to (→ Installation / Page 351) DMS main covers.

7.3.5 Startup

7.3.5.1 Mechanical

1. Remove the data ring protections
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.



Rotation may start during init for positioning!

7.3.5.2 Software

1. Update the system configuration with the new DMS information:
Local Service -> Configuration -> Local Hosts -> Serial.

7.3.5.3 Tune-up

1. Run the tune-up sequence **Local Service -> Tune-Up -> FRU Replace.**
2. Close the gantry covers when needed (table generations)



Disable the gantry rotation using S301 when performing mechanical adjustments inside the gantry during tune-up!

7.3.5.4 Testing

1. Run Quality Constancy.
2. Perform a backup of the new settings.

7.3.6 Final steps

Prepare the DMS transport crate

1. Mount the transport stabilizer to the transport device and tighten the screws with a torque of **30 Nm**.
2. Mount the transport protection to the transport device and tighten the screws with a torque of **30 Nm**.
3. Complete the DMS return form.
4. Reassemble the DMS transport crate, add the service tools and DMS return form.
5. Remove the transport brackets from the top cover of the DMS box.
6. Mount the transport brackets into the box and close the top cover.

7.4 Modules

7.4.1 Prerequisites

7.4.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

7.4.1.2 Tools and auxiliary equipment

- Standard tool kit
- Torque wrench 1 Nm, 1.4 Nm, 5 Nm, 14 Nm, 31 Nm

7.4.1.3 Time and manpower

- Estimated total repair time: 1.5 - 3 hours depending on the position of the module

7.4.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.



The sagittal light marker has to be removed during this procedure. Mark the position of the laser as a reference on the tabletop, using a piece of tape and a pen.

2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Turn the gantry by hand until DMS A is in the 12 o'clock position.
7. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)

8. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

7.4.3 Removing



Follow the ESD guidelines when handling the DMS



Slot 1 and Slot 42 of the DMS are used for air slice normalization. If modules have to be swapped for troubleshooting purposes, do not swap the modules to slot 1 and slot 42.

1. Remove the sagittal light marker, the metal and the plastic DMS front covers. Refer to the (→ Removing / Page 350) DMS main covers.
2. Locate the defective module using the engraved numbers on the DMS frame.



If the DMS is in 12 o'clock position, channel 1 is on the left side.

Channels	Slot	Channels	Slot
1 - 16	1	337 - 352	22
17 - 32	2	353 - 368	23
33 - 48	3	369 - 384	24
49 - 64	4	385 - 400	25
65 - 80	5	401 - 416	26
81 - 96	6	417 - 432	27
97 - 112	7	433 - 448	28
113 - 128	8	449 - 464	29
129 - 144	9	465 - 480	30
145 - 160	10	481 - 496	31
161 - 176	11	497 - 512	32
177 - 192	12	513 - 528	33
193 - 208	13	529 - 544	34
209 - 224	14	545 - 560	35
225 - 240	15	561 - 576	36
241 - 256	16	577 - 592	37
257 - 272	17	593 - 608	38

Channels	Slot	Channels	Slot
273 - 288	18	609 - 624	39
289 - 304	19	625 - 640	40
305 - 320	20	641 - 656	41
321 - 336	21	657 - 672	42

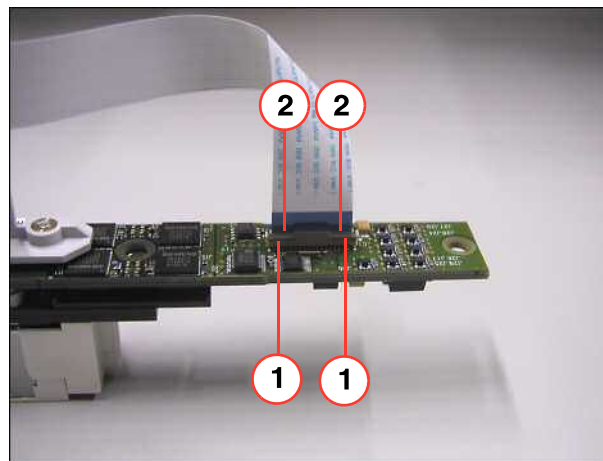


3. Remove module cable from DMS backplane

Hint: To unlock the plug, pull the locking device. To close the plug, press down on the locking device.

The example shows how to open the connector at a module. The connectors at the backplane are the same.

Fig. 316: Module cable connector



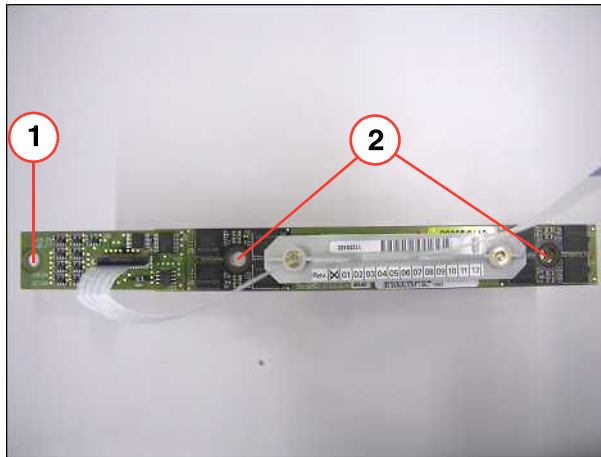
- To unlock the plug, pull on both sides of the brown locking device (item 1).
- To close the plug, press on both sides of the brown locking device (item 2).

4. Remove the module.

For slots 14 to 29, remove the DAS Controller for easier access.

For slots 11,12 or 31,32, remove protection tube for easier access. To remove the protection tube, use a screwdriver and turn it counterclockwise.

Fig. 317: Removed Detector Module



- Remove the front screw (M3x8) (item 1) using a T10 Torx key.
- Remove the 2 screws (M4x20) (item 2) next to the plastic handle using a T20 Torx key.
- Remove the old module by pulling on the plastic handle.

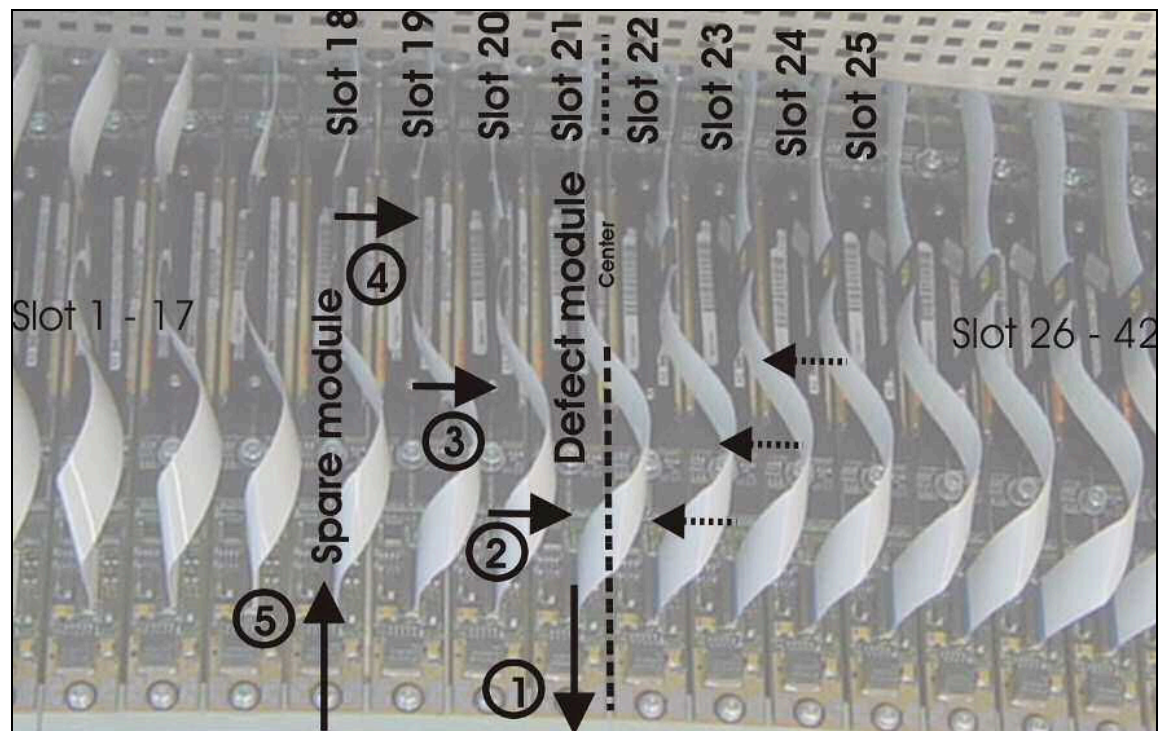


Do not put stress the module and do not touch any part of the PCB.

Replacement of a Center Module

In case of a defective center module (slot 19 -24), not the entire six pack will be replaced. However, to ensure the best image quality, first move the existing good modules towards the center to fill the empty slot where the bad module has been removed. Then place the new module outside of the center six slots (slot 18 or 25). This method keeps the center of the detector homogeneous.

Fig. 318: Center module replacement



The example shows a defective module in the center slot 21.

The following steps have to be performed:

- 1 remove the defective module in slot 21.
- 2 move slot 20 to 21
- 3 move slot 19 to 20
- 4 move slot 18 to 19
- 5 insert spare module in slot 18

Modules in slots 19 to 24 have to be rearranged according to this scheme.

7.4.4 Installation



1. Insert the new module and align it with the rotating plane.
2. While tightening the 3 screws, press the module toward the rotating plane.
Tighten the screw in front with a torque of **1.0 Nm**, and the screws next to the handle with a torque of **1.4 Nm**.
3. Connect the module cable at the backplane
4. Reinstall if necessary:
 - DAS controller. Refer to (→ Installation / Page 371) of the DAS controller.
 - Protective tube, and tighten with a torque of **5 Nm**

5. Install the metal and plastic DMS front covers and the sagittal light marker. Refer to the (→ Installation / Page 351) DMS main covers.

7.4.5 Startup

7.4.5.1 Mechanical

1. Remove the data ring protections (delivered with the system)
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.



Rotation may start during init for positioning!

6. Disable gantry rotation using rotation enable switch S301.
7. Check the adjustment of light marker and correct if necessary. Refer to (→ Adjustment of the sagittal Light Marker / Page 122)
8. Enable gantry rotation using rotation enable switch S301.

7.4.5.2 Tune-up

1. Run the tune-up sequence as described under **Local Service -> Tune Up -> FRU replacement**.
2. Close gantry covers when instructed during the tune-up procedure.



Disable gantry rotation using S301 when performing mechanical adjustments inside the gantry during tune-up!

7.4.5.3 Testing

1. Run Quality Constancy.
2. Perform a backup of the new settings.

7.4.6 Final steps

n.a.

7.5 DAS Controller

7.5.1 Prerequisites

7.5.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

7.5.1.2 Tools and auxiliary equipment

- Standard tool kit
- Torque wrench 1.0 Nm, 1.4 Nm, 3 Nm, 5 Nm, 14 Nm, 31 Nm

7.5.1.3 Time and manpower

- Estimated total repair time: 1.0 hours

7.5.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.



The sagittal light marker has to be replaced during this procedure. Mark the position of the laser as a reference on the tabletop, using a piece of tape and a pen before switch-off.

2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Turn the gantry by hand until DMS A is in the 12 o'clock position.
7. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)

8. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

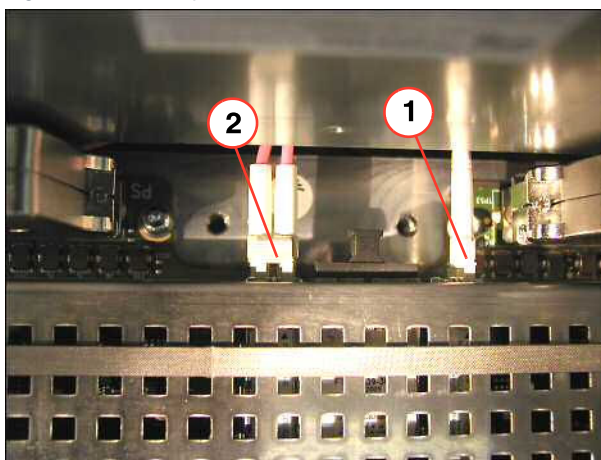
7.5.3 Removing



Follow the ESD guidelines when handling the DMS

1. Remove the sagittal light marker, the metal and the plastic DMS front covers. Refer to the (→ Removing / Page 350) DMS main covers.
2. Remove the fiber optic cables.

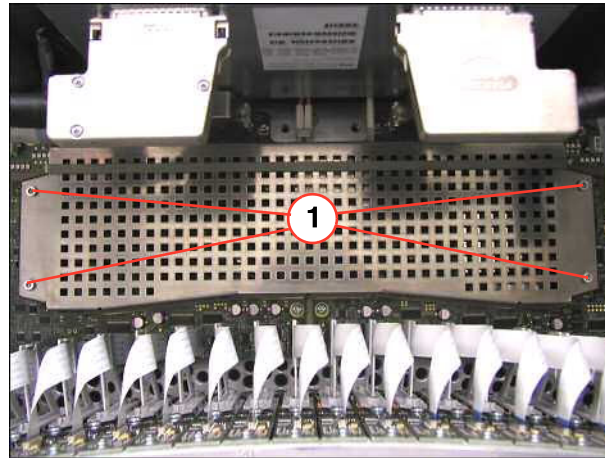
Fig. 319: Fiber Optic Cables



- Disconnect plug FODL1 (item 1) at the DAS Controller.
- Disconnect plug FODL2 (item 2) at the DAS Controller.
- Remove the rubber protective cover (plugged in at FODL3) on the DAS Controller.

3. Remove the DAS Controller cover.

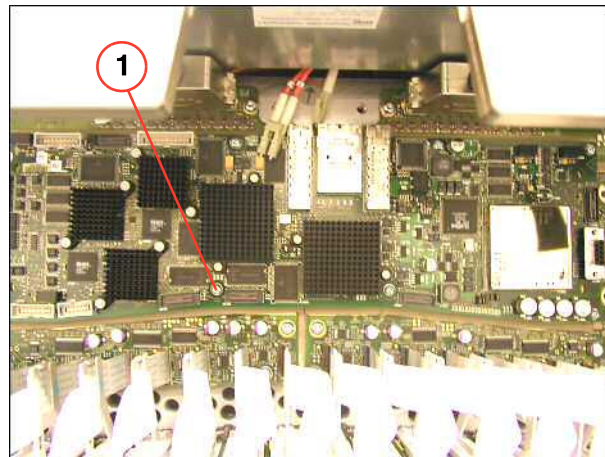
Fig. 320: DAS Controller Cover



- Remove the 4 screws (M4x45) from the DAS Controller cover using a T20 Torx key.
- Remove the cover.

4. Remove the DAS Controller.

Fig. 321: DAS Controller



- Remove the 8 screws (M3x8) (e.g. item 1) holding the DAS controller at the backplanes using a T10 Torx key.
- Remove the DAS Controller by pulling both sides at once.

7.5.4 Installation

1. To align the DAS controller board and the two backplanes, loosen the 18 Torx screws at the left backplane. Use a T20 Torx key to loosen the screws.
2. Install the DAS Controller.
3. Tighten all accessible screws of the left backplane with a torque of **1.4 Nm**.
4. Remove the DAS Controller.



5. Tighten the remaining screws of the left backplane with a torque of **1.4 Nm**.
6. Install the DAS Controller.
7. Tighten the 8 screws (M3x8) of the DAS Controller with a torque of **1.0 Nm**.
8. Install the DAS Controller cover and tighten the 4 (M4x45) screws with a torque of **3 Nm**.
9. Install the metal and plastic DMS front covers and the sagittal light marker. Refer to the (→ Installation / Page 351) DMS main covers.

7.5.5 Startup

7.5.5.1 Mechanical

1. Remove the data ring protections (delivered with the system)
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.



Rotation may start during during init for positioning!

6. Disable gantry rotation using rotation enable switch S301.
7. Check the adjustment of light marker and correct if necessary. Refer to (→ Adjustment of the sagittal Light Marker / Page 122)
8. Enable gantry rotation using rotation enable switch S301.
9. Install all covers.

7.5.5.2 Software

1. If the application was not restarted after replacing the DAS controller, restart the application now to ensure that the firmware is actualized.



FW loading is finished when the system is in standby. Wait for standby, after a successful FW loading message appears at the screen.

7.5.5.3 Tune-up

n.a.

7.5.5.4 Testing

1. Run the disk test.

2. Run spiral and sequence scans.
3. Run Quality Constancy.

7.5.6 Final steps

n.a.

7.6 DMS backplanes

7.6.1 Prerequisites

7.6.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

7.6.1.2 Tools and auxiliary equipment

- Standard tool kit
- Torque wrenches 1.0 Nm, 1.4 Nm, 3 Nm, 5 Nm, 14 Nm, 31 Nm

7.6.1.3 Time and manpower

- Estimated total repair time: 2 hours

7.6.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.



The sagittal light marker has to be replaced during this procedure. Mark the position of the laser as a reference on the tabletop, using a piece of tape and a pen before switch-off.

2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Turn the gantry by hand until DMS A is in the 12 o'clock position.
7. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)

8. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

7.6.3 Removing



Follow the ESD guidelines when handling the DMS

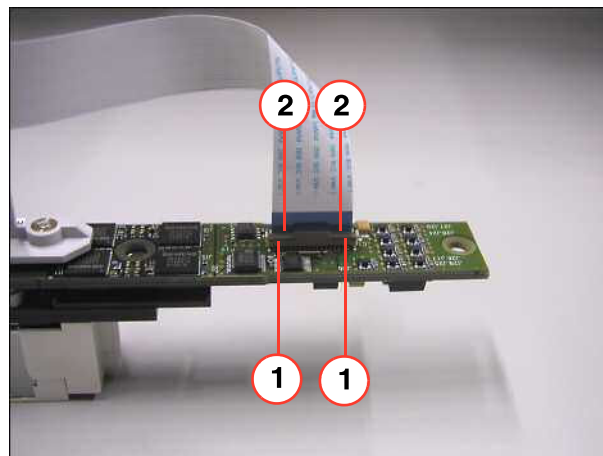


1. Remove the sagittal light marker, the metal and the plastic DMS front covers. Refer to the (→ Removing / Page 350) DMS main covers.
2. Remove the DAS Controller. Refer to (→ Removing / Page 370) DAS Controller.
3. Remove all module cables from the defective backplane.

Hint: To unlock the plug, pull the locking device. To close the plug, press down on the locking device.

The example shows how to open the connector at a module. The connectors at the backplane are the same.

Fig. 322: Module cable connector



- To unlock the plug, pull on both sides of the brown locking device (item 1).
 - To close the plug, press on both sides of the brown locking device (item 2).
4. Remove the cables connected to the defective backplane.
 - left backplane: X20 and X30
 - right backplane: X31
 5. Remove the 18 holding screws (M4x10) from the defective backplane using a T20 Torx key.
 6. Remove the defective backplane.

7. Only for backplane type 8429800 (introduced app. 07/2007):

The id- dongle (10023379) is a plug that has to be moved from the old to the new backplane.

(The previous backplane type 10023388 doesn't have a dongle).

7.6.4 Installation

1. Insert the new backplane and the 18 Torx screws (M4x10), but do not tighten the screws yet.
2. Only for backplane type 8429800 (introduced app. 07/2007):
Plug in the id- dongle (10023379) that was removed from the old backplane.
3. Insert the DAS Controller and tighten some of the 18 screws to align the positions of the backplane and the DAS Controller.
4. Tighten all accessible screws of the backplane with a torque of **1.4 Nm**.
5. Remove the DAS controller and tighten the remaining backplane screws with a torque of **1.4 Nm**.
6. Reconnect all module cables to the backplane.
7. Reconnect the cables to the new backplane.
 - left backplane: X20 and X30
 - right backplane: X31
8. Reinstall the DAS Controller. Refer to (→ Installation / Page 371) DAS Controller.
9. Install the metal and plastic DMS front covers and the sagittal light marker. Refer to the (→ Installation / Page 351) DMS main covers.



7.6.5 Startup

7.6.5.1 Mechanical

1. Remove the data ring protections (delivered with the system)
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.



Rotation may start during init for positioning!

6. Disable gantry rotation using rotation enable switch S301.
7. Check the adjustment of light marker and correct if necessary. Refer to (→ Adjustment of the sagittal Light Marker / Page 122)

8. Enable gantry rotation using rotation enable switch S301.
9. Install all covers.

7.6.5.2 Tune-up

n.a.

7.6.5.3 Testing

1. Run the disk test.
2. Run spiral and sequence scans.
3. Run Quality Constancy.

7.6.6 Final steps

n.a.

7.7 DMS Power supply

7.7.1 Prerequisites

7.7.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

7.7.1.2 Tools and auxiliary equipment

- Standard tool kit
- Torque wrenches 10 Nm, 14 Nm, 31 Nm

7.7.1.3 Time and manpower

- Estimated total repair time: 1.0 hours

7.7.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.
2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Turn the gantry by hand until the part to be replaced is in the 6 o'clock position.
6. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)
7. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

7.7.3 Removing



Follow the ESD guidelines when handling the DMS

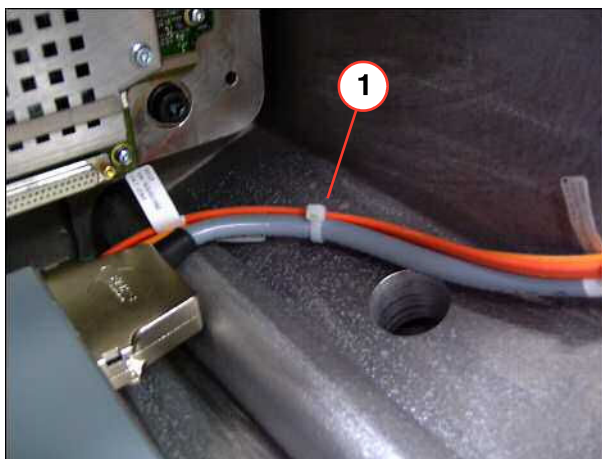
1. Remove the metal DMS front cover. Refer to (→ Removing / Page 350) DMS main covers.
2. Remove the grounding strap.

Fig. 323: Grounding strap on power supply



- Remove the grounding strap at the left side of the power supply using a 10 mm nut (item 1).
3. Remove the cable ties

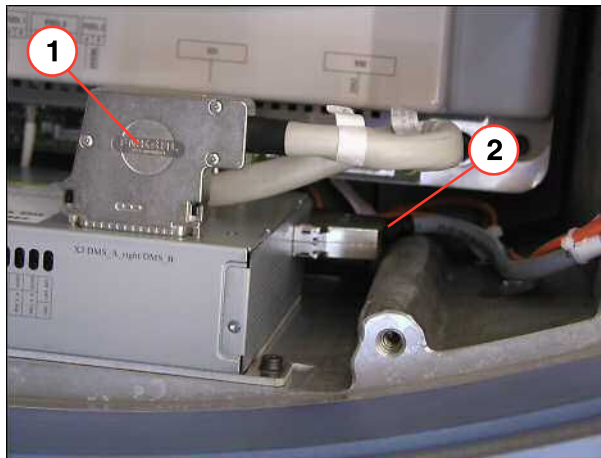
Fig. 324: Power connector X1 with cable tie



- Remove the cable tie (item 1)
- Remove the cable ties on top of the power supply.

4. Remove cables at the power supply.

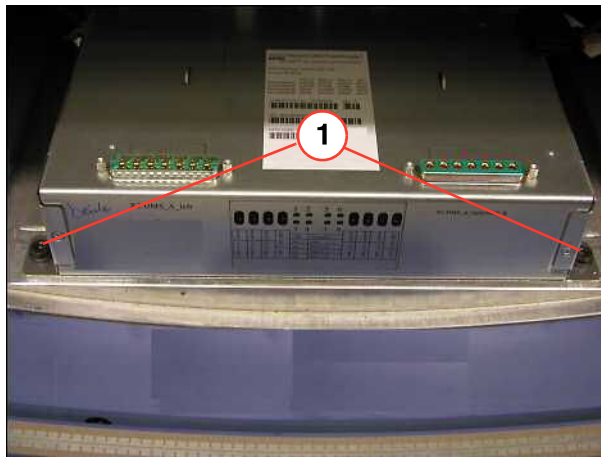
Fig. 325: DMS power supply connections



- Remove connectors X 2 and X3 on top of the DMS power supply (e.g. item 1).
- To remove X1 at the right side of the power supply (item 2), the power supply has to be pulled out first. Refer to next step.

5. Remove the power supply.

Fig. 326: Screws for power supply

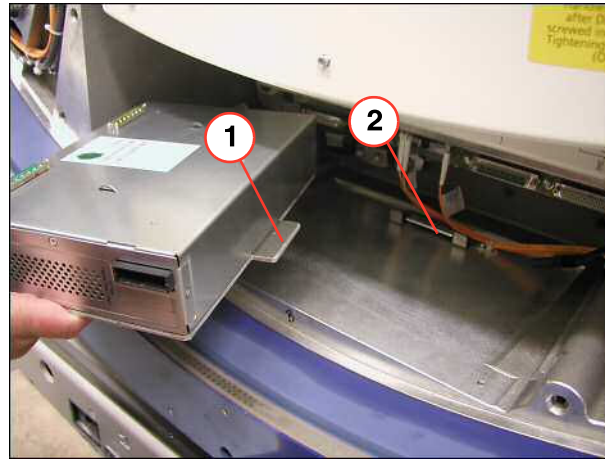


- Remove the 2 (M6x16) Torx screws using a T30 Torx key (item 1).
- Remove the power supply by pulling it straight out and remove connector X1.

7.7.4 Installation

1. Install the power supply.

Fig. 327: Installing the power supply



- Connect X1 to the right side of the power supply.
 - Push the guide plate (item 1) of the power supply into the clamp (item 2).
 - Install the power supply with 2 (M6x16) Torx screws and tighten with a torque of **14 Nm**.
2. Reconnect connector X1.
 3. Install the power cables according to the original configuration and secure them with cable ties.
 4. Connect the grounding strap using a 10 mm nut and tighten with a torque of **10 Nm**.
 5. Install the metal DMS front cover. Refer to (→ Installation / Page 351) DMS main covers.

7.7.5 Startup

7.7.5.1 Mechanical

1. Remove the data ring protections (delivered with the system)
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Install all covers.
5. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
6. Switch on the system (**SYSTEM/ON**) at the control box.

7.7.5.2 Tune-up

n.a.

7.7.5.3 Testing

1. Run sequence and spiral scans to check for proper functioning.

7.7.6 Final steps

n.a.

7.8 Complete UHR assembly

7.8.1 Prerequisites

7.8.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

7.8.1.2 Tools and auxiliary equipment

- Standard tool kit
- Torque wrench 1.4Nm, 3 Nm, 14 Nm

7.8.1.3 Time and manpower

- Estimated time to repair: 1.5 hours

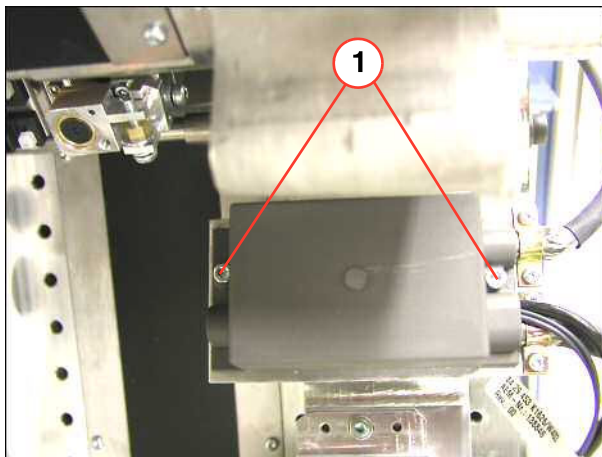
7.8.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.
2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Turn the gantry by hand until the part to be replaced is in the 6 o'clock position.
6. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)
7. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

7.8.3 Removing

1. Remove the cover of PHI/-Z Control board.

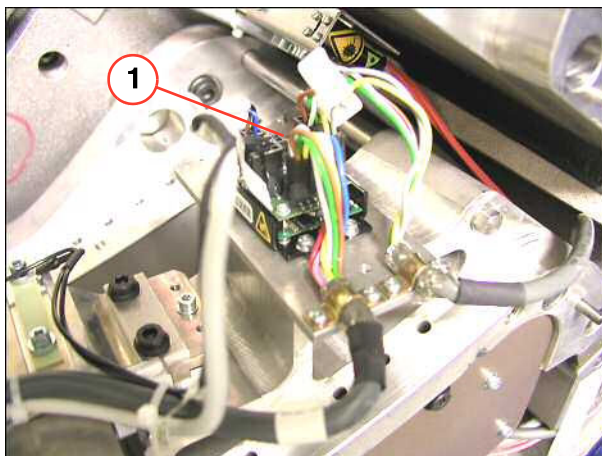
Fig. 328: PHI/Z Control Cover



- Remove the 2 screws (M4x16) (item 1) from cover using a T20 Torx key.

2. Remove the connectors (item 1) at the PHI/-Z Control board.

Fig. 329: PHI/Z Control Board

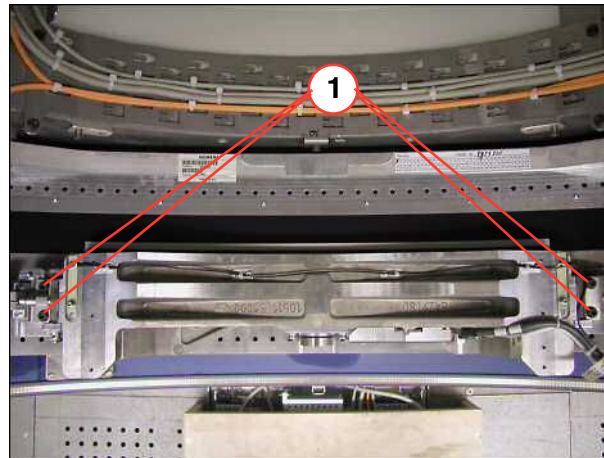


- Remove the **home** connector at the PHI/-Z Control.
- Remove the **motor** connector at the PHI/-Z Control.
- Remove the **end** connector at the PHI/-Z Control.
- Remove the left cable clamp using a T10 Torx key.

3. Remove cable ties as necessary.

4. Remove the UHR assembly

Fig. 330: UHR comb screws

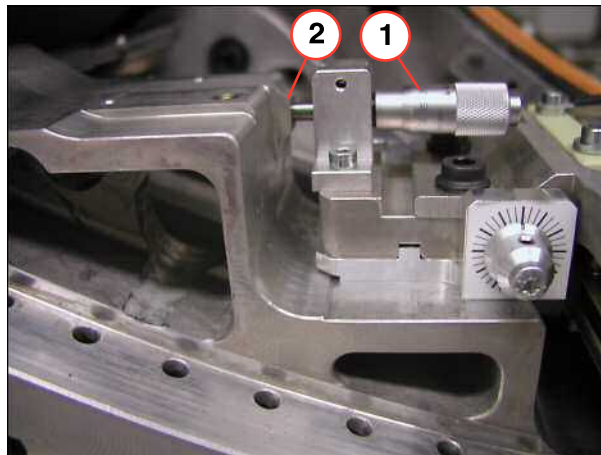


- Remove the 4 (M6x35mm) Torx screws (item 1) using a T30 Torx key.
- Remove the UHR assembly from the DMS.

7.8.4 Installation

1. Attach the new UHR assembly in reverse order.
 - insert the 4 holding screws but do not tighten the screws yet.
2. Install the UHR assembly

Fig. 331: Micrometer gauge



- The micrometer gauge (item 1) is preadjusted to 5.00mm; if not, turn it to adjust it.
- Carefully push the assembly against the DMS frame (item 2) and tighten the 4 screws with a torque of **14 Nm** using a T30 Torx key.



5 mm is a reference value for the following steps (tune-up). This value has to be adjusted prior to tightening the screws.



4. Reconnect all connectors according to the original configuration
5. Reinstall the cable clamp and tighten the 2 screws (M3x8) with a torque of **1.4 Nm**.
6. Install the cover of the PHI-/Z Control board and tighten the screws (M4x16) with a torque of **3 Nm**.

Be careful not to damage the cables when closing the cover.

7.8.5 Startup

7.8.5.1 Mechanical

1. Remove the data ring protections (delivered with the system)
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.



Rotation may start during during init for positioning!

7.8.5.2 Tune-up

1. Run tune up sequence =>**guided tour for comb replacement**.
2. Close the gantry covers if the adjustment is successful.



Disable gantry rotation using S301 when performing mechanical adjustments inside the gantry during tune-up!

7.8.5.3 Testing

1. Run some modes using UHR to check for proper function.
2. Run Quality Constancy.

7.8.6 Final steps

n.a.

7.9 UHR motor

7.9.1 Prerequisites

7.9.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

7.9.1.2 Tools and auxiliary equipment

- Standard tool kit
- Loctite 243
- Torque wrench 1.4 Nm, 3.0 Nm, 14 Nm

7.9.1.3 Time and manpower

Estimated time to repair: 1.0 hours

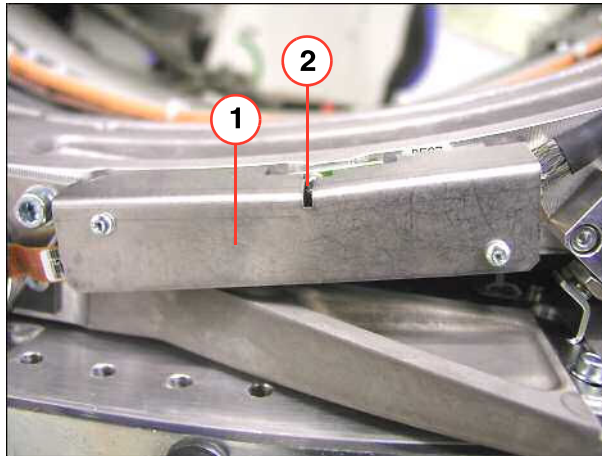
7.9.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.
2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Turn the gantry by hand until the part to be replaced is in the 6 o'clock position.
6. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)
7. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

7.9.3 Removing

1. Remove connector X1 on D587.

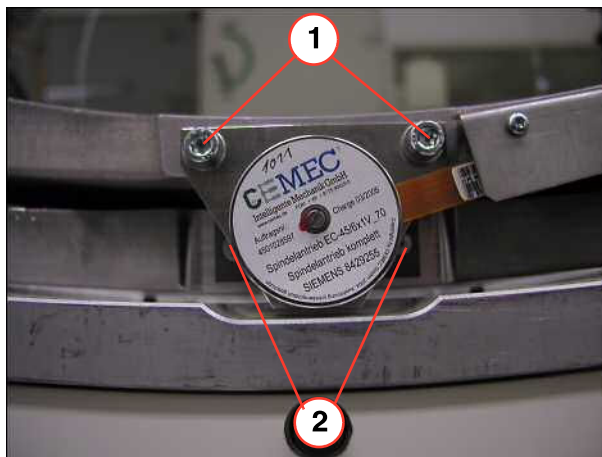
Fig. 332: Cover on D587



- Remove the cover (item 1) from adapter D587 (item 2) using a T10 Torx key.
- At adapter D587, pull out the cable interlocks.
- Remove the printed cable X1 by pulling it out.

2. Remove the UHR motor.

Fig. 333: UHR motor



- Remove the 2 holding screws (M6x20) (item 1) using a T30 Torx key.
- Remove the 2 holding screws (M4x12) (item 2) using a T20 Torx key.
- Remove the motor by pulling it out.

7.9.4 Installation

1. Insert the motor carefully, do not touch or damage the thread.



The spindle must be installed centrally. Check smooth running by rotating the motor by hand.



2. Connect plug X1 and lock it (at adapter D587).
3. Tighten the 2 upper screws (holding the motor) with a torque of **14 Nm**.
Tighten the 2 lower screws (holding the comb) with a torque of **3 Nm**.
4. Install the cover (D587), use Loctite 243 and tighten the screws with a torque of **1.4 Nm**.

7.9.5 Startup

7.9.5.1 Mechanical

1. Remove the data ring protections (delivered with the system)
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Install all covers.
5. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
6. Switch on the system (**SYSTEM/ON**) at the control box.

7.9.5.2 Tune-up

n.a.

7.9.5.3 Testing

1. Run some modes using UHR to check for proper function.
2. Run a MTF measurement, if out of tolerance run tune up **comb ali**.

7.9.6 Final steps

n.a.

7.10 Home Sensor

7.10.1 Prerequisites

7.10.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage of parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the “General” section of this document and/or “Product-specific safety notes”.

7.10.1.2 Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 3 Nm, 5 Nm
- Loctite 243

7.10.1.3 Time and manpower

Estimated total repair time: 1.0 hours

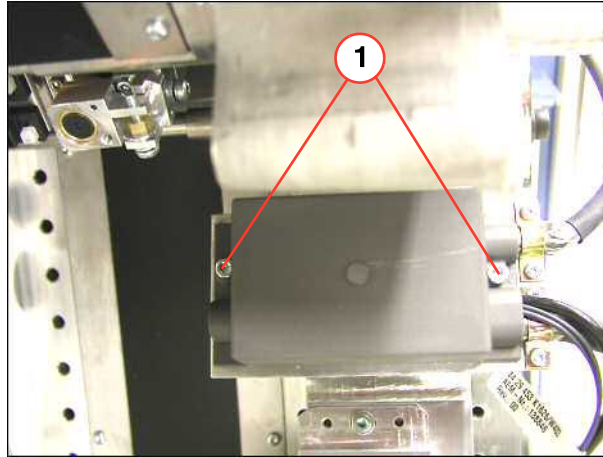
7.10.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.
2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Turn the gantry by hand until the part to be replaced is in the 6 o'clock position.
6. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)
7. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

7.10.3 Removing

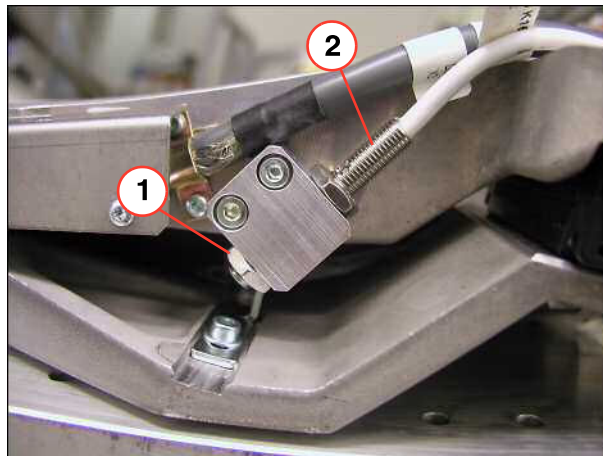
1. Remove cover from PHI/Z Control board

Fig. 334: PHI/Z Control Cover



- Remove the 2 screws (M4x16) (item1) from cover using a T20 Torx key.
 - Remove the cover (item 2).
2. Remove the **Home** plug on the PHI/Z Control board.
 3. Remove cable ties as necessary.
 4. Remove the Home Sensor.

Fig. 335: Home sensor



- Remove the union nut (item 1) and pull out the sensor (item 2).

7.10.4 Installation

1. Insert the new sensor.
2. Position the new sensor so that the LED in the shaft is visible.



3. Adjust the gap between the Sensor and the flag of the moving part to 0.25 mm +/- 0.05 mm.

Use **Loctite 243** at both nuts of the sensor and tighten with a torque of **5 Nm**.

4. Reconnect the **Home** plug on PHI/Z Control board.
5. Reattach the PHI/Z Control cover and tighten the 2 screws (M4x16) with a torque of **3 Nm**.

Be careful not to damage the cables when closing the cover.

6. Install all cable ties that were removed.

7.10.5 Startup

7.10.5.1 Mechanical

1. Remove the data ring protections (delivered with the system)
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Install all covers.
5. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
6. Switch on the system (**SYSTEM/ON**) at the control box.

7.10.5.2 Tune-up

n.a.

7.10.5.3 Testing

1. Run some modes using UHR to check for proper function.
2. Run a MTF measurement, if out of tolerance run tune up **comb ali**.

7.10.6 Final steps

n.a.

7.11 PHI/-Z Control

7.11.1 Prerequisites

7.11.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product-specific safety notes".

7.11.1.2 Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 1.4 Nm, 3 Nm

7.11.1.3 Time and manpower

Estimated total repair time: 1.0 hours

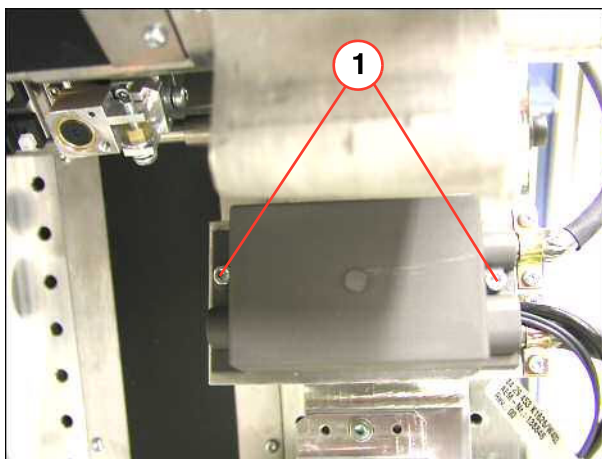
7.11.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.
2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Turn the gantry by hand until the part to be replaced is in the 6 o'clock position.
6. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)
7. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

7.11.3 Removing

1. Remove the cover from the PHI/-Z Control

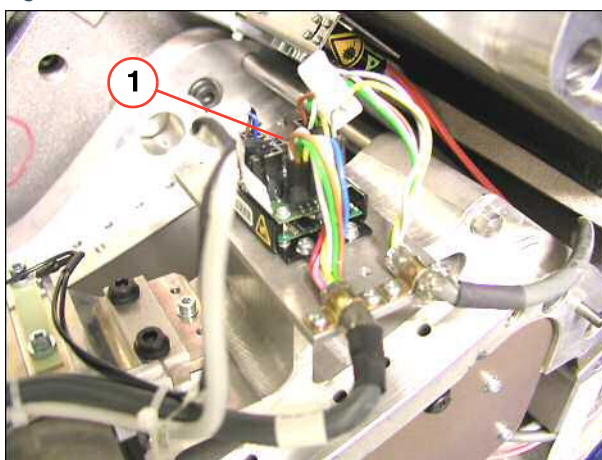
Fig. 336: PHI/Z Control Cover



- Remove the 2 screws (M4x16) (item 1) from cover using a T20 Torx key.
- Remove the cover.

2. Remove all connectors (item 1) at the PHI/-Z Control board (D586).

Fig. 337: PHI/Z Control Board



- Remove Connector **RS232** at the PHI/-Z-Control.
- Remove the **Power** connector at the PHI/-Z Control.
- Remove the **home** connector at the PHI/-Z Control.
- Remove the **motor** connector at the PHI/-Z Control.
- Remove the **end** connector at the PHI/-Z Control.

3. Remove the PHI/Z Control board (D586).
 - Remove the 2 cable clamps using a T10 Torx key.
 - Remove the 4 holding screws using a T10 Torx key.

- Remove the PHI/Z Control board (D586) from the DMS.

7.11.4 Installation

1. Insert the new PHI/Z Control board.
2. Tighten the 4 screws (M3x8) of the PHI/Z control board with a torque of **1.4 Nm**.
3. Reconnect all plugs according to the original configuration.
4. Install cable clamps and tighten the screws (M3x8) with a torque of **1.4 Nm**.
5. Install the cover of PHI/-Z Control and tighten the 2 screws with a torque of **3 Nm**.

7.11.5 Startup

7.11.5.1 Mechanical

1. Remove the data ring protections (delivered with the system)
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Install all covers.
5. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
6. Switch on the system (**SYSTEM/ON**) at the control box.

7.11.5.2 Tune-up

n.a.

7.11.5.3 Testing

1. Run some modes using UHR to check for proper function.
2. Run a MTF measurement, if out of tolerance run tune up **comb ali**.

7.11.6 Final steps

n.a.

8.1 Overview

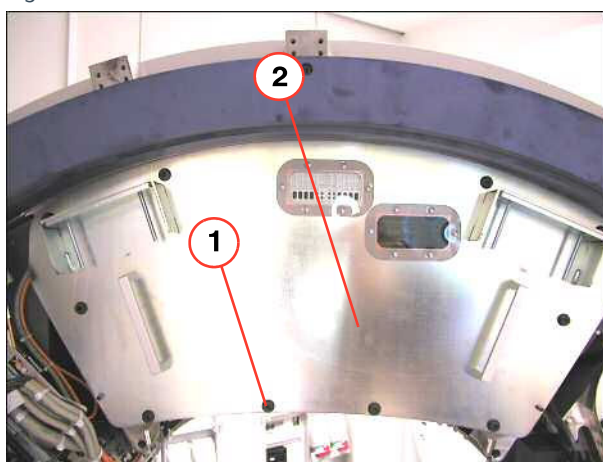
- The first figures show an overview of all DMS B parts involved in the Data Measurement and their location inside the gantry. (stationary light markers removed)
The next sections contain detailed descriptions of how to exchange all major parts of the DMS B.



Follow the ESD guidelines when handling the DMS

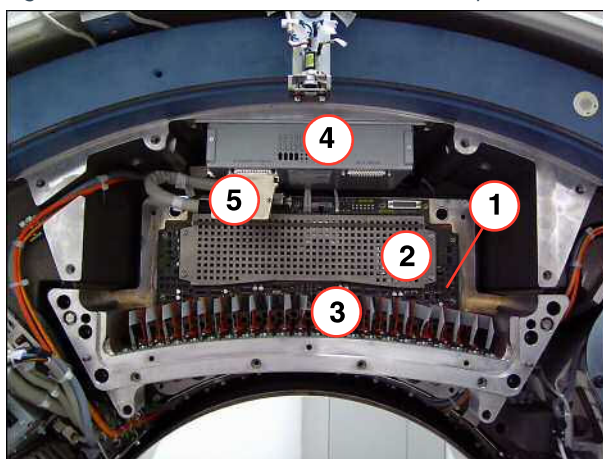
- This figure shows the DMS B, including the metal front cover and the stationary light markers removed.

Fig. 338: DMS metal front cover



- DMS B with metal front cover (item 2) and cover screws (item 1)
- This figure shows the DMS B without covers (metal and plastic cover removed)

Fig. 339: DMS B without covers (metal and plastic covers removed)



- DMS B replaceable parts
 - Backplane (item 1)
 - DAS controller (item 2)
 - Modules (item 3)

- Power supply (item 4)
- Power cable (item 5)

8.2 DMS Main Covers

8.2.1 Prerequisites

8.2.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the “General” section of this document and/or the “Product-specific safety notes”.



Follow the ESD guidelines when handling the DMS

8.2.1.2 Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 5 Nm, 14 Nm, 31 Nm
- Loctite 243

8.2.1.3 Time and manpower

Estimated total repair time: 0.5 hours

8.2.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.
2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Turn the gantry by hand until the DMS B is in the 6 o'clock or 10 o'clock position.

7. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)
8. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

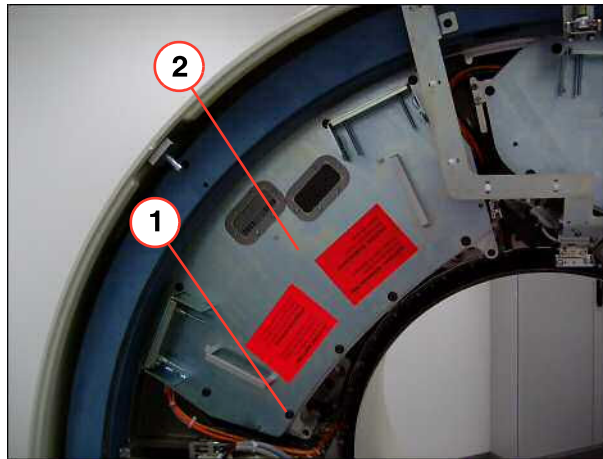
8.2.3 Removing



This chapter describes how to remove the 2 main covers of DMS B. For some steps, the DMS has to be in the 6 o'clock position before removing the covers.

1. Remove the metal DMS front cover

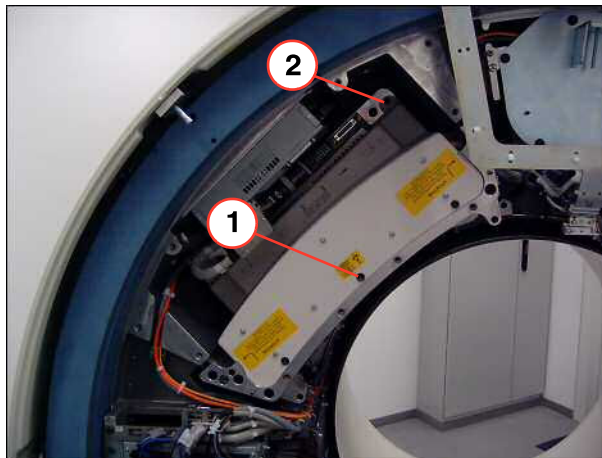
Fig. 340: Screws for metal DMS cover



- Remove the 10 (M8x20) Torx screws (e.g. item 1) using a T40 Torx key. **Hold the cover (item 2) when removing the last screws!**
- Remove the metal DMS front cover using the handles.

2. Remove the plastic DMS cover.

Fig. 341: Screws for plastic DMS cover



- Remove the 3 (M6x35) screws in front (e.g. item 1) using a T30 Torx key.
- Remove the 2 (M6x35) screws at the rear (e.g. item 2) using a T30 Torx key.
- Remove the plastic DMS cover by pulling outwards and lifting up to clear the power supply.

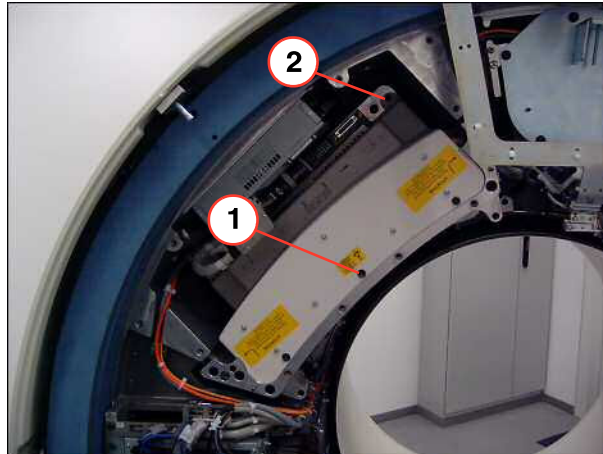
8.2.4 Installation

1. Install the plastic DMS cover.



If screws with lacquer-coated thread were installed, clean the thread prior to installing the screws (e.g. use a wire brush)

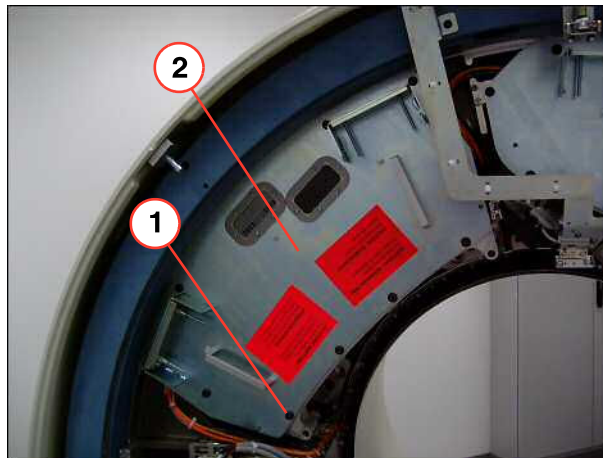
Fig. 342: Screws for plastic DMS cover



- Insert the 5 screws. Use Loctite 243
- Tighten the 3 (M6x35) screws in front (e.g. item 1) with a torque of **14 Nm** using a T30 Torx key.
- Tighten the 2 (M6x35) screws at the rear (e.g. item 2) with a torque of **5 Nm** using a T30 Torx key.

2. Install the metal DMS front cover

Fig. 343: Screws for metal DMS cover



- Install the 10 (M8x20) Torx screws (e.g. item 1) using a T40 Torx key. **Hold the cover until the first screws are installed!**
- Move the cover (item 2) upwards while tightening the screws to close the slot between the cover and gantry frame (not necessary in 6 o'clock position).
- Tighten the 10 screws with a torque of **31 Nm**.

8.2.5 Startup

8.2.5.1 Mechanical

1. Remove the data ring protections (delivered with the system)
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Install all covers.
5. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
6. Switch on the system (**SYSTEM/ON**) at the control box.

8.2.5.2 Tune-up

n.a.

8.2.5.3 Testing

n.a.

8.2.6 Final steps

n.a.

8.3 Complete DMS

8.3.1 Prerequisites

8.3.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

8.3.1.2 Tools and auxiliary equipment

- Standard tool kit
- Torque wrench 10 Nm, 14 Nm, 30 Nm, 31 Nm, 62 Nm
- Tools delivered with the spare part:
 - Strap, 2 m with 2 carabiners
 - Torx T50 3/8"
 - Extension 3/8"
 - Reduction 1/2" to 3/8"
 - Pad for tabletop
 - Set of new screws
 - DMS guide pins
 - Wooden block
 - Handle

8.3.1.3 Time and manpower

- Estimated total repair time: 5 hours
One person is required to replace the DMS
A second person is required only for opening and closing the cover of the DMS transport box.

8.3.2 Preliminary Steps

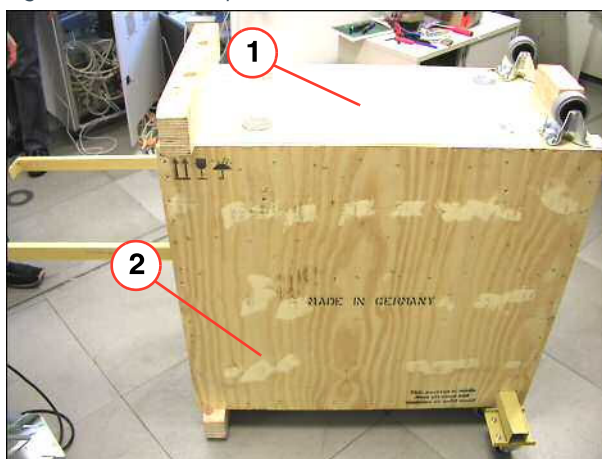
1. Allow at least five minutes cooling time after the last scan. During the cooling time the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may be up to 130 degree Celsius.

2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Press the blue Reset button on the MCB of the PDC_XGS_Control (piggyback board of the D700 main board) in the PDC-A to enable service power to the PHS.
5. Remove the bottom front ring segment. (→ Removing / Page 32)
6. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
7. Turn the gantry by hand until DMS B is in the 6 o'clock position.
8. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)
9. Install the data ring protection. (→ Data ring protection / Page 23)

8.3.2.1 Prepare the transport crate

1. Open the transport crate.

Fig. 344: DMS transport crate



- Open and remove the top cover (item 1) of the transport crate.
 - Remove the service tools.
 - Remove the side covers (item.2) of the transport crate by lifting it up.
2. Attach the transport brackets (delivered with the part) to the top cover of the transport crate using a 17mm hex socket.

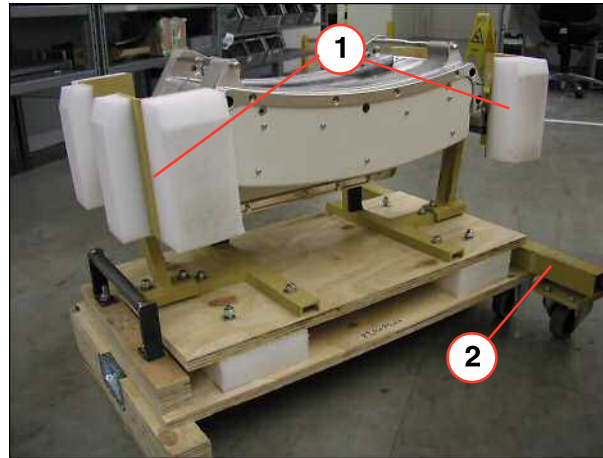


The top of the crate will serve as transport device for the old DMS. For orientation of the transport brackets, refer to the transport device on the new DMS.

4. Tighten the 2 nuts holding the transport brackets with a torque of **30 Nm**.

5. Remove the transport protection. (The figure below is an example. The DMS might look different.)

Fig. 345: Transport protection and stabilizer



- Remove the transport protection (item 1) from the transport device using an 8 mm Allen key.
- Remove the transport stabilizer (item 2) from the transport device using a 17 mm hex socket.
- Mount the wooden block (delivered with the part) to the transport device instead of the transport stabilizer, and tighten the 2 nuts using a 17 mm hex socket.

8.3.3 Removing

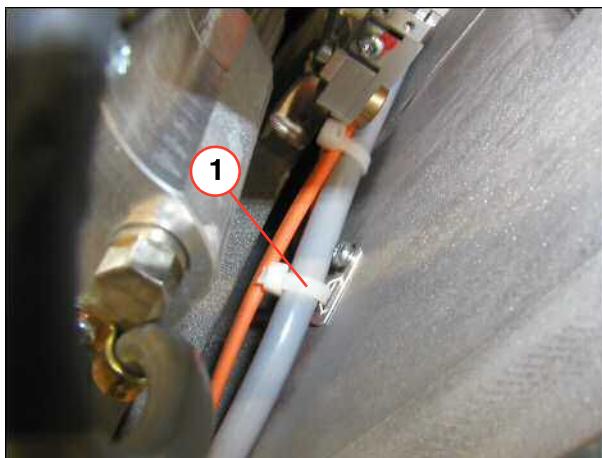


The metal front cover may be heavy because it contains balancing weights. The entire procedure has to be performed in the 6 o'clock position with the safety bolt installed to prevent unintended gantry rotation with the metal front cover removed.

1. Remove the metal DMS front cover. Refer to (→ Removing / Page 399) DMS main covers.
2. Remove DMS power supply. Refer to (→ Removing / Page 426) power supply.

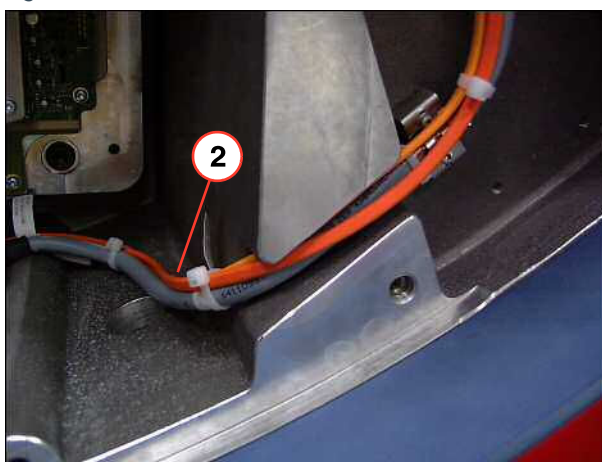
3. Remove the cable tie at the clamp on the left side of the DMS

Fig. 346: Clamp with cable tie



- Remove the cable tie (item 1).
4. Disconnect the fiber optic cables at the DAS controller.
 5. Remove the grounding strap of DMS B at the **gantry frame** using a T30 Torx key
 6. Remove the power cable (X31) from the DMS backplane.
 7. Pull out the cables from the DMS.

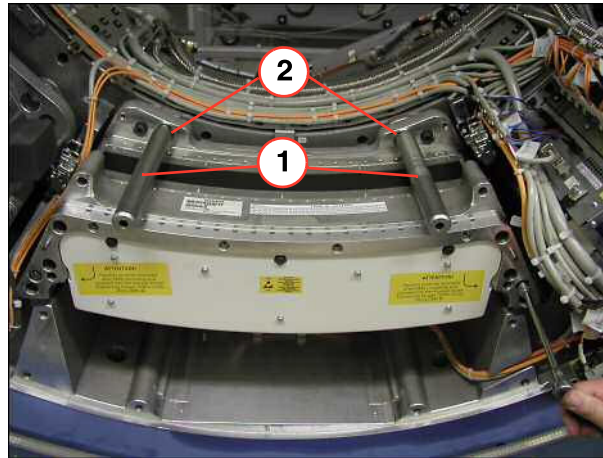
Fig. 347: Cables to be moved out



- Remove cable ties as necessary. Be careful not to damage the cables.
- To prevent the cables on the right side of the DMS (item2) from getting damaged, the cables can be pulled out.

8. Install the DMS guide pins.

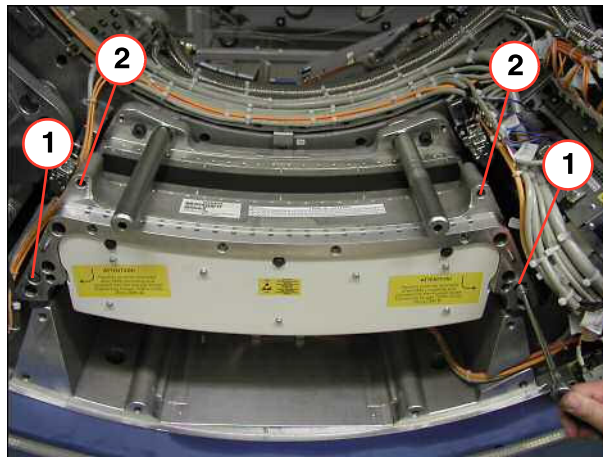
Fig. 348: Guide pins



- Install the 2 DMS guide pins (item 1) through the holes in the DMS (item 2).
- Tighten the guide pins by hand.

9. Install the handles.

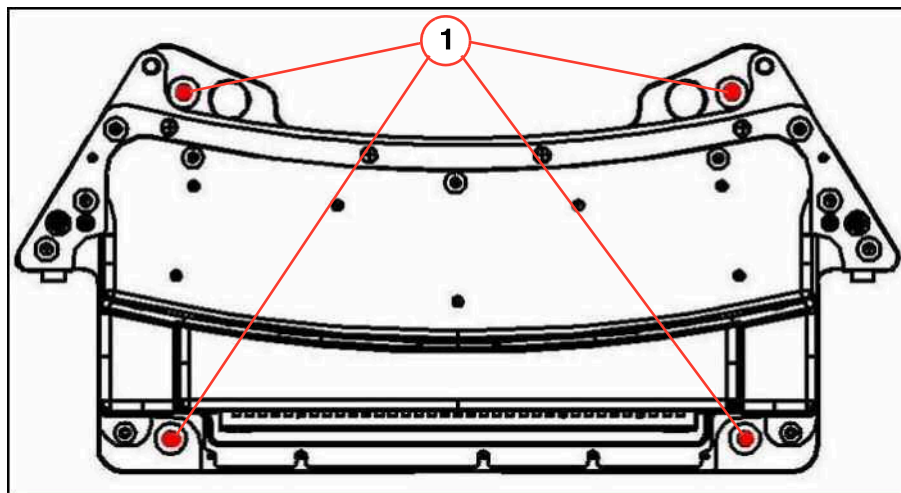
Fig. 349: DMS handles



- Remove the 2 handles from park position at the right and left front side of the DMS (item 1) using a 10 mm hex socket.
- Install the 2 handles in the holes at the top of DMS (item 2) and tighten the handles by hand.

10. Remove the DMS holding screws and position the transport device.

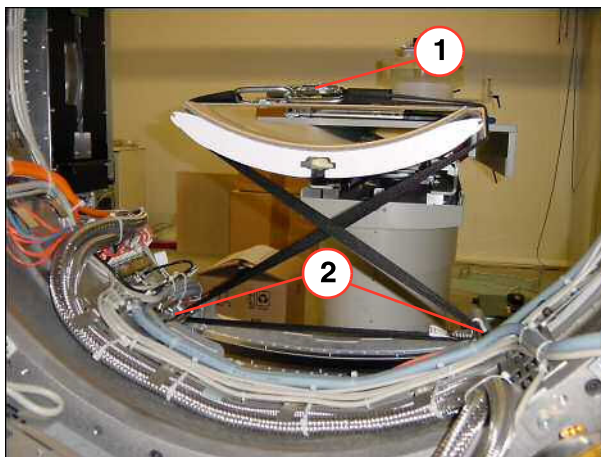
Fig. 350: DMS B holding screws



- Remove the 4 holding screws M10x40 (item 1) using a T50 Torx key.
- Position the transport device (top of the DMS box) between the gantry and the PHS. Refer to the new DMS on its transport device for positioning.

11. Remove the DMS.

Fig. 351: Removing the DMS



- Position the tabletop above the DMS using the table control on the left side of PHS.
- Place the pad on the tabletop.
- Attach the strap through the handles (item 2) and around the pad on the tabletop using **2 carabiners** (item 1).

Do not mix up the handles (item 2) with the guide pins.

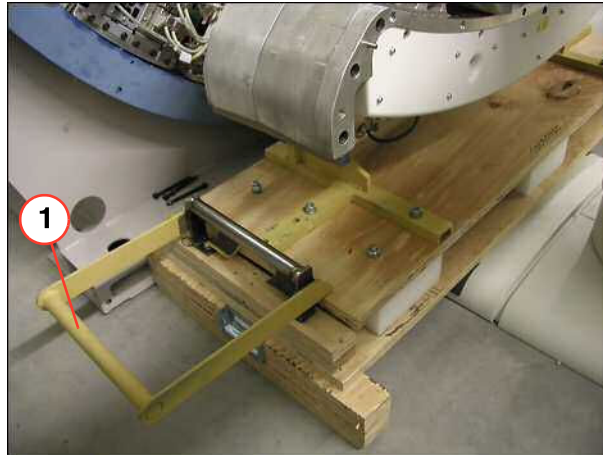
- Gently move the table up using table control, until the DMS is at the correct height to be moved out via the guide pins.
- Gently move the table out using the table control. **Move the tabletop support only!**

12. Position the DMS on the transport device.

- Position the transport device underneath the DMS between the gantry and the table. For orientation, refer to the new DMS on its transport device
- Position the DMS onto the transport device using the table control.
- Secure the DMS with 2 screws, tighten the screws with a torque of **30 Nm**, and remove the strap

13. Remove the transport device. (The figure below is an example. The DMS might look different.)

Fig. 352: Handle on transport device



- Remove the pad from the tabletop and move the table out for better access.
- Remove the transport device, including the old DMS, from between the gantry and table.
- Use the handle (item 1) to lift and move the transport device

8.3.4 Installation

1. Remove the grounding strap from old DMS and attach it to the new DMS. Tighten the M6 screw with a torque of **10 Nm**.



The grounding strap has to be mounted to the DMS before installing the DMS. Otherwise, you will not be able to access the screw with a torque wrench.

2. Position the transport device with the new DMS between the Gantry and the PHS.
3. Position the pad on the tabletop and move the table above the DMS.
4. Attach the strap through lifting bolts and around pad on the tabletop.
5. Gently move the table up using the table control to add tension to the strap.
6. Remove the 2 holding screws from the transport device and remove the transport device.
7. Gently move the table up using table control, until the DMS is at the correct height to be moved in via the guide pins.

8. Gently move the table inwards using table control until the DMS is completely in.
9. Lower the patient table.
 - Remove the strap.
 - Remove the pad from the tabletop.
 - Move the table out.
10. Insert the 4 holding screws M10x40 (delivered with new DMS) and tighten with a torque of **62 Nm**.
11. Remove the DMS guide pins.
12. Remove the handles and install them in the park position; tighten the handles with a torque of **10 Nm**.



It is important to reinstall the lifting handles in the park position. They are in the field of view and appear as artifacts if forgotten!

13. Reinstall the connectors according to the original configuration.
14. Install the grounding strap and tighten the screw with a torque of **10 Nm**.
15. Reinstall cable ties where removed.
16. Install the DMS Power supply. Refer to (→ Installation / Page 428) power supply.
17. Install the metal DMS front cover. Refer to (→ Installation / Page 400) DMS main covers.

8.3.5 Startup

8.3.5.1 Mechanical (rotating Part without closing Covers)

1. Remove the data ring protections
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.



Rotation may start during during init for positioning!

8.3.5.2 Software

1. Update the system configuration with the new DMS information:
Local Service -> Configuration -> Local Hosts -> Serial.

8.3.5.3 Tune-up

1. Run the tune-up sequence **Local Service -> Tune-Up -> FRU Replace**.
2. Close the gantry covers when needed (table generations)



Disable the gantry rotation using S301 when performing mechanical adjustments inside the gantry during tune-up!

8.3.5.4 Testing

1. Run Quality Constancy.
2. Perform a backup of the new settings.

8.3.6 Final steps

Prepare the DMS transport crate

1. Mount the transport stabilizer to the transport device and tighten the screws with a torque of **30 Nm**.
2. Mount the transport protection to the DMS and tighten the screws with a torque of **30 Nm**.
3. Complete the DMS return form.
4. Reassemble the DMS transport crate, add the service tools and DMS return form.
5. Remove the transport brackets from the top cover of the DMS box.
6. Mount the transport brackets into the box and close the top cover.

8.4 Modules

8.4.1 Prerequisites

8.4.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

8.4.1.2 Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 1.0 Nm, 1.4 Nm, 5 Nm, 14 Nm, 31 Nm

8.4.1.3 Time and manpower

- Estimated total repair time: 1.5 - 3 hours depending on the position of the module

8.4.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may be up to 130 degree Celsius.
2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Turn the gantry by hand until DMS B is in the 10 o'clock position.
7. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)
8. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

8.4.3 Removing



Follow the ESD guidelines when handling the DMS

1. Remove the metal and the plastic DMS front covers. Refer to the (→ Removing / Page 399) DMS main covers.
2. Locate the defective module by its label at the corresponding plug on the DMS back-planes.



If the DMS is in 10 o'clock position, channel 1 is on the left side.

Channels	Slot
1 - 16	1
17 - 32	2
33 - 48	3
49 - 64	4
65 - 80	5
81 - 96	6
97 - 112	7
113 - 128	8
129 - 144	9
145 - 160	10
161 - 176	11
177 - 192	12
193 - 208	13
209 - 224	14
225 - 240	15
241 - 256	16
257 - 272	17
273 - 288	18
289 - 304	19
305 - 320	20

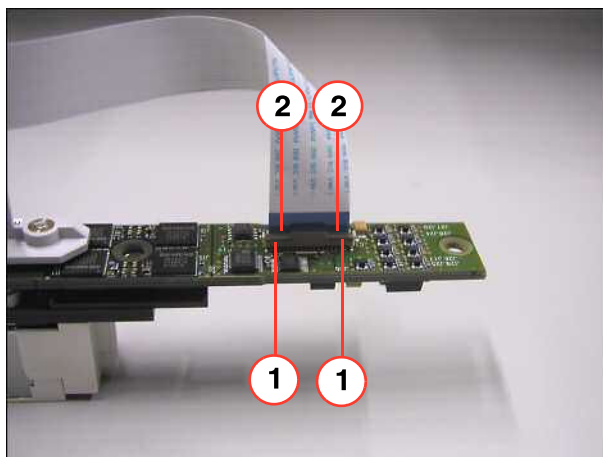
Channels	Slot
321 - 336	21
337 - 352	22



3. Remove module cable from DMS backplane

Hint: **The example shows how to open the connector on a module. The connectors on the backplane are the same. To unlock the connector, pull the locking device. To close the connector, press down on the locking device.**

Fig. 353: Module cable connector

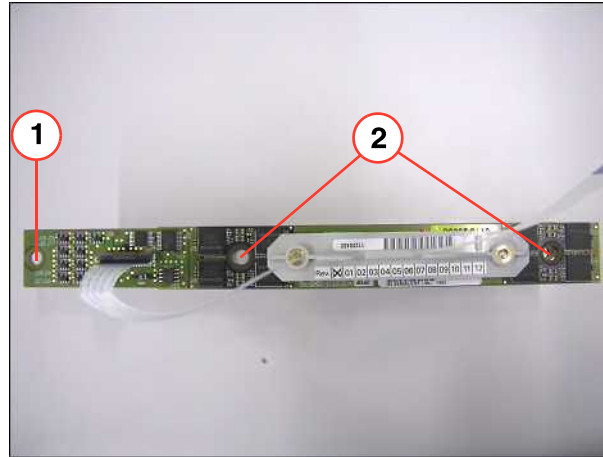


- To unlock the connector, pull on both sides of the brown locking device (item 1).
- To close the connector, press on both sides of the brown locking device (item 2).

4. Remove the module.

For slots 4 to 19, remove the DAS Controller for easier access. Refer to (→ Removing / Page 419) DAS Controller.

Fig. 354: Removed Detector Module



- Remove the front screw (M3x8) (item 1) using a T10 Torx key.
- Remove the 2 screws (M4x20) (item 2) next to the plastic handle using a T20 Torx key.
- Remove the old module by pulling on the plastic handle.



Do not stress the module and do not touch any part of the PCB.

Replacement of a Center Module

In case of a defective center module (slot 9 - 14), the entire six pack will not be replaced. However, to ensure the best image quality, place the new module outside of the center six slots (slot 8 or 15). Then move the existing good modules towards the center to fill the bad slot. This method keeps the center of the detector homogeneous.

The following is an example for a defective module in the center **slot 11**:

1. Remove defective module from slot 11.
2. move the module known to be good from slot 10 to slot 11.
3. move the module known to be good from slot 9 to slot 10.
4. move the module known to be good from slot 8 to slot 9.
5. insert the spare module (the one you ordered) in slot 8.

Slots 9 to 14 have to be exchanged according to this scheme.



This shifting method is used for the center modules in DMS A and DMS B. For a graphical overview, refer to replacement of a center module in DMS A.

8.4.4 Installation



1. Insert the new module and align it toward the rotating plane.
2. While tightening the 3 screws, press the module toward the rotating plane.
Tighten the screws next to the handle with a torque of **1.4 Nm**, and the one in front with **1.0 Nm**.
3. Connect the module cable at the backplane
4. Install the DAS Controller if necessary. Refer to (→ Installation / Page 420) DAS Controller.
5. Install the metal and plastic DMS front covers. Refer to (→ Installation / Page 400) DMS main covers.

8.4.5 Startup

8.4.5.1 Mechanical

1. Remove the data ring protections (delivered with the system)
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Switch on the system (**SYSTEM/ON**) at the control box.



Rotation may start during during init for positioning!

8.4.5.2 Tune-up

1. Run the tune-up sequence as described under **Local Service -> Tune Up -> FRU replacement**.
2. Close gantry covers when instructed during the tune-up procedure.



Disable gantry rotation using S301 when performing mechanical adjustments inside the gantry during tune-up!

8.4.5.3 Testing

1. Run Quality Constancy.
2. Perform a backup of the new settings.

8.4.6 Final steps

n.a.

8.5 DAS Controller

8.5.1 Prerequisites

8.5.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the "General" section of this document and/or the "Product specific safety notes".

8.5.1.2 Tools and auxiliary equipment

- Standard tool kit
- Torque wrench 1.0 Nm, 3 Nm, 5 Nm, 14 Nm, 31 Nm

8.5.1.3 Time and manpower

- Estimated total repair time: 1 hours

8.5.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may be up to 130 degree Celsius.
2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Turn the gantry by hand until DMS B is in the 10 o'clock position.
7. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)
8. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

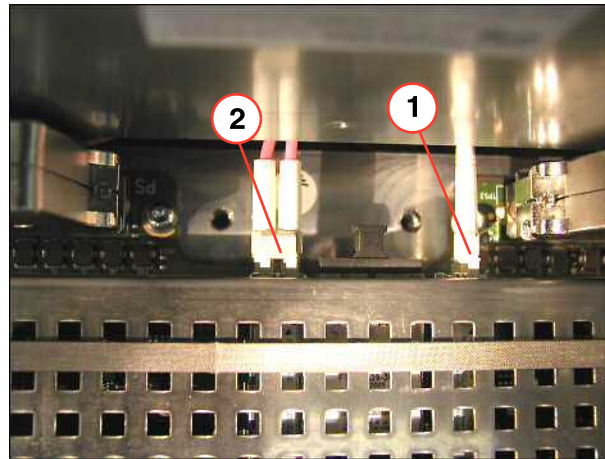
8.5.3 Removing



Follow the ESD guidelines when handling the DMS

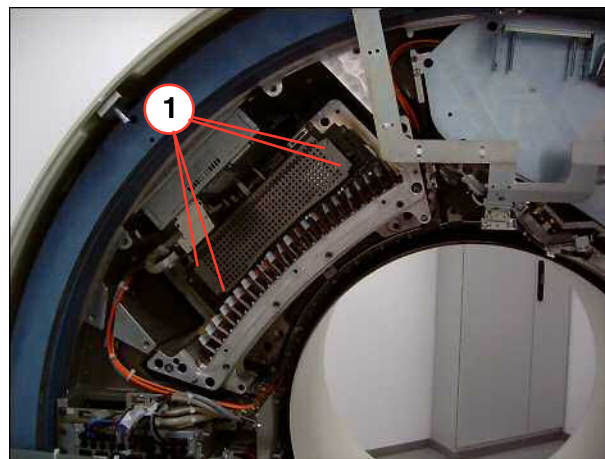
1. Remove the metal and the plastic DMS front covers. Refer to the (→ Removing / Page 399) DMS main covers.
2. Remove the fiber optic cables.

Fig. 355: Fiber Optic Cables



- Disconnect plug FODL1 (item 1) at the DAS Controller.
 - Disconnect plug FODL2 (item 2) at the DAS Controller.
3. Remove the DAS Controller cover.

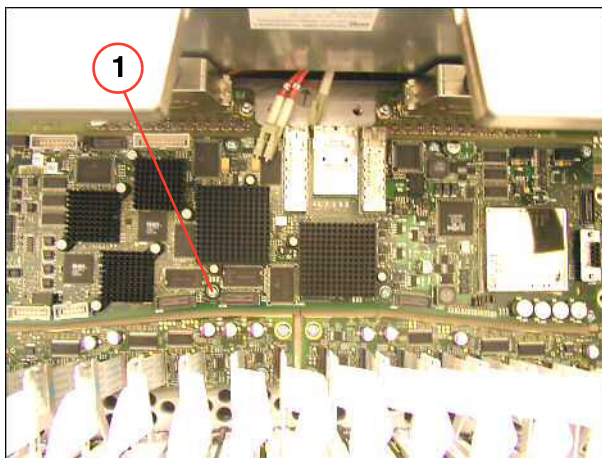
Fig. 356: DAS controller cover



- Remove the 4 screws (item 1) from the DAS Controller cover using a T20 Torx key.
- Remove the cover.

4. Remove the DAS Controller.

Fig. 357: DAS Controller



- Remove the 8 screws holding the DAS controller at the backplane (e.g. item 1).
- Remove the DAS Controller by pulling both sides at once.

8.5.4 Installation



1. Plug the DAS Controller to the backplane.
2. Insert the 8 screws (M3x8) of the DAS Controller and tighten with a torque of **1.0 Nm**.
3. Install the DAS Controller cover and tighten the 4 screws with a torque of **3 Nm**.
4. Reconnect fiber optic cables FODL1 and FODL2.
5. Install the metal and plastic DMS front covers. Refer to (→ Installation / Page 400) DMS main covers.

8.5.5 Startup

8.5.5.1 Mechanical

1. Remove the data ring protections (delivered with the system)
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Install all covers.
5. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
6. Switch on the system (**SYSTEM/ON**) at the control box.

8.5.5.2 Software

1. The Firmware of the DAS controller is checked and loaded automatically.



FW loading is finished when the system is in standby. Wait for standby, after a successful FW loading message appears at the screen.

8.5.5.3 Tune-up

n.a.

8.5.5.4 Testing

1. Run the disk test.
2. Run spiral and sequence scans.
3. Run constancy.

8.5.6 Final steps

n.a.

8.6 DMS backplane

8.6.1 Prerequisites

8.6.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the “General” section of this document and/or the “Product specific safety notes”.

8.6.1.2 Tools and auxiliary equipment

- Standard tool kit
- Torque wrench 1.0 Nm, 1.4 Nm, 3 Nm, 5 Nm, 14 Nm, 31 Nm

8.6.1.3 Time and manpower

- Estimated time to repair: 1.5 hours

8.6.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.
2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Remove or open (depending on cover type) the gantry top front ring segment. Refer to (→ Opening/closing the top front ring segment cover / Page 25).
6. Turn the gantry by hand until DMS B is in the 10 o'clock position.
7. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)
8. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

8.6.3 Removing



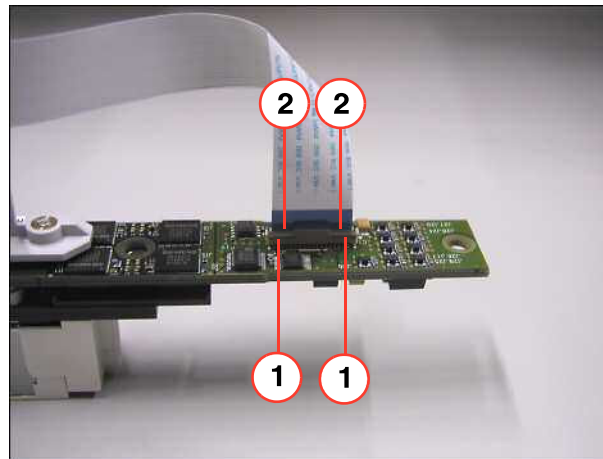
Follow the ESD guidelines when handling the DMS



1. Remove the the metal and the plastic DMS front covers. Refer to the (→ Removing / Page 399) DMS main covers.
2. Remove the DAS Controller. Refer to (→ Removing / Page 419) DAS Controller.
3. Remove all module cables from the defective backplane.

Hint: The example shows how to open the connector on a module. The connectors on the backplane are the same. To unlock the connector, pull the locking device. To close the connector, press down on the locking device.

Fig. 358: Module cable connector



- To unlock the connector, pull on both sides of the brown locking device (item 1).
 - To close the connector, press on both sides of the brown locking device (item 2).
4. Remove the power cable (X31) from the backplane.
 5. Remove the 14 holding screws (M4x10) from the defective backplane using a T20 Torx key.
 6. Remove the defective backplane.
 7. Only for backplane type 8429750 (introduced app. 07/2007):
The id- dongle (10023379) is a plug that has to be moved from the old to the new backplane.
(The previous backplane type 10023390 doesn't have a dongle.)

8.6.4 Installation

1. Insert the new backplane and tighten the 14 screws (M4x10) with a torque of **1.4 Nm**.

2. Only for backplane type 8429750:
Install the id dongle (10023379).
3. Reconnect all module cables to the backplane.
4. Reconnect power cable X31 to the backplane.
5. Install the DAS Controller. Refer to (→ Installation / Page 420) DAS Controller.
6. Install the metal and plastic DMS front covers. Refer to (→ Installation / Page 400) DMS main covers.

8.6.5 Startup

8.6.5.1 Mechanical

1. Remove the data ring protections (delivered with the system)
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
5. Install all covers.
6. Switch on the system (**SYSTEM/ON**) at the control box.

8.6.5.2 Tune-up

n.a.

8.6.5.3 Testing

1. Run the disk test.
2. Run spiral and sequence scans.
3. Run quality constancy.

8.6.6 Final steps

n.a.

8.7 DMS power supply

8.7.1 Prerequisites

8.7.1.1 Safety



WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage to parts.

Risk of accident and injury!

- » Read and observe the safety information contained in the “General” section of this document and/or the “Product specific safety notes”.

8.7.1.2 Tools and auxiliary equipment

- Standard tool kit
- Torque wrenches 10 Nm, 14 Nm, 31 Nm

8.7.1.3 Time and manpower

- Estimated total repair time: 1.0 hours

8.7.2 Preliminary Steps

1. Allow for at least five minutes cooling time after the last scan. During the cooling time, the system must be switched on.
 - » Be aware that some system components may be very hot, e.g., the Straton tube may reach up to 130 degrees Celsius.
2. Switch system to status **COMP/ON** at the control box.
3. Switch off the gantry power using service push-button S1 in the PDC A (→ Service Push-Button S1 in PDC-A / Page 20).
4. Remove the bottom front ring segment. (→ Removing / Page 32)
5. Turn the gantry by hand until the part to be replaced is in the 6 o'clock position.
6. Secure against unintended rotation with the front safety bolt. (→ Safety bolts / Page 21)
7. Install the data ring protections (delivered with the system) (→ Data ring protection / Page 23)

8.7.3 Removing



Follow the ESD guidelines when handling the DMS

1. Remove the metal DMS front cover. Refer to (→ Removing / Page 399) DMS main covers.
2. Remove the grounding strap.

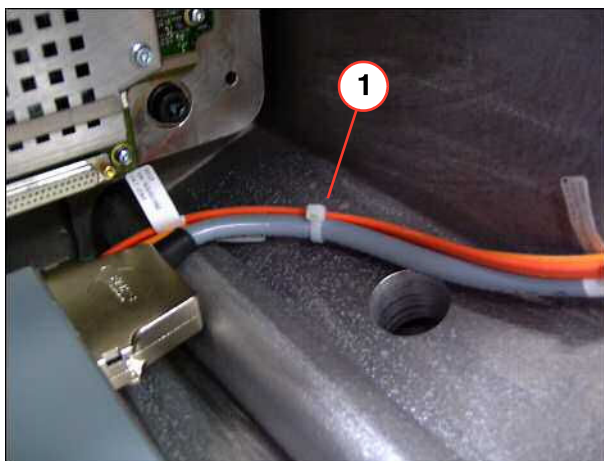
Fig. 359: Grounding strap on power supply



- Remove the grounding strap at the left side of the power supply using a 10 mm nut (item 1).

3. Remove the cable ties

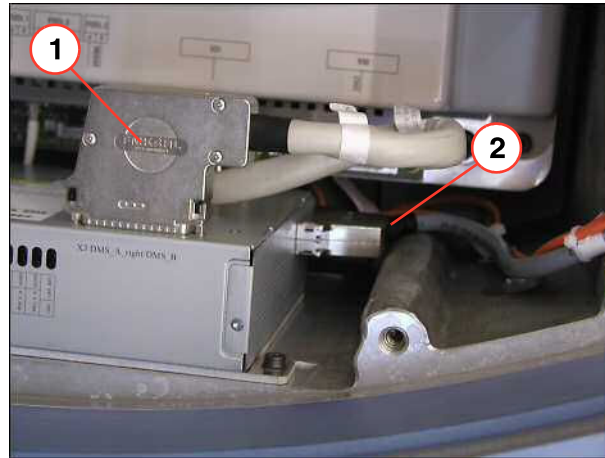
Fig. 360: Power connector X1 with cable tie



- Remove the cable tie (item 1)
- Remove the cable tie on top of the power supply.

4. Remove the cables at the power supply.

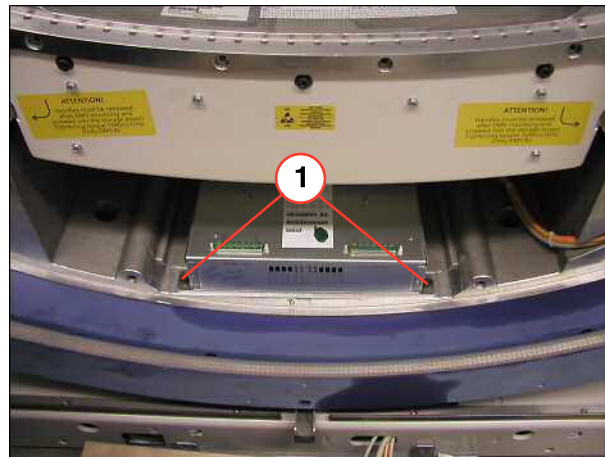
Fig. 361: DMS power supply connections



- Remove connector X3 on top of the DMS power supply. (item 1)
- To remove X1 at the right side of the power supply (item 2), the power supply has to be pulled out first. Refer to next step.

5. Remove the power supply

Fig. 362: Screws for power supply

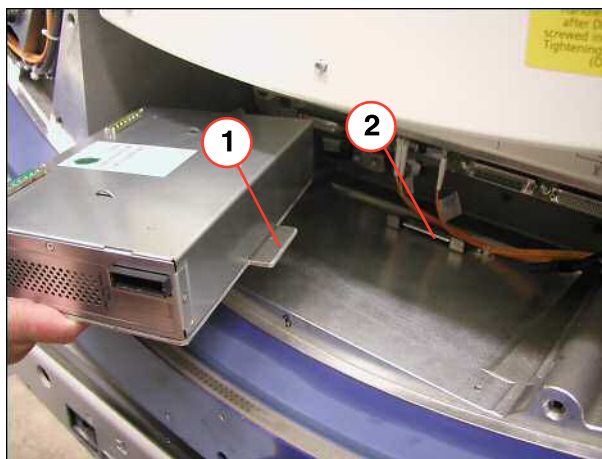


- Remove the 2 (M6x16) Torx screws using a T30 Torx key (item 1).
- Remove the power supply by pulling it straight out and remove connector X1.

8.7.4 Installation

1. Install the power supply.

Fig. 363: Installing the power supply



- Connect X1 to the right side of the power supply.
 - Push the guide plate (item 1) of the power supply into the clamp (item 2).
 - Install the power supply with 2 (M6x12) Torx screws and tighten with a torque of **14 Nm**.
2. Install the power cable according to the original configuration.
 3. Install cable ties, where removed.
 4. Connect the grounding strap using a 10 mm nut and tighten with a torque of **10 Nm**.
 5. Install the metal DMS cover. Refer to (→ Installation / Page 400) DMS main covers.

8.7.5 Startup

8.7.5.1 Mechanical

1. Remove the data ring protections (delivered with the system)
2. Unlock the safety bolt for gantry rotation.
3. Rotate the gantry by hand and check for unusual noise.
4. Install all covers.
5. Switch on breakers F2, F3, F5, F6, F7, and F11 in the PDC A and F2 in the PDC B.
 - » First switch the F2, F3, F5, F6 breakers to the "0" position and then to the "1" position.
6. Switch on the system (**SYSTEM/ON**) at the control box.

8.7.5.2 Tune-up

n.a.

8.7.5.3 Testing

1. Run sequence and spiral scans to check for proper function.

8.7.6 Final steps

n.a.

Chapter	Change	Reason
5	Instructions for braking resistor modified	Change ID: 612469

There are no Hazard IDs in this document.

H	hz_serdoc_F13G01U01M08	18
	hz_serdoc_F13G01U02M01	28
	hz_serdoc_F13G01U05M03	17, 202, 208, 213, 216
	hz_serdoc_F13G01U12M03 ...	31, 34, 37, 42, 45, 47, 50, 53, 57, 61, 64, 67, 70, 73, 76, 79, 81, 84, 87, 92, 97, 102, 104, 111, 116, 120, 124, 128, 132, 145, 158, 167, 170, 173, 176, 179, 182, 189, 194, 197, 201, 207, 213, 216, 221, 225, 228, 230, 233, 235, 237, 241, 254, 257, 260, 264, 267, 271, 274, 280, 305, 313, 320, 331, 344, 349, 354, 363, 369, 374, 378, 383, 387, 390, 393, 398, 403, 412, 418, 422, 425
	hz_serdoc_F13G01U13M02	17
	hz_serdoc_F13G02U01M01	283, 306, 332
	hz_serdoc_F13G02U01M04	18
	hz_serdoc_F13G05U01M02	18
	hz_serdoc_F13G05U01M03	17
	hz_serdoc_F13G05U01M04	18
	hz_serdoc_F13G07U01M01	16

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